

US Army Corps
of Engineers ®
New Orleans District

IFB NO. W912P825BA016

Mississippi River Outlets, Vicinity of Venice, Louisiana

Tiger Pass Bar Channel, Maintenance Dredging

Plaquemines Parish, Louisiana

Construction Solicitation
and Specifications

July 2025



DEPARTMENT OF THE ARMY
U.S. ARMY CORPS OF ENGINEERS, NEW ORLEANS DISTRICT
7400 LEAKE AVENUE
NEW ORLEANS, LA 70118-3651

SOLICITATION: W912P825BA016

July 2025

FOR: Mississippi River Outlets, Vicinity of Venice, Louisiana, Tiger Pass Bar Channel, Maintenance Dredging, Plaquemines Parish, Louisiana

TO OPEN:

- I. NOTE THE AFFIRMATIVE ACTION PROGRAM REQUIREMENT OF THE EQUAL OPPORTUNITY CLAUSE WHICH MAY APPLY TO THE CONTRACT RESULTING FROM THIS SOLICITATION.
- II. NOTE THE CERTIFICATION OF NONSEGREGATED FACILITIES IN THIS SOLICITATION. *Bidders, offerors and applicants are cautioned to note the "Certification of Non-segregated Facilities" in the solicitation. Failure of a bidder or offeror to agree to the certification will render his bid or offer non-responsive to the terms of solicitations involving awards of contracts exceeding \$10,000 which are not exempt from the provisions of the Equal Opportunity clause.*
- III. Prospective contractors must register in the System for Award Management (SAM). See FAR Clause 52.204-7 for required information. The website for SAM is <https://www.sam.gov>. You will be required to provide your company's Dun and Bradstreet (D&B) D-U-N-S number. If you do not already have a D&B D-U-N-S number, one can be requested at <http://fedgov.dnb.com/webform>.

BIDDERS MUST PROVIDE FULL, ACCURATE AND COMPLETE INFORMATION AS REQUIRED BY THIS SOLICITATION AND ITS ATTACHMENTS. THE PENALTY FOR MAKING FALSE STATEMENTS IN BIDS IS PRESCRIBED IN 18 U.S.C. 1001. (FAR 52.214-4 APR 1984)

DESCRIPTION AND MAGNITUDE OF WORK: The work consists of the removal and satisfactory disposal of dredged material excavated from the Tiger Pass Bar Channel, within the limits shown on the contract drawings.

CAUTION TO BIDDERS: In delivery of hand-carried bids, bidders are cautioned to allow sufficient time for delays which may be encountered as a result of frequent trains which are subject to block all access roads to place of bid opening for various lengths of time and for visitor background checks by MVN security when entering the installation at place of bid opening. *Such delays DO NOT permit acceptance or consideration of late bids.*

NOTE: ALL WORK UNDER THESE SPECIFICATIONS SHALL BE PERFORMED IN ACCORDANCE WITH THE PROVISIONS OF EM 385-1-1 "CORPS OF ENGINEERS SAFETY AND HEALTH REQUIREMENTS MANUAL", DATED MARCH 2024.

ATTENTION: Contractors must complete, and submit with their bid, the provisions located in Section 00600. A bidder may also submit, in lieu of the completed Section 00600, a complete hard copy of their Online Representations and Certifications Applications (ORCA) registration with an expiration date on or after the date of award.

ALL INQUIRIES REGARDING THIS SOLICITATION
SHOULD BE MADE TO THE FOLLOWING:

ANGELA C. DAY
CONTRACT SPECIALIST

U.S. ARMY CORPS OF ENGINEERS, NEW ORLEANS DISTRICT

E-MAIL: ANGELA.C.DAY@USACE.ARMY.MIL

PHONE: (504) 862-1047

COLLECT CALLS NOT ACCEPTED

NOTE: FOR THE MOST PROMPT RESPONSE, PLEASE SEND ALL QUESTIONS VIA E-MAIL.

PROJECT TABLE OF CONTENTS

Division 00 – PROCUREMENT AND CONTRACT REQUIREMENTS

00010	Solicitation, Offer and Award (SF-1442)
00100	Bidding Schedule
00600	Instructions to Bidders
00700	Representations and Certifications
	Contract Clauses

Division 01 - GENERAL REQUIREMENTS

01100	General Provisions
01 33 00	Submittal Procedures
01 35 26	Governmental Safety Requirements
01 42 00	Sources for Reference Publications
01 45 00.15 10	Resident Management System Contractor Mode (RMS CM)
01 45 04.00 10	Contractor Quality Control
01 57 20.01 12	Environmental Protection
01 78 02.00 10	Closeout Submittals

Division 02 – EXISTING CONDITIONS

02 21 10.00 12	Surveys
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Division 35 – WATERWAY AND MARINE CONSTRUCTION

35 20 23.00 12	Dredging
35 20 23.33	National Dredging Quality Management Program

SOLICITATION, OFFER, AND AWARD <i>(Construction, Alteration, or Repair)</i>		1. SOLICITATION NUMBER	2. TYPE OF SOLICITATION ☐ SEALED BID (IFB) INVITATION FOR BID ☐ NEGOTIATED (RFP) REQUEST FOR PROPOSAL	3. DATE ISSUED	PAGE OF PAGES
IMPORTANT - The "offer" section on the reverse must be fully completed by offeror.					
4. CONTRACT NUMBER		5. REQUISITION/PURCHASE REQUEST NUMBER		6. PROJECT NUMBER	
7. ISSUED BY		CODE	8. ADDRESS OFFER TO		
9. FOR INFORMATION CALL: 		a. NAME		b. TELEPHONE NUMBER <i>(Include area code) (NO COLLECT CALLS)</i>	

SOLICITATION

NOTE: In sealed bid solicitations "offer" and "offeror" mean "bid and "bidder".

10. THE GOVERNMENT REQUIRES PERFORMANCE OF THE WORK DESCRIBED IN THESE DOCUMENTS *(Title, identifying number, date)*

11. The contractor shall begin performance within _____ calendar days and complete it within _____ calendar days after receiving
☐ award, ☐ notice to proceed. This performance period is ☐ mandatory ☐ negotiable. **(See _____).**

12a. THE CONTRACTOR MUST FURNISH ANY REQUIRED PERFORMANCE AND PAYMENT BONDS?
(If "YES", indicate within how many calendar days after award in Item 12b.)
☐ YES ☐ NO

12b. CALENDAR DAYS

13. ADDITIONAL SOLICITATION REQUIREMENTS:

a. Sealed offers in original and _____ copies to perform the work required are due at the place specified in Item 8 by _____ **(hour)**
local time _____ (date). If this is a sealed bid solicitation, offers will be publicly opened at that time. Sealed envelopes
containing offers shall be marked to show the offeror's name and address, the solicitation number, and the date and time offers are due.

b. An offer guarantee ☐ is, ☐ is not required.

c. All offers are subject to the (1) work requirements, and (2) other provisions and clauses incorporated in the solicitation in full text or by reference.

d. Offers providing less than _____ calendar days for Government acceptance after the date offers are due will not be considered and will be rejected.

OFFER (Must be fully completed by offeror)

14. NAME AND ADDRESS OF OFFEROR (Include ZIP Code)		15. TELEPHONE NUMBER (Include area code)	
		16. REMITTANCE ADDRESS (Include only if different than Item 14.)	
CODE	FACILITY CODE		

17. The offeror agrees to perform the work required at the prices specified below in strict accordance with the terms of this solicitation, if this offer is accepted by the Government in writing within _____ calendar days after the date offers are due. (Insert any number equal to or greater than the minimum requirement stated in Item 13d. Failure to insert any number means the offeror accepts the minimum in Item 13d.)

AMOUNTS



18. The offeror agrees to furnish any required performance and payment bonds.

19. ACKNOWLEDGMENT OF AMENDMENTS

(The offeror acknowledges receipt of amendments to the solicitation -- give number and date of each)

AMENDMENT NUMBER										
DATE										

20a. NAME AND TITLE OF PERSON AUTHORIZED TO SIGN OFFER (Type or print)

20b. SIGNATURE

20c. OFFER DATE

AWARD (To be completed by Government)

21. ITEMS ACCEPTED:

22. AMOUNT		23. ACCOUNTING AND APPROPRIATION DATA	
24. SUBMIT INVOICES TO ADDRESS SHOWN IN (4 copies unless otherwise specified)	ITEM	25. OTHER THAN FULL AND OPEN COMPETITION PURSUANT TO THE UNITED STATES CODE AT ☐ 10 U.S.C. 3204(a) () ☐ 41 U.S.C. 3304(a) ()	
26. ADMINISTERED BY		27. PAYMENT WILL BE MADE BY	

CONTRACTING OFFICER WILL COMPLETE ITEM 28 OR 29 AS APPLICABLE

☐ 28. NEGOTIATED AGREEMENT (Contractor is required to sign this document and return _____ copies to issuing office.) Contractor agrees to furnish and deliver all items or perform all work requirements identified on this form and any continuation sheets for the consideration stated in this contract. The rights and obligations of the parties to this contract shall be governed by (a) this contract award, (b) the solicitation, and (c) the clauses, representations, certifications, and specifications incorporated by reference in or attached to this contract.		☐ 29. AWARD (Contractor is not required to sign this document.) Your offer on this solicitation is hereby accepted as to the items listed. This award consummates the contract, which consists of (a) the Government solicitation and your offer, and (b) this contract award. No further contractual document is necessary.	
30a. NAME AND TITLE OF CONTRACTOR OR PERSON AUTHORIZED TO SIGN (Type or print)		31a. NAME OF CONTRACTING OFFICER (Type or print)	
30b. SIGNATURE	30c. DATE	31b. UNITED STATES OF AMERICA BY	31c. DATE

SECTION 00010 – BIDDING SCHEDULE

Mississippi River Outlets
Vicinity of Venice, Louisiana
Tiger Pass Bar Channel,
Maintenance Dredging
Plaquemines Parish, LA

Item	DESCRIPTION	Estimated Quantity	Unit	Unit Price	Estimated Amount
BASE WORK					
0001	Mobilization and Demobilization	1	JOB		
0002	Dredging				
0002AA	First 1,650,000 CY	1,650,000	CY		
0002AB	All Over 1,650,000 CY	600,000	CY		
TOTAL BASE WORK:					\$
OPTIONAL WORK – (OW)					
0003	Bird Nesting Prevention and Avoidance Measures - OW	60	DAY		
TOTAL OPTIONAL WORK:					\$
TOTAL BASE WORK + OPTIONAL WORK:					\$

Award will be made as a whole to one bidder.

NOTE 1: The unit “Job” as used in this Bidding Schedule is synonymous with the term “Lump Sum” used elsewhere within these plans and specifications.

NOTE 2: Bidders shall furnish unit prices for each item listed in the Schedule requiring a unit price. If the bidder fails to insert a unit price in the appropriate blank for required item(s), but does furnish an extended total, or an estimated amount for such item(s), the Government shall deem the unit price to be the quotient obtained by dividing the extended amount for that line item by the quantity. IF A BIDDER OMITS BOTH THE UNIT PRICE AND THE EXTENDED TOTAL OR ESTIMATED AMOUNT FOR ANY ITEM, ITS BID SHALL BE DECLARED NON-RESPONSIVE AND THEREFORE INELIGIBLE FOR AWARD.

NOTE 3: Any bid may be rejected if the Contracting Officer determines in writing that it is unreasonable as to price. Unreasonableness of price includes not only total price of bid, but the price for individual line items as well. Any bid may be rejected if the prices for any line items or subline items are materially unbalanced (See [FAR 14.404-2, Rejection of Individual Bids](#)).

NOTE 4: THE NOTICE TO PROCEED (NTP): The successful bidder is advised that performance and payment bonds shall be submitted in accordance with the time frame in block 12B of SF 1442 after Notice of Award. The NTP will be issued immediately after verification of acceptable performance and payment bonds. Within 24 hours after issuance of the NTP, the Contractor shall initiate a meeting to discuss the submittal

process with the Area or Resident Engineer or his authorized representative. Physical work cannot start until the Accident Prevention Program, Contractor Quality Control Plan, and other submittals which may be required, have been submitted and approved and all preliminary meetings called for under the contract, have been conducted.

NOTE 5: RAPID VENDOR PAYMENT. Web based instructions for the submission of invoices may be found at the following website:

<http://www.mvn.usace.army.mil/BusinessWithUs/Contracting/RapidVendorPayment>

NOTE 6: EVALUATION OF OPTIONS (FAR 52.217-5 JUL 1990). Except when it is determined in accordance with FAR 17.206(b) not to be in the Government's best interests, the government will evaluate offers for award purposes by adding the total price for all options to the total price for the basic requirement. Evaluation of options will not obligate the government to exercise the option(s).

NOTE 7: OPTIONAL WORK ITEM: The Government may require the delivery of, in whole or part, the Option Item. The Government may exercise the Optional Item by written notice to the Contractor. If the Government activates an Optional Item, the activation will occur while the Contractor's plant is within the limits of work. No additional mobilization/ demobilization will be paid to the Contractor for the Optional work. Bidders shall bid on all items including Optional Items (see "Evaluation of Options" FAR 52.217-5). Failure to bid on all items will result in a non-responsive bid.

NOTE 8: TIME ADJUSTMENT IN THE EVENT OF THE EXERCISE OF OPTION WORK. In the event that the OPTIONAL WORK item is exercised under the terms of the contract, the time required for completion of the contract will not be adjusted but shall be performed concurrent with the original contract duration of 155 days established for the BASE WORK items specified in the Contract Clause entitled "COMMENCEMENT, PROCECUTION, AND COMPLETION OF WORK (FAR 52-211-10 – APR 1984)".

EVALUATION OF SUBLINE ITEMS (Dec 2019 - AFARS 5152.211-9000).

Item No. 0002 is subdivided into two or more estimated quantities and is to be separately priced. The Government will evaluate each of these items on the basis of total price of its subline items.

(End of Clause)

VARIATIONS IN ESTIMATED QUANTITIES - SUBLINE ITEMS (Dec 2019) - AFARS 5152.211-9001).

This variation in estimated quantities clause is applicable only to item No. 0002.

(a) Variation from the estimated quantity in the actual work performed under any second or subsequent subline item or elimination of all work under such a second or subsequent subline item will not be the basis for an adjustment in contract unit price.

(b) Where the actual quantity of work performed for item No. 0002 is less than 85% of the quantity of the first subline item listed under such item, the contractor will be paid at the contract unit price for that subline item for the actual quantity of

work performed and, in addition, an equitable adjustment shall be made in accordance with the clause at Federal Acquisition Regulation (FAR) 52.211-18, VARIATION IN ESTIMATED QUANTITY.

(c) If the actual quantity of work performed under item No. 0002 exceeds 115% or is less than 85% of the total estimated quantity of the subline item under that item and/or if the quantity of the work performed under the second subline item or any subsequent subline item under item Nos. NONE exceeds 115% or is less than 85% of the estimated quantity of any such subline item, and if such variation causes an increase or a decrease in the time required for performance of this contract the contract completion time will be adjusted in accordance with the clause FAR 52.211-18, VARIATION IN ESTIMATED QUANTITY.

(End of Clause)

Bid Bond (See Instructions on Page 3)		Date Bond Executed (Must Not Be Later Than Bid Opening Date)	OMB Control Number: 9000-0001 Expiration Date: 1/31/2027
Principal (Legal Name And Business Address)		Type Of Organization ("X" One) ☐ Individual ☐ Partnership ☐ Joint Venture ☐ Corporation ☐ Other (Specify)	
		State Of Incorporation	
Surety(ies) (Name And Business Address)			

Penal Sum Of Bond					Bid Identification	
Percent Of Bid Price	Amount Not To Exceed				Bid Date	Invitation Number
	Million(s)	Thousand(s)	Hundred(s)	Cents		
						For (Construction, Supplies Or Services)

Obligation:

We, the Principal and Surety(ies) are firmly bound to the United States of America (hereinafter called the Government) in the above penal sum. For payment of the penal sum, we bind ourselves, our heirs, executors, administrators, and successors, jointly and severally. However, where the Sureties are corporations acting as co-sureties, we, the Sureties, bind ourselves in such sum "jointly and severally" as well as "severally" only for the purpose of allowing a joint action or actions against any or all of us. For all other purposes, each Surety binds itself, jointly and severally with the Principal, for the payment of the sum shown opposite the name of the Surety. If no limit of liability is indicated, the limit of liability is the full amount of the penal sum.

Conditions:

The Principal has submitted the bid identified above.

Therefore:

The above obligation is void if the Principal - (a) upon acceptance by the Government of the bid identified above, within the period specified therein for acceptance (sixty (60) days if no period is specified), executes the further contractual documents and gives the bond(s) required by the terms of the bid as accepted within the time specified (ten (10) days if no period is specified) after receipt of the forms by the Principal; or (b) in the event of failure to execute such further contractual documents and give such bonds, pays the Government for any cost of procuring the work which exceeds the amount of the bid.

Each Surety executing this instrument agrees that its obligation is not impaired by any extension(s) of the time for acceptance of the bid that the Principal may grant to the Government. Notice to the Surety(ies) of extension(s) is waived. However, waiver of the notice applies only to extensions aggregating not more than sixty (60) calendar days in addition to the period originally allowed for acceptance of the bid.

Witness:

The Principal and Surety(ies) executed this bid bond and affixed their seals on the above date.

Principal

Signature(s)	1. (Seal)	2. (Seal)	3. (Seal)	Corporate Seal
Name(s) And Title(s) (Typed)	1.	2.	3.	

Individual Surety(ies)

Signature(s)	1. (Seal)	2. (Seal)
Name(s) (Typed)	1.	2.

Corporate Surety(ies)

Surety A	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety B	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety C	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety D	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		

Surety E	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety F	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		
Surety G	Name And Address		State Of Incorporation	Liability Limit (\$)	Corporate Seal
	Signature(s)	1.	2.		
	Name(s) And Title(s) (Typed)	1.	2.		

Instructions

1. This form is authorized for use when a bid guaranty is required. Any deviation from this form will require the written approval of the Administrator of General Services.
2. Insert the full legal name and business address of the Principal in the space designated "Principal" on the face of the form. An authorized person shall sign the bond. Any person signing in a representative capacity (e.g., an attorney-in-fact) must furnish evidence of authority if that representative is not a member of the firm, partnership, or joint venture, or an officer of the corporation involved.
3. The bond may express penal sum as a percentage of the bid price. In these cases, the bond may state a maximum dollar limitation (e.g., 20% of the bid price but the amount not to exceed _____ dollars).
4. (a) Corporations executing the bond as sureties must appear on the Department of the Treasury's list of approved sureties and must act within the limitations listed therein. The value put into the Liability Limit block is the penal sum (i.e., the face value) of the bond, unless a co-surety arrangement is proposed.

(b) When multiple corporate sureties are involved, their names and addresses shall appear in the spaces (Surety A, Surety B, etc.) headed "Corporate Surety(ies)." In the space designated "Surety(ies)" on the face of the form, insert only the letter identifier corresponding to each of the sureties. Moreover, when co-surety arrangements exist, the parties may allocate their respective limitations of liability under the bond, provided that the sum total of their liability equals 100% of the bond penal sum.

(c) When individual sureties are involved, a completed [Affidavit of Individual Surety \(Standard Form 28\)](#) for each individual surety, shall accompany the bond. The Government may require the surety to furnish additional substantiating information concerning its financial capability.

5. Corporations executing the bond shall affix their corporate seals. Individuals shall execute the bond opposite the word "Corporate Seal"; and shall affix an adhesive seal if executed in Maine, New Hampshire, or any other jurisdiction requiring adhesive seals.
6. Type the name and title of each person signing this bond in the space provided.
7. In its application to negotiated contracts, the terms "bid" and "bidder" shall include "proposal" and "offeror."

Paperwork Reduction Act Statement - This information collection meets the requirements of 44 USC § 3507, as amended by section 2 of the Paperwork Reduction Act of 1995. You do not need to answer these questions unless we display a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 9000-0001. We estimate that it will take 1 hour to read the instructions, gather the facts, and answer the questions. Send only comments relating to our time estimate, including suggestions for reducing this burden, or any other aspects of this collection of information to: General Services Administration, Regulatory Secretariat Division (MVCB), 1800 F Street, NW, Washington, DC 20405.

Section 00 21 00 - Instructions

FAR Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation	Variation Effective Date
52.204-7	System for Award Management.	Nov 2024		
52.204-22	Alternative Line Item Proposal.	Jan 2017		
52.222-5	Construction Wage Rate Requirements-Secondary Site of the Work.	May 2014		

DFARS Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation	Variation Effective Date
252.204-7019	Notice of NIST SP 800-171 DoD Assessment Requirements.	Nov 2023		

FAR Clauses Incorporated by Full Text

52.233-2 Service of Protest. (Sep 2006)

Service of Protest (Sept 2006)

(a) Protests, as defined in section 33.101 of the Federal Acquisition Regulation, that are filed directly with an agency, and copies of any protests that are filed with the Government Accountability Office (GAO), shall be served on the Contracting Officer (addressed as follows) by obtaining written and dated acknowledgment of receipt from Karen D. Hargrave US Army Corps of Engineers 7400 Leake Ave, New Orleans, LA 70118.

(b) The copy of any protest shall be received in the office designated above within one day of filing a protest with the GAO.

(End of provision)

52.236-27 Site Visit (Construction). (Feb 1995)

Site Visit (Construction) (Feb 1995)

(a) The clauses at 52.236-2, Differing Site Conditions, and 52.236-3, Site Investigations and Conditions Affecting the Work, will be included in any contract awarded as a result of this solicitation. Accordingly, offerors or quoters are urged and expected to inspect the site where the work will be performed.

(b) Site visits may be arranged during normal duty hours by contacting:

Name: Charles "Chuck" Freeman

Address: 43020 LA Hwy 23, Venice, LA 70091

Telephone: 504-862-1554

(End of Provision)

52.252-1 Solicitation Provisions Incorporated by Reference. (Feb 1998)

Solicitation Provisions Incorporated by Reference (Feb 1998)

This solicitation incorporates one or more solicitation provisions by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. The offeror is cautioned that the listed provisions may include blocks that must be completed by the offeror and submitted with its quotation or offer. In lieu of submitting the full text of those provisions, the offeror may identify the provision by paragraph identifier and provide the appropriate information with its quotation or offer. Also, the full text of a solicitation provision may be accessed electronically at this/these address(es):

<https://www.acquisition.gov/>

(End of provision)

Authorized Deviations in Provisions (Nov 2020)

(a) The use in this solicitation of any Federal Acquisition Regulation (48 CFR Chapter 1) provision with an authorized deviation is indicated by the addition of "(DEVIATION)" after the date of the provision.

(b) The use in this solicitation of any Defense Federal Acquisition Regulation (48 CFR Chapter2) provision with an authorized deviation is indicated by the addition of "(DEVIATION)" after the name of the regulation.

(End of provision)

Section 00 21 13 - Instructions to Bidders

FAR Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation	Variation Effective Date
52.214-3	Amendments to Invitations for Bids.	Dec 2016		
52.214-4	False Statements in Bids.	Apr 1984		
52.214-5	Submission of Bids.	Dec 2016		
52.214-6	Explanation to Prospective Bidders.	Apr 1984		
52.214-7	Late Submissions, Modifications, and Withdrawals of Bids.	Nov 1999		
52.214-18	Preparation of Bids-Construction.	Apr 1984		
52.214-19	Contract Award-Sealed Bidding-Construction.	Aug 1996		

Section 00 21 16 - Instructions to Proposers

FAR Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation	Variation Effective Date
52.204-16	Commercial and Government Entity Code Reporting.	Aug 2020		

Section 00 22 00 - Supplementary Instructions

FAR Clauses Incorporated by Full Text

52.252-3 Alterations in Solicitation. (Apr 1984)

Alterations in Solicitation (Apr 1984)

Portions of this solicitation are altered as follows:

(End of clause)

Section 00 45 00 - Representations and Certifications

FAR Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation	Variation Effective Date
52.204-24	Representation Regarding Certain Telecommunications and Video Surveillance Services or Equipment.	Nov 2021		
52.209-7	Information Regarding Responsibility Matters.	Oct 2018		

DFARS Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation	Variation Effective Date
252.203-7005	Representation Relating to Compensation of Former DoD Officials.	Sep 2022		
252.204-7008	Compliance with Safeguarding Covered Defense Information Controls.	Oct 2016		
252.204-7017	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services-Representation.	May 2021		
252.225-7055	Representation Regarding Business Operations with the Maduro Regime.	May 2022		

FAR Clauses Incorporated by Full Text

52.204-8 Annual Representations and Certifications. (Jan 2025)

Annual Representations and Certifications (Jan 2025)

(a)

(1) The North American Industry Classification System (NAICS) code for this acquisition is ____237990.

(2) The small business size standard is ____\$37,000,000.00.

(3) The small business size standard for a concern that submits an offer, other than on a construction or service acquisition, but proposes to furnish an end item that it did not itself manufacture, process, or produce is 500 employees, or 150 employees for information technology value-added resellers under NAICS code 541519 if the acquisition-

(i) Is set aside for small business and has a value above the simplified acquisition threshold;

(ii) Uses the HUBZone price evaluation preference regardless of dollar value, unless the offeror waives the price evaluation preference; or

(iii) Is an 8(a), HUBZone, service-disabled veteran-owned, economically disadvantaged women-owned, or women-owned small business set-aside or sole-source award regardless of dollar value.

(b)

(1) If the provision at 52.204-7, System for Award Management, is included in this solicitation, paragraph (d) of this provision applies.

(2) If the provision at 52.204-7, System for Award Management, is not included in this solicitation, and the Offeror has an active registration in the System for Award Management (SAM), the Offeror may choose to use paragraph (d) of this provision instead of completing the corresponding individual representations and certifications in the solicitation. The Offeror shall indicate which option applies by checking one of the following boxes:

(i) ☐ Paragraph (d) applies.

(ii) ☐ Paragraph (d) does not apply and the offeror has completed the individual representations and certifications in the solicitation.

(c)

(1) The following representations or certifications in SAM are applicable to this solicitation as indicated:

(i) 52.203-2, Certificate of Independent Price Determination. This provision applies to solicitations when a firm-fixed-price contract or fixed-price contract with economic price adjustment is contemplated, unless-

- (A) The acquisition is to be made under the simplified acquisition procedures in part 13;
 - (B) The solicitation is a request for technical proposals under two-step sealed bidding procedures; or
 - (C) The solicitation is for utility services for which rates are set by law or regulation.
- (ii) 52.203-11, Certification and Disclosure Regarding Payments to Influence Certain Federal Transactions. This provision applies to solicitations expected to exceed \$150,000.
- (iii) 52.203-18, Prohibition on Contracting with Entities that Require Certain Internal Confidentiality Agreements or Statements-Representation. This provision applies to all solicitations.
- (iv) 52.204-3, Taxpayer Identification. This provision applies to solicitations that do not include the provision at 52.204-7, System for Award Management.
- (v) 52.204-5, Women-Owned Business (Other Than Small Business). This provision applies to solicitations that-
- (A) Are not set aside for small business concerns;
 - (B) Exceed the simplified acquisition threshold; and
 - (C) Are for contracts that will be performed in the United States or its outlying areas.
- (vi) 52.204-26, Covered Telecommunications Equipment or Services-Representation. This provision applies to all solicitations.
- (vii) 52.209-2, Prohibition on Contracting with Inverted Domestic Corporations-Representation.
- (viii) 52.209-5, Certification Regarding Responsibility Matters. This provision applies to solicitations where the contract value is expected to exceed the simplified acquisition threshold.
- (ix) 52.209-11, Representation by Corporations Regarding Delinquent Tax Liability or a Felony Conviction under any Federal Law. This provision applies to all solicitations.
- (x) 52.214-14, Place of Performance-Sealed Bidding. This provision applies to invitations for bids except those in which the place of performance is specified by the Government.
- (xi) 52.215-6, Place of Performance. This provision applies to solicitations unless the place of performance is specified by the Government.
- (xii) 52.219-1, Small Business Program Representations (Basic, Alternates I, and II). This provision applies to solicitations when the contract is for supplies to be delivered or services to be performed in the United States or its outlying areas, or when the contracting officer has applied part 19 in accordance with 19.000(b)(1)(ii).
- (A) The basic provision applies when the solicitations are issued by other than DoD, NASA, and the Coast Guard.
 - (B) The provision with its Alternate I applies to solicitations issued by DoD, NASA, or the Coast Guard.
 - (C) The provision with its Alternate II applies to solicitations that will result in a multiple-award contract with more than one NAICS code assigned.
- (xiii) 52.219-2, Equal Low Bids. This provision applies to solicitations when contracting by sealed bidding and the contract is for supplies to be delivered or services to be performed in the United States or its outlying areas, or when the contracting officer has applied part 19 in accordance with 19.000(b)(1)(ii).
- (xiv) 52.222-22, Previous Contracts and Compliance Reports. This provision applies to solicitations that include the clause at 52.222-26, Equal Opportunity.
- (xv) 52.222-25, Affirmative Action Compliance. This provision applies to solicitations, other than those for construction, when the solicitation includes the clause at 52.222-26, Equal Opportunity.
- (xvi) 52.222-38, Compliance with Veterans' Employment Reporting Requirements. This provision applies to solicitations when it is anticipated the contract award will exceed the simplified acquisition threshold and the contract is not for acquisition of commercial products or commercial services.
- (xvii) 52.223-1, Biobased Product Certification. This provision applies to solicitations that require the delivery or specify the use of biobased products in USDA-designated product categories; or include the clause at 52.223-2, Reporting of Biobased Products Under Service and Construction Contracts.
- (xviii) 52.223-4, Recovered Material Certification. This provision applies to solicitations that are for, or specify the use of, EPA-designated items.
- (xix) 52.223-22, Public Disclosure of Greenhouse Gas Emissions and Reduction Goals-Representation. This provision applies to solicitations that include the provision at 52.204-7.
- (xx) 52.225-2, Buy American Certificate. This provision applies to solicitations containing the clause at 52.225-1.

(xxi) 52.225-4, Buy American-Free Trade Agreements-Israeli Trade Act Certificate. (Basic, Alternates II and III.) This provision applies to solicitations containing the clause at 52.225-3.

(A) If the acquisition value is less than \$50,000, the basic provision applies.

(B) If the acquisition value is \$50,000 or more but is less than \$100,000, the provision with its Alternate II applies.

(C) If the acquisition value is \$100,000 or more but is less than \$102,280, the provision with its Alternate III applies.

(xxii) 52.225-6, Trade Agreements Certificate. This provision applies to solicitations containing the clause at 52.225-5.

(xxiii) 52.225-20, Prohibition on Conducting Restricted Business Operations in Sudan-Certification. This provision applies to all solicitations.

(xxiv) 52.225-25, Prohibition on Contracting with Entities Engaging in Certain Activities or Transactions Relating to Iran-Representation and Certifications. This provision applies to all solicitations.

(xxv) 52.226-2, Historically Black College or University and Minority Institution Representation. This provision applies to solicitations for research, studies, supplies, or services of the type normally acquired from higher educational institutions.

(2) The following representations or certifications are applicable as indicated by the Contracting Officer:

[Contracting Officer check as appropriate.]

☐ (i) 52.204-17, Ownership or Control of Offeror

☐ (ii) 52.204-20, Predecessor of Offeror.

☐ (iii) 52.222-18, Certification Regarding Knowledge of Child Labor for Listed End Products.

☐ (iv) 52.222-48, Exemption from Application of the Service Contract Labor Standards to Contracts for Maintenance, Calibration, or Repair of Certain Equipment- Certification.

☐ (v) 52.222-52, Exemption from Application of the Service Contract Labor Standards to Contracts for Certain Services-Certification.

☐ (vi) 52.227-6, Royalty Information.

☐ (A) Basic.

☐ (B) Alternate I.

☐ (vii) 52.227-15, Representation of Limited Rights Data and Restricted Computer Software.

(d) The offeror has completed the annual representations and certifications electronically in SAM website accessed through <https://www.sam.gov>. After reviewing the SAM information, the offeror verifies by submission of the offer that the representations and certifications currently posted electronically that apply to this solicitation as indicated in paragraph (c) of this provision have been entered or updated within the last 12 months, are current, accurate, complete, and applicable to this solicitation (including the business size standard applicable to the NAICS code referenced for this solicitation), as of the date of this offer and are incorporated in this offer by reference (see FAR 4.1201); except for the changes identified below [offeror to insert changes, identifying change by clause number, title, date]. These amended representation(s) and/or certification(s) are also incorporated in this offer and are current, accurate, and complete as of the date of this offer.

FAR Clause # Title Date Change

Any changes provided by the offeror are applicable to this solicitation only, and do not result in an update to the representations and certifications posted on SAM.

(End of provision)

DFARS Clauses Incorporated by Full Text

252.204-7007 Alternate A, Annual Representations and Certifications.

(Oct 2024) Alternate A (Oct 2024)

ALTERNATE A, ANNUAL REPRESENTATIONS AND CERTIFICATIONS (OCT 2024)

Substitute the following paragraphs (b), (d), and (e) for paragraphs (b) and (d) of the provision at FAR 52.204-8:

(b)(1) If the provision at 52.204-7, System for Award Management, is included in this solicitation, paragraph (e) of this provision applies.

(2) If the provision at 52.204-7, System for Award Management, is not included in this solicitation, and the Offeror has an active registration in the System for Award Management (SAM), the Offeror may choose to use paragraph (e) of this provision instead of completing the corresponding individual representations and certifications in the solicitation. The Offeror shall indicate which option applies by checking one of the following boxes:

☐ (i) Paragraph (e) applies.

☐ (ii) Paragraph (e) does not apply and the Offeror has completed the individual representations and certifications in the solicitation.

(d)(1) The following representations or certifications in the SAM database are applicable to this solicitation as indicated:

(i) 252.204-7016, Covered Defense Telecommunications Equipment or Services-Representation. Applies to all solicitations.

(ii) 252.216-7008, Economic Price Adjustment-Wage Rates or Material Prices Controlled by a Foreign Government. Applies to solicitations for fixed-price supply and service contracts when the contract is to be performed wholly or in part in a foreign country, and a foreign government controls wage rates or material prices and may during contract performance impose a mandatory change in wages or prices of materials.

(iii) 252.225-7042, Authorization to Perform. Applies to all solicitations when performance will be wholly or in part in a foreign country.

(iv) 252.225-7049, Prohibition on Acquisition of Certain Foreign Commercial Satellite Services-Representations. Applies to solicitations for the acquisition of commercial satellite services.

(v) 252.225-7050, Disclosure of Ownership or Control by the Government of a Country that is a State Sponsor of Terrorism. Applies to all solicitations expected to result in contracts of \$150,000 or more.

(vi) 252.229-7012, Tax Exemptions (Italy)-Representation. Applies to solicitations and contracts when contract performance will be in Italy.

(vii) 252.229-7013, Tax Exemptions (Spain)-Representation. Applies to solicitations and contracts when contract performance will be in Spain.

(2) The following representations or certifications in SAM are applicable to this solicitation as indicated by the Contracting Officer: [Contracting Officer check as appropriate.]

☐ (i) 252.209-7002, Disclosure of Ownership or Control by a Foreign Government.

☐ (ii) 252.225-7000, Buy American-Balance of Payments Program Certificate.

☐ (iii) 252.225-7020, Trade Agreements Certificate.

☐ Use with Alternate I.

☐ (iv) 252.225-7031, Secondary Arab Boycott of Israel.

☐ (v) 252.225-7035, Buy American-Free Trade Agreements-Balance of Payments Program Certificate.

☐ Use with Alternate I.

☐ Use with Alternate II.

☐ Use with Alternate III.

☐ Use with Alternate IV.

☐ Use with Alternate V.

[](vi) 252.226-7002, Representation for Demonstration Project for Contractors Employing Persons with Disabilities.

[](vii) 252.232-7015, Performance-Based Payments-Representation.

(e) The Offeror has completed the annual representations and certifications electronically via the SAM website at <https://www.sam.gov>. After reviewing the SAM database information, the Offeror verifies by submission of the offer that the representations and certifications currently posted electronically that apply to this solicitation as indicated in FAR 52.204-8(c) and paragraph (d) of this provision have been entered or updated within the last 12 months, are current, accurate, complete, and applicable to this solicitation (including the business size standard applicable to the NAICS code referenced for this solicitation), as of the date of this offer, and are incorporated in this offer by reference (see FAR 4.1201); except for the changes identified below [Offeror to insert changes, identifying change by provision number, title, date]. These amended representation(s) and/or certification(s) are also incorporated in this offer and are current, accurate, and complete as of the date of this offer.

FAR/DFARS Provision #	Title	Date	Change
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Any changes provided by the Offeror are applicable to this solicitation only, and do not result in an update to the representations and certifications located in the SAM database.

(End of provision)

Section 00 70 00 - Conditions of the Contract

FAR Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation	Variation Effective Date
52.203-3	Gratuities.	Apr 1984		
52.203-5	Covenant Against Contingent Fees.	May 2014		
52.203-6	Restrictions on Subcontractor Sales to the Government.	Jun 2020		
52.203-7	Anti-Kickback Procedures.	Jun 2020		
52.203-8	Cancellation, Rescission, and Recovery of Funds for Illegal or Improper Activity.	May 2014		
52.203-10	Price or Fee Adjustment for Illegal or Improper Activity.	May 2014		
52.203-12	Limitation on Payments to Influence Certain Federal Transactions.	Jun 2020		
52.203-13	Contractor Code of Business Ethics and Conduct.	Nov 2021		
52.203-19	Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements.	Jan 2017		
52.204-10	Reporting Executive Compensation and First-Tier Subcontract Awards.	Jun 2020		
52.204-13	System for Award Management Maintenance.	Oct 2018		
52.204-18	Commercial and Government Entity Code Maintenance.	Aug 2020		
52.204-19	Incorporation by Reference of Representations and Certifications.	Dec 2014		
52.204-23	Prohibition on Contracting for Hardware, Software, and Services Developed or Provided by Kaspersky Lab Covered Entities.	Dec 2023		
52.204-25	Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment.	Nov 2021		
52.204-27	Prohibition on a ByteDance Covered Application.	Jun 2023		
52.204-29	Federal Acquisition Supply Chain Security Act Orders-Representation and Disclosures.	Dec 2023		
52.209-6	Protecting the Government's Interest When Subcontracting With Contractors Debarred, Suspended, Proposed for Debarment, or Voluntarily Excluded	Jan 2025		
52.209-9	Updates of Publicly Available Information Regarding Responsibility Matters.	Oct 2018		
52.209-10	Prohibition on Contracting with Inverted Domestic Corporations.	Nov 2015		
52.209-13	Violation of Arms Control Treaties or Agreements-Certification.	Nov 2021		
52.210-1	Market Research.	Nov 2021		
52.211-18	Variation in Estimated Quantity.	Apr 1984		
52.214-12	Preparation of Bids.	Apr 1984		
52.214-26	Audit and Records-Sealed Bidding.	Jun 2020		
52.214-27	Price Reduction for Defective Certified Cost or Pricing Data-Modifications-Sealed Bidding.	Jun 2020		
52.214-28	Subcontractor Certified Cost or Pricing Data-Modifications-Sealed Bidding.	Jun 2020		
52.214-29	Order of Precedence-Sealed Bidding.	Jan 1986		
52.214-34	Submission of Offers in the English Language.	Apr 1991		
52.214-35	Submission of Offers in U.S. Currency.	Apr 1991		
52.217-5	Evaluation of Options	Jul 1990		
52.219-8	Utilization of Small Business Concerns.	Jan 2025		
52.219-9	Small Business Subcontracting Plan. (Alternate I)	Jan 2025	Alternate I	Nov 2016
52.219-16	Liquidated Damages-Subcontracting Plan.	Sep 2021		
52.222-3	Convict Labor.	Jun 2003		
52.222-6	Construction Wage Rate Requirements.	Aug 2018		
52.222-7	Withholding of Funds.	May 2014		
52.222-8	Payrolls and Basic Records.	Jul 2021		
52.222-9	Apprentices and Trainees.	Jul 2005		
52.222-10	Compliance with Copeland Act Requirements.	Feb 1988		
52.222-11	Subcontracts (Labor Standards).	May 2014		
52.222-12	Contract Termination-Debarment.	May 2014		
52.222-13	Compliance with Construction Wage Rate Requirements and Related Regulations.	May 2014		
52.222-14	Disputes Concerning Labor Standards.	Feb 1988		
52.222-15	Certification of Eligibility.	May 2014		
52.222-30	Construction Wage Rate Requirements-Price Adjustment (None or Separately Specified Method).	Aug 2018		
52.222-35	Equal Opportunity for Veterans.	Jun 2020		
52.222-36	Equal Opportunity for Workers with Disabilities.	Jun 2020		
52.222-37	Employment Reports on Veterans.	Jun 2020		
52.222-50	Combating Trafficking in Persons.	Nov 2021		
52.222-54	Employment Eligibility Verification.	Jan 2025		
52.222-55	Minimum Wages for Contractor Workers Under Executive Order 14026.	Jan 2022		
52.222-62	Paid Sick Leave Under Executive Order 13706.	Jan 2022		
52.223-2	Reporting of Biobased Products Under Service and Construction Contracts	May 2024		
52.223-21	Foams.	May 2024		
52.223-23	Sustainable Products and Services.	May 2024		
52.225-12	Notice of Buy American Requirement-Construction Materials Under Trade Agreements.	May 2014		
52.225-12	Notice of Buy American Requirement-Construction Materials Under Trade Agreements. (Alternate I)	May 2014	Alternate I	May 2014

52.225-13	Restrictions on Certain Foreign Purchases.	Feb 2021		
52.226-7	Drug-Free Workplace.	May 2024		
52.226-8	Encouraging Contractor Policies to Ban Text Messaging While Driving.	May 2024		
52.228-2	Additional Bond Security.	Oct 1997		
52.228-11	Individual Surety-Pledge of Assets.	Feb 2021		
52.228-12	Prospective Subcontractor Requests for Bonds.	Dec 2022		
52.228-14	Irrevocable Letter of Credit.	Nov 2014		
52.228-17	Individual Surety-Pledge of Assets (Bid Guarantee).	Feb 2021		
52.229-3	Federal, State, and Local Taxes.	Feb 2013		
52.229-11	Tax on Certain Foreign Procurements-Notice and Representation.	Jun 2020		
52.232-15	Progress Payments Not Included.	Apr 1984		
52.232-17	Interest.	May 2014		
52.232-23	Assignment of Claims.	May 2014		
52.232-33	Payment by Electronic Funds Transfer-System for Award Management.	Oct 2018		
52.232-39	Unenforceability of Unauthorized Obligations.	Jun 2013		
52.232-40	Providing Accelerated Payments to Small Business Subcontractors.	Mar 2023		
52.233-3	Protest after Award.	Aug 1996		
52.233-4	Applicable Law for Breach of Contract Claim.	Oct 2004		
52.236-2	Differing Site Conditions.	Apr 1984		
52.236-3	Site Investigation and Conditions Affecting the Work.	Apr 1984		
52.236-5	Material and Workmanship.	Apr 1984		
52.236-6	Superintendence by the Contractor.	Apr 1984		
52.236-7	Permits and Responsibilities.	Nov 1991		
52.236-8	Other Contracts.	Apr 1984		
52.236-9	Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements.	Apr 1984		
52.236-10	Operations and Storage Areas.	Apr 1984		
52.236-11	Use and Possession Prior to Completion.	Apr 1984		
52.236-12	Cleaning Up.	Apr 1984		
52.236-13	Accident Prevention.	Nov 1991		
52.236-13	Accident Prevention. (Alternate I)	Nov 1991	Alternate I	Nov 1991
52.236-21	Specifications and Drawings for Construction.	Feb 1997		
52.242-5	Payments to Small Business Subcontractors.	Jan 2017		
52.242-13	Bankruptcy.	Jul 1995		
52.242-14	Suspension of Work.	Apr 1984		
52.243-4	Changes.	Jun 2007		
52.244-6	Subcontracts for Commercial Products and Commercial Services.	Jan 2025		
52.246-12	Inspection of Construction.	Aug 1996		
52.249-2	Termination for Convenience of the Government (Fixed-Price).	Apr 2012		
52.249-4	Termination for Convenience of the Government (Services) (Short Form).	Apr 1984		
52.249-10	Default (Fixed-Price Construction).	Apr 1984		

DFARS Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation
252.201-7000	Contracting Officer's Representative.	Dec 1991	
252.203-7000	Requirements Relating to Compensation of Former DoD Officials.	Sep 2011	
252.203-7001	Prohibition on Persons Convicted of Fraud or Other Defense-Contract-Related Felonies.	Jan 2023	
252.203-7002	Requirement to Inform Employees of Whistleblower Rights.	Dec 2022	
252.203-7003	Agency Office of the Inspector General.	Aug 2019	
252.203-7004	Display of Hotline Posters.	Jan 2023	
252.204-7003	Control of Government Personnel Work Product.	Apr 1992	
252.204-7012	Safeguarding Covered Defense Information and Cyber Incident Reporting.	May 2024	
252.204-7018	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services.	Jan 2023	
252.204-7020	NIST SP 800-171 DoD Assessment Requirements.	Nov 2023	
252.204-7024	Notice on the Use of the Supplier Performance Risk System.	Mar 2023	
252.205-7000	Provision of Information to Cooperative Agreement Holders.	Oct 2024	
252.209-7004	Subcontracting with Firms that are Owned or Controlled by the Government of a Country that is a State Sponsor of Terrorism.	May 2019	
252.215-7013	Supplies and Services Provided by Nontraditional Defense Contractors.	Jan 2023	
252.219-7003	Small Business Subcontracting Plan (DoD Contracts).	Dec 2019	
252.222-7006	Restrictions on the Use of Mandatory Arbitration Agreements.	Jan 2023	
252.225-7048	Export-Controlled Items.	Jun 2013	
252.225-7056	Prohibition Regarding Business Operations with the Maduro Regime.	Jan 2023	
252.225-7057	Preaward Disclosure of Employment of Individuals Who Work in the People's Republic of China.	Aug 2022	
	Postaward Disclosure of Employment of Individuals Who Work in the People's Republic of China.	Jan 2023	
252.225-7058	Prohibition on Certain Procurements from the Xinjiang Uyghur Autonomous Region- Representation.	Jan 2023	
252.225-7059			
252.225-7060	Prohibition on Certain Procurements from the Xinjiang Uyghur Autonomous Region.	Jun 2023	
252.226-7001	Utilization of Indian Organizations, Indian-Owned Economic Enterprises, and Native Hawaiian Small Business Concerns.	Jan 2023	
252.226-7003	Drug-Free Work Force.	Aug 2024	
252.232-7003	Electronic Submission of Payment Requests and Receiving Reports.	Dec 2018	
252.232-7010	Levies on Contract Payments.	Dec 2006	
252.236-7000	Modification Proposals--Price Breakdown.	Dec 1991	
252.236-7007	Additive or Deductive Items.	Dec 1991	
252.236-7008	Contract Prices--Bidding Schedules.	Dec 1991	
252.243-7001	Pricing of Contract Modifications.	Dec 1991	
252.243-7002	Requests for Equitable Adjustment.	Dec 2022	
252.244-7000	Subcontracts for Commercial Products or Commercial Services.	Nov 2023	

FAR Clauses Incorporated by Full Text

52.204-1 Approval of Contract.

(Dec 1989)

Approval of Contract (Dec 1989)

This contract is subject to the written approval of Karen D. Hargrave[identify title of designated agency official here] and shall not be binding until so approved.

(End of clause)

52.211-10 Commencement, Prosecution, and Completion of Work.

(Apr 1984)

Commencement, Prosecution, and Completion of Work (Apr 1984)

The Contractor shall be required to (a) commence work under this contract within 10 calendar days[Contracting Officer insert number] calendar days after the date the Contractor receives the notice to proceed, (b) prosecute the work diligently, and (c) complete the entire work ready for use not later than 155 calendar days after the date of receipt by him of notice to proceed.* The time stated for completion shall include final cleanup of the premises.

* The Contracting Officer shall specify either a number of days after the date the contractor receives the notice to proceed, or a calendar date.

(End of clause)

NOTE: The term “work” includes tasks such as submission of submittals, performance of preliminary and/or BD surveys, and does not have to be dredging.

52.211-12 Liquidated Damages-Construction.

(Sep 2000)

Liquidated Damages-Construction (Sept 2000)

(a) If the Contractor fails to complete the work within the time specified in the contract, the Contractor shall pay liquidated damages to the Government in the amount of \$3,415.00[Contracting Officer insert amount] for each calendar day of delay until the work is completed or accepted.

(b) If the Government terminates the Contractor's right to proceed, liquidated damages will continue to accrue until the work is completed. These liquidated damages are in addition to excess costs of repurchase under the Termination clause.

(End of clause)

52.216-1 Type of Contract.

(Apr 1984)

Type of Contract (Apr 1984)

The Government contemplates award of a firm fixed-price[Contracting Officer insert specific type of contract] contract resulting from this solicitation.

(End of provision)

52.217-4 Evaluation of Options Exercised at Time of Contract Award.

(Apr 1984)

Except when it is determined in accordance with FAR 17.206(b) not to be in the Government's best interests, the Government will evaluate the total price for the basic requirement together with any option(s) exercised at the time of award.

(End of provision)

52.217-7 Option for Increased Quantity – Separately Priced Line Item

(Jun 1988)

The Government may require the delivery of the numbered line item, identified in the Schedule as an option item, in the quantity and at the price stated in the Schedule. The Contracting Officer may exercise the option by written notice to the Contractor at any time after Notice to Proceed and up to three days prior to the estimated completion date of the basic line item or any previously exercised option. Delivery of added items shall continue at the same rate that like items are called for under the contract, unless the parties otherwise agree.

(End of clause)

Postaward Small Business Program Rerepresentation (Jan 2025)

(a) Definitions. As used in this clause-

Long-term contract means a contract of more than five years in duration, including options. However, the term does not include contracts that exceed five years in duration because the period of performance has been extended for a cumulative period not to exceed six months under the clause at 52.217-8, Option to Extend Services, or other appropriate authority.

Small business concern-

(1) Means a concern, including its affiliates, that is independently owned and operated, not dominant in its field of operation, and qualified as a small business under the criteria in 13 CFR part 121 and the size standard in paragraph (d) of this clause.

(2) Affiliates, as used in this definition, means business concerns, one of whom directly or indirectly controls or has the power to control the others, or a third party or parties control or have the power to control the others. In determining whether affiliation exists, consideration is given to all appropriate factors including common ownership, common management, and contractual relationships. SBA determines affiliation based on the factors set forth at 13 CFR 121.103.

(b) If the Contractor represented that it was any of the small business concerns identified in 19.000(a)(3) prior to award of this contract, the Contractor shall rerepresent its size and socioeconomic status according to paragraph (f) of this clause or, if applicable, paragraph (h) of this clause, upon occurrence of any of the following:

(1) Within 30 days after execution of a novation agreement or within 30 days after modification of the contract to include this clause, if the novation agreement was executed prior to inclusion of this clause in the contract.

(2) Within 30 days after a merger or acquisition that does not require a novation or within 30 days after modification of the contract to include this clause, if the merger or acquisition occurred prior to inclusion of this clause in the contract.

(3) For long-term contracts-

(i) Within 60 to 120 days prior to the end of the fifth year of the contract; and

(ii) Within 60 to 120 days prior to the date specified in the contract for exercising any option thereafter.

(c) If the Contractor represented its status as any of the small business concerns identified at 19.000(a)(3) prior to award of this contract, the Contractor shall rerepresent its size and socioeconomic status according to paragraph (f) of this clause or, if applicable, paragraph (h) of this clause, for the NAICS code assigned to an order (except that paragraphs (c)(1) through (3) of this clause do not apply to an order issued under a Federal Supply Schedule contract at subpart 8.4)-

(1) Set aside exclusively for a small business concern identified at 19.000(a)(3) that is issued under an unrestricted multiple-award contract, unless the order is issued under the reserved portion of an unrestricted multiple-award contract (e.g., an order set aside for a woman-owned small business under a multiple-award contract that is not set-aside, unless the order is issued under the reserved portion of the multiple-award contract);

(2) Issued under a multiple-award contract set aside for small businesses that is further set aside for a specific socioeconomic category that differs from the underlying multiple-award contract (e.g., an order set aside for a HUBZone small business concern under a multiple-award contract that is set aside for small businesses);

(3) Issued under the part of the multiple-award contract that is set aside for small businesses that is further set aside for a specific socioeconomic category that differs from the underlying set-aside part of the multiple-award contract (e.g., an order set aside for a WOSB concern under the part of the multiple-award contract that is partially set aside for small businesses); and

(4) When the Contracting Officer explicitly requires it for an order issued under a multiple-award contract, including for an order issued under a Federal Supply Schedule contract (see 8.405-5(b) and 19.301-2(b)(2)).

(d) The Contractor shall rerepresent its size status in accordance with the size standard in effect at the time of this rerepresentation that corresponds to the North American Industry Classification System (NAICS) code(s) assigned to this contract. The small business size standard corresponding to this NAICS code(s) can be found at <https://www.sba.gov/document/support--table-size-standards>.

(e) The small business size standard for a Contractor providing an end item that it does not manufacture, process, or produce itself, for a contract other than a construction or service contract, is 500 employees, or 150 employees for information technology value-added resellers under NAICS code 541519, if the acquisition-

(1) Was set aside for small business and has a value above the simplified acquisition threshold;

(2) Used the HUBZone price evaluation preference regardless of dollar value, unless the Contractor waived the price evaluation preference; or

(3) Was an 8(a), HUBZone, service-disabled veteran-owned, economically disadvantaged women-owned, or women-owned small business set-aside or sole-source award regardless of dollar value.

(f) Except as provided in paragraph (h) of this clause, the Contractor shall make the representation(s) required by paragraphs (b) and (c) of this clause by validating or updating all its representations in the Representations and Certifications section of the System for Award Management (SAM) and its other data in SAM, as necessary, to ensure that they reflect the Contractor's current status. The Contractor shall notify the contracting officer in writing within the timeframes specified in paragraph (b) of this clause, or with its offer for an order (see paragraph (c) of this clause), that the data have been validated or updated, and provide the date of the validation or update.

(g) If the Contractor represented that it was other than a small business concern prior to award of this contract, the Contractor may, but is not required to, take the actions required by paragraphs (f) or (h) of this clause.

(h) If the Contractor does not have representations and certifications in SAM, or does not have a representation in SAM for the NAICS code applicable to this contract, the Contractor is required to complete the following rerepresentation and submit it to the contracting office, along with the contract number and the date on which the rerepresentation was completed:

(1) The Contractor represents that it ☐ is, ☐ is not a small business concern under ____NAICS Code assigned to ____contract number.

(2) [Complete only if the Contractor represented itself as a small business concern in paragraph (h)(1) of this clause.] The Contractor represents that it ☐ is, ☐ is not, a small disadvantaged business concern as defined in 13 CFR 124.1001.

(3) [Complete only if the Contractor represented itself as a small business concern in paragraph (h)(1) of this clause.] The Contractor represents that it ☐ is, ☐ is not a women-owned small business concern.

(4) Women-owned small business (WOSB) joint venture eligible under the WOSB Program. The Contractor represents that it ☐ is, ☐ is not a joint venture that complies with the requirements of 13 CFR 127.506(a) through (c). [____The Contractor shall enter the name and unique entity identifier of each party to the joint venture: ____.]

(5) Economically disadvantaged women-owned small business (EDWOSB) joint venture. The Contractor represents that it ☐ is, ☐ is not a joint venture that complies with the requirements of 13 CFR 127.506(a) through (c). [____The Contractor shall enter the name and unique entity identifier of each party to the joint venture: ____.]

(6) [Complete only if the Contractor represented itself as a small business concern in paragraph (h)(1) of this clause.] The Contractor represents that it ☐ is, ☐ is not a veteran-owned small business concern.

(7) [Complete only if the Contractor represented itself as a veteran-owned small business concern in paragraph (h)(6) of this clause.] The Contractor represents that it ☐ is, ☐ is not a service-disabled veteran-owned small business concern.

(8) Service-disabled veteran-owned small business (SDVOSB) joint venture eligible under the SDVOSB Program. The Contractor represents that it ☐ is, ☐ is not an SDVOSB joint venture eligible under the SDVOSB Program that complies with the requirements of 13 CFR 128.402. [____The Contractor shall enter the name and unique entity identifier of each party to the joint venture: ____.]

(9) [Complete only if the Contractor represented itself as a small business concern in paragraph (h)(1) of this clause.] The Contractor represents that-

(i) It ☐ is, ☐ is not a HUBZone small business concern listed, on the date of this representation, on the List of Qualified HUBZone Small Business Concerns maintained by the Small Business Administration, and no material changes in ownership and control, principal office, or HUBZone employee percentage have occurred since it was certified in accordance with 13 CFR part 126; and

(ii) It ☐ is, ☐ is not a HUBZone joint venture that complies with the requirements of 13 CFR part 126, and the representation in paragraph (h)(8)(i) of this clause is accurate for each HUBZone small business concern participating in the HUBZone joint venture. [____The Contractor shall enter the names of each of the HUBZone small business concerns participating in the HUBZone joint venture: .] Each HUBZone small business concern participating in the HUBZone joint venture shall submit a separate signed copy of the HUBZone representation.

[____Contractor to sign and date and insert authorized signer's name and title.]

(End of clause)

52.219-28 Postaward Small Business Program Rerepresentation. (Alternate I)

(Jan 2025) Alternate I (Mar 2020)

Alternate I (Mar 2020). As prescribed in 19.309 (c)(2), substitute the following paragraph (h)(1) for paragraph (h)(1) of the basic clause:

(h)(1) The Contractor represents its small business size status for each one of the NAICS codes assigned to this contract.

NAICS Code	Small business concern (yes/no)
237990	

(End Clause)

52.222-40 Notification of Employee Rights Under the National Labor Relations Act.

(Dec 2010)

Notification of Employee Rights Under the National Labor Relations Act (Dec 2010)

(a) During the term of this contract, the Contractor shall post an employee notice, of such size and in such form, and containing such content as prescribed by the Secretary of Labor, in conspicuous places in and about its plants and offices where employees covered by the National Labor Relations Act engage in activities relating to the performance of the contract, including all places where notices to employees are customarily posted both physically and electronically, in the languages employees speak, in accordance with 29 CFR 471.2 (d) and (f).

(1) Physical posting of the employee notice shall be in conspicuous places in and about the Contractor's plants and offices so that the notice is prominent and readily seen by employees who are covered by the National Labor Relations Act and engage in activities related to the performance of the contract.

(2) If the Contractor customarily posts notices to employees electronically, then the Contractor shall also post the required notice electronically by displaying prominently, on any website that is maintained by the Contractor and is customarily used for notices to employees about terms and conditions of employment, a link to the Department of Labor's website that contains the full text of the poster. The link to the Department's website, as referenced in (b)(3) of this section, must read, "Important Notice about Employee Rights to Organize and Bargain Collectively with Their Employers."

(b) This required employee notice, printed by the Department of Labor, may be-

(1) Obtained from the Division of Interpretations and Standards, Office of Labor-Management Standards, U.S. Department of Labor, 200 Constitution Avenue, NW., Room N-5609, Washington, DC 20210, (202) 693-0123, or from any field office of the Office of Labor-Management Standards or Office of Federal Contract Compliance Programs;

(2) Provided by the Federal contracting agency if requested;

(3) Downloaded from the Office of Labor-Management Standards Web site at <http://www.dol.gov/olms/regs/compliance/EO13496.htm>; or

(4) Reproduced and used as exact duplicate copies of the Department of Labor's official poster.

(c) The required text of the employee notice referred to in this clause is located at Appendix A, Subpart A, 29 CFR Part 471.

(d) The Contractor shall comply with all provisions of the employee notice and related rules, regulations, and orders of the Secretary of Labor.

(e) In the event that the Contractor does not comply with the requirements set forth in paragraphs (a) through (d) of this clause, this contract may be terminated or suspended in whole or in part, and the Contractor may be suspended or debarred in accordance with 29 CFR 471.14 and subpart 9.4. Such other sanctions or remedies may be imposed as are provided by 29 CFR part 471, which implements Executive Order 13496 or as otherwise provided by law.

(f) Subcontracts.

(1) The Contractor shall include the substance of this clause, including this paragraph (f), in every subcontract that exceeds \$10,000 and will be performed wholly or partially in the United States, unless exempted by the rules, regulations, or orders of the Secretary of Labor issued pursuant to section 3 of Executive Order 13496 of January 30, 2009, so that such provisions will be binding upon each subcontractor.

(2) The Contractor shall not procure supplies or services in a way designed to avoid the applicability of Executive Order 13496 or this clause.

(3) The Contractor shall take such action with respect to any such subcontract as may be directed by the Secretary of Labor as a means of enforcing such provisions, including the imposition of sanctions for noncompliance.

(4) However, if the Contractor becomes involved in litigation with a subcontractor, or is threatened with such involvement, as a result of such direction, the Contractor may request the United States, through the Secretary of Labor, to enter into such litigation to protect the interests of the United States.

(End of clause)

52.225-11 Buy American-Construction Materials under Trade Agreements.

(Nov 2023)

Buy American-Construction Materials under Trade Agreements (Nov 2023)

(a) Definitions. As used in this clause-

Caribbean Basin country construction material means a construction material that-

(1) Is wholly the growth, product, or manufacture of a Caribbean Basin country; or

(2) In the case of a construction material that consists in whole or in part of materials from another country, has been substantially transformed in a Caribbean Basin country into a new and different construction material distinct from the materials from which it was transformed.

Commercially available off-the-shelf (COTS) item-

(1) Means any item of supply (including construction material) that is-

- (i) A commercial product (as defined in paragraph (1) of the definition of "commercial product" at Federal Acquisition Regulation (FAR) 2.101);
 - (ii) Sold in substantial quantities in the commercial marketplace; and
 - (iii) Offered to the Government, under a contract or subcontract at any tier, without modification, in the same form in which it is sold in the commercial marketplace; and
- (2) Does not include bulk cargo, as defined in 46 U.S.C.40102(4), such as agricultural products and petroleum products.

Component means an article, material, or supply incorporated directly into a construction material.

Construction material means an article, material, or supply brought to the construction site by the Contractor or subcontractor for incorporation into the building or work. The term also includes an item brought to the site preassembled from articles, materials, or supplies. However, emergency life safety systems, such as emergency lighting, fire alarm, and audio evacuation systems, that are discrete systems incorporated into a public building or work and that are produced as complete systems, are evaluated as a single and distinct construction material regardless of when or how the individual parts or components of those systems are delivered to the construction site. Materials purchased directly by the Government are supplies, not construction material.

Cost of components means-

- (1) For components purchased by the Contractor, the acquisition cost, including transportation costs to the place of incorporation into the construction material (whether or not such costs are paid to a domestic firm), and any applicable duty (whether or not a duty-free entry certificate is issued); or
- (2) For components manufactured by the Contractor, all costs associated with the manufacture of the component, including transportation costs as described in paragraph (1) of this definition, plus allocable overhead costs, but excluding profit. Cost of components does not include any costs associated with the manufacture of the construction material.

Critical component means a component that is mined, produced, or manufactured in the United States and deemed critical to the U.S. supply chain. The list of critical components is at FAR 25.105.

Critical item means a domestic construction material or domestic end product that is deemed critical to U.S. supply chain resiliency. The list of critical items is at FAR 25.105.

Designated country means any of the following countries:

- (1) A World Trade Organization Government Procurement Agreement (WTO GPA) country (Armenia, Aruba, Australia, Austria, Belgium, Bulgaria, Canada, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hong Kong, Hungary, Iceland, Ireland, Israel, Italy, Japan, Korea (Republic of), Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Moldova, Montenegro, Netherlands, New Zealand, North Macedonia, Norway, Poland, Portugal, Romania, Singapore, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Taiwan, Ukraine, or United Kingdom);
- (2) A Free Trade Agreement (FTA) country (Australia, Bahrain, Chile, Colombia, Costa Rica, Dominican Republic, El Salvador, Guatemala, Honduras, Korea (Republic of), Mexico, Morocco, Nicaragua, Oman, Panama, Peru, or Singapore);
- (3) A least developed country (Afghanistan, Angola, Bangladesh, Benin, Bhutan, Burkina Faso, Burundi, Cambodia, Central African Republic, Chad, Comoros, Democratic Republic of Congo, Djibouti, Equatorial Guinea, Eritrea, Ethiopia, Gambia, Guinea, Guinea-Bissau, Haiti, Kiribati, Laos, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mozambique, Nepal, Niger, Rwanda, Samoa, Sao Tome and Principe, Senegal, Sierra Leone, Solomon Islands, Somalia, South Sudan, Tanzania, Timor-Leste, Togo, Tuvalu, Uganda, Vanuatu, Yemen, or Zambia); or
- (4) A Caribbean Basin country (Antigua and Barbuda, Aruba, Bahamas, Barbados, Belize, Bonaire, British Virgin Islands, Curacao, Dominica, Grenada, Guyana, Haiti, Jamaica, Montserrat, Saba, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines, Sint Eustatius, Sint Maarten, or Trinidad and Tobago).

Designated country construction material means a construction material that is a WTO GPA country construction material, an FTA country construction material, a least developed country construction material, or a Caribbean Basin country construction material.

Domestic construction material means-

- (1) For construction material that does not consist wholly or predominantly of iron or steel or a combination of both-
 - (i) An unmanufactured construction material mined or produced in the United States; or
 - (ii) A construction material manufactured in the United States, if-
 - (A) The cost of its components mined, produced, or manufactured in the United States exceeds 60 percent of the cost of all its components, except that the percentage will be 65 percent for items delivered in calendar years 2024 through 2028 and 75 percent for items delivered starting in calendar year 2029.

(B) The construction material is a COTS item; or

(2) For construction material that consists wholly or predominantly of iron or steel or a combination of both, a construction material manufactured in the United States if the cost of foreign iron and steel constitutes less than 5 percent of the cost of all components used in such construction material. The cost of foreign iron and steel includes but is not limited to the cost of foreign iron or steel mill products (such as bar, billet, slab, wire, plate, or sheet), castings, or forgings utilized in the manufacture of the construction material and a good faith estimate of the cost of all foreign iron or steel components excluding COTS fasteners. Iron or steel components of unknown origin are treated as foreign. If the construction material contains multiple components, the cost of all the materials used in such construction material is calculated in accordance with the definition of "cost of components".

Fastener means a hardware device that mechanically joins or affixes two or more objects together. Examples of fasteners are nuts, bolts, pins, rivets, nails, clips, and screws.

Foreign construction material means a construction material other than a domestic construction material.

Foreign iron and steel means iron or steel products not produced in the United States. Produced in the United States means that all manufacturing processes of the iron or steel must take place in the United States, from the initial melting stage through the application of coatings, except metallurgical processes involving refinement of steel additives. The origin of the elements of the iron or steel is not relevant to the determination of whether it is domestic or foreign.

Free Trade Agreement country construction material means a construction material that-

(1) Is wholly the growth, product, or manufacture of a Free Trade Agreement (FTA) country; or

(2) In the case of a construction material that consists in whole or in part of materials from another country, has been substantially transformed in a FTA country into a new and different construction material distinct from the materials from which it was transformed.

Least developed country construction material means a construction material that-

(1) Is wholly the growth, product, or manufacture of a least developed country; or

(2) In the case of a construction material that consists in whole or in part of materials from another country, has been substantially transformed in a least developed country into a new and different construction material distinct from the materials from which it was transformed.

Predominantly of iron or steel or a combination of both means that the cost of the iron and steel content exceeds 50 percent of the total cost of all its components. The cost of iron and steel is the cost of the iron or steel mill products (such as bar, billet, slab, wire, plate, or sheet), castings, or forgings utilized in the manufacture of the product and a good faith estimate of the cost of iron or steel components excluding COTS fasteners.

Steel means an alloy that includes at least 50 percent iron, between 0.02 and 2 percent carbon, and may include other elements.

United States means the 50 States, the District of Columbia, and outlying areas.

WTO GPA country construction material means a construction material that-

(1) Is wholly the growth, product, or manufacture of a WTO GPA country; or

(2) In the case of a construction material that consists in whole or in part of materials from another country, has been substantially transformed in a WTO GPA country into a new and different construction material distinct from the materials from which it was transformed.

(b) Construction materials.

(1) This clause implements 41 U.S.C. chapter 83, Buy American, by providing a preference for domestic construction material. In accordance with 41 U.S.C. 1907, the domestic content test of the Buy American statute is waived for construction material that is a COTS item, except that for construction material that consists wholly or predominantly of iron or steel or a combination of both, the domestic content test is applied only to the iron and steel content of the construction material, excluding COTS fasteners. (See FAR 12.505(a)(2)). In addition, the Contracting Officer has determined that the WTO GPA and Free Trade Agreements (FTAs) apply to this acquisition. Therefore, the Buy American restrictions are waived for designated country construction materials.

(2) The Contractor shall use only domestic or designated country construction material in performing this contract, except as provided in paragraphs (b)(3) and (b)(4) of this clause.

(3) The requirement in paragraph (b)(2) of this clause does not apply to information technology that is a commercial product or to the construction materials or components listed by the Government as follows:

_____ [Contracting Officer to list applicable excepted materials or indicate "none"]

(4) The Contracting Officer may add other foreign construction material to the list in paragraph (b)(3) of this clause if the Government determines that-

(i) The cost of domestic construction material would be unreasonable.

(A) For domestic construction material that is not a critical item or does not contain critical components.

(1) The cost of a particular domestic construction material subject to the restrictions of the Buy American statute is unreasonable when the cost of such material exceeds the cost of foreign material by more than 20 percent;

(2) For construction material that is not a COTS item and does not consist wholly or predominantly of iron or steel or a combination of both, if the cost of a particular domestic construction material is determined to be unreasonable or there is no domestic offer received, and the low offer is for foreign construction material that does not exceed 55 percent domestic content, the Contracting Officer will treat the lowest offer of foreign construction material that is manufactured in the United States and exceeds 55 percent domestic content as a domestic offer and determine whether the cost of that offer is unreasonable by applying the evaluation factor listed in paragraph (b)(4)(i)(A)(1) of this clause.

(3) The procedures in paragraph (b)(4)(i)(A)(2) of this clause will no longer apply as of January 1, 2030.

(B) For domestic construction material that is a critical item or contains critical components.

(1) The cost of a particular domestic construction material that is a critical item or contains critical components, subject to the requirements of the Buy American statute, is unreasonable when the cost of such material exceeds the cost of foreign material by more than 20 percent plus the additional preference factor identified for the critical item or construction material containing critical components listed at FAR 25.105.

(2) For construction material that does not consist wholly or predominantly of iron or steel or a combination of both, if the cost of a particular domestic construction material is determined to be unreasonable or there is no domestic offer received, and the low offer is for foreign construction material that does not exceed 55 percent domestic content, the Contracting Officer will treat the lowest offer of foreign construction material that is manufactured in the United States and exceeds 55 percent domestic content as a domestic offer, and determine whether the cost of that offer is unreasonable by applying the evaluation factor listed in paragraph (b)(4)(i)(B)(1) of this clause.

(3) The procedures in paragraph (b)(4)(i)(B)(2) of this clause will no longer apply as of January 1, 2030.

(ii) The application of the restriction of the Buy American Act to a particular construction material would be impracticable or inconsistent with the public interest; or

(iii) The construction material is not mined, produced, or manufactured in the United States in sufficient and reasonably available commercial quantities of a satisfactory quality.

(c) Request for determination of inapplicability of the Buy American statute.

(1)

(i) Any Contractor request to use foreign construction material in accordance with paragraph (b)(4) of this clause shall include adequate information for Government evaluation of the request, including-

(A) A description of the foreign and domestic construction materials;

(B) Unit of measure;

(C) Quantity;

(D) Price;

(E) Time of delivery or availability;

(F) Location of the construction project;

(G) Name and address of the proposed supplier; and

(H) A detailed justification of the reason for use of foreign construction materials cited in accordance with paragraph (b)(3) of this clause.

(ii) A request based on unreasonable cost shall include a reasonable survey of the market and a completed price comparison table in the format in paragraph (d) of this clause.

(iii) The price of construction material shall include all delivery costs to the construction site and any applicable duty (whether or not a duty-free certificate may be issued).

(iv) Any Contractor request for a determination submitted after contract award shall explain why the Contractor could not reasonably foresee the need for such determination and could not have requested the determination before contract award. If the Contractor does not submit a satisfactory explanation, the Contracting Officer need not make a determination.

(2) If the Government determines after contract award that an exception to the Buy American statute applies and the Contracting Officer and the Contractor negotiate adequate consideration, the Contracting Officer will modify the contract to allow use of the foreign construction material. However, when the basis for the exception is the unreasonable price of a domestic construction material, adequate consideration is not less than the

differential established in paragraph (b)(4)(i) of this clause.

(3) Unless the Government determines that an exception to the Buy American statute applies, use of foreign construction material is noncompliant with the Buy American statute.

(d) Data. To permit evaluation of requests under paragraph (c) of this clause based on unreasonable cost, the Contractor shall include the following information and any applicable supporting data based on the survey of suppliers:

Foreign and Domestic Construction Materials Price Comparison			
Construction Material Description	Unit of Measure	Quantity	Price (Dollars)*
Item1			
Foreign construction material	_____	_____	_____
Domestic construction material	_____	_____	_____
Item1			
Foreign construction material	_____	_____	_____
Domestic construction material	_____	_____	_____

[* Include all delivery costs to the construction site and any applicable duty (whether or not a duty-free entry certificate is issued)].

[List name, address, telephone number, and contact for suppliers surveyed. Attach copy of response; if oral, attach summary.]

[Include other applicable supporting information.]

(End of clause)

52.228-1 Bid Guarantee.

(Sep 1996)

Bid Guarantee (Sept 1996)

(a) Failure to furnish a bid guarantee in the proper form and amount, by the time set for opening of bids, may be cause for rejection of the bid.

(b) The bidder shall furnish a bid guarantee in the form of a firm commitment, e.g., bid bond supported by good and sufficient surety or sureties acceptable to the Government, postal money order, certified check, cashier's check, irrevocable letter of credit, or, under Treasury Department regulations, certain bonds or notes of the United States. The Contracting Officer will return bid guarantees, other than bid bonds-

(1) To unsuccessful bidders as soon as practicable after the opening of bids; and

(2) To the successful bidder upon execution of contractual documents and bonds (including any necessary coinsurance or reinsurance agreements), as required by the bid as accepted.

(c) The amount of the bid guarantee shall be 20% percent of the bid price or \$3,000,000.00 whichever is less.

(d) If the successful bidder, upon acceptance of its bid by the Government within the period specified for acceptance, fails to execute all contractual documents or furnish executed bond(s) within 10 days after receipt of the forms by the bidder, the Contracting Officer may terminate the contract for default.

(e) In the event the contract is terminated for default, the bidder is liable for any cost of acquiring the work that exceeds the amount of its bid, and the bid guarantee is available to offset the difference.

(End of clause)

52.228-15 Performance and Payment Bonds-Construction.

(Jun 2020)

Performance and Payment Bonds-Construction (Jun 2020)

(a) Definitions. As used in this clause-

Original contract price means the award price of the contract; or, for requirements contracts, the price payable for the estimated total quantity; or, for indefinite-quantity contracts, the price payable for the specified minimum quantity. Original contract price does not include the price of any options, except those options exercised at the time of contract award.

(b) Amount of required bonds. Unless the resulting contract price is valued at or below the threshold specified in Federal Acquisition Regulation 28.102-1(a) on the date of award of this contract, the successful offeror shall furnish performance and payment bonds to the Contracting Officer as follows:

(1) Performance bonds (Standard Form 25). The penal amount of performance bonds at the time of contract award shall be 100 percent of the original contract price.

(2) Payment Bonds (Standard Form 25A). The penal amount of payment bonds at the time of contract award shall be 100 percent of the original contract price.

(3) Additional bond protection.

(i) The Government may require additional performance and payment bond protection if the contract price is increased. The increase in protection generally will equal 100 percent of the increase in contract price.

(ii) The Government may secure the additional protection by directing the Contractor to increase the penal amount of the existing bond or to obtain an additional bond.

(c) Furnishing executed bonds. The Contractor shall furnish all executed bonds, including any necessary reinsurance agreements, to the Contracting Officer, within the time period specified in the Bid Guarantee provision of the solicitation, or otherwise specified by the Contracting Officer, but in any event, before starting work.

(d) Surety or other security for bonds. The bonds shall be in the form of firm commitment, supported by corporate sureties whose names appear on the list contained in Treasury Department Circular 570, individual sureties, or by other acceptable security such as postal money order, certified check, cashier's check, irrevocable letter of credit, or, in accordance with Treasury Department regulations, certain bonds or notes of the United States. Treasury Circular 570 is published in the Federal Register or may be obtained from the:

U.S. Department of the Treasury,

Financial Management,

Service Surety Bond Branch,

3700 East West Highway,

Room 6 F01,

Hyattsville, MD 20782.

Or via the internet at <http://www.fms.treas.gov/c570/>.

(e) Notice of subcontractor waiver of protection (40 U.S.C. 3133(c)). Any waiver of the right to sue on the payment bond is void unless it is in writing, signed by the person whose right is waived, and executed after such person has first furnished labor or material for use in the performance of the contract.

(End of clause)

52.229-12 Tax on Certain Foreign Procurements.

(Feb 2021)

(a) Definitions. As used in this clause--

Foreign person means any person other than a United States person.

United States person, as defined in 26 U.S.C. 7701(a)(30), means--

(1) A citizen or resident of the United States;

(2) A domestic partnership;

(3) A domestic corporation;

(4) Any estate (other than a foreign estate, within the meaning of 26 U.S.C. 7701(a)(31)); and

(5) Any trust if--

(i) A court within the United States is able to exercise primary supervision over the administration of the trust; and

(ii) One or more United States persons have the authority to control all substantial decisions of the trust.

(b) This clause applies only to foreign persons. It implements 26 U.S.C. 5000C and its implementing regulations at 26 CFR 1.5000C-1 through 1.5000C-7.

(c)(1) If the Contractor is a foreign person and has only a partial or no exemption to the withholding, the Contractor shall include the Department of the Treasury Internal Revenue Service Form W-14, Certificate of Foreign Contracting Party Receiving Federal Procurement Payments, with each voucher or invoice submitted under this contract throughout the period in which this status is applicable. The excise tax withholding is applied at the payment level, not at the contract level. The Contractor should revise each IRS Form W-14 submission to reflect the exemption (if any) that applies to that particular invoice, such as a different exemption applying. In the absence of a completed IRS Form W-14 accompanying a payment request, the default withholding percentage is 2 percent for the section 5000C withholding for that payment request. Information about IRS Form W-14 and its separate instructions is available via the internet at www.irs.gov/w14.

(2) If the Contractor is a foreign person and has indicated in its offer in the provision 52.229-11, Tax on Certain Foreign Procurements--Notice and Representation, that it is fully exempt from the withholding, and certified the full exemption on the IRS Form W-14, and if that full exemption no longer applies due to a change in circumstances during the performance of the contract that causes the Contractor to become subject to the withholding for the 2 percent excise tax then the Contractor shall--

(i) Notify the Contracting Officer within 30 days of a change in circumstances that causes the Contractor to be subject to the excise tax withholding under 26 U.S.C. 5000C; and

(ii) Comply with paragraph (c)(1) of this clause.

(d) The Government will withhold a full 2 percent of each payment unless the Contractor claims an exemption. If the Contractor enters a ratio in Line 12 of the IRS Form W-14, the result of Line 11 divided by Line 10, the Government will withhold from each payment an amount equal to 2 percent multiplied by the contract ratio. If the Contractor marks box 9 of the IRS Form W-14 (rather than completes Lines 10 through 12), the Contractor must identify and enter the specific exempt and nonexempt amounts in Line 15 of the IRS Form W-14; the Government will then withhold 2 percent only from the nonexempt amount. See the IRS Form W-14 and its instructions.

(e) Exemptions from the withholding under this clause are described at 26 CFR 1.5000C-1(d)(5) through (7). Any exemption claimed and self-certified on the IRS Form W-14 is subject to audit by the IRS. Any disputes regarding the imposition and collection of the 26 U.S.C. 5000C tax are adjudicated by the IRS as the 26 U.S.C. 5000C tax is a tax matter, not a contract issue.

(f) Taxes imposed under 26 U.S.C. 5000C may not be--

(1) Included in the contract price; nor

(2) Reimbursed.

(g) A taxpayer may, for a fee, seek advice from the Internal Revenue Service (IRS) as to the proper tax treatment of a transaction. This is called a private letter ruling. Also, the IRS may publish a revenue ruling, which is an official interpretation by the IRS of the Internal Revenue Code, related statutes, tax treaties, and regulations. A revenue ruling is the conclusion of the IRS on how the law is applied to a specific set of facts. For questions relating to the interpretation of the IRS regulations go to <https://www.irs.gov/help/tax-law-questions>.

(End of clause)

52.232-5 Payments under Fixed-Price Construction Contracts.

(May 2014)

Payments under Fixed-Price Construction Contracts (May 2014)

(a) Payment of price. The Government shall pay the Contractor the contract price as provided in this contract.

(b) Progress payments. The Government shall make progress payments monthly as the work proceeds, or at more frequent intervals as determined by the Contracting Officer, on estimates of work accomplished which meets the standards of quality established under the contract, as approved by the Contracting Officer.

(1) The Contractor's request for progress payments shall include the following substantiation:

(i) An itemization of the amounts requested, related to the various elements of work required by the contract covered by the payment requested.

(ii) A listing of the amount included for work performed by each subcontractor under the contract.

(iii) A listing of the total amount of each subcontract under the contract.

(iv) A listing of the amounts previously paid to each such subcontractor under the contract.

(v) Additional supporting data in a form and detail required by the Contracting Officer.

(2) In the preparation of estimates, the Contracting Officer may authorize material delivered on the site and preparatory work done to be taken into consideration. Material delivered to the Contractor at locations other than the site also may be taken into consideration if-

(i) Consideration is specifically authorized by this contract; and

(ii) The Contractor furnishes satisfactory evidence that it has acquired title to such material and that the material will be used to perform this contract.

(c) Contractor certification. Along with each request for progress payments, the Contractor shall furnish the following certification, or payment shall not be made: (However, if the Contractor elects to delete paragraph (c)(4) from the certification, the certification is still acceptable.)

I hereby certify, to the best of my knowledge and belief, that-

(1) The amounts requested are only for performance in accordance with the specifications, terms, and conditions of the contract;

(2) All payments due to subcontractors and suppliers from previous payments received under the contract have been made, and timely payments will be made from the proceeds of the payment covered by this certification, in accordance with subcontract agreements and the requirements of Chapter 39 of Title 31, United States Code;

(3) This request for progress payments does not include any amounts which the prime contractor intends to withhold or retain from a subcontractor or supplier in accordance with the terms and conditions of the subcontract; and

(4) This certification is not to be construed as final acceptance of a subcontractor's performance.

____(Name)

____(Title)

____(Date)

(d) Refund of unearned amounts. If the Contractor, after making a certified request for progress payments, discovers that a portion or all of such request constitutes a payment for performance by the Contractor that fails to conform to the specifications, terms, and conditions of this contract (hereinafter referred to as the "unearned amount"), the Contractor shall-

(1) Notify the Contracting Officer of such performance deficiency; and

(2) Be obligated to pay the Government an amount (computed by the Contracting Officer in the manner provided in paragraph (j) of this clause) equal to interest on the unearned amount from the 8th day after the date of receipt of the unearned amount until-

(i) The date the Contractor notifies the Contracting Officer that the performance deficiency has been corrected; or

(ii) The date the Contractor reduces the amount of any subsequent certified request for progress payments by an amount equal to the unearned amount.

(e) Retainage. If the Contracting Officer finds that satisfactory progress was achieved during any period for which a progress payment is to be made, the Contracting Officer shall authorize payment to be made in full. However, if satisfactory progress has not been made, the Contracting Officer may retain a maximum of 10 percent of the amount of the payment until satisfactory progress is achieved. When the work is substantially complete, the Contracting Officer may retain from previously withheld funds and future progress payments that amount the Contracting Officer considers adequate for protection of the Government and shall release to the Contractor all the remaining withheld funds. Also, on completion and acceptance of each separate building, public work, or other division of the contract, for which the price is stated separately in the contract, payment shall be made for the completed work without retention of a percentage.

(f) Title, liability, and reservation of rights. All material and work covered by progress payments made shall, at the time of payment, become the sole property of the Government, but this shall not be construed as-

(1) Relieving the Contractor from the sole responsibility for all material and work upon which payments have been made or the restoration of any damaged work; or

(2) Waiving the right of the Government to require the fulfillment of all of the terms of the contract.

(g) Reimbursement for bond premiums. In making these progress payments, the Government shall, upon request, reimburse the Contractor for the amount of premiums paid for performance and payment bonds (including coinsurance and reinsurance agreements, when applicable) after the Contractor has furnished evidence of full payment to the surety. The retainage provisions in paragraph (e) of this clause shall not apply to that portion of progress payments attributable to bond premiums.

(h) Final payment. The Government shall pay the amount due the Contractor under this contract after-

(1) Completion and acceptance of all work;

(2) Presentation of a properly executed voucher; and

(3) Presentation of release of all claims against the Government arising by virtue of this contract, other than claims, in stated amounts, that the Contractor has specifically excepted from the operation of the release. A release may also be required of the assignee if the Contractor's claim to amounts payable under this contract has been assigned under the Assignment of Claims Act of 1940 (31 U.S.C.3727 and 41 U.S.C.6305).

(i) Limitation because of undefinitized work. Notwithstanding any provision of this contract, progress payments shall not exceed 80 percent on work accomplished on undefinitized contract actions. A "contract action" is any action resulting in a contract, as defined in FAR subpart 2.1, including contract modifications for additional supplies or services, but not including contract modifications that are within the scope and under the terms of the contract, such as contract modifications issued pursuant to the Changes clause, or funding and other administrative changes.

(j) Interest computation on unearned amounts. In accordance with 31 U.S.C.3903(c)(1), the amount payable under paragraph (d)(2) of this clause shall be-

(1) Computed at the rate of average bond equivalent rates of 91-day Treasury bills auctioned at the most recent auction of such bills prior to the date the Contractor receives the unearned amount; and

(2) Deducted from the next available payment to the Contractor.

(End of clause)

52.232-8 Discounts for Prompt Payment.

(Feb 2002)

Discounts for Prompt Payment (Feb 2002)

(a) Discounts for prompt payment will not be considered in the evaluation of offers. However, any offered discount will form a part of the award, and will be taken if payment is made within the discount period indicated in the offer by the offeror. As an alternative to offering a discount for prompt payment in conjunction with the offer, offerors awarded contracts may include discounts for prompt payment on individual invoices.

(b) In connection with any discount offered for prompt payment, time shall be computed from the date of the invoice. If the Contractor has not placed a date on the invoice, the due date shall be calculated from the date the designated billing office receives a proper invoice, provided the agency annotates such invoice with the date of receipt at the time of receipt. For the purpose of computing the discount earned, payment shall be considered to have been made on the date that appears on the payment check or, for an electronic funds transfer, the specified payment date. When the discount date falls on a Saturday, Sunday, or legal holiday when Federal Government offices are closed and Government business is not expected to be conducted, payment may be made on the following business day.

(End of clause)

52.232-27 Prompt Payment for Construction Contracts.

(Jan 2017)

Prompt Payment for Construction Contracts (Jan 2017)

Notwithstanding any other payment terms in this contract, the Government will make invoice payments under the terms and conditions specified in this clause. The Government considers payment as being made on the day a check is dated or the date of an electronic funds transfer. Definitions of pertinent terms are set forth in sections 2.101, 32.001, and 32.902 of the Federal Acquisition Regulation. All days referred to in this clause are calendar days, unless otherwise specified. (However, see paragraph (a)(3) concerning payments due on Saturdays, Sundays, and legal holidays.)

(a) Invoice payments-

(1) Types of invoice payments. For purposes of this clause, there are several types of invoice payments that may occur under this contract, as follows:

(i) Progress payments, if provided for elsewhere in this contract, based on Contracting Officer approval of the estimated amount and value of work or services performed, including payments for reaching milestones in any project.

(A) The due date for making such payments is 14 days after the designated billing office receives a proper payment request. If the designated billing office fails to annotate the payment request with the actual date of receipt at the time of receipt, the payment due date is the 14th day after the date of the Contractor's payment request, provided the designated billing office receives a proper payment request and there is no disagreement over quantity, quality, or Contractor compliance with contract requirements.

(B) The due date for payment of any amounts retained by the Contracting Officer in accordance with the clause at 52.232-5, Payments Under Fixed-Price Construction Contracts, is as specified in the contract or, if not specified, 30 days after approval by the Contracting Officer for release to the Contractor.

(ii) Final payments based on completion and acceptance of all work and presentation of release of all claims against the Government arising by virtue of the contract, and payments for partial deliveries that have been accepted by the Government (e.g., each separate building, public work, or other division of the contract for which the price is stated separately in the contract).

(A) The due date for making such payments is the later of the following two events:

(1) The 30th day after the designated billing office receives a proper invoice from the Contractor.

(2) The 30th day after Government acceptance of the work or services completed by the Contractor. For a final invoice when the payment amount is subject to contract settlement actions (e.g., release of claims), acceptance is deemed to occur on the effective date of the contract settlement.

(B) If the designated billing office fails to annotate the invoice with the date of actual receipt at the time of receipt, the invoice payment due date is the 30th day after the date of the Contractor's invoice, provided the designated billing office receives a proper invoice and there is no disagreement over quantity, quality, or Contractor compliance with contract requirements.

(2) Contractor's invoice. The Contractor shall prepare and submit invoices to the designated billing office specified in the contract. A proper invoice must include the items listed in paragraphs (a)(2)(i) through (a)(2)(xi) of this clause. If the invoice does not comply with these requirements, the designated billing office must return it within 7 days after receipt, with the reasons why it is not a proper invoice. When computing any interest penalty owed the Contractor, the Government will take into account if the Government notifies the Contractor of an improper invoice in an untimely manner.

(i) Name and address of the Contractor.

(ii) Invoice date and invoice number. (The Contractor should date invoices as close as possible to the date of mailing or transmission.)

(iii) Contract number or other authorization for work or services performed (including order number and line item number).

(iv) Description of work or services performed.

(v) Delivery and payment terms (e.g., discount for prompt payment terms).

(vi) Name and address of Contractor official to whom payment is to be sent (must be the same as that in the contract or in a proper notice of assignment).

(vii) Name (where practicable), title, phone number, and mailing address of person to notify in the event of a defective invoice.

(viii) For payments described in paragraph (a)(1)(i) of this clause, substantiation of the amounts requested and certification in accordance with the requirements of the clause at 52.232-5, Payments Under Fixed-Price Construction Contracts.

(ix) Taxpayer Identification Number (TIN). The Contractor shall include its TIN on the invoice only if required elsewhere in this contract.

(x) Electronic funds transfer (EFT) banking information.

(A) The Contractor shall include EFT banking information on the invoice only if required elsewhere in this contract.

(B) If EFT banking information is not required to be on the invoice, in order for the invoice to be a proper invoice, the Contractor shall have submitted correct EFT banking information in accordance with the applicable solicitation provision (e.g., 52.232-38, Submission of Electronic Funds Transfer Information with Offer), contract clause (e.g., 52.232-33, Payment by Electronic Funds Transfer-System for Award Management, or 52.232-34, Payment by Electronic Funds Transfer-Other Than System for Award Management), or applicable agency procedures.

(C) EFT banking information is not required if the Government waived the requirement to pay by EFT.

(xi) Any other information or documentation required by the contract.

(3) Interest penalty. The designated payment office will pay an interest penalty automatically, without request from the Contractor, if payment is not made by the due date and the conditions listed in paragraphs (a)(3)(i) through (a)(3)(iii) of this clause are met, if applicable. However, when the due date falls on a Saturday, Sunday, or legal holiday, the designated payment office may make payment on the following working day without incurring a late payment interest penalty.

(i) The designated billing office received a proper invoice.

(ii) The Government processed a receiving report or other Government documentation authorizing payment and there was no disagreement over quantity, quality, Contractor compliance with any contract term or condition, or requested progress payment amount.

(iii) In the case of a final invoice for any balance of funds due the Contractor for work or services performed, the amount was not subject to further contract settlement actions between the Government and the Contractor.

(4) Computing penalty amount. The Government will compute the interest penalty in accordance with the Office of Management and Budget prompt payment regulations at 5 CFR Part 1315.

(i) For the sole purpose of computing an interest penalty that might be due the Contractor for payments described in paragraph (a)(1)(ii) of this clause, Government acceptance or approval is deemed to occur constructively on the 7th day after the Contractor has completed the work or services in accordance with the terms and conditions of the contract. If actual acceptance or approval occurs within the constructive acceptance or approval period, the Government will base the determination of an interest penalty on the actual date of acceptance or approval. Constructive acceptance or constructive approval requirements do not apply if there is a disagreement over quantity, quality, or Contractor compliance with a contract provision. These requirements also do not compel Government officials to accept work or services, approve Contractor estimates, perform contract administration functions, or make payment prior to fulfilling their responsibilities.

(ii) The prompt payment regulations at 5 CFR 1315.10(c) do not require the Government to pay interest penalties if payment delays are due to disagreement between the Government and the Contractor over the payment amount or other issues involving contract compliance, or on amounts temporarily withheld or retained in accordance with the terms of the contract. The Government and the Contractor shall resolve claims involving disputes, and any interest that may be payable in accordance with the clause at FAR 52.233-1, Disputes.

(5) Discounts for prompt payment. The designated payment office will pay an interest penalty automatically, without request from the Contractor, if the Government takes a discount for prompt payment improperly. The Government will calculate the interest penalty in accordance with the prompt payment regulations at 5 CFR Part 1315.

(6) Additional interest penalty.

(i) The designated payment office will pay a penalty amount, calculated in accordance with the prompt payment regulations at 5 CFR Part 1315 in addition to the interest penalty amount only if-

(A) The Government owes an interest penalty of \$1 or more;

(B) The designated payment office does not pay the interest penalty within 10 days after the date the invoice amount is paid; and

(C) The Contractor makes a written demand to the designated payment office for additional penalty payment, in accordance with paragraph (a)(6)(ii) of this clause, postmarked not later than 40 days after the date the invoice amount is paid.

(ii)

(A) The Contractor shall support written demands for additional penalty payments with the following data. The Government will not request any additional data. The Contractor shall-

(1) Specifically assert that late payment interest is due under a specific invoice, and request payment of all overdue late payment interest penalty and such additional penalty as may be required;

(2) Attach a copy of the invoice on which the unpaid late payment interest was due; and

(3) State that payment of the principal has been received, including the date of receipt.

(B) If there is no postmark or the postmark is illegible-

(1) The designated payment office that receives the demand will annotate it with the date of receipt provided the demand is received on or before the 40th day after payment was made; or

(2) If the designated payment office fails to make the required annotation, the Government will determine the demand's validity based on the date the Contractor has placed on the demand, provided such date is no later than the 40th day after payment was made.

(b) Contract financing payments. If this contract provides for contract financing, the Government will make contract financing payments in accordance with the applicable contract financing clause.

(c) Subcontract clause requirements. The Contractor shall include in each subcontract for property or services (including a material supplier) for the purpose of performing this contract the following:

(1) Prompt payment for subcontractors. A payment clause that obligates the Contractor to pay the subcontractor for satisfactory performance under its subcontract not later than 7 days from receipt of payment out of such amounts as are paid to the Contractor under this contract.

(2) Interest for subcontractors. An interest penalty clause that obligates the Contractor to pay to the subcontractor an interest penalty for each payment not made in accordance with the payment clause-

(i) For the period beginning on the day after the required payment date and ending on the date on which payment of the amount due is made; and

(ii) Computed at the rate of interest established by the Secretary of the Treasury, and published in the Federal Register, for interest payments under 41 U.S.C. 7109 in effect at the time the Contractor accrues the obligation to pay an interest penalty.

(3) Subcontractor clause flow down. A clause requiring each subcontractor to-

(i) Include a payment clause and an interest penalty clause conforming to the standards set forth in paragraphs (c)(1) and (c)(2) of this clause in each of its subcontracts; and

(ii) Require each of its subcontractors to include such clauses in their subcontracts with each lower-tier subcontractor or supplier.

(d) Subcontract clause interpretation. The clauses required by paragraph (c) of this clause shall not be construed to impair the right of the Contractor or a subcontractor at any tier to negotiate, and to include in their subcontract, provisions that-

(1) Retainage permitted. Permit the Contractor or a subcontractor to retain (without cause) a specified percentage of each progress payment otherwise due to a subcontractor for satisfactory performance under the subcontract without incurring any obligation to pay a late payment interest penalty, in accordance with terms and conditions agreed to by the parties to the subcontract, giving such recognition as the parties deem appropriate to the ability of a subcontractor to furnish a performance bond and a payment bond;

(2) Withholding permitted. Permit the Contractor or subcontractor to make a determination that part or all of the subcontractor's request for payment may be withheld in accordance with the subcontract agreement; and

(3) Withholding requirements. Permit such withholding without incurring any obligation to pay a late payment penalty if-

(i) A notice conforming to the standards of paragraph (g) of this clause previously has been furnished to the subcontractor; and

(ii) The Contractor furnishes to the Contracting Officer a copy of any notice issued by a Contractor pursuant to paragraph (d)(3)(i) of this clause.

(e) Subcontractor withholding procedures. If a Contractor, after making a request for payment to the Government but before making a payment to a subcontractor for the subcontractor's performance covered by the payment request, discovers that all or a portion of the payment otherwise due such subcontractor is subject to withholding from the subcontractor in accordance with the subcontract agreement, then the Contractor shall-

(1) Subcontractor notice. Furnish to the subcontractor a notice conforming to the standards of paragraph (g) of this clause as soon as practicable upon ascertaining the cause giving rise to a withholding, but prior to the due date for subcontractor payment;

(2) Contracting Officer notice. Furnish to the Contracting Officer, as soon as practicable, a copy of the notice furnished to the subcontractor pursuant to paragraph (e)(1) of this clause;

(3) Subcontractor progress payment reduction. Reduce the subcontractor's progress payment by an amount not to exceed the amount specified in the notice of withholding furnished under paragraph (e)(1) of this clause;

(4) Subsequent subcontractor payment. Pay the subcontractor as soon as practicable after the correction of the identified subcontract performance deficiency, and-

(i) Make such payment within-

(A) Seven days after correction of the identified subcontract performance deficiency (unless the funds therefor must be recovered from the Government because of a reduction under paragraph (e)(5)(i)) of this clause; or

(B) Seven days after the Contractor recovers such funds from the Government; or

(ii) Incur an obligation to pay a late payment interest penalty computed at the rate of interest established by the Secretary of the Treasury, and published in the Federal Register, for interest payments under 41 U.S.C. 7109 in effect at the time the Contractor accrues the obligation to pay an interest penalty;

(5) Notice to Contracting Officer. Notify the Contracting Officer upon-

(i) Reduction of the amount of any subsequent certified application for payment; or

(ii) Payment to the subcontractor of any withheld amounts of a progress payment, specifying-

(A) The amounts withheld under paragraph (e)(1) of this clause; and

(B) The dates that such withholding began and ended; and

(6) Interest to Government. Be obligated to pay to the Government an amount equal to interest on the withheld payments (computed in the manner provided in 31 U.S.C.3903(c)(1)), from the 8th day after receipt of the withheld amounts from the Government until-

(i) The day the identified subcontractor performance deficiency is corrected; or

(ii) The date that any subsequent payment is reduced under paragraph (e)(5)(i) of this clause.

(f) Third-party deficiency reports-

(1) Withholding from subcontractor. If a Contractor, after making payment to a first-tier subcontractor, receives from a supplier or subcontractor of the first-tier subcontractor (hereafter referred to as a "second-tier subcontractor") a written notice in accordance with 40 U.S.C. 3133, asserting a deficiency in such first-tier subcontractor's performance under the contract for which the Contractor may be ultimately liable, and the Contractor determines that all or a portion of future payments otherwise due such first-tier subcontractor is subject to withholding in accordance with the subcontract agreement, the Contractor may, without incurring an obligation to pay an interest penalty under paragraph (e)(6) of this clause-

(i) Furnish to the first-tier subcontractor a notice conforming to the standards of paragraph (g) of this clause as soon as practicable upon making such determination; and

(ii) Withhold from the first-tier subcontractor's next available progress payment or payments an amount not to exceed the amount specified in the notice of withholding furnished under paragraph (f)(1)(i) of this clause.

(2) Subsequent payment or interest charge. As soon as practicable, but not later than 7 days after receipt of satisfactory written notification that the identified subcontract performance deficiency has been corrected, the Contractor shall-

(i) Pay the amount withheld under paragraph (f)(1)(ii) of this clause to such first-tier subcontractor; or

(ii) Incur an obligation to pay a late payment interest penalty to such first-tier subcontractor computed at the rate of interest established by the Secretary of the Treasury, and published in the Federal Register, for interest payments under 41 U.S.C. 7109 in effect at the time the Contractor accrues the obligation to pay an interest penalty.

(g) Written notice of subcontractor withholding. The Contractor shall issue a written notice of any withholding to a subcontractor (with a copy furnished to the Contracting Officer), specifying-

(1) The amount to be withheld;

(2) The specific causes for the withholding under the terms of the subcontract; and

(3) The remedial actions to be taken by the subcontractor in order to receive payment of the amounts withheld.

(h) Subcontractor payment entitlement. The Contractor may not request payment from the Government of any amount withheld or retained in accordance with paragraph (d) of this clause until such time as the Contractor has determined and certified to the Contracting Officer that the subcontractor is entitled to the payment of such amount.

(i) Prime-subcontractor disputes. A dispute between the Contractor and subcontractor relating to the amount or entitlement of a subcontractor to a payment or a late payment interest penalty under a clause included in the subcontract pursuant to paragraph (c) of this clause does not constitute a dispute to which the Government is a party. The Government may not be interpleaded in any judicial or administrative proceeding involving such a dispute.

(j) Preservation of prime-subcontractor rights. Except as provided in paragraph (i) of this clause, this clause shall not limit or impair any contractual,

administrative, or judicial remedies otherwise available to the Contractor or a subcontractor in the event of a dispute involving late payment or nonpayment by the Contractor or deficient subcontract performance or nonperformance by a subcontractor.

(k) Non-recourse for prime contractor interest penalty. The Contractor's obligation to pay an interest penalty to a subcontractor pursuant to the clauses included in a subcontract under paragraph (c) of this clause shall not be construed to be an obligation of the Government for such interest penalty. A cost-reimbursement claim may not include any amount for reimbursement of such interest penalty.

(l) Overpayments. If the Contractor becomes aware of a duplicate contract financing or invoice payment or that the Government has otherwise overpaid on a contract financing or invoice payment, the Contractor shall-

(1) Remit the overpayment amount to the payment office cited in the contract along with a description of the overpayment including the-

(i) Circumstances of the overpayment (e.g., duplicate payment, erroneous payment, liquidation errors, date(s) of overpayment);

(ii) Affected contract number and delivery order number if applicable;

(iii) Affected line item or subline item, if applicable; and

(iv) Contractor point of contact.

(2) Provide a copy of the remittance and supporting documentation to the Contracting Officer.

(End of clause)

52.236-1 Performance of Work by the Contractor.

(Apr 1984)

The Contractor shall perform on the site, and with its own organization, work equivalent to at least twenty percent (20%) of the total amount of work to be performed under the contract. This percentage may be reduced by a supplemental agreement to this contract if, during performing the work, the Contractor requests a reduction and the Contracting Officer determines that the reduction would be to the advantage of the Government.

(End of clause)

52.236-4 Physical Data.

(Apr 1984)

Physical Data (Apr 1984)

Data and information furnished or referred to below is for the Contractor's information. The Government shall not be responsible for any interpretation of or conclusion drawn from the data or information by the Contractor.

(a) The indications of physical conditions on the drawings and in the specifications are the result of site investigations by surveys.

(b) Weather Conditions. Data on weather conditions may be obtained from the National Weather Service.

(c) Transportation Facilities. Floating plant may access the site of work via the Mississippi River and/or Breton Sound.

(d) Condition of Channel. Depths within the channel may be inadequate for the movement of floating plant due to serious shoaling. However, the Contractor should verify the required dimensions. Cross sections and/or dredging history of the channel are available for inspection at the Corps of Engineers District Office, New Orleans, Louisiana.

(e) Channel Traffic. Traffic in the channels consists of tugs, barge tows, commercial fishing boats, oil field access and small pleasure craft.

(f) Obstruction of Channel. The Government will not undertake to keep the channel free from vessels or other obstructions, except to the extent of such regulations, if any, as may be prescribed by the Secretary of the Army, in accordance with the provisions of Section 7 of the River and Harbor Act approved 8 August 1917. The Contractor shall conduct the work in such manner as to obstruct navigation as little as possible, and in case the Contractor's plant so obstructs the channel as to make difficult or endanger the passage of vessels, said plant shall be promptly moved on the approach of any vessel to such an extent as may be necessary to afford a practicable passage. Upon completion of the work, the Contractor shall promptly remove its plant, including ranges, buoys, piles and other markers placed by him/her under the contract in navigable water or on shore.

(g) Estimates of quantities involved in certain items of work for which bids are being solicited on a lump sum or job basis have been made for the use of the Government. Copies of these quantity estimates may be viewed or obtained by contacting the U.S. Army Engineer District, New Orleans, Corps of Engineers, Attention: Contracting Officer, CEMVN-CT, 7400 Leake Ave, New Orleans, Louisiana 70118-3651. It is expressly understood that the accuracy of these estimates is in no way warranted and that the furnishing of this information to a bidder will not relieve him of his responsibility to estimate the quantities involved.

(h) Stage Hydrograph. Real time stage hydrograph data of the Mississippi River Basin may be obtained from the New Orleans District website at:

(<http://rivergages.mvr.usace.army.mil/WaterControl/new/layout.cfm>)

The specific gages are as follows:

- Mississippi River at Venice – Gage # 01480
- Mississippi River at New Orleans – Gage # 01300

Historic hydrographic data provided does not constitute a prediction of future water stages.

(End of clause)

52.236-15 Schedules for Construction Contracts.

Apr 1984

(a) The Contractor shall, within five (5) days after the work commences on the contract or another period of time determined by the Contracting Officer, prepare and submit to the Contracting Officer for approval an electronic copy of a practicable schedule showing the order in which the Contractor proposes to perform the work, and the dates on which the Contractor contemplates starting and completing the several salient features of the work (including acquiring materials, plant, and equipment). The schedule shall be in the form of a progress chart of suitable scale to indicate appropriately the percentage of work scheduled for completion by any given date during the period. If the Contractor fails to submit a schedule within the time prescribed, the Contracting Officer may withhold approval of progress payments until the Contractor submits the required schedule.

(b) The Contractor shall enter the actual progress on the chart as directed by the Contracting Officer, and upon doing so shall immediately deliver three (3) copies of the annotated schedule to the Contracting Officer. If, in the opinion of the Contracting Officer, the Contractor falls behind the approved schedule, the Contractor shall take steps necessary to improve its progress, including those that may be required by the Contracting Officer, without additional cost to the Government. In this circumstance, the Contracting Officer may require the Contractor to increase the number of shifts, overtime operations, days of work, and/or the amount of construction plant, and to submit for approval any supplementary schedule or schedules in chart form as the Contracting Officer deems necessary to demonstrate how the approved rate of progress will be regained.

(c) Failure of the Contractor to comply with the requirements of the Contracting Officer under this clause shall be grounds for a determination by the Contracting Officer that the Contractor is not prosecuting the work with sufficient diligence to ensure completion within the time specified in the contract. Upon making this determination, the Contracting Officer may terminate the Contractor's right to proceed with the work, or any separable part of it, in accordance with the default terms of this contract.

(End of Clause)

NOTE: An electronic copy of the schedule as discussed in subparagraph (a) above, that is compatible with Primavera 6, shall also be provided.

52.236-16 ALTI Quantity Surveys

Apr 1984

(a) Quantity surveys shall be conducted, and the data derived from these surveys shall be used in computing the quantities of work performed and the actual construction completed and in place.

(b) The Contractor shall conduct the original and final surveys and surveys for any periods for which progress payments are requested. All these surveys shall be conducted under the direction of a representative of the Contracting Officer, unless the Contracting Officer waives this requirement in a specific instance. The Government shall make such computations as are necessary to determine the quantities of work performed or finally in place. The Contractor shall make the computations based on the surveys for any periods for which progress payments are requested.

(c) Promptly upon completing a survey, the Contractor shall furnish the originals of all field notes and all other records relating to the survey or to the layout of the work to the Contracting Officer, who shall use them as necessary to determine the amount of progress payments. The Contractor shall retain copies of all such material furnished to the Contracting Officer.

(End of Clause)

Note: In subparagraph (c), the word "Promptly" is defined as "within three (3) calendar days."

52.236-17 Layout of Work.

Apr 1984

The Contractor shall lay out its work from Government established baselines and bench marks indicated on the drawings, and shall be responsible for all measurements in connection with the layout. The Contractor shall furnish, at its own expense, all stakes, templates, platforms, equipment, tools, materials, and labor required to lay out any part of the work. The Contractor shall be responsible for executing the work to the lines and grades that may be established or indicated by the Contracting Officer. The Contractor shall also be responsible for maintaining and preserving all stakes and other marks established by the Contracting Officer until authorized to remove them. If such marks are destroyed by the Contractor or through its negligence before their removal is authorized, the Contracting Officer may replace them and deduct the expense of replacement from any amounts due or to become due to the Contractor.

(End of Clause)

NOTE: For clarification purposes, the term “baselines and benchmarks” shall refer to “centerline and gage(s)” to be used in this contract for horizontal and vertical control respectively. Gages established by the Government shall be re-established by the Contractor as deemed necessary in order to layout and control the work.

52.248-3 Value Engineering-Construction.

(Oct 2020)

Value Engineering-Construction (Oct 2020)

(a) General. The Contractor is encouraged to develop, prepare, and submit value engineering change proposals (VECP's) voluntarily. The Contractor shall share in any instant contract savings realized from accepted VECP's, in accordance with paragraph (f) of this clause.

(b) Definitions. "Collateral costs," as used in this clause, means agency costs of operation, maintenance, logistic support, or Government-furnished property.

Collateral savings, as used in this clause, means those measurable net reductions resulting from a VECP in the agency's overall projected collateral costs, exclusive of acquisition savings, whether or not the acquisition cost changes.

Contractor's development and implementation costs, as used in this clause, means those costs the Contractor incurs on a VECP specifically in developing, testing, preparing, and submitting the VECP, as well as those costs the Contractor incurs to make the contractual changes required by Government acceptance of a VECP.

Government costs, as used in this clause, means those agency costs that result directly from developing and implementing the VECP, such as any net increases in the cost of testing, operations, maintenance, and logistic support. The term does not include the normal administrative costs of processing the VECP.

Instant contract savings, as used in this clause, means the estimated reduction in Contractor cost of performance resulting from acceptance of the VECP, minus allowable Contractor's development and implementation costs, including subcontractors' development and implementation costs (see paragraph (h) of this clause).

Value engineering change proposal (VECP) means a proposal that-

(1) Requires a change to this, the instant contract, to implement; and

(2) Results in reducing the contract price or estimated cost without impairing essential functions or characteristics; provided, that it does not involve a change-

(i) In deliverable end item quantities only; or

(ii) To the contract type only.

(c) VECP preparation. As a minimum, the Contractor shall include in each VECP the information described in paragraphs (c)(1) through (7) of this clause. If the proposed change is affected by contractually required configuration management or similar procedures, the instructions in those procedures relating to format, identification, and priority assignment shall govern VECP preparation. The VECP shall include the following:

(1) A description of the difference between the existing contract requirement and that proposed, the comparative advantages and disadvantages of each, a justification when an item's function or characteristics are being altered, and the effect of the change on the end item's performance.

(2) A list and analysis of the contract requirements that must be changed if the VECP is accepted, including any suggested specification revisions.

(3) A separate, detailed cost estimate for (i) the affected portions of the existing contract requirement and (ii) the VECP. The cost reduction associated with the VECP shall take into account the Contractor's allowable development and implementation costs, including any amount attributable to subcontracts under paragraph (h) of this clause.

(4) A description and estimate of costs the Government may incur in implementing the VECP, such as test and evaluation and operating and support costs.

(5) A prediction of any effects the proposed change would have on collateral costs to the agency.

(6) A statement of the time by which a contract modification accepting the VECP must be issued in order to achieve the maximum cost reduction, noting any effect on the contract completion time or delivery schedule.

(7) Identification of any previous submissions of the VECP, including the dates submitted, the agencies and contract numbers involved, and previous Government actions, if known.

(d) Submission. The Contractor shall submit VECP's to the Resident Engineer at the worksite, with a copy to the Contracting Officer.

(e) Government action.

(1) The Contracting Officer will notify the Contractor of the status of the VECP within 45 calendar days after the contracting office receives it. If additional time is required, the Contracting Officer will notify the Contractor within the 45-day period and provide the reason for the delay and the expected date of the decision. The Government will process VECP's expeditiously; however, it will not be liable for any delay in acting upon a VECP.

(2) If the VECP is not accepted, the Contracting Officer will notify the Contractor in writing, explaining the reasons for rejection. The Contractor may withdraw any VECP, in whole or in part, at any time before it is accepted by the Government. The Contracting Officer may require that the Contractor provide written notification before undertaking significant expenditures for VECP effort.

(3) Any VECP may be accepted, in whole or in part, by the Contracting Officer's award of a modification to this contract citing this clause. The Contracting Officer may accept the VECP, even though an agreement on price reduction has not been reached, by issuing the Contractor a notice to proceed with the change. Until a notice to proceed is issued or a contract modification applies a VECP to this contract, the Contractor shall perform in accordance with the existing contract. The decision to accept or reject all or part of any VECP is a unilateral decision made solely at the discretion of the Contracting Officer.

(f) Sharing-

(1) Rates. The Government's share of savings is determined by subtracting Government costs from instant contract savings and multiplying the result by-

(i) 45 percent for fixed-price contracts; or

(ii) 75 percent for cost-reimbursement contracts.

(2) Payment. Payment of any share due the Contractor for use of a VECP on this contract shall be authorized by a modification to this contract to-

(i) Accept the VECP;

(ii) Reduce the contract price or estimated cost by the amount of instant contract savings; and

(iii) Provide the Contractor's share of savings by adding the amount calculated to the contract price or fee.

(g) Collateral savings. If a VECP is accepted, the Contracting Officer will increase the instant contract amount by 20 percent of any projected collateral savings determined to be realized in a typical year of use after subtracting any Government costs not previously offset. However, the Contractor's share of collateral savings will not exceed the contract's firm-fixed-price or estimated cost, at the time the VECP is accepted, or \$100,000, whichever is greater. The Contracting Officer is the sole determiner of the amount of collateral savings.

(h) Subcontracts. The Contractor shall include an appropriate value engineering clause in any subcontract of \$75,000 or more and may include one in subcontracts of lesser value. In computing any adjustment in this contract's price under paragraph (f) of this clause, the Contractor's allowable development and implementation costs shall include any subcontractor's allowable development and implementation costs clearly resulting from a VECP accepted by the Government under this contract, but shall exclude any value engineering incentive payments to a subcontractor. The Contractor may choose any arrangement for subcontractor value engineering incentive payments; provided, that these payments shall not reduce the Government's share of the savings resulting from the VECP.

(i) Data. The Contractor may restrict the Government's right to use any part of a VECP or the supporting data by marking the following legend on the affected parts:

These data, furnished under the Value Engineering-Construction clause of contract ____, shall not be disclosed outside the Government or duplicated, used, or disclosed, in whole or in part, for any purpose other than to evaluate a value engineering change proposal submitted under the clause. This restriction does not limit the Government's right to use information contained in these data if it has been obtained or is otherwise available from the Contractor or from another source without limitations.

If a VECP is accepted, the Contractor hereby grants the Government unlimited rights in the VECP and supporting data, except that, with respect to data qualifying and submitted as limited rights technical data, the Government shall have the rights specified in the contract modification implementing the VECP and shall appropriately mark the data. (The terms "unlimited rights" and "limited rights" are defined in part 27 of the Federal Acquisition Regulation.)

(End of clause)

52.252-2 Clauses Incorporated by Reference.

(Feb 1998)

Clauses Incorporated By Reference (Feb 1998)

This contract incorporates one or more clauses by reference, with the same force and effect as if they were given in full text. Upon request, the Contracting Officer will make their full text available. Also, the full text of a clause may be accessed electronically at this/these address(es):

<https://www.acquisition.gov/> _____ [Insert one or more Internet addresses]

(End of clause)

52.252-4 Alterations in Contract.

(Apr 1984)

Alterations in Contract (Apr 1984)

Portions of this contract are altered as follows:

(End of clause)

52.252-6 Authorized Deviations in Clauses.

(Nov 2020)

Authorized Deviations in Clauses (Nov 2020)

(a) The use in this solicitation or contract of any Federal Acquisition Regulation (48 CFR Chapter 1) clause with an authorized deviation is indicated by the addition of "(DEVIATION)" after the date of the clause.

(b) The use in this solicitation or contract of any Defense Federal Acquisition Regulation[insert regulation name] (48 CFR 2) clause with an authorized deviation is indicated by the addition of "(DEVIATION)" after the name of the regulation.

(End of clause)

DFARS Clauses Incorporated by Full Text

252.215-7008 Only One Offer.

(Dec 2022)

ONLY ONE OFFER (DEC 2022)

(a) Cost or pricing data requirements. After initial submission of offers, if the Contracting Officer notifies the Offeror that only one offer was received, the Offeror agrees to-

(1) Submit any additional cost or pricing data that is required in order to determine whether the price is fair and reasonable (10 U.S.C. 3705) or to comply with the statutory requirement for certified cost or pricing data (10 U.S.C. 3702 and FAR 15.403-3); and

(2) Except as provided in paragraph (b) of this provision, if the acquisition exceeds the certified cost or pricing data threshold and an exception to the requirement for certified cost or pricing data at FAR 15.403-1(b)(2) through (5) does not apply, certify all cost or pricing data in accordance with paragraph (c) of DFARS provision 252.215-7010, Requirements for Certified Cost or Pricing Data and Data Other Than Certified Cost or Pricing Data, of this solicitation.

(b) Canadian Commercial Corporation. If the Offeror is the Canadian Commercial Corporation, certified cost or pricing data are not required. If the Contracting Officer notifies the Canadian Commercial Corporation that additional data other than certified cost or pricing data are required in accordance with DFARS 225.870-4(c), the Canadian Commercial Corporation shall obtain and provide the following:

(1) Profit rate or fee (as applicable).

(2) Analysis provided by Public Works and Government Services Canada to the Canadian Commercial Corporation to determine a fair and reasonable price (comparable to the analysis required at FAR 15.404-1).

(3) Data other than certified cost or pricing data necessary to permit a determination by the U.S. Contracting Officer that the proposed price is fair and reasonable ____ [U.S. Contracting Officer to provide description of the data required in accordance with FAR 15.403-3(a)(1) with the notification].

(4) As specified in FAR 15.403-3(a)(4), an offeror who does not comply with a requirement to submit data that the U.S. Contracting Officer has deemed necessary to determine price reasonableness or cost realism is ineligible for award unless the head of the contracting activity determines that it is in the best interest of the Government to make the award to that offeror.

(c) Subcontracts. Unless the Offeror is the Canadian Commercial Corporation, the Offeror shall insert the substance of this provision, including this paragraph (c), in all subcontracts exceeding the simplified acquisition threshold defined in FAR part 2.

(End of provision)

252.236-7001 Contract Drawings and Specifications.

(Aug 2000)

CONTRACT DRAWINGS AND SPECIFICATIONS (AUG 2000)

(a) The Government will provide to the Contractor, without charge, one set of contract drawings and specifications, except publications incorporated into the technical provisions by reference, in electronic or paper media as chosen by the Contracting Officer.

(b) The Contractor shall-

- (1) Check all drawings furnished immediately upon receipt;
- (2) Compare all drawings and verify the figures before laying out the work;
- (3) Promptly notify the Contracting Officer of any discrepancies;
- (4) Be responsible for any errors that might have been avoided by complying with this paragraph (b); and
- (5) Reproduce and print contract drawings and specifications as needed.

(c) In general--

- (1) Large-scale drawings shall govern small-scale drawings; and
- (2) The Contractor shall follow figures marked on drawings in preference to scale measurements.

(d) Omissions from the drawings or specifications or the misdescription of details of work that are manifestly necessary to carry out the intent of the drawings and specifications, or that are customarily performed, shall not relieve the Contractor from performing such omitted or misdescribed details of the work. The Contractor shall perform such details as if fully and correctly set forth and described in the drawings and specifications.

(e) The work shall conform to the specifications and the contract drawings identified on the following index of drawings:

Title	File	Drawing No.
Mississippi River	H-16- 48731	G-01 thru G-03;
Outlets Vicinity of		C-01 thru C-11
Venice, LA Tiger Pass Bar Channel		
Maintenance Dredging Plaquemines Parish, LA		

(End of clause)

252.236-7002 Obstruction of Navigable Waterways.

Dec 1991

(a) The Contractor shall:

- (1) Promptly recover and remove any material, plant, machinery, or appliance which the Contractor loses, dumps, throws overboard, sinks, or misplaces, and which, in the opinion of the Contracting Officer, may be dangerous to or obstruct navigation;
- (2) Give immediate notice, with description and locations of any such obstructions, to the Contracting Officer; and
- (3) When required by the Contracting Officer, mark or buoy such obstructions until the same are removed.

(b) The Contracting Officer may:

- (1) Remove the obstructions by contract or otherwise should the Contractor refuse, neglect, or delay compliance with paragraph (a) of this clause; and
- (2) Deduct the cost of removal from any monies due or to become due to the Contractor; or
- (3) Recover the cost of removal under the Contractor's bond.

(c) The Contractor's liability for the removal of a vessel wrecked or sunk without fault or negligence is limited to that provided in Sections 15, 19 and 20 of the River and Harbor Act of March 3, 1899 (33 U.S.C.410 et.seq.).

(End of clause)

252.236-7004 Payment for Mobilization and Demobilization.

(Dec 1991)

PAYMENT FOR MOBILIZATION AND DEMOBILIZATION (DEC 1991)

(a) The Government will pay all costs for the mobilization and demobilization of all of the Contractor's plant and equipment at the contract lump sum price for this item.

(1) Sixty percent (60%) percent of the lump sum price upon completion of the contractor's mobilization at the work site.

(2) The remaining forty percent (40%) percent upon completion of demobilization.

(b) The Contracting Officer may require the Contractor to furnish cost data to justify this portion of the bid if the Contracting Officer believes that the percentages in paragraphs (a)(1) and (2) of this clause do not bear a reasonable relation to the cost of the work in this contract.

(1) Failure to justify such price to the satisfaction of the Contracting Officer will result in payment, as determined by the Contracting Officer, of-

(i) Actual mobilization costs at completion of mobilization;

(ii) Actual demobilization costs at completion of demobilization; and

(iii) The remainder of this item in the final payment under this contract.

(2) The Contracting Officer's determination of the actual costs in paragraph (b)(1) of this clause is not subject to appeal.

(End of clause)

5152.211-9000 Evaluation of Subline Items.

(Dec 2019)

Item No. 0002 are subdivided into two or more estimated quantities and are to be separately priced. The Government will evaluate each of these items on the basis of total price of its subline items.

(End of provision)

5152.211-9001 Variations in Estimated Quantities—Subline Items.

(Dec 2019)

This variation in estimated quantities clause is applicable only to item No. 0002.

(a) Variation from the estimated quantity in the actual work performed under any second or subsequent subline item or elimination of all work under such a second or subsequent subline item will not be the basis for an adjustment in contract unit price.

(b) Where the actual quantity of work performed for item No. 0002 is less than 85% of the quantity of the first subline item listed under such item, the contractor will be paid at the contract unit price for that subline item for the actual quantity of work performed and, in addition, an equitable adjustment shall be made in accordance with the clause at Federal Acquisition Regulation (FAR) 52.211-18, Variation in Estimated Quantities.

(c) If the actual quantity of work performed under item No. 0002 exceeds 115% or is less than 85% of the total estimated quantity of the subline item under that item and/or if the quantity of the work performed under the second subline item or any subsequent subline item under item Nos. NONE exceeds 115% or is less than 85% of the estimated quantity of any such subline item, and if such variation causes an increase or a decrease in the time required for performance of this contract the contract completion time will be adjusted in accordance with the clause FAR 52.211-18, Variation in Estimated Quantities.

(End of clause)

Section 00 71 00 - Contracting Definitions

FAR Clauses Incorporated by Reference

Number	Title	Effective Date	Alternate/ Deviation	Variation Effective Date
52.202-1	Definitions.	Jun 2020		

Section 00 73 00 - Supplementary Conditions

Overall Contract Delivery Period

From date of lead time event to beginning of performance

10 Calendar Days

Date of Notice to Proceed Receipt

From date of lead time event to completion of performance

155 Calendar Days

Date of Notice to Proceed Receipt

State: Louisiana

Construction Type: Heavy Dredging

Counties: Louisiana Statewide.

DREDGING PROJECTS ALONG THE GULF COAST AREA INCLUDING THE MISSISSIPPI RIVER AND ITS TRIBUTARIES TO THE OHIO RIVER

Note: Contracts subject to the Davis-Bacon Act are generally required to pay at least the applicable minimum wage rate required under Executive Order 14026 or Executive Order 13658. Please note that these Executive Orders apply to covered contracts entered into by the federal government that are subject to the Davis-Bacon Act itself, but do not apply to contracts subject only to the Davis-Bacon Related Acts, including those set forth at 29 CFR 5.1(a)(1).

If the contract is entered into on or after January 30, 2022, or the contract is renewed or extended (e.g., an option is exercised) on or after January 30, 2022:	◆ Executive Order 14026 generally applies to the contract. ◆ The contractor must pay all covered workers at least \$17.75 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025.
If the contract was awarded on or between January 1, 2015 and January 29, 2022, and the contract is not renewed or extended on or after January 30, 2022:	◆ Executive Order 13658 generally applies to the contract. ◆ The contractor must pay all covered workers at least \$13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours performing on that contract in 2025.

The applicable Executive Order minimum wage rate will be adjusted annually. If this contract is covered by one of the Executive Orders and a classification considered necessary for performance of work on the contract does not appear on this wage determination, the contractor must still submit a conformance request.

Additional information on contractor requirements and worker protections under the Executive Orders is available at <http://www.dol.gov/whd/govcontracts>.

Modification Number Publication Date
0 02/07/2025

SULA1994-001 04/01/1994

The following rates were revised pursuant to 29 CFR 1.6(c)(1) on 02/07/2025.

	Rates	Fringes
Derrick Operator.....	\$ 16.85	**
Dozer Operator.....	\$ 16.85	**
Dredge 16'" and Over		
Deckhand.....	\$ 16.85	**
Dredge tender operator.....	\$ 16.85	**
Fireman.....	\$ 16.85	**
First assistant engineer....	\$ 16.85	**
Leverman.....	\$ 16.85	**
Oiler.....	\$ 16.85	**
Second assistant engineer...	\$ 16.85	**
Shoreman.....	\$ 16.85	**
Third assistant engineer....	\$ 16.85	**
Truck driver.....	\$ 16.85	**
Welder.....	\$ 16.85	**
Dredge Under 16'"		
Deckhand.....	\$ 16.85	**
Dredge tender operator.....	\$ 16.85	**
Leverman.....	\$ 16.85	**
Oiler.....	\$ 16.85	**
Welder.....	\$ 16.85	**
Hydraulic Dredging		
First cook.....	\$ 16.85	**
Handyman.....	\$ 16.85	**
Janitor, cabin person.....	\$ 16.85	**
Second cook.....	\$ 16.85	**
Marsh Buggy Dragline, Oiler.....	\$ 16.85	**
Marsh Buggy Dragline, Operator...	\$ 16.85	**
Self-Propelled Hopper Dredge, Drag Tender.....	\$ 22.55	8.02+a

FOOTNOTE: Fourteen paid vacation days and eight paid holidays: New Year's Day, Good Friday, Memorial Day, Independence Day, Labor Day, Veterans' Day, Thanksgiving Day & Christmas Day provided the employee has one year of service.

WELDERS - Receive rate prescribed for craft performing operation to which welding is incidental.

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** Workers in this classification may be entitled to a higher minimum wage under Executive Order 14026 (\$17.75) or 13658 (\$13.30). Please see the Note at the top of the wage determination for more information. Please also note that the minimum wage requirements of Executive Order 14026 are not currently being enforced as to any contract or subcontract to which the states of Texas, Louisiana, or Mississippi, including their agencies, are a party.

Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at <https://www.dol.gov/agencies/whd/government-contracts>.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than ""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE: UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024

in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007 01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

----- WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

- a) a survey underlying a wage determination
- b) an existing published wage determination
- c) an initial WHD letter setting forth a position on a wage determination matter
- d) an initial conformance (additional classification and rate) determination

On survey related matters, initial contact, including requests for summaries of surveys, should be directed to the WHD Branch of Wage Surveys. Requests can be submitted via email to davisbaconinfo@dol.gov or by mail to:

Branch of Wage Surveys
Wage and Hour Division

U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

Regarding any other wage determination matter such as conformance decisions, requests for initial decisions should be directed to the WHD Branch of Construction Wage Determinations. Requests can be submitted via email to BCWD-Office@dol.gov or by mail to:

Branch of Construction Wage Determinations
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

2) If an initial decision has been issued, then any interested party (those affected by the action) that disagrees with the decision can request review and reconsideration from the Wage and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7). Requests for review and reconsideration can be submitted via email to dba.reconsideration@dol.gov or by mail to:

Wage and Hour Administrator
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

The request should be accompanied by a full statement of the interested party's position and any information (wage payment data, project description, area practice material, etc.) that the requestor considers relevant to the issue.

3) If the decision of the Administrator is not favorable, an interested party may appeal directly to the Administrative Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

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END OF GENERAL DECISION"

Section Table of Contents

SECTION 01100 - GENERAL PROVISIONS

1.	DAMAGE TO WORK	1
2.	SAFETY PROVISIONS	1
3.	SIGNAL LIGHTS	6
4.	CONTINUITY OF WORK	6
5.	INSPECTION	7
6.	SHOALING	7
7.	ACCOMMODATIONS AND MEALS FOR INSPECTORS	7
8.	SEAWORTHINESS CERTIFICATION	8
9.	ENVIRONMENTAL LITIGATION	9
10.	STATE TAXES	9
11.	FUEL CONSUMPTION REPORTING REQUIREMENTS	10
12.	RIGHTS-OF-WAY	10
13.	CONTRACT COORDINATION	11
14.	CONTRACTOR'S RESPONSIBILITY	12
15.	MOBILIZATION OF ATTENDANT PLANT	12
16.	ACCESS PLAN	12
17.	SUPERVISION	13
18.	PRE-BID SITE VISIT	13
19.	WORK IN THE VICINITY OF OTHER GOVERNMENT CONTRACTORS	13
20.	CONTRACTOR PERFORMANCE EVALUATIONS – CONSTRUCTION	13
21.	SECURITY REQUIREMENTS	14
22.	REGULATORY REQUIREMENTS	14
23.	EQUIPMENT OWNERSHIP AND OPERATING EXPENSE SCHEDULE	15
24.	DATUM RELATIONSHIPS	15
25.	PROJECT SIGN	17

SECTION 01100 - GENERAL PROVISIONS

1. DAMAGE TO WORK

The responsibility for damage to any part of the permanent work shall be as set forth in the Contract Clause in Section 00700 entitled PERMITS AND RESPONSIBILITIES (FAR 52.236-7). However, if, in the judgment of the Contracting Officer, any part of the permanent work performed by the Contractor is damaged by flood, earthquake, hurricane, or tornado which damage is not due to the failure of the Contractor to take reasonable precautions or to exercise sound engineering and construction practices in the conduct of the work, the Contractor shall make the repairs as ordered by the Contracting Officer and full compensation for such repairs will be made at the applicable contract unit price or Job prices as fixed and established in the contract. If, in the opinion of the Contracting Officer, there is no contract unit or Job prices applicable to any part of such work, an equitable adjustment shall be made pursuant to the Contract Clause in Section 00700 entitled CHANGES (FAR 52.243-4). Except as herein provided; damage to all work (including temporary construction), utilities, materials, equipment and plant shall be repaired to the satisfaction of the Contracting Officer at the Contractor's expense, regardless of the cause of such damage.

2. SAFETY PROVISIONS

The safety provisions as specified herein refer to the USACE Safety and Occupational Health Requirements EM 385-1-1 dated 15 March 2024. The most current edition can be found at:

<https://www.publications.usace.army.mil/Portals/76/EM%20385-1-1%20EFFECTIVE%2015March2024.pdf>

- (a) Accident Investigation and Reporting. The reporting and associated investigation of accidents and near misses is considered a leading indicator. Utilize ENG Form 3394, U.S. Army Corps of Engineers Mishap Notification and Investigation, to report accidents and near misses. Link to ENG Form 3394:

https://www.publications.usace.army.mil/Portals/76/Publications/EngineerForms/Eng_Form_3394_2021Aug.pdf.

Contractors must investigate, analyze, and report all accidents and near misses occurring incidentally to an operation, project, or facility according to EM 385-1-1, Table 2-1 Accident Reporting Required Timeline.

- (b) Accident Prevention Plan (APP). (See the Contract Clause in Section 00700, entitled ACCIDENT PREVENTION (FAR 52.236-13).) Within 15 days after receipt of Notice of Award of the contract, and at least 7 days prior to the prework conference, an electronic copy and one hard copy of the Accident Prevention Plan (APP) Worksheet (ENG FORM 6293) shall be submitted to

the Contracting Officer for review and acceptance. Link to ENG form 6293 (APP):

https://www.publications.usace.army.mil/LinkClick.aspx?fileticket=LmW_eJDoLpl%3d&tabid=16438&portalid=76&mid=43543

The Plan shall be prepared in the following format:

- (1) Activity Hazard Analysis (AHA) Requirements. The contractor shall complete the Activity Hazard Analysis (AHA) Requirements on ENG form 6206 in accordance with EM 385-1-1, Chapter 2-6. Specific AHA requirements are identified in paragraph 6 of each chapter of EM 385-1-1. Link to fillable ENG form 6206 (AHA form):
https://www.publications.usace.army.mil/Portals/76/Eng_Form_6206_2023Aug24.pdf
 - (2) The contractor shall submit ENG Form 6282 (Site Safety and Health Officer Designation Letter). A Designation Letter is required for all levels of onsite SSHOs. The contractor shall maintain a copy of the accepted form at the site of work. Link to ENG form 6282:
https://www.publications.usace.army.mil/Portals/76/Eng_Form_6282_2023Aug28.pdf?ver=1GgPT1ZS1AFX9P8AKU5og%3d%3d.
 - (3) Site Safety and Health Officer (SSHO). The contractor must employ a minimum of one (1) Level 1 Full-time Site Safety and Health Officer to function as the primary Level 1 SSHO responsible for ensuring all site personnel are implementing Safety and Occupational Health requirements according to the contract, EM 385-1-1, applicable federal, state, and local requirements. A designated Level 1 SSHO must be at the site of work whenever work is performed. The SSHO shall meet the training and experience requirements of EM 385-1-1, Chapter 2-3.b. Level 1 SSHOs are not permitted to supervise non-safety personnel at the site of work.
 - (4) A copy of company policy statement regarding accident prevention.
 - (5) When marine plant and equipment are in use under a contract, the method of fuel oil transfer shall be included on MVN Form 385-10R (Latest Edition), Fuel Oil Transfer (attached at the end of this section), (Refer to 33 CFR 156).
- (c) Commencement of Work. The contractor shall not commence physical work at the site until the Contracting Officer, or his/her authorized representative has accepted the Accident Prevention Plan. The contractor may submit its Accident Prevention Plan only for the first phase of construction provided that it is accompanied by an outline of the remaining phases of construction. All

remaining phases shall be submitted and accepted prior to the beginning of work in each phase. Also refer to EM 385-1-1, Chapter 2.

- (d) Hazard Communication (HazCom) Program. The contractor shall develop, implement, and maintain at the workplace a written, Hazard Communication Program (see EM 385-1-1, Chapter 6-7) that includes identification of potential hazards as prescribed in 29 CFR Part 1910.1200 and/or 1926.59, effects of exposure and control measures to be used for chemical products and physical agents that may be encountered during the performance of work on this contract, provisions for container labeling, Safety Data Sheets, and employee training program, and other criteria in accordance with 29 CFR Part 1910.1200 and/or 1926.59. Training shall include communication methods and systems to be used (i.e., voice, hand signals, radios or other means), and training in the use and understanding of safety data sheets and chemical product hazard warning labels. Prior to bringing hazardous substances, as defined in 29 CFR 1910.1200 and/or 1926.59, onto the job site, a copy of the Hazard Communication Program and the Safety Data Sheets of each substance shall be submitted to the Contracting Officer and made available to the contractor's employees as part of its Accident Prevention Plan. A site map shall be attached to the inventory showing where the inventoried hazardous substances are stored. The inventory list and site map shall be updated monthly to assure accuracy.
- (e) Daily Inspections. The contractor shall perform daily safety inspections and record them on the forms approved by the Contracting Officer. Reports of daily inspections shall be maintained at the jobsite in accordance with Section 01 45 04.00 10, CONTRACTOR QUALITY CONTROL. The reports shall be records of the daily inspections and resulting actions. Each report shall include, as a minimum, the following:
 - (1) Phase(s) of construction underway during the inspection.
 - (2) Locations of areas where inspections were made.
 - (3) Results of inspections, including nature of deficiencies observed and corrective actions taken, or to be taken, date, and signature of the person responsible for its contents.
- (f) Safety Sign. The contractor shall furnish, erect, and maintain a safety sign at the site where indicated by the Contracting Officer. The sign shall conform to the requirements of this paragraph and the drawing included at the end of this section. The lettering shall be black, the safety circle and cross green, and the background white. When placed on a floating plant, the sign may be half size. The sign shall be erected as soon as practicable, but not later than 15 calendar days after the date established for commencement of work. The

data required shall be current. The sign coordinator is Timothy Lacoste@504.862.2663.

- (g) Ground Fault Protection Equipment. Electrical equipment used on this contract shall be equipped with ground fault circuit interrupters in accordance with EM 385-1-1, Chapter 11-8.
- (h) Means of Escape for Personnel Quartered, or Working on Floating Plant. Two means of escape shall be provided for assembly, sleeping, and messing areas on floating plants. For areas involving 10 or more persons, both means of egress shall be through standard size doors opening to different exit routes. Where nine or fewer persons are involved, one of the means of escape may be a window (minimum dimensions 24-inches by 36-inches), which leads to a different exit route. Refer to EM 385-1-1, Chapter 19-8.h.
- (i) Emergency Alarms and Signals.
 - (1) Alarms. Emergency alarms shall be installed and maintained on all floating plant requiring a crew where it is possible for either a passenger or crewman to be out of sight or hearing from any other person. The alarm system shall be operated from the primary electrical system with standby batteries on trickle charge that will automatically furnish the required energy during an electrical-system failure. A sufficient number of signaling devices shall be placed on each deck so that the sound can be heard distinctly at any point above the usual background noise. All signaling devices shall be so interconnected that actuation can occur from at least one strategic point on each deck.
 - (2) Signals.
 - (a) Fire Alarm Signals. The general fire alarm signal shall be in accordance with 46 CFR Ch. I; Subpart E.109.503 of the Coast Guard Rules and Regulations for Cargo and Miscellaneous Vessels, Sub-Chapter I & Ia.
 - (b) Abandon Ship Signals. The signal for abandon ship shall be in accordance with paragraph 109.503(b) of the reference cited in subparagraph (2)(a) above.
 - (c) Man-Overboard Signal. Hail and pass the word to the bridge. All personnel and vessels capable of rendering assistance shall respond.
- (j) Hurricane Plan. A detailed plan for protection and evacuation of personnel and the construction site, in the event of an impending hurricane or storm, is

required as an enclosure to the contractor's Accident Prevention Plan. This plan shall be submitted to the Contracting Officer, or his/her representative, for review and approval prior to the preconstruction conference. Work being performed to satisfy the Hurricane Plan will not be measured for payment. Payment for all work associated with the Hurricane Plan, and providing the equipment required for the duration specified, shall be distributed amongst the existing bid items. The Hurricane Plan shall include at least the following:

- (1) The time each phase of the plan will be put in effect. The time shall be the number of hours remaining for the storm to reach the worksite if it continues at the predicted speed and direction.
- (2) The estimated time necessary to secure and evacuate the site including any emergency flood protection.
- (3) The safe harbor for personnel and plant specifically identified.
- (4) The name of the boat which will be used to move the plant, its type, capacity, speed, and availability.
- (5) The estimated time necessary to move the plant to the safe harbor after movement is started.

NOTE: Floating plant, boat, or other marine activities must submit an Emergency Plan for Marine Activities which includes a Severe Weather Plan for Marine Activities per EM 385-1-1, Chapters 19-7.a and 19-7.b.

- (k) Hazardous Energy Control Program. The contractor shall develop, implement and maintain at the workplace, a written Hazardous Energy Control Program (HECP). Refer to EM 385-1-1, Chapter 12-7.
- (l) Cranes. The contractor (including subcontractors) shall have cage boom guards, insulating links, or proximity warning devices on cranes that will be working adjacent to power lines. These devices shall not alter the requirements of any other regulation of this part - even if such device is required by law or other regulation. Insulating links shall be rated for the energized lines they are working near (EM 385-1-1, Chapter 11-8.t.6.d). Calibration records or stamped date of required manufacturer inspection of proximity warning devices shall be kept on the crane. Additionally, prior to any work commencing an Activity Hazard Analysis) identifying and satisfying EM 385-1-1, Chapter 2-6, requirements shall be submitted and accepted by the Contracting Officer.
- (m) Crane/Derrick and Dragline Certification. The contractor shall submit a copy of the certification and performance test in accordance with EM-385-1-1, Chapter 16-5.

- (n) Emergency Drills. The Contractor shall conduct its drills in accordance with EM 385-1-1, Chapter 19-8.b.
- (o) Equipment Operator Authorization. The contractor shall submit a list of designated personnel qualified and authorized to operate machinery and mechanized equipment in accordance with EM 385-1-1, Chapter 16-3.
- (p) Dive Plan. The contractor shall submit a Dive Plan in accordance with EM 385-1-1, Chapter 30-7.
- (q) Radiation. If a production meter that uses nuclear materials is being used aboard the dredge, the contractor shall perform following requirements. The production meter nuclear device system designer and installer shall be qualified in these fields of expertise by the Nuclear Regulatory Commission (NRC). The contractor shall obtain licensing and training as required by the NRC for their personnel aboard the dredge for the use of those components of the production meter containing or are affected by the nuclear source. The contractor shall implement A Nuclear Device Awareness Program as required by the NRC for all personnel aboard the dredge not directly involved in the activities of the nuclear device. The contractor shall submit a Nuclear Device Safety Plan to the Government within 24 hours after receipt of Notice to Proceed by the contractor. While a nuclear device is present aboard the dredge, the contractor shall strictly adhere to all applicable NRC rules and regulations.
- (r) Overhead Power Lines. In locations where trucks and other equipment will be passing beneath overhead power lines, the contractor shall install at least 2 signs on either side of the crossings warning drivers of the overhead lines. The signs shall face both directions of travel. The signs shall have arrows pointing upward and state "Overhead Electric". The contractor's proposed sign design shall be submitted to the Contracting Officer for approval."

3. SIGNAL LIGHTS

The Contractor shall display signal lights and conduct his/her operations in accordance with U. S. Coast Guard regulations governing lights and day signals to be displayed, as set forth in Commandant, U. S. Coast Guard Instruction M16672.2C, Navigation Rules, International - Inland (COMDTINST M16672); 33 CFR 81, Appendix A (International); and 33 CFR 84 through 33 CFR 90 (Inland) as applicable.

4. CONTINUITY OF WORK

No payment will be made for work done in any area designated by the Contracting Officer until the full dimensions required under the contract are secured in the whole of such area, nor will payment be made for excavation in any area not adjacent to and

in prolongation of areas where full dimensions have been secured except by decision of the Contracting Officer. Should any such non-adjacent area be excavated to full dimensions during the operations carried on under the contract, payment for all work therein may be deferred until the required dimensions have been made in the area intervening. The Contractor may be required to suspend dredging at any time when for any reason the gages or ranges cannot be seen or properly followed.

5. INSPECTION

(a) The presence of the Inspector shall not relieve the Contractor of the responsibility for the proper execution of the work in accordance with the specifications. The Contractor shall furnish on the request of the Contracting Officer or any inspector:

(1) The use of such boats, boatmen, laborers and material forming a part of the ordinary and usual equipment and crew of the dredging plant as may be reasonably necessary in inspecting and supervising the work.

(2) Suitable transportation from all points on shore designated by the Contracting Officer to and from the various pieces of plant, and to and from the disposal areas.

(b) When the Contractor elects to work on weekends, holidays or nights, notice shall be given to the Contracting Officer at least 24 hours in advance thereof. Adequate lighting for thorough inspection of night operations shall be provided by the Contractor at his/her expense.

(c) Should the Contractor refuse, neglect, or delay compliance with these requirements, the specific facilities may be furnished and maintained by the Contracting Officer, and the cost thereof will be deducted from any amounts due or to become due to the Contractor.

6. SHOALING

If, before the contract is completed, shoaling occurs in any section previously accepted, including shoaling in the finished channel, because of the natural lowering of the side slopes, re-dredging at contract price, within the limit of available funds, may be done if agreeable to both the Contractor and the Contracting Officer.

7. ACCOMMODATIONS AND MEALS FOR INSPECTORS

(a) The Contractor shall furnish regularly to inspectors on board the dredge or other craft upon which they are employed a suitable separate room approximately 150 square feet in size for a Corps of Engineers field office and sleeping purposes. The room shall be fully equipped and maintained to the satisfaction of the Contracting Officer; it shall be properly heated, ventilated, air

conditioned, and lighted, shall have sufficient electrical outlets with power surge protection to power a laptop computer and printer, shall have a desk which can be locked, and a drafting table measuring at least four (4) feet by six (6) feet, a comfortable chair for each inspector, washing conveniences and daily janitorial services. If the dredge does not have suitable sleeping quarters, the Contractor may elect to obtain a suitable sleeping facility on shore, such as hotel, motel, etc. This will not be a separate cost and will be included in the price of dredging. Should the Contractor have a photocopy machine on the jobsite, the Contractor may allow the use of his own in lieu of providing a separate machine for use by the Government inspector. The entire cost to the Contractor for furnishing, equipping, and maintaining the foregoing accommodations, shall be distributed amongst the existing bid items. If the Contractor fails to meet these requirements, the facilities referred to above will be secured by the Contracting Officer, and the cost thereof will be deducted from payments to the Contractor.

(b) If the Contractor maintains on this work an establishment for the subsistence of his/her own employees, he/she shall, when required, furnish to inspectors employed on the work, and to all Government agents who may visit the work on official business, meals of quality satisfactory to the Contracting Officer. All meals shall be available for purchase, by the Government's inspectors and agents, at the cost not to exceed \$3.25 per person for each meal.

(c) The Contractor shall provide the inspector's field office with a cellular wireless Broadband / WiFi 802.11n capable router that accepts cellular data signals through Express Card / Air Card or USB modems. This service must be capable of providing adequate connection to allow the inspectors to import / export files through RMS, namely 10 Mbps download speed and 1.5 Mbps upload speed. The Contractor must field verify that the service provider chosen has adequate continuous coverage at the construction site. The Contractor shall be responsible for the installation, maintenance of, and the monthly service fees necessary to provide continuous high speed internet service for the duration of the contract.

8. SEAWORTHINESS CERTIFICATION

EM 385-1-1, Chapter 19-5. All dredges and quarter boats not subject to USCG inspection and certification or not having a current American Bureau of Shipping (ABS) classification shall be inspected in the working mode annually by a marine surveyor accredited by the National Association of Marine Surveyors (NAMS) or Society of Accredited Marine Surveyors (SAMS) and having at least five (5) years' experience in commercial marine plant and equipment. All other plant shall be inspected annually by a qualified person. The inspection shall be documented, and a copy of the most recent inspection report shall be posted in a public area on board the vessel and a copy shall be furnished to the designated authority upon request. The inspection shall be appropriate for the intended use of the plant and shall, as a

minimum, evaluate structural integrity and compliance with NFPA 302, Fire Protection Standard for Pleasure and Commercial Motor Craft.

9. ENVIRONMENTAL LITIGATION

(a) The term "environmental litigation", as used herein, means a lawsuit alleging that the work has an adverse effect on the environment or that the Government has not duly considered, either substantively or procedurally, the effect of the work on the environment.

(b) If the performance of all or any part of the work is suspended, delayed, or interrupted due to an order of a court of competent jurisdiction as a result of environmental litigation, as defined above, the Contracting Officer, at the request of the Contractor, shall determine whether the order is due in any part to the acts or omissions of the Contractor or a Subcontractor at any tier and required by the terms of this contract. If the order is not due in any part to acts or omissions of the Contractor (or a Subcontractor at any tier) other than as required by this contract, such suspension, delay, or interruption shall be as if ordered by the Contracting Officer under the Contract Clause in Section 00700 entitled, SUSPENSION OF WORK (FAR 52.242-14). The period of such suspension, delay or interruption shall be considered unreasonable, and an adjustment shall be made for any increase in the cost of performance of this contract (excluding profit) as provided in that clause, subject to all the provisions thereof.

10. STATE TAXES

(a) The bid submitted in response to this Invitation shall not include any amount whatever for payment of any of the following taxes, fees or charges:

(1) The Louisiana "Severance Tax" imposed by LSA R.S. 47:631 and made applicable to the dredging of fill material from rivers and bodies of water within the State of Louisiana by the Severance Tax Regulations promulgated by the Collector of Revenue dated 31 March 1968.

(2) Any amounts claimed by the Louisiana Department of Wildlife and Fisheries for the privilege of removing fill from the water bottoms of the State of Louisiana.

(b) If the Contractor is required to pay or bear the burden of any tax, fee or charge described in paragraphs (a)(1) and/or (a)(2) above, the contract prices shall be increased by that amount which the Contractor is required to pay to the State of Louisiana; provided, however, that no increase in contract price shall be made for any liability the Contractor may incur as a result of his fault or negligence or his failure to follow the instructions of the Contracting Officer.

(c) The Contractor shall promptly notify the Contracting Officer of all matters pertaining to taxes, fees, or charges as described herein which reasonably may be expected to affect the contract price and shall at all times follow the directions and instructions of the Contracting Officer in regard to the payment of such taxes, fees, or charges.

(d) Before any increase in contract price becomes effective in accordance with the provisions of this clause, the Contractor shall warrant in writing that no amount of such taxes, fees or charges was included in the contract price as a contingency reserve or otherwise.

11. FUEL CONSUMPTION REPORTING REQUIREMENTS

On the first day of each month, the Contractor shall furnish, to the Government Inspector, a report of the quantities of fuel consumed during the previous month in execution of the work covered by the contract. The quantities reported shall include fuel consumed by the Contractor and all of his/her Subcontractors for the main plant and all support plant during the preceding month. This information may be consolidated and shall be included in the Report of Operations-Pipelines, Dipper or Bucket Dredges, ENG Form 4267; or in the Report of Operations-Hopper Dredges, ENG Form No. 27A (costs), as applicable.

12. RIGHTS-OF-WAY

(a) The rights of entry required for the work to be constructed under this contract, within the rights-of-way limits indicated on the drawings, have been obtained by the Government and are provided without cost to the Contractor. The Contractor shall make its own investigations to determine the conditions, restrictions, and difficulties which may be encountered in the transportation of equipment and material to and from the work site. The proposed work, including rights-of-way, as defined by these specifications and as shown on the drawings, is in compliance with all applicable Federal and state environmental laws and regulations. Upon completion of the Contractor's work, rights-of-way furnished by the Government shall be returned to its original condition prior to construction unless otherwise noted.

(b) If the Contractor proposes a deviation from the Government furnished rights-of-way for his convenience, the Contractor shall notify the Contracting Officer or its representative in writing. Contractor shall not provide any permanent rights-of-way for the project. The Contractor is cautioned that any deviation to the Government furnished rights-of-way is subject to all applicable Federal and state environmental laws and regulations. Compliance with these environmental laws and regulations may require additional National Environmental Policy Act (NEPA) documents, cultural resources surveys, coordination with the Louisiana State Historical Preservation Officer, water quality certification, modification of the Federal consistency determination, etc. The Government is ultimately

responsible for environmental compliance; therefore, the Government will determine the additional environmental coordination and documentation necessary for a proposed deviation to the Government furnished rights-of-way. For any environmental investigations the Government is to perform on areas outside of Government furnished rights-of-way, the Contractor shall provide sufficient rights of entry to the Government. The Contracting Officer will advise the Contractor of the additional environmental coordination and documentation that must be completed. The Government will be responsible for any additional environmental compliance; however, the Contractor may conduct specific tasks identified by the Government. The Government will offer advice and assistance to the Contractor in conducting these tasks. Depending on the environmental impact of the proposed deviation, obtaining the coordination and documentation may not be approved or could take as much as 180 calendar days for approval by the Government. The Government must review, approve and ensure distribution of all environmental compliance documentation and ensure all comments on the same have been resolved before any utilization of any areas outside of the Government furnished rights-of-way. The Contractor shall reimburse the Government for actual expenses incurred for assistance in completing or attempting to complete additional environmental coordination and documentation, which expenses will not exceed one hundred thousand (\$100,000.00) dollars. There is no guarantee that environmental compliance will be obtained; therefore, the Contractor shall assume all risks and liabilities associated with pursuing a deviation. Any delays resulting from the deviation and/or the environmental coordination and documentation shall not be made the basis of any Contractor claim for increase in the contract cost and/or increase in contract time. Deviations will be at Contractor's sole risk and liability, including, but not limited to, such liabilities associated with items such as hazardous substances regulated under the Comprehensive Environmental Response, Compensation, and Liability Act (42 U.S.C. 9601 et. seq.), and at no cost to the Government. Government assistance in obtaining additional environmental clearances does not relieve the Contractor of responsibility for complying with other Federal, state or local licenses and permits.

(c) The Contractor is not allowed to utilize the Corps of Engineers Venice Sub-office or harbor for parking and/or getting on and off of boats.

13. CONTRACT COORDINATION

The Contractor shall assist the Government with available on-site plant and manpower in monitoring the water quality aspects of the dredging and disposal operations. This assistance shall consist of, but not be limited to, furnishing boat transportation, temporary storage of samples, etc.

14. CONTRACTOR'S RESPONSIBILITY

The Contractor shall be responsible for ensuring that all its employees strictly comply with all laws that may apply to operations under this contract. The Contractor assumes full responsibility for the safety of its employees, plant, and materials and for any damage or injury done by or to them from any source or cause, except damage caused by acts of the Government, its officers, agents or employees. Such damages will be the responsibility of the Government in accordance with applicable Federal laws. The terms "officer", "agent", and "employee" of the Government do not include persons in the employment of the Contractor and whose services have been furnished to the Government.

15. MOBILIZATION OF ATTENDANT PLANT

Pursuant to the Contract Clause in Section 00700 entitled COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK (FAR 52.211-10), mobilization of all attendant plant, if required, shall be concurrent with dredge mobilization. Failure to timely mobilize such auxiliary/attendant plant may result in one (1) or more of the following actions by the Contracting Officer: reasonable suspension (without Government cost) of work until required plant is provided; formulation of credit to offset deficient plant; imposition of liquidated damages for late overall completion of the contract after excusable delays, if any. The Government's rights under any other Contract Clause are preserved.

16. ACCESS PLAN

The Contractor shall submit an Access Plan to be reviewed and approved by the Contracting Officer to include, as a minimum, the following:

- (a) Layout drawings showing the location of all equipment, office structures, toilets, and storage areas for materials.
- (b) Show mobilization and demobilization routing and locations of large equipment, such as draglines, cranes, and etc. while on the jobsite.
- (c) Show waterway channels or canals used to mobilize and demobilize equipment and materials and show access routes and docking areas of all marine equipment with respect to the jobsite.
- (d) Airboats and small outboards shall be used whenever practical to reduce the usage of marsh buggies. Existing trails and canals shall be utilized whenever possible. Marsh buggy use shall be limited to the construction limits of the project features.

17. SUPERVISION

At all times during performance of this contract and until the work is completed and accepted by the Government, the Contractor shall have on the worksite a competent superintendent who is satisfactory to the Contracting Officer and has authority to act for the Contractor. Inspectors appointed by the Contracting Officer will enforce strict compliance with the terms of the contract. The inspectors will keep a record of the work done, but neither the presence nor absence of inspectors shall relieve the Contractor of responsibility for the proper execution of the work in accordance with the contract and directives issued by the Contracting Officer.

18. PRE-BID SITE VISIT

Prior to the submission of any bids, all bidders are highly encouraged to visit the project site location to become familiar with the project requirements. Failure to visit the project site will not disqualify a bid; however, the bidder is required to comply with the terms and conditions of any resultant contract by reason of such failure. In no event will a failure to inspect the site constitute grounds for a claim after contract award.

19. WORK IN THE VICINITY OF OTHER GOVERNMENT CONTRACTORS

The Contractor shall coordinate his/her operations, through the Contracting Officer's Representative, with any other Government Contractors who may be working in the vicinity. The Contractor shall coordinate his staging operations, access routes, and construction requirements with any Contractor currently performing work in the general vicinity. Any disputes or disagreements arising from area use shall be brought to the attention of the Contracting Officer, who will then make a determination for operational procedures. Determinations made by the Contracting Officer shall be binding on all parties concerned.

20. CONTRACTOR PERFORMANCE EVALUATIONS – CONSTRUCTION

In accordance with the provisions of Subpart 36.201 (Evaluation of Contractor Performance) of the Federal Acquisition Regulation (FAR), construction Contractor's performance shall be evaluated throughout the performance of the contract. The United States Army Corps of Engineers (USACE) follows the procedures outlined in Engineering Regulation 415-1-17 to fulfill this FAR requirement. For construction contracts awarded at or above the Simplified Acquisition Threshold, the USACE will evaluate Contractor's performance and prepare a performance report using the Contractor Performance Assessment Reporting System (CPARS), which is now a web-based system. After an evaluation (interim or final) is written up by the USACE, the Contractor shall have the ability to access, review and comment on the evaluation for a period of 30 days. Accessing and using CPARS requires specific software, called PKI certification, which is installed on the user's computer. The certification is a Department of Defense requirement and was implemented to provide security in

electronic transactions. The certification software could cost approximately \$110 - \$125 per certificate per year and is purchased from an External Certificate Authorities (ECA) vendor. Current information about the PKI certification process and for contacting vendors can be found on the web site: <http://www.cpars.gov>. If the Contractor wishes to participate in the performance evaluation process, access to CPARS and PKI certification is the sole responsibility of the Contractor.

21. SECURITY REQUIREMENTS

a. Suspicious Activity Reporting Training (e.g. iWATCH, CorpsWatch, or See Something, Say Something). The Contractor shall not have access to the CORPS network. This is an unclassified contract and the Contractor shall not have access to critical information. The Contractor and all associated Sub-contractors shall receive locally developed training provided by the New Orleans District Security Office on the Local Suspicious Activity Reporting Program. This training will be used to inform employees of the types of behavior to watch for and instruct employees to report suspicious activity relating to the project manager, security representative or law enforcement entity. The Contractor shall provide local background checks to New Orleans District Security Office before performing work. Point of contact is Rhonda Brown, 251.694.4436. This training shall be completed within 30 calendar days of contract award and within 30 calendar days of new employees commencing performance. The results of this training shall be reported to the COR within 5 calendar days after the completion of the training.

b. Pre-Screen Candidates Using E-Verify Program. The Contractor shall pre-screen Candidates using the E-verify Program (<https://www.uscis.gov/e-verify>) website to meet the established employment eligibility requirements. The Contractor shall ensure that the Candidate has two valid forms of Government issued identification prior to enrollment to ensure the correct information is entered into the E-verify system. An initial list of verified/eligible Candidates shall be provided to the COR no later than 3 business days after the initial contract award. When contracts are with individuals, the individuals shall complete a Form I-9, Employment Eligibility Verification, with the designated Government representative. The completed Form I-9 shall be provided to the Contracting Officer and shall become part of the official contract file.

22. REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this contract and the requirements as defined within USACE EM 385-1-1, the Contractor shall comply with all pertinent federal, state, and local laws, ordinances, criteria, rules, and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations and referenced documents vary, the most stringent requirements shall apply.

23. EQUIPMENT OWNERSHIP AND OPERATING EXPENSE SCHEDULE

(a) This clause does not apply to terminations.

(b) Allowable costs for construction and marine plant and equipment in sound workable condition owned or controlled and furnished by a Contractor or Subcontractor at any tier shall be based on actual cost data for each piece of equipment or groups of similar serial and series for which the Government can determine both ownership and operating costs from the Contractor's accounting records. When both ownership and operating costs cannot be determined for any piece of equipment or groups of similar serial or series equipment from the Contractor's accounting records, costs for that equipment shall be based upon the applicable provisions of Engineering Pamphlet (EP) 1110-1-8, "Construction Equipment Ownership and Operating Expense Schedule," Region III. Working conditions shall be considered to be average for determining equipment rates using the schedule unless specified otherwise by the Contracting Officer. For equipment not included in the schedule, rates for comparable pieces of equipment may be used or a rate may be developed using the formula provided in the schedule. For forward pricing, the schedule in effect at the time of negotiations shall apply. For retroactive pricing, the schedule in effect at the time the work was performed shall apply.

(c) Equipment rental costs are allowable, subject to the provisions of FAR 31.105(d) (ii) and FAR 31.205-36. Rates for equipment rented from an organization under common control, lease-purchase arrangements, and sale-leaseback arrangements, will be determined using the schedule, except that actual rates will be used for equipment leased from an organization under common control that has an established practice of leasing the same or similar equipment to unaffiliated lessees.

(d) When actual equipment costs are proposed and the total amount of the pricing action exceeds the simplified acquisition threshold (SAT), the Contracting Officer will request the Contractor to submit either certified cost or pricing data, or partial/limited data, as appropriate.

NOTE: A copy of the "EQUIPMENT OWNERSHIP AND OPERATING EXPENSE SCHEDULE" for Region III can be obtained from the following website:

https://www.publications.usace.army.mil/Portals/76/Users/182/86/2486/EP%201110-1-8_RV1-12.pdf?ver=x6psY-gBPGg5dSOqQdjEMg%3d%3d

24. DATUM RELATIONSHIPS

The following information is provided for informational use only and is not to be used for this contract. The purpose of this paragraph is to provide datum relationships

between Mean Lower Low Water (MLLW) datum and Mean Low Gulf (MLG) datum. This information is provided to comply with the requirements outlined in Engineering Manual, EM 1110-2-6056, Standards and Procedures for Referencing Project Elevation Grades to Nationwide Vertical Datums. This engineering manual was released in response to IPET Lessons Learned, in order to address datum and project control issues.

This datum relationship must be identified on the plans, in order to identify the adjustment from MLG to NAVD88 and Mean Lower Low Water (MLLW).

→ Tiger Pass

0.0' MLLW (2012-2016) = 0.0' NAVD88 (2009.55) = 3.0' MLG

PROJECT CONTROL POINTS

The following information describing the project control points (PCPs) must be included in the specifications. These benchmarks have been incorporated in the NSRS database as required by EM 1110-2-6056.

→ DESIGNATION: 876 0849A TIDAL
PID: AT1390

STATE/COUNTY: LA/PLAQUEMINES
LAT/LON: N 29°16'00.52795" W 089°21'10.35485"
NAD83 (2011)
NORTH/EAST: N 284,325.15' E 3,912,287.82' SPC LA S (US feet)
ELEVATION: 1.53 feet NAVD88 (2009.55)
MARKER: BENCH MARK DISK
STAMPING: 0849 A 1985
DESCRIPTION: THIS MONUMENT IS TIED TO THE OD VENICE STAFF GAGE. IT IS LOCATED IN VENICE, AT THE END OF HWY 23, 80 FEET NORTH FROM THE NORTH 1 OF 2 YELLOW KEYPAD SUPPORT POSTS, 45 FEET EAST FROM A RED FIRE HYDRANT, 26 FEET NORTHEAST FROM THE CENTER OF A GRAVEL ROAD, 3.3 FEET NORTHEAST FROM A METAL POST WITH WITNESS SIGN ATTACHED, 1.6 FEET FROM A CARSONITE WITNESS POST, AND 1 FOOT BELOW THE LEVEL OF THE GROUND.

→ DESIGNATION: GPS TIGER
PID: BBCY07

STATE/COUNTY: LA/PLAQUEMINES
LAT/LON: N 29°09'02.45047" W 089°24'02.39074"
NAD83 (2011)
NORTH/EAST: N 241,839.06' E 3,897,764.72' SPC LA S (US feet)
ELEVATION: 4.55 feet NAVD88 (2009.55)

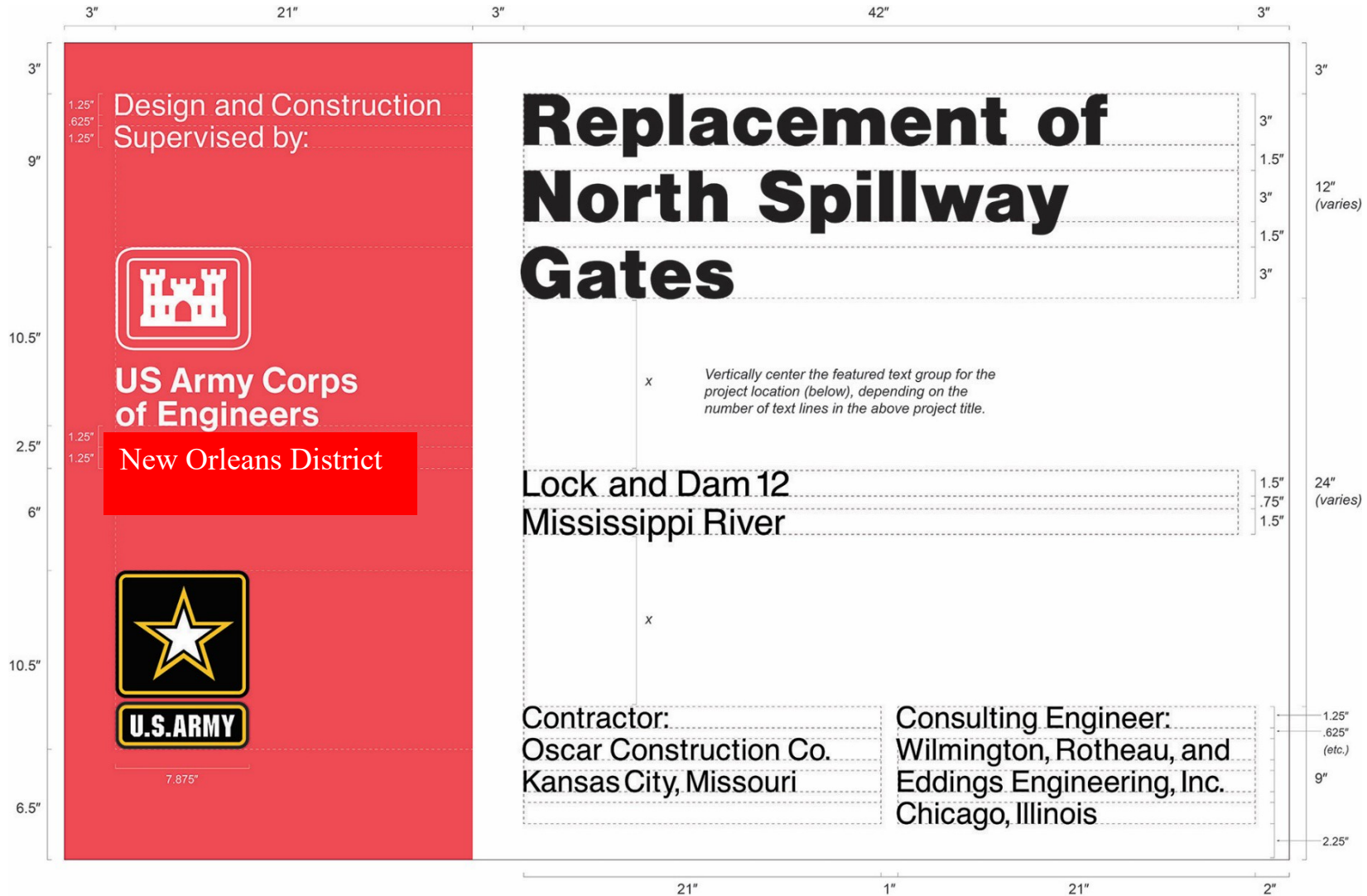
MARKER: BENCH MARK DISK
STAMPING: GPS TIGER 2013
DESCRIPTION: THIS MONUMENT IS TIED TO THE STAFF GAGES SET ON DM10 AND DM17 (SGU 01535). IT IS LOCATED IN TIGER PASS, SOUTH OF VENICE, AT THE SOUTHWEST CORNER OF A CONCRETE SLAB AT HILLCORP ENERGY COMPANY WEST BAY FIELD PRODUCTION FACILITY #1. LOCATED ON THE SOUTH SIDE OF THE CHANNEL AT RED BUOY #20. MONUMENT IS 4.3 FEET FROM THE SOUTH EDGE OF SLAB.

25. PROJECT SIGN

Prior to commencement of work, the Contractor shall construct a project sign at the site of the work at a location directed by the Contracting Officer. The sign, which will identify the work with the Corps of Engineers shall be 4 feet by 6 feet in size and shall conform to the requirements of the PROJECT SIGN drawing and installation instructions attached at the end of this section. The lettering for the 2 feet by 4 feet section of the sign with the Corps logo and Army Star shall be white; all other lettering shall be black. Lettering for the project name shall be Helvetica Bold, all other lettering shall be Helvetica Regular. The sign coordinator is Timothy Lacoste at (504) 862-2663. No separate payment will be made for construction and erection of the project sign and all costs in connection therewith will be considered an incidental obligation of the Contractor. Upon completion of the work, the sign shall become the property of the Contractor and shall be removed from the job site.

ACCIDENT PREVENTION PLAN
FUEL OIL TRANSFER -- FLOATING PLANT
 U.S. Army Engineer District, New Orleans
 EM 385-1-1, Chapter 19-8.d

1) Contractor		2) Contract Name & Number		3) Date			
4) Officer in Charge of Fuel Transfer		4a) Name of Vessel		4b) Fuel to be Transferred			
5) Name of Vessel		5a) Names of Qualified Tankermen		5b) Type of Certification and expiration date			
6) Name of Vessel	6a) Type of fill nozzle or connection on Vessel	6b) Location of fill pipes openings	6c) Location of vents openings				
7) Type, number, and size of fire fighting equipment to be available during fuel transfer operations.							
8) Sequential steps to be followed when taking on fuel.							
Contractor's Signature Date C.O. or C.O.R. Signature Date							



SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 33 00

SUBMITTAL PROCEDURES

12/2021

PART 1 GENERAL

- 1.1 MEASUREMENT AND PAYMENT
- 1.2 DEFINITIONS
 - 1.2.1 Submittal Descriptions (SD)
 - 1.2.2 Approving Authority
 - 1.2.3 Work
- 1.3 SUBMITTALS
- 1.4 SUBMITTAL CLASSIFICATION
 - 1.4.1 Government Approved
 - 1.4.2 Information Only
- 1.5 APPROVED SUBMITTALS
- 1.6 DISAPPROVED SUBMITTALS
- 1.7 GENERAL
- 1.8 SUBMITTAL REGISTER
- 1.9 SCHEDULING
- 1.10 TRANSMITTAL FORM (ENG FORM 4025)
- 1.11 SUBMITTAL PROCEDURES
 - 1.11.1 Procedures
 - 1.11.2 Deviations
- 1.12 CONTROL OF SUBMITTALS
- 1.13 GOVERNMENT APPROVED SUBMITTALS
- 1.14 INFORMATION ONLY SUBMITTALS
- 1.15 STAMPS

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 33 00

SUBMITTAL PROCEDURES
12/2021

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for submittal requirements as specified herein. Payment for the work covered under this section shall be distributed throughout the existing bid items. Payment for materials incorporated in the work will not be made if required approvals have not been obtained.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittals requirements are specified in the technical sections. Submittals are identified by SD numbers and titles as follows.

SD-01 Preconstruction Submittals

- Certificates of insurance.
- Surety bonds.
- List of proposed subcontractors.
- List of proposed products.
- Construction Progress Schedule.
- Submittal register.
- Schedule of prices.
- Health and safety plan.
- Work plan.
- Quality control plan.
- Environmental protection plan.
- Traffic Control Plan.

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a

material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. (Testing must have been within three years of date of contract award for the project.)

Report which includes findings of a test required to be performed by the Contractor on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report which includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports.

Daily logs and checklists.

Final acceptance test and operational test procedure.

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that product, system or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or subcontractor through Contractor, the purpose of which is to further quality of orderly progression of a portion of the work by documenting procedures, acceptability of methods or personnel qualifications.

Confined space entry permits.

Text of posted operating instructions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and must state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

1.2.2 Approving Authority

Office or designated person authorized to approve submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only or as otherwise designated. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.4 SUBMITTAL CLASSIFICATION

Submittals are classified as follows:

1.4.1 Government Approved

Governmental approval is required for extensions of design, critical materials, deviations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Contracting Officer. Within the terms of the Clause in Section 00700 CONTRACT CLAUSES, entitled SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION (FAR 52.236-21), they are considered to be "shop drawings". Any reference to Government approval by the Contracting Officer (CO) includes the approving authority of the CO, the Administrative Contracting Officer (ACO), or the Contracting Officer's Representative (COR).

1.4.2 Information Only

All submittals not requiring Government approval will be for information only. They are not considered to be "shop drawings" within the terms of the Contract Clause referred to above.

1.5 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals shall not be construed as a complete check, but will indicate only that the general method of construction, materials, detailing and other information are satisfactory. Approval will not relieve the Contractor of the responsibility for any error which may exist, as the Contractor under the Contractor Quality Control (CQC) requirements of this contract is responsible for dimensions, the design of adequate connections and details, and the satisfactory construction of all work. After submittals have been approved by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.6 DISAPPROVED SUBMITTALS

The Contractor shall respond to all concerns expressed by the Contracting

Officer and promptly make any corrections necessary to address those concerns. The Contractor shall promptly furnish a corrected submittal in the form and number of copies specified for the initial submittal. If the Contractor considers any correction indicated on the submittals to constitute a change to the contract, a notice in accordance with the Clause in Section 00700 CONTRACT CLAUSE, entitled CHANGES (FAR 52.243-4), shall be given promptly to the Contracting Officer.

1.7 GENERAL

The Contractor shall submit all items listed on the Submittal Register (ENG Form 4288) or specified in the other sections of these specifications. The Contractor shall make submittals as required by the specifications. The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Units of weights and measures used on all submittals shall be the same as those used in the contract drawings. Each submittal shall be complete and in sufficient detail to allow ready determination of compliance with contract requirements. Each submittal shall be submitted, with its complete backup material, in both paper and electronic form. Prior to submittal, all items shall be checked and approved by the Contractor's Quality Control (CQC) representative and each item shall be stamped, signed, and dated by the CQC representative indicating action taken. Proposed deviations from the contract requirements shall be clearly identified. Submittals shall include items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals. Submittals requiring Government approval shall be scheduled and made prior to the acquisition of the material or equipment covered thereby. Samples remaining upon completion of the work shall be picked up and disposed of in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.8 SUBMITTAL REGISTER

At the end of this section is a submittal register showing items of equipment and materials for which submittals are required by the specifications; this submittal register may not be all inclusive and additional submittals may be required. The Contractor shall maintain a submittal register for the project in accordance with Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM). The Government will provide the initial submittal register in electronic format. Thereafter, the Contractor shall maintain a complete list of all submittals, including completion of all data columns. Dates on which submittals are received and returned by the Government will be included in its export file to the Contractor. The Contractor shall track all submittals.

1.9 SCHEDULING

Submittals covering component items forming a system or items that are interrelated shall be scheduled to be coordinated and submitted concurrently. Certifications to be submitted with the pertinent drawings shall be so scheduled. Adequate time (a minimum of 30 calendar days exclusive of mailing time) shall be allowed and shown on the register for review and approval. No delay damages or time extensions will be allowed for time lost in late submittals.

1.10 TRANSMITTAL FORM (ENG FORM 4025)

The sample transmittal form (ENG Form 4025) attached to this section shall be used for submitting both Government approved and information only submittals in accordance with the instructions on the reverse side of the form. These forms are included in the QCS software that the Contractor is required to use for this contract. This form shall be properly completed by filling out all the heading blank spaces and identifying each item submitted. Special care shall be exercised to ensure proper listing of the specification paragraph and/or sheet number of the contract drawings pertinent to the data submitted for each item.

1.11 SUBMITTAL PROCEDURES

Submittals shall be made as follows:

1.11.1 Procedures

Procedures for submittals will be stipulated by the Contracting Officer at the preconstruction conference.

1.11.2 Deviations

For submittals which include proposed deviations requested by the Contractor, the column "variation" of ENG Form 4025 shall be checked. The Contractor shall set forth in writing the reason for any deviations and annotate such deviations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted deviations.

1.12 CONTROL OF SUBMITTALS

The Contractor shall carefully control his procurement operations to ensure that each individual submittal is made on or before the Contractor scheduled submittal date shown on the approved "Submittal Register".

1.13 GOVERNMENT APPROVED SUBMITTALS

All submittals shall be received through RMS, including any attachments. Upon completion of review of submittals requiring Government approval, the submittals will be identified as having received approval by being so stamped and dated. The Contractor shall provide at least one (1) hard copy to the Government Inspector of all submittals. This copy will be retained by the Contracting Officer. If the Contractor requires the return of any hard copies for their records, these copies shall be provided in addition to the one (1) Government copy at the time of submittal.

1.14 INFORMATION ONLY SUBMITTALS

Normally submittals for information only will not be returned. Approval of the Contracting Officer is not required on information only submittals. The Government reserves the right to require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so

prescribe. The Contractor shall also submit electronic copies (i.e., .pdf files) of all submittals to expedite the review and approval process.

1.15 STAMPS

Stamps used by the Contractor on the submittal data to certify that the submittal meets contract requirements shall be similar to the following:

CONTRACTOR
(Firm Name)
_____ Approved
_____ Approved with corrections as noted on submittal data and/or attached sheets(s).
SIGNATURE: _____
TITLE: _____
DATE: _____

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

U.S. Army Corps of Engineers (USACE) TRANSMITTAL OF SHOP DRAWINGS, EQUIPMENT DATA, MATERIAL SAMPLES, OR MANUFACTURER'S CERTIFICATES OF COMPLIANCE For use of this form, see ER 415-1-10; the proponent agency is CECW-CE.					DATE		TRANSMITTAL NO.	
SECTION I - REQUEST FOR APPROVAL OF THE FOLLOWING ITEMS <i>(This section will be initiated by the contractor)</i>								
TO:			FROM:		CONTRACT NO.		CHECK ONE: ☐ THIS IS A NEW TRANSMITTAL ☐ THIS IS A RESUBMITTAL OF TRANSMITTAL _____	
SPECIFICATION SEC. NO. <i>(Cover only one section with each transmittal)</i>			PROJECT TITLE AND LOCATION			THIS TRANSMITTAL IS FOR: <i>(Check one)</i> ☐ FIO ☐ GA ☐ DA ☐ CR ☐ DA/CR ☐ DA/GA		
ITEM NO. <i>(See Note 3)</i> a.	DESCRIPTION OF SUBMITTAL ITEM <i>(Type size, model number/etc.)</i> b.	SUBMITTAL TYPE CODE <i>(See Note 8)</i> c.	NO. OF COPIES d.	CONTRACT DOCUMENT REFERENCE		CONTRACTOR REVIEW CODE g.	VARIATION Enter "Y" if requesting a variation <i>(See Note 6)</i> h.	USACE ACTION CODE <i>(Note 9)</i> i.
e.	f.							
REMARKS				I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.				
				NAME OF CONTRACTOR			SIGNATURE OF CONTRACTOR	
SECTION II - APPROVAL ACTION								
ENCLOSURES RETURNED <i>(List by item No.)</i>			NAME AND TITLE OF APPROVING AUTHORITY			SIGNATURE OF APPROVING AUTHORITY		DATE

INSTRUCTIONS

1. Section I will be initiated by the Contractor in the required number of copies.
2. Each Transmittal shall be numbered consecutively. The Transmittal Number typically includes two parts separated by a dash (-). The first part is the specification section number. The second part is a sequential number for the submittals under that spec section. If the Transmittal is a resubmittal, then add a decimal point to the end of the original Transmittal Number and begin numbering the resubmittal packages sequentially after the decimal.
3. The "Item No." for each entry on this form will be the same "Item No." as indicated on ENG FORM 4288-R.
4. Submittals requiring expeditious handling will be submitted on a separate ENG Form 4025-R.
5. Items transmitted on each transmittal form will be from the same specification section. Do not combine submittal information from different specification sections in a single transmittal.
6. If the data submitted are intentionally in variance with the contract requirements, indicate a variation in column h, and enter a statement in the Remarks block describing the detailed reason for the variation.
7. ENG Form 4025-R is self-transmitting - a letter of transmittal is not required.
8. When submittal items are transmitted, indicate the "Submittal Type" (*SD-01 through SD-11*) in column c of Section I.
 Submittal types are the following:

SD-01 - Preconstruction	SD-02 - Shop Drawings	SD-03 - Product Data	SD-04 - Samples	SD-05 - Design Data	SD-06 - Test Reports
SD-07 - Certificates	SD-08 - Manufacturer's Instructions	SD-09 - Manufacturer's Field Reports	SD-10 - O&M Data	SD-11 - Closeout	
9. For each submittal item, the Contractor will assign Submittal Action Codes in column g of Section I. The U.S. Army Corps of Engineers approving authority will assign Submittal Action Codes in column i of Section I. The Submittal Action Codes are:

A -- Approved as submitted. B -- Approved, except as noted on drawings. Resubmission not required. C -- Approved, except as noted on drawings. Refer to attached comments. Resubmission required. D -- Will be returned by separate correspondence. E -- Disapproved. Refer to attached comments.	F -- Receipt acknowledged. X -- Receipt acknowledged, does not comply with contract requirements, as noted. G -- Other action required (<i>Specify</i>) K -- Government concurs with intermediate design. (<i>For D-B contracts</i>) R -- Design submittal is acceptable for release for construction. (<i>For D-B contracts</i>)
--	---
10. Approval of items does not relieve the contractor from complying with all the requirements of the contract.

SUBMITTAL REGISTER											CONTRACT NO.						
TITLE AND LOCATION Tiger Pass Maintenance Dredging						CONTRACTOR											
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION REVIEWER	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		APPROVING AUTHORITY						REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.7	G												
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.8	G CT												
			Dive Operations Plan	1.15	G CT												
			SSHO Designation Letter	1.7.1.2	G CT												
			SD-06 Test Reports														
			Accident Reports	1.12.2	G												
			LHE Inspection Reports	1.12.3													
			Monthly Exposure Reports	1.5	G												
			SD-07 Certificates														
			Crane Operators/Riggers														
			Activity Hazard Analysis (AHA)	1.9	G CT												
			Certificate of Compliance	1.12.4													
			Hot Work Permit	1.13													
			License Certificates														
			Standard Lift Plan	1.12.4	G												
		01 45 04.00 10	SD-01 Preconstruction Submittals														
			Contractor Quality Control Plan	3.2	G CD												
		01 57 20.01 12	SD-01 Preconstruction Submittals														
			Environment Pollution Control	1.5.1	G CD												
			Plan														
			Bird Nesting Prevention Plan		G ODT												
		01 78 03.00 10	SD-02 Shop Drawings														
			'As-Built' Drawings	1.3.1	G EDC												
		02 21 10.00 12	SD-02 Shop Drawings														

SUBMITTAL REGISTER											CONTRACT NO.						
TITLE AND LOCATION Tiger Pass Maintenance Dredging						CONTRACTOR											
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION OR REVIEW NUMBER	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		APPROVING AUTHORITY						REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		02 21 10.00 12	Preliminary Field Surveys	3.1.3	G EDC												
			Preliminary Field Surveys	3.1.3	G EDC												
			'Before and After' Dredging Cross Sections		G EDC												
			Disposal Area Surveys														
			SD-05 Design Data														
			Compiled Survey Data CD Disk														
			Acceptance Profiles		G												
			SD-06 Test Reports														
			Calibration Checks	1.6.1.2	G												
			SD-09 Manufacturer's Field Reports														
			Survey Equipment	1.6	G CD												
		35 20 23.00 12	SD-01 Preconstruction Submittals														
			Dredge Plant Data	1.7.2													
			Plan of Operation	3.2.2													
			Access Plan		G EDC												
			Construction Schedule		G CD												
			Disposal of Dredged Material	3.3	G CD												
			SD-07 Certificates														
			Certified Letters	3.2.2	G EDC												
			SD-11 Closeout Submittals														
			Report of Operations		G EDC												
		35 20 23.33	SD-01 Preconstruction Submittals														
			Dredge Plant Instrumentation		G OM												
			Plan Revisions														

CONTRACT NO.

CONTRACTOR

[illegible]

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

05/24

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Site Safety and Health Officer (SSHO)
 - 1.2.1.1 Level One SSHO
 - 1.2.1.2 Level Two SSHO
 - 1.2.1.3 Level Three SSHO
 - 1.2.1.4 Alternate SSHO
 - 1.2.2 Competent Person (CP)
 - 1.2.3 Qualified Person (QP)
- 1.3 SUBMITTALS
- 1.4 PUBLIC HEALTH EMERGENCIES
- 1.5 MONTHLY EXPOSURE REPORTS
- 1.6 REGULATORY REQUIREMENTS
- 1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS
 - 1.7.1 Site Safety and Health Officer (SSHO)
 - 1.7.1.1 Qualifications of SSHO
 - 1.7.1.2 Duties of SSHO
 - 1.7.1.3 Safety Meetings
 - 1.7.2 Roles and Responsibilities of Prime Contractor and SSHO
 - 1.7.3 Dredging Contract Site Safety and Health Officer(s) (SSHOs) Requirements
 - 1.7.3.1 Dredging SSHO Personnel Requirements
 - 1.7.4 Preconstruction Conference
- 1.8 ACCIDENT PREVENTION PLAN (APP)
 - 1.8.1 Accident Prevention Plan (APP)
 - 1.8.2 Names and Qualifications
- 1.9 ACTIVITY HAZARD ANALYSIS (AHA)
 - 1.9.1 AHA Management
 - 1.9.2 AHA Signature Log
- 1.10 SITE SAFETY REFERENCE MATERIALS
- 1.11 EMERGENCY MEDICAL TREATMENT
- 1.12 NOTIFICATIONS AND REPORTS
 - 1.12.1 Accident Notification
 - 1.12.2 Accident Reports
 - 1.12.3 LHE Inspection Reports
 - 1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging
- 1.13 HOT WORK PERMIT
 - 1.13.1 Permit and Personnel Requirements
 - 1.13.2 Work Around Flammable Materials
- 1.14 SEVERE WEATHER PLAN FOR MARINE ACTIVITIES (SWPMA)

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

3.1.1 Worksite Communication

3.2 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

3.2.1 Lockout/Tagout Isolation

3.2.2 Lockout/Tagout Removal

-- End of Section Table of Contents --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS
05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.44 (2020) Control of Energy Sources
(Lockout/Tagout) for Construction and
Demolition Operations

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1 (2014; R 2019) American National Standard
for Industrial Head Protection

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 51B (2024) Standard for Fire Prevention During
Welding, Cutting, and Other Hot Work

NFPA 70 (2023; ERTA 7 2023; TIA 23-15) National
Electrical Code

NFPA 70E (2024) Standard for Electrical Safety in
the Workplace

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Health
Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1915 Confined and Enclosed Spaces and Other
Dangerous Atmospheres in Shipyard
Employment

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further

clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G, CT

Dive Operations Plan; G, CT

SSHO Designation Letter; G, CT

SD-06 Test Reports

Accident Reports; G

LHE Inspection Reports

Monthly Exposure Reports; G

SD-07 Certificates

Crane Operators/Riggers

Activity Hazard Analysis (AHA); G, CT

Certificate of Compliance

Hot Work Permit

License Certificates

Standard Lift Plan; G

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth of each month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the progress payment.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and the applicable federal, state, and local laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Site Safety and Health Officer (SSHO)

1.7.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1 and this Contract.

1.7.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1. All SSHOs will be designated on an ENG Form 6282, provided by the Contracting Officer (SSHO Designation Letter). Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs will:

- (1) Are designated on an ENG Form 6282.
- (2) Meet minimum training and experience requirements identified in EM 385-1-1.
- (3) Execute roles and responsibilities identified in EM 385-1-1.

1.7.3 Dredging Contract Site Safety and Health Officer(s) (SSHOs) Requirements

1.7.3.1 Dredging SSHO Personnel Requirements

- a. Provide a minimum of one primary Level One SSHO assigned for the primary shift.

Note: Hopper Dredges with U.S. Coast Guard, credentialed crews the prime contractor may designate a Level Two SSHO in lieu of having a Level one SSHO onboard.

- b. For a project involving multiple work shifts, provide a minimum of a Level One SSHO for each additional shift.
- c. For individual dredging projects the prime contractor will designate additional Level Three SSHOs at locations where the primary Level One SSHO is not located (example: on dredge, tug, material placement site).

Examples of one dredging project site is reflected in each of the following:

- (1) a mechanical dredge, tug(s) and scow(s), scow route, and material placement site; or
 - (2) a hydraulic pipeline dredge, attendant plant, and material placement site; or,
 - (3) a hopper dredge (include land-based material placement site - if applicable.)
- d. Designated SSHOs must be present at the project site, located so that they have full mobility and reasonable access to all major work operations and must be available during their shift for immediate verbal consultation and notification.
 - e. Designated Level One and Two SSHOs must have direct report authority to a senior project (or corporate) management official.
 - f. Designated Level Three SSHOs must report potential safety and occupational health hazards, incidents, and concerns to the Level One or Two SSHO on shift.
 - g. Level One and Level Two SSHOs for dredging must have a minimum of 3 years experience in one of the following areas:
 - (1) Supervising/managing dredging activities.
 - (2) Supervising/managing marine construction activities.
 - (3) Supervising/managing land-based construction activities.
 - (4) Work managing safety programs or processes.
 - (5) Conducting hazard analyses and developing controls in activities or environments with similar hazards.

1.7.4 Preconstruction Conference

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP and Schedule will be created after being given Notice to Proceed.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Accident Prevention Plan (APP)

Submit the Accident Prevention Plan (APP), an electronic copy and one hard copy, for review and acceptance by the Government within 15 calendar days after being given Notice to Proceed, and at least 7 days prior to the prework conference. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with the APP and ENG Form 6293 Accident Prevention Plan Worksheet. The SSO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1 or as directed by the Contracting Officer. Submit the AHA for review prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in

accordance with EM 385-1-1. The Government has no responsibility to provide emergency medical treatment.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Accident Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

Table Accident Reporting Required Timeline		
Accident Type	Notify KO or COR	Complete Final Accident Report on ENG 3394 and provide to KO or COR
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over \$600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.12.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. All accidents are reportable, regardless of whether or not it is recordable. Complete the applicable USACE Accident Report, ENG Form 3394, and provide the report to the Contracting Officer within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms. All accidents are reportable, regardless of whether or not it is recordable.
- b. Near Misses: For Army projects, report all "Near Misses" to the Contracting Officer, using local accident reporting procedures, within 24 hours. The Contracting Officer will provide the Contractor the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as

specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1 and using Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.13 HOT WORK PERMIT

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e., welding or cutting) or operating other flame-producing/spark producing devices, from the SSHO. Contractors are required to meet all criteria before a permit is issued. Provide at least two 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of 1 hour after completion of the task or as specified on the hot work permit.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1.

1.14 SEVERE WEATHER PLAN FOR MARINE ACTIVITIES (SWPMA)

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

- a. Head Protection that meets ANSI/ISEA Z89.1
- b. Long Pants
- c. Appropriate Safety Footwear
- d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.2 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

3.2.1 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1.

3.2.2 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal

notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

07/06

PART 1 GENERAL

1.1 REFERENCES

1.2 ORDERING INFORMATION

-- End of Section Table of Contents --

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

07/06

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization, (e.g. ASTM B 564 Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided. Documents listed in the specifications with numbers which were not assigned by the standards producing organization should be ordered from the source by title rather than by number.

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: <http://www.astm.org>

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)
1901 North Moore Street
Arlington, VA 22209-1762
Ph: 703-525-1695
Fax: 703-528-2148
E-mail: isea@safetyequipment.org
Internet: <http://www.safetyequipment.org/>

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
445 Hoes Lane
Piscataway, NJ 08855-1331
Ph: 732-981-0060
Fax: 732-981-1712
E-mail: customer-services@ieee.org
Internet: <http://www.ieee.org>

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
1 Batterymarch Park
Quincy, MA 02169-7471
Ph: 617-770-3000
Fax: 617-770-0700

E-mail: webmaster@nfpa.org
Internet: <http://www.nfpa.org>

U.S. ARMY CORPS OF ENGINEERS (USACE)
Order CRD-C DOCUMENTS from:
U.S. Army Engineer Research & Development Center
ATTN: Materials Testing Center
Building 6000
3909 Halls Ferry Road
Vicksburg, MS 39180-6199
Ph: 601-634-2496
Fax: 601-634-3242
E-mail: mtc-info@erdc.usace.army.mil
Internet: <http://www.wes.army.mil/SL/MTC/handbook.htm>

Order Other Documents from:
USACE Publications Depot
Attn: CEHEC-IM-PD
2803 52nd Avenue
Hyattsville, MD 20781-1102
Ph: 301-394-0081
Fax: 301-394-0084
E-mail: mtc-info@erdc.usace.army.mil
Internet: <http://gsl.erdc.usace.army.mil/SL/MTC/>

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)
Ariel Rios Building
1200 Pennsylvania Avenue, N.W.
Washington, DC 20460
Ph: 202-272-0167
Internet: <http://www.epa.gov>

--- Some EPA documents are available only from:
National Technical Information Service (NTIS)
5285 Port Royal Road
Springfield, VA 22161
Ph: 703-605-6585
Fax: 703-605-6900
E-mail: info@ntis.gov
Internet: <http://www.ntis.gov>

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
8601 Adelphi Road
College Park, MD 20740-6001
Ph: 866-272-6272
Fax: 301-837-0483
Internet: <http://www.archives.gov>

Order documents from:
Superintendent of Documents
U.S. Government Printing Office (GPO)
732 North Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800
Fax: 202-512-2104
E-mail: contactcenter@gpo.gov
Internet: <http://www.gpoaccess.gov>

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

11/16

PART 1 GENERAL

- 1.1 MEASUREMENT AND PAYMENT
- 1.2 REFERENCES
- 1.3 CONTRACT ADMINISTRATION
 - 1.3.1 Correspondence and Electronic Communications
 - 1.3.2 Other Factors
- 1.4 RMS SOFTWARE
- 1.5 SYSTEM REQUIREMENTS
- 1.6 CONTRACT DATABASE - GOVERNMENT
- 1.7 CONTRACT DATABASE - CONTRACTOR
 - 1.7.1 Administration
 - 1.7.1.1 Contractor Information
 - 1.7.1.2 Subcontractor Information
 - 1.7.1.3 Correspondence
 - 1.7.1.4 Equipment
 - 1.7.1.5 Reports
 - 1.7.1.6 Request For Information (RFI)
 - 1.7.2 Finances
 - 1.7.2.1 Pay Activity Data
 - 1.7.2.2 Payment Requests
 - 1.7.3 Quality Control (QC)
 - 1.7.3.1 Quality Control (QC) Reports
 - 1.7.3.2 Deficiency Tracking
 - 1.7.3.3 Three-Phase Control Meetings
 - 1.7.3.4 Labor and Equipment Hours
 - 1.7.3.5 Accident/Safety Reporting
 - 1.7.3.6 Definable Features of Work
 - 1.7.3.7 Activity Hazard Analysis
 - 1.7.4 Submittal Management
 - 1.7.5 Schedule
 - 1.7.6 Closeout
- 1.8 IMPLEMENTATION
- 1.9 NOTIFICATION OF NONCOMPLIANCE

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
11/16

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for resident management system requirements as specified herein. Payment for the work covered under this section shall be distributed throughout the existing bid items.

1.2 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Health
Requirements Manual

RMS CM Guide (2021) Resident Management System (RMS)
User Manual For Contractors

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Contractor accesses the system using the Contractor Mode (RMS CM) module. The term RMS will be used in the remainder of this section for both RMS and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

Administration
Finances
Quality Control
Submittal Monitoring
Scheduling
Closeout
Import/Export of Data

For assistance in providing contract-required data to the Government, the Contractor is directed to the following website for guidance:
https://rms.usace.army.mil/datafiles/rms_qcs_manuals/qcs_manual_2_38.pdf

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Some correspondence, including pay requests and payrolls, are also to be provided in paper format with original signatures. Paper documents will govern, in the event of discrepancy with the electronic version.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to Contract Clause, (FAR 52.236-15) "SCHEDULES FOR CONSTRUCTION CONTRACTS"; Contract Clause, (FAR 52.232-27) "PROMPT PAYMENT FOR CONSTRUCTION CONTRACTS"; Contract Clause, (FAR 52.232-5) "PAYMENTS UNDER FIXED-PRICED CONSTRUCTION CONTRACTS"; Section 01 33 00 SUBMITTAL PROCEDURES; and Section 01 45 04.00 10 CONTRACTOR QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a Windows-based program that can be run on a Windows-based PC meeting the requirements as specified in paragraph SYSTEM REQUIREMENTS. Download, install and be able to utilize the latest version of the RMS software within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals (RMS CM Guide), access and installation instructions, program updates and training information are available from the RMS website (<https://rms.usace.army.mil/datafiles/rmsdocwebsite/default.html>). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 SYSTEM REQUIREMENTS

The following is the recommended system configuration to run the Contractor Mode RMS for full utilization of all features for all types and sizes of contracts. Smaller, less complicated, projects may not require the configuration levels described below. Required configuration also noted below.

Minimum RMS System Requirements	
Hardware	
Windows-based PC	Intel i3 or- AMD A6 3650; 1.7 GHz or higher processor (REQUIRED)
RAM	8 GB
Hard drive disk	100 GB space for sole use by RMS system
Monitor	Screen resolution 1366 x 768
Mouse or other pointing device	
Windows compatible printer	Laser printer must have 4 MB+ of RAM
Connection to the Internet	minimum 10 Mbs per user
Software	

Minimum RMS System Requirements	
MS Windows	Windows 10 64-bit (RMS requires a 64-bit OS) or newer (REQUIRED)
Word Processing software	Viewer for MS Word 2013, MS Excel 2013 or newer (REQUIRED)
E-mail	MAPI compatible (REQUIRED)
Virus protection software	Regularly upgraded with all issued Manufacturer's updates and is able to detect most zero day viruses (REQUIRED)

1.6 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance (QA) comments, as well as other miscellaneous administrative information.

1.7 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The contractor responsibilities for entries in RMS typically include the following items:

1.7.1 Administration

1.7.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.7.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within 7 calendar days of receiving access to the contract in RMS or within 7 calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.7.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.7.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates.

1.7.1.5 Reports

Track the status of the project utilizing the reports available in RMS. The value of these reports is reflective of the quality of the data input. These reports include the Progress Payment Request worksheet, Quality Control (QC) comments, Submittal Register Status, and Three-Phase Control worksheets.

1.7.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response.

1.7.2 Finances

1.7.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule. The sum of pay activities equals the total contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.7.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data either electronically or by hard copy. Unless waived by the Contracting Officer, a signed paper copy of the approved payment certification and request is also required and will govern in the event of discrepancy with the electronic version.

1.7.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Insure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments of same within 7 calendar days of Government acceptance of the CQC Plan.

1.7.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may

be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report. Provide the Government a printed signed copy of the QC Daily Report, unless waived by the Contracting Officer.

1.7.3.2 Deficiency Tracking

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.7.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS. Worksheets for the three-phase control meetings are generated within RMS.

1.7.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.7.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor will monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1 and as required by any other applicable Federal, State or local agencies.

1.7.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.7.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager.

1.7.4 Submittal Management

Enter all current submittal register data and information into RMS within 7 calendar days of receiving access to the contract in RMS. The information shown on the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES will already be entered into the RMS database when access is granted. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, spare parts, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the contract. Submit hard copies of all submittals unless waived by the Contracting

Officer.

1.7.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file.

1.7.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.8 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.9 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within 7 calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 04.00 10

CONTRACTOR QUALITY CONTROL

03/14

PART 1 GENERAL

- 1.1 MEASUREMENT AND PAYMENT
- 1.2 SUBMITTALS
- 1.3 ELECTRONIC TEST REPORT DATA

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 GENERAL REQUIREMENTS
- 3.2 CONTRACTOR QUALITY CONTROL (CQC) PLAN
 - 3.2.1 Content of the CQC Plan
 - 3.2.2 Acceptance of Plan
 - 3.2.3 Notification of Changes
- 3.3 PRE-WORK MEETINGS
- 3.4 COORDINATION MEETING
 - 3.4.1 Pre-Work Coordination Meeting
- 3.5 QUALITY CONTROL ORGANIZATION
 - 3.5.1 Personnel Requirements
 - 3.5.2 CQC System Manager
 - 3.5.3 CQC Personnel
 - 3.5.4 Additional Requirement
 - 3.5.5 Organizational Changes
- 3.6 SUBMITTALS AND DELIVERABLES
- 3.7 CONTROL
 - 3.7.1 Preparatory Phase
 - 3.7.2 Initial Phase
 - 3.7.3 Follow-up Phase
 - 3.7.4 Additional Preparatory and Initial Phases
- 3.8 COMPLETION INSPECTION
 - 3.8.1 Punch-Out Inspection
 - 3.8.2 Pre-Final Inspection
 - 3.8.3 Final Acceptance Inspection
- 3.9 DOCUMENTATION
- 3.10 SAMPLE FORMS
- 3.11 NOTIFICATION OF NONCOMPLIANCE

-- End of Section Table of Contents --

SECTION 01 45 04.00 10

CONTRACTOR QUALITY CONTROL
03/14

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for providing and maintaining an effective Contractor Quality Control program, and all costs associated therewith shall be included in the applicable contract unit or job prices contained in the Bidding Schedule.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor Quality Control Plan; G, CD

1.3 ELECTRONIC TEST REPORT DATA

As part of the Contractor Quality Control program, his/her selected QC laboratory shall provide electronic transmission of the test report data in the prescribed formats with the original hard copy test report data to the government. The New Orleans District Construction Control Manual (NODCC Manual) specifies the minimum number of tests to be made, and includes forms which shall be used to report the number of tests and test data. The technical specification sections may include testing and/or frequency requirements other than those listed in the NODCC manual. These additional requirements shall be followed in addition to the aforementioned. A copy of the NODCC manual is attached at the end of this section. Test results shall be emailed to mvn-cd-q-testresults@usace.army.mil and to the Government Project Engineer. In addition, all test results shall be uploaded to the Resident Management System Contractor Mode (RMS CM).

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

The Contractor is responsible for quality control and shall establish and maintain an effective quality control system in compliance with the Contract Clause in Section 00700, entitled "INSPECTION OF CONSTRUCTION" (FAR 52.246-12). The quality control system shall consist of plans, procedures, and organization necessary to produce an end product which complies with the contract requirements. The system shall cover all construction operations, both onsite and offsite, and shall be keyed to the

proposed construction sequence. The site project superintendent and Quality Control Manager will be held responsible for the quality of work on the job and is subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the contract. The site project superintendent in this context shall be the highest level manager responsible for the overall construction activities at the site, including quality and production. The site project superintendent shall maintain a physical presence at the site at all times, except as otherwise acceptable to the Contracting Officer, and shall be responsible for all construction and construction related activities at the site.

3.2 CONTRACTOR QUALITY CONTROL (CQC) PLAN

The Contractor shall furnish for review by the Government, not later than 15 days after receipt of notice of award, the Contractor Quality Control Plan proposed to implement the requirements of the Contract Clause in Section 00700, entitled INSPECTION OF CONSTRUCTION (FAR 52.246-12). The plan shall identify personnel, procedures, control, instructions, tests, records, and forms to be used. The Government will consider an interim plan for the first 30 days of operation. Construction will be permitted to begin only after acceptance of the CQC Plan or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the features of work included in an accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional features of work to be started. A sample CQC Plan is attached at the end of the section.

3.2.1 Content of the CQC Plan

The CQC Plan shall include, as a minimum, the following to cover all operations, both onsite and offsite, including work by subcontractors, fabricators, suppliers, and purchasing agents:

- a. A description of the quality control organization, including a chart showing lines of authority and acknowledgment that the CQC staff shall implement the three phase control system for all aspects of the work specified. The staff shall include a CQC System Manager who shall report to the project superintendent.
- b. The name, qualifications (in resume format), duties, responsibilities, and authorities of each person assigned a CQC function.
- c. A copy of the letter to the CQC System Manager signed by an authorized official of the firm which describes the responsibilities and delegates sufficient authorities to adequately perform the functions of the CQC System Manager, including authority to stop work which is not in compliance with the contract. The CQC System Manager shall issue letters of direction to all other various quality control representatives outlining duties, authorities, and responsibilities. Copies of these letters shall also be furnished to the Government.
- d. Procedures for scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. These procedures shall be in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.
- e. Control, verification, and acceptance testing procedures for each

specific test to include the test name, specification paragraph requiring test, feature of work to be tested, test frequency, and person responsible for each test. (Laboratory facilities approved by the Contracting Officer shall be used.)

- f. Procedures for tracking preparatory, initial, and follow-up control phases and control, verification, and acceptance tests including documentation.
- g. Procedures for tracking construction deficiencies from identification through acceptable corrective action. These procedures shall establish verification that identified deficiencies have been corrected.
- h. Reporting procedures, including proposed reporting formats.
- i. A list of the definable features of work. A definable feature of work is a task which is separate and distinct from other tasks, has separate control requirements, and may be identified by different trades or disciplines, or it may be work by the same trade in a different environment. Although each section of the specifications may generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. This list will be agreed upon during the coordination meeting.

3.2.2 Acceptance of Plan

Acceptance of the Contractor's plan is required prior to the start of construction. Acceptance is conditional and will be predicated on satisfactory performance during the construction. The Government reserves the right to require the Contractor to make changes in his/her CQC Plan and operations including removal of personnel, as necessary, to obtain the quality specified.

3.2.3 Notification of Changes

After acceptance of the CQC Plan, the Contractor shall notify the Contracting Officer in writing of any proposed change. Proposed changes are subject to acceptance by the Contracting Officer.

3.3 PRE-WORK MEETINGS

Prior to the start of construction, the contractor shall meet with the Contracting Officer or his/her authorized representative in a Pre-Construction Conference and a Pre-Work Coordination Meeting. At least 14 calendar days prior to the Preconstruction Conference, the contractor shall submit to the government for review and approval his/her Contractor Accident Prevention Plan and Contractor Quality Control Plan.

3.4 COORDINATION MEETING

3.4.1 Pre-Work Coordination Meeting

After the Preconstruction Conference, before start of construction, the contractor shall meet with the Contracting Officer, or his/her authorized representative, in a Pre-Work Coordination Meeting to discuss the contractor's Quality Control Plan. During this meeting, a mutual understanding of the system details shall be developed, including the forms

for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting shall be prepared by the Government and signed by both the Contractor and the Contracting Officer. The minutes shall become a part of the contract file. There may be occasions when subsequent conferences will be called by either party to reconfirm mutual understandings and/or address deficiencies in the CQC system or procedures which may require corrective action by the Contractor.

3.5 QUALITY CONTROL ORGANIZATION

3.5.1 Personnel Requirements

The requirements for the CQC organization are a CQC System Manager and sufficient number of additional qualified personnel to ensure safety and contract compliance. The Safety and Health Manager shall serve as a member of the CQC staff. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly will also be included as part of the CQC organization. The Contractor's CQC staff shall maintain a presence at the site at all times during progress of the work and have complete authority and responsibility to take any action necessary to ensure contract compliance. The CQC staff shall be subject to acceptance by the Contracting Officer. The Contractor shall provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Complete records of all letters, material submittals, shop drawing submittals, schedules and all other project documentation shall be promptly furnished to the CQC organization by the Contractor. The CQC organization shall be responsible to maintain these documents and records at the site at all times, except as otherwise acceptable to the Contracting Officer.

3.5.2 CQC System Manager

The Contractor shall identify as CQC System Manager an individual within the onsite work organization who shall be responsible for overall management of CQC and have the authority to act in all CQC matters for the Contractor. The CQC System Manager shall be a construction person with a minimum of 3 years (full time) experience in related equivalent work. This CQC System Manager shall be on the site at all times during construction and shall be employed by the prime Contractor. The CQC System Manager may not have any other duties than quality control. When the alternate is activated, they shall also have no other duties than quality control. An alternate for the CQC System Manager shall be identified in the Contractor's CQC Plan to serve in the event of the System Manager's absence. The requirements for the alternate shall be the same as for the designated CQC System Manager if he is called upon to act in that capacity.

3.5.3 CQC Personnel

In addition to CQC personnel specified elsewhere in the contract, the Contractor shall provide as part of the CQC organization specialized personnel to assist the CQC System Manager for the following areas: civil, environmental, submittals clerk. These individuals may be employees of the prime or subcontractor; be responsible to the CQC System Manager; be physically present at the construction site during work on their areas of responsibility; have the necessary education and/or experience in accordance with the experience matrix listed herein. These individuals may perform other duties but must be allowed sufficient time to

perform their assigned quality control duties as described in the Quality Control Plan. A single person may cover more than one area provided that they are qualified to perform QC activities in each designated field of expertise and if his/her workload allows.

Experience Matrix

	Area	Qualifications
a.	Civil	Graduate Civil Engineer or Construction Manager with 2 years experience in the type of work being performed on this project or technician with 5 yrs related experience
b.	Environmental	Graduate Environmental Engineer with 3 yrs experience
c.	Submittals	Submittal Clerk with 1 yr experience

3.5.4 Additional Requirement

In addition to the above experience and education requirements the CQC System Manager, and his/her alternate, shall have completed the course entitled "Construction Quality Management for Contractors" within the past 3 years. This course is periodically offered at the New Orleans District and other Corps of Engineers districts.

3.5.5 Organizational Changes

The Contractor shall maintain the CQC staff at full strength at all times. When it is necessary to make changes to the CQC staff, the Contractor shall revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

3.6 SUBMITTALS AND DELIVERABLES

Submittals, if needed, shall be made as specified in Section 01 33 00 SUBMITTAL PROCEDURES. The CQC organization shall be responsible for certifying that all submittals and deliverables are in compliance with the contract requirements.

3.7 CONTROL

Contractor Quality Control is the means by which the Contractor ensures that the construction, to include that of subcontractors and suppliers, complies with the requirements of the contract. At least three phases of control shall be conducted by the CQC System Manager for each definable feature of the construction work as follows:

3.7.1 Preparatory Phase

This phase shall be performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. This phase shall include:

- a. A review of each paragraph of applicable specifications
- b. A review of the contract drawings.
- c. A check to assure that all materials and/or equipment have been tested, submitted, and approved.
- d. Review of provisions that have been made to provide required control inspection and testing.
- e. Examination of the work area to assure that all required preliminary work has been completed and is in compliance with the contract.
- f. A physical examination of required materials, equipment, and sample work to assure that they are on hand, conform to approved shop drawings or submitted data, and are properly stored.
- g. A review of the appropriate activity hazard analysis to assure safety requirements are met.
- h. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.
- i. A check to ensure that the portion of the plan for the work to be performed has been accepted by the Contracting Officer.
- j. Discussion of the initial control phase.
- k. The Government Quality Assurance personnel shall be notified at least 48 hours in advance of beginning the preparatory control phase. The Contractor shall submit a written agenda of the topics to be discussed at the preparatory meeting on the day prior to the meeting date. This phase shall include a meeting conducted by the CQC System Manager and attended by the superintendent, other CQC personnel (as applicable), Government Quality Assurance personnel, and the foreman responsible for the definable feature. The results of the preparatory phase actions shall be documented by separate minutes prepared by the CQC System Manager and attached to the daily CQC report. The Contractor shall instruct applicable workers as to the acceptable level of workmanship required in order to meet contract specifications.

3.7.2 Initial Phase

This phase shall be accomplished at the beginning of a definable feature of work. The following shall be accomplished:

- a. A check of work to ensure that it is in full compliance with contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full contract compliance. Verify required control inspection and testing.
- c. Establish level of workmanship and verify that it meets minimum acceptable workmanship standards. Compare with required sample panels as appropriate.

- d. Resolve all differences.
- e. Check safety to include compliance with and upgrading of the safety plan and activity hazard analysis. Review the activity analysis with each worker.
- f. The Government shall be notified at least 24 hours in advance of beginning the initial phase. Separate minutes of this phase shall be prepared by the CQC System Manager and attached to the daily CQC report. Exact location of initial phase shall be indicated for future reference and comparison with follow-up phases.
- g. The initial phase should be repeated for each new crew to work onsite, or any time acceptable specified quality standards are not being met.

3.7.3 Follow-up Phase

Daily checks shall be performed to assure control activities, including control testing, are providing continued compliance with contract requirements, until completion of the particular feature of work. The checks shall be made a matter of record in the CQC documentation. Final follow-up checks shall be conducted and all deficiencies corrected prior to the start of additional features of work which may be affected by the deficient work. The Contractor shall not build upon nor conceal non-conforming work.

3.7.4 Additional Preparatory and Initial Phases

Additional preparatory and initial phases shall be conducted on the same definable features of work if: the quality of on-going work is unacceptable; if there are changes in the applicable CQC staff, onsite production supervision or work crew; if work on a definable feature is resumed after a substantial period of inactivity; or if other problems develop.

3.8 COMPLETION INSPECTION

3.8.1 Punch-Out Inspection

Near the end of the work, or any increment of the work established by a time stated in the Contract Clause in Section 00700, entitled "COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK (FAR 52.211-10)" or stated elsewhere in the specifications, the CQC Manager shall conduct an inspection of the work. A punch list of items which do not conform to the approved drawings and specifications shall be prepared and included in the CQC documentation, as required by paragraph DOCUMENTATION. The list of deficiencies shall include the estimated date by which the deficiencies will be corrected. The CQC System Manager or staff shall make a second inspection to ascertain that all deficiencies have been corrected. Once this is accomplished, the Contractor shall notify the Government that the facility is ready for the Government Pre-Final inspection.

3.8.2 Pre-Final Inspection

The Government will perform the pre-final inspection to verify that the work is complete. A Government Pre-Final Punch List may be developed as a result of this inspection. The Contractor's CQC System Manager shall

ensure that all items on this list have been corrected before notifying the Government, so that a Final inspection with the customer can be scheduled. Any items noted on the Pre-Final inspection shall be corrected in a timely manner. These inspections and any deficiency corrections required by this paragraph shall be accomplished within the time slated for completion of the entire work or any particular increment of the work if the project is divided into increments by separate completion dates.

3.8.3 Final Acceptance Inspection

The Contractor's Quality Control Inspection personnel, plus the superintendent or other primary management person, and the Contracting Officer's Representative shall be in attendance at this inspection. Additional Government personnel including, but not limited to, those from the New Orleans District, Mississippi Valley Division, and local interest may also be in attendance. The final acceptance inspection will be formally scheduled by the Contracting Officer based upon results of the Pre-Final inspection. Notice shall be given to the Contracting Officer at least 14 days prior to the final acceptance inspection and shall include the Contractor's assurance that all specific items previously identified to the Contractor as being unacceptable, along with all remaining work performed under the contract, will be complete and acceptable by the date scheduled for the final acceptance inspection. Failure of the Contractor to have all contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause in Section 00700 entitled INSPECTION OF CONSTRUCTION (FAR 52.246-12).

3.9 DOCUMENTATION

The Contractor shall maintain current records providing factual evidence that required quality control activities and/or tests have been performed. These records shall include the work of subcontractors and suppliers and shall be on an acceptable form that includes, as a minimum, the following information:

- a. Contractor/subcontractor and their area of responsibility.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When Network Analysis (NAS) is used, identify each phase of work performed each day by NAS activity number.
- d. Test and/or control activities performed with results and references to specifications/drawings requirements. The control phase shall be identified (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action.
- e. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications/drawings requirements.
- f. Submittals and deliverables reviewed, with contract reference, by whom, and action taken.
- g. Offsite surveillance activities, including actions taken.

- h. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- i. Instructions given/received and conflicts in plans and/or specifications.
- j. Contractor's verification statement.

These records shall indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. These records shall cover both conforming and deficient features and shall include a statement that equipment and materials incorporated in the work and workmanship comply with the contract. The original and one copy of these records in report form shall be furnished to the Government daily within 12 hours after the date covered by the report, except that reports need not be submitted for days on which no work is performed. As a minimum, one report shall be prepared and submitted for every 7 days of no work and on the last day of a no work period. All calendar days shall be accounted for throughout the life of the contract. The first report following a day of no work shall be for that day only. Reports shall be signed and dated by the CQC System Manager. The report from the CQC System Manager shall include copies of test reports and copies of reports prepared by all subordinate quality control personnel.

3.10 SAMPLE FORMS

Sample forms for guidance in preparing the CQC Plan are enclosed at the end of this section.

3.11 NOTIFICATION OF NONCOMPLIANCE

The Contracting Officer will notify the Contractor of any detected noncompliance with the foregoing requirements. The Contractor shall take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, shall be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders shall be made the subject of claim for extension of time or for excess costs or damages by the Contractor.

-- End of Section --



**US Army Corps
of Engineers ®**
New Orleans District

Construction Control Manual

**Sampling & Testing Construction Materials
Reporting Test Results**

**CEMVN CD 415-Q-11
14 March 2016**



DEPARTMENT OF THE ARMY
NEW ORLEANS DISTRICT, CORPS OF ENGINEERS
7400 LEAKE AVENUE
NEW ORLEANS, LOUISIANA 70118

CEMVDN-CD

Pamphlet
Number CEMVN-CD-415-Q-11

14 March 2016

Construction

CONSTRUCTION CONTROL MANUAL

1. Purpose. This manual describes the means and methods for the Contractor Quality Control (QC) and Government Quality Assurance (QA) testing of some of the more common construction materials incorporated into New Orleans District projects. Information is given on sampling, the test required, testing frequency, reporting requirements, and database maintenance. This manual only describes a minimum testing program on a limited number of common construction materials and the specifications may require additional tests that demonstrate compliance with the contract documents.

2. Applicability. This manual applies to all New Orleans District elements having responsibility for the design and construction of assigned projects.

3. Scope of the Manual. This manual is intended to guide the Quality Control and Quality Assurance process and provide for the construction of a project whose quality and durability is a direct reflection of the Contractor's and the Government's efforts in meeting the project's goals and objectives. If there is a conflict between this manual and the technical specification sections, the most stringent requirements shall govern.

DEPARTMENT OF THE ARMY
U.S. ARMY ENGINEER DISTRICT, NEW ORLEANS
CORPS OF ENGINEERS

Number CEMVN-CD-415-Q-11

Table of Contents

Chapter 1 Introduction	1
1. General:.....	1
2. Definitions:	1
3. Responsibility, Compilation, and Submittal of Test Results:	1
Chapter 2 Soils.....	4
1. Scope:.....	4
2. Samples:.....	4
3. Testing Personnel:.....	4
4. Typical Test Requirements:	5
5. Sampling and Testing of Compacted Fill:	6
6. Sampling and Testing of Un-Compacted Berm Material:	7
7. Compilation of Test Data for Submittal:	7
8. Soil Electronic Conductivity (EC) and Total Soluble Salt Analysis:	9
9. Field and Laboratory Determination of Non-Soil Volume for Levee Fill:.....	10
10. Additional Testing:.....	14
Chapter 3 Concrete	15
1. Scope:.....	15
2. Samples:	15
3. Testing Personnel:.....	15
4. Typical Test Requirements:	16
5. Compilation of Test Data for Submittal:	17
Chapter 4 Welding Inspection	20
1. Scope:.....	20
2. Definitions:	20
3. Testing Personnel:.....	20
4. Visual Inspection Requirements:	21
5. Nondestructive Testing Requirements:.....	21
6. Acceptance Criteria:.....	21
7. Frequency of Testing:	21
8. Compilation of Test Data for Submittal:	22

Appendix A – Test Form Management	23
1. Report Numbering:	23
2. Naming the Test Form Files:	23
3. Submitting Test Forms:.....	24
4. How to Access USACE QACC SharePoint Site	25
5. Uploading test forms.....	25
6. Revisions and Special Naming Considerations:	25
Appendix B - Material Testing Laboratory Requirements	26
1. Purpose:.....	26
2. Applicability:	26
3. References:.....	26
4. Responsibilities:.....	26
5. Procedures:.....	26
6. Records:	27
Appendix C - Filling Out Test Form Templates.....	28
1. Test Form Templates:	28
2. MVNQ Terms Document:	28

Chapter 1 Introduction

1. General:

This manual describes the means and methods for the Contractor Quality Control and Government Quality Assurance testing of construction materials incorporated into the New Orleans District (CEMVN) projects. Information is given on sampling, the test required, testing frequency, reporting requirements, and database maintenance. This manual only describes a minimum testing program on a limited number of common construction materials and the specifications may require additional tests that demonstrate compliance with the contract documents. If there is a conflict between this manual and the technical specification sections, the most stringent requirements shall govern. The most recent version of this manual at the time of contract solicitation will supplement the construction material control requirements for a specific contract unless noted otherwise.

The Contractor shall only use those laboratories, including his own that have been validated by an inspection or audit performed by the USACE Materials Testing Center, Vicksburg, MS.

2. Definitions:

- a. **Quality Management System.** Quality management is defined as all control, inspection, and other assurance activities instituted to achieve the product quality established by the contract plans and specifications.
- b. **Contractor Quality Control.** Contractor Quality Control (QC) is that part of the system by which the Contractor regulates, tests and inspects their own, suppliers, and sub-Contractors procedures, equipment, materials, and personnel so that the completed product will comply with the requirements of the project's contract documents.
- c. **Government Quality Assurance.** Government Quality Assurance (QA) is that part of the system by which the Government verifies or assures that the Contractor's Quality Control system is performing properly and the completed product conforms to the contract documents. The number of QC test observed by QA personnel should be generally related to the consistency in QC and QA test results.

3. Responsibility, Compilation, and Submittal of Test Results:

- a. The Contractor is responsible for complying with the contract documents in the performance of all required tests and the preparation, submittal, and maintenance of those test reports outlined in this manual and the contract specifications. The test results from QC and QA testing shall be compiled separately as outlined in this manual.
- b. The Contractors' QC Laboratory shall appoint a Registered Professional Civil Engineer to certify QC inspections and test results prior to the start of work. The certification shall state that the tests and observations were performed by or under the direct supervision of the Registered Professional Civil Engineer and that the results are representative of the

materials and conditions being certified by the tests. The certification shall be submitted within two weeks after final inspections and testing is complete. The certification shall be submitted to USACE for the referenced project in accordance with the New Orleans Construction Control Manual, Appendix A. Failure to submit certifications as stated may result in nonpayment for related work performed and disapproval of the QC test facility for this contract.

- c. Acceptance of the Contractors' QC plan is required prior to the start of construction. Acceptance is conditional and will be predicated on satisfactory performance during the construction. The Government reserves the right to require the Contractor to make changes to the QC Plan and operations including removal of personnel and QC Laboratory, as necessary, to obtain the quality specified.
- d. All test results will be entered into the CEMVN Quality Assurance Control Center (QACC) construction material testing database as described in Appendix A by the QC laboratory performing the testing. Test results will be entered into the testing database within 48 hours from sampling. Payment for any material placed, as well as for any subsequent construction, will not be made until test results are entered into the database and analyzed by Quality Assurance personnel. The Contractor shall maintain a hard copy of the materials testing log, test reports and control charts at the Contractor's field office. These records will be available at all times for review by Government personnel. The original test report will be distributed to the Administrative Contracting Officer (ACO) within 48 of completion of the test. This original test report (supporting documentation) submission is in addition to any required electronic submission.
- e. Any tests not conforming to the contract documents will be immediately reported to the Administrative Contracting Officer along with the recommended corrective action to bring the work into complete compliance with the specifications. The Administrative Contracting Officer may designate additional re-sampling or retesting to verify the work represented by the failing test. This testing is at the Contractor's expense.
- f. Reference to standard test methods and testing procedures for sampling and testing of common construction materials are given in each chapter of this manual. Additional testing may also be required in the contract documents.
- g. Laboratory Facilities. For work that involves aggregates, concrete, masonry, rock or soil the QC Laboratory shall, at its own expense, obtain and maintain validation as an approved testing laboratory by the Materials Testing Center (MTC) of the Engineering Research and Development Center (ERDC). This shall be done in accordance with ER 1110-1- 8100 and ER 1110-1-261. Appendix B further describes this requirement. Refer to Chapter 4 for welding laboratories.

For work that involves vibration, steel, steel reinforcing bars, coatings inspections and other specialized construction material testing and inspection the QC Laboratory shall maintain personnel, procedures and equipment that meet applicable industry standards.

- h. Field sampling and testing locations shall be recorded using Latitude/Longitude coordinates reported in decimal degree format to the millionth decimal and be surveyed using techniques to achieve ± 10 feet accuracy.

Report Form input example: 29.934003, -90.133745

Chapter 2 Soils

1. Scope:

This chapter specifies methods and procedures for the Contractor Quality Control (QC) and Government Quality Assurance (QA) testing of materials used, but not limited to, compacted levee embankments, compacted berms, un-compacted berms, ramps, and structural backfill. The Government will also perform checks, and assurance testing of control testing required by the Contractor.

2. Samples:

Samples shall be collected and secured in accordance applicable ASTM testing procedures.

3. Testing Personnel:

The individuals who inspect, monitor, sample and test Embankment construction as required in this specification shall meet the following minimum criteria of certification and/or documented experience. Work experience shall be related to the field for which the inspector is being qualified and may be obtained by working either for an inspection/testing agency or engineering firm as a technician, inspector or engineer.

- Current NICET Level II certification in Geotechnical Engineering technology/construction, or
- Current ICC Soils Special Inspector with one year related experience, or
- Geologist-in-Training with one year related experience, or
- Engineer Intern with one year related experience, or
- Registered Geologist, or
- Registered Professional Engineer.

The Contractors' QC laboratory shall submit certification and/or documentation to provide evidence of qualification. The appointed Registered Professional Civil Engineer, identified in Chapter 1, Section 3.b to certify inspections and test results, remains responsible for compliance of all inspection and testing activities.

All Laboratory facilities, personnel and equipment used to test soils as required in this specification shall be part of a Laboratory that has been validated by the USACE Materials Testing Center, Vicksburg, MS.

4. Typical Test Requirements:

Testing and reporting shall be performed in accordance with the latest American Society of Testing and Materials (ASTM) Standard, as indicated in Table 2-1.

Table 2-1
ASTM References

Gradation	
ASTM C 117	Materials Finer than No. 200 Sieve in Mineral Aggregates by Washing
ASTM C 136	Sieve Analysis of Fine and Course Aggregates
ASTM D 1140	Amount of Material in Soils Finer than No. 200 (75- μ m) Sieve
ASTM D 6913	Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis
Moisture Content	
ASTM D 2216	Laboratory Determination of Water, (Moisture) Content of Soil and Rock by Mass (Method B)
ASTM D 4643	Determination of Water (Moisture) Content of Soil by Microwave Method
Moisture/Density Relationship	
ASTM D 698	Laboratory Compaction Characteristics of Soil Using Standard Efforts (12,400ft lbs/ft ³ (6000KN))
ASTM D 1557	Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft ³ (2,700 kN-m/m ³))
Field Density	
ASTM D 1556	Density and Unit Weight of Soil in Place by the Sand-Cone Method
ASTM D 6938	In-Place Density and Water Content of Soil and Soil-Aggregate Nuclear Methods (Shallow Depth)
Materials Classification	
ASTM D 2487	Classification of Soils for Engineering Purposes
ASTM D 4318	Liquid Limit (One-Point Method B), Plastic Limit, and Plasticity Index of Soils
Organic Content	
ASTM D 2974	Moisture, Ash, and Organic Matter of Peat and Other Organic Soils (Method C)
Unconfined Compressive Strength	
ASTM D 1633	Compressive Strength of Molded Soil-Cement Cylinders
ASTM D 2166	Unconfined Compressive Strength of Cohesive Soil

5. Sampling and Testing of Compacted Fill:

This sampling and testing shall be in accordance with the standard procedures referred to in this manual. The minimum number of QC tests to be performed shall be as indicated in Table 2-2.

The Government will also perform checks, and assurance testing of the other control testing required by the Contractor.

Table 2-2
Type of Tests and Frequency of Testing
Compacted Embankments and Berms, Ramps, and Structural Backfill Material

Property	Form	Minimum Frequency	Standard
Nuclear Field Density	MVNQS11	One test per 1,500 cubic yards of compacted fill placed per lift, but not less than one density test per 500 linear feet per lift. A lift placed on any one side of an existing embankment will be considered as a separate lift. At least one test shall be performed in any shift that compacted fill is placed.	ASTM D 1556 or ASTM D 6938
Nuclear Field Density Relative Density	MVNQS12	Used to record test results from testing uncohesive material. One density test per lift per 150 linear feet of the base course. Isolated repairs (less than 150 linear feet) must have at least one density test per isolated area per lift.	ASTM D 6938
Sand Cone Field Density	MVNQS03	One test to be obtained for every ten (10) Nuclear Field Density locations to verify Nuclear Field Density.	ASTM D 1556
Compaction Control Curve	MVNQS02	Control Compaction Curves shall be established in accordance with ASTM D 698 - Laboratory Compaction Characteristics of Soil Using Standard Effort. A Compaction Control Curve will be required for each type of material from each source or a minimum of one Compaction Control Curve every 25,000 cubic yards of compacted fill placement. Where construction operations result in the blending of material, two representative Compaction Control Curves will be required for each resulting blend of material. The samples collected for the resultant blended material shall be collected from separate locations. If the borrow or source of fill material changes, new Compaction Control Curves shall be performed. Material test samples for Compaction Control Curve shall be prepared by air-dry, rewet, and cured.	ASTM D 698
One-Point Proctor Verification	MVNQS02	One test to be obtained for every five (5) field density locations.	ASTM D 698 (modified)
Moisture Content	MVNQS11	One test at each field density test location.	ASTM D 2216 or ASTM D 4643
Organic Content	MVNQS07	One test at each field density test location.	ASTM D 2974 (Method C)
Materials Classification	MVNQS06	One test obtained for each Control Compaction Curve and one test for each field density test. Determine Atterberg Limits (LL One-Point Method B), minus #200 and Sand Content.	ASTM D 2487 ASTM D 1140 ASTM D 4318
Unconfined Compressive (UC) Strength	MVNQS05	For Deep Soil Mixing (DSM) QC operations a minimum of three percent of the DSM columns per site will be drilled and three UC strength samples collected and tested at each test column.	ASTM D 2166 ASTM D 1633

6. Sampling and Testing of Un-Compacted Berm Material:

This sampling and testing shall be in accordance with the standard procedures referred to in this manual. The minimum number of QC tests to be performed shall be as indicated in Table 2-3. The Government will also perform check and assurance testing of the other control testing required by the Contractor.

Table 2-3
Type of Tests and Frequency of Testing
Un-Compacted Berm Material

Property	Form	Frequency	Standard
Organic Content	MVNQS07	One test at materials classification test location.	ASTM D 2974 (Method C)
Materials Classification	MVNQS06	One test per 3,000 cubic yards of un-compacted fill placed, but not less than one test per 1,000 linear feet of un-compacted fill placed. At least one test shall be performed in any shift that un-compacted fill is placed. Determine Atterberg Limits (LL One-Point Method B), minus #200 and Sand Content.	ASTM D 2487 ASTM D 1140 ASTM D 4318

7. Compilation of Test Data for Submittal:

The results of the test and inspections shall be recorded in the MVN database as directed in Appendix. Samples of the reporting forms and instruction for each form are provided on the MVN SharePoint site; **Test Form Examples** and are described as follows. The latest forms should be referenced on the MVN SharePoint site and described in Appendices. All data is to be submitted electronically **within 24 hours of completion of the tests by the laboratory performing the testing.**

- a. **MVNQS01** Sieve Analysis – ASTM C 117, ASTM C 136 and ASTM D 1140. This form is to be used in reporting the material finer than No 200 sieve and a sieve analysis of coarse grain material.
- b. **MVNQS02 (Compaction Control Curve)** ASTM D 698. This form is to be used in reporting the determination of the optimum moisture content and the maximum dry density. The moisture-density curve shall be plotted based on a minimum of five compaction test specimens. A one-point Proctor test – ASTM D 698 (modified, Figure 2-2) shall be obtained for every five (5) field density test locations, and reported with same. The soil One-Point proctor result obtained from the in-place density test location will serve as the basis for determining the applicable compaction control curve.

- c. **MVNQS03** (Field Density Sand Cone Method) ASTM D 1556. This form is to be used in reporting the determination of the degree of compaction and moisture content. Contract specifications shall govern the required compaction effort.
- d. **MVNQS05** (Unconfined Compressive Strength) ASTM D 2166. This form is to be used to report the compressive strength of an intact, remolded or reconstituted cohesive soil, using a strain-controlled application of the axial load. Contract specifications shall govern the acceptable strength requirements.
- e. **MVNQS06** (Unified Soil Classification System) ASTM D 2487. This form is to be used to report the determination of the liquid limit (One-point Method B), plastic limit, plasticity index, % sand content and % fines. MVNQS01 Sieve Analysis – ASTM C 117 and ASTM C 136 is to be used to report the results of gradation tests of the material if a granular material is specified. The final soil classification in accordance with ASTM D 2487 shall be stated on the same forms. Contract specifications shall govern the acceptable Atterberg limits, gradation limits, and material classification. If the Nuclear Method (ASTM D 6938) is used for field density determinations, the soil sample utilized for material classification shall come from within a radius of 12 inches of the center of the in-place density test site. The soil classification obtained from in-place density test location will serve as a basis for determining the applicable compaction control curves.
- f. **MVNQS07** (Moisture, Ash, and Organic Content Determination) ASTM D 2974 (Method C). This form is to be used in reporting the determination of the organic content of the material. Determination of organic content shall be performed in accordance with ASTM D 2974; Method C. Contract specifications shall govern the acceptable limits of organic content.
- g. **MVNQS09** (Moisture Content Determination) ASTM D 2216, ASTM D 4643 and ASTM D 6938. This form is to be used in reporting the determination of the moisture content of the in-place material when ASTM D 2216, ASTM D 4643 or ASTM D 6938 is the test method utilized. This form is not to be used when performing Field Density Test Nuclear Method with Moisture Content Determination. Contract specifications shall govern the acceptable limits of moisture content.
- h. **MVNQS11** (Field Density Test Nuclear Method). This form is to be used in reporting the determination of the degree of compaction and moisture content by oven, microwave or nuclear gauge. Contract specifications shall govern the required compaction effort and moisture range. If the nuclear method is selected for field density testing, the Sand-Cone Method shall be used to confirm the accuracy of the Nuclear Method. This shall be accomplished by performing an initial comparison test of the two methods when a nuclear gage is brought on-site for the first time. If the Nuclear Method wet density is within 3 percent of the Sand Cone Method, no correction of the Nuclear Method wet density will be required and the testing may continue with the Nuclear Method. The Nuclear Method wet density shall be verified throughout the project at a rate of one Sand-Cone test for every ten nuclear tests per nuclear gage thereafter. If the variance at any time between the Nuclear Method and the Sand Cone Method exceeds 3 percent, testing

with the Nuclear Method shall stop until the Contractor provides a Root Cause Analysis and five consecutive comparison tests are performed as evidence that Corrective Actions will provide results within 3 percent. For comparison purposes, the nuclear and sand-cone wet densities should represent the same layer thickness within the testing area selected. When a nuclear density result is in doubt, the sand-cone density test shall be used for acceptance.

- i. **MVNQS12** (Field Density (Relative Density) Nuclear Method). This form is to be used in reporting the determination of the Relative degree of compaction as determined based on relationship of the Minimum Dry density and Maximum Dry density. Contract specifications shall govern the required Relative Density.

8. Soil Electronic Conductivity (EC) and Total Soluble Salt Analysis:

The following test method shall be used for determining the Total Soluble Salt (Total Salinity) of Embankment soils. This method shall be followed when testing embankment soil salinity levels. Sampling of materials shall be performed by a USACE Validated Laboratory.

- A. Sampling; Sampling shall consist of one 12,500 gram composite sample per 1,000 linear feet per lift. A Composite soil sample is defined as 5 separate representative 2,500 gram samples taken randomly at relatively evenly spaced intervals within the 1,000 linear foot. A lift on any one side of the levee will be considered one lift. The locations of the samples shall be as directed by the Contracting Officer. When a composite soil sample is collected, it should be handled in accordance with ASTM D 4220, Group B Standard Practices for Preserving and Transporting Soil Samples.

As directed by the Contracting Officer, when samples are to be split for replicate testing, the entire composite sample shall be processed over a No. 4 (4.75 mm) sieve by the contractors QC laboratory. The material passing the No. 4 sieve shall be thoroughly mixed and split in accordance with ASTM C 702 Standard Practice for Reducing Samples of Aggregate to Testing Size.

- B. Sample Preparation; Composite soil samples passing a No. 4 sieve are to be thoroughly remixed and reduced to a minimum 200 g sample for testing in accordance with ASTM C 702 Standard Practice for Reducing Samples of Aggregate to Testing Size.

The reduced composite soil sample is air dried at a temperature not to exceed 140° F for a minimum of 18 hours. After the sample is air dried, process and collect material passing No. 10 (2 mm) sieve. Material retained on the No. 10 sieve will be discarded.

- C. Procedure; (EC 1:2 preparation) To determine soil EC, collect a representative 20 gram sample from the sieved air-dried material and mix with 40 mL deionized water in a 125 mL Erlenmeyer flask.

The container is sealed and the mixture is either agitated for 1 hour in a mechanical shaker or mixed by hand every 30 minutes for 3 hours.

The mixture is filtered through a Whatman 42 filter paper. EC (dS/m) of the filtrate is determined immediately using a standard conductivity meter. Follow manufacture's direction for standard conductivity meter operations and temperature corrections.

- D. Reporting; The directly-measured EC 1:2 is converted to Saturated Extract-Equivalent EC (EC_e) by multiplying by a factor of 2. (Southern Cooperative Series Bulletin No. 419 ISBN# 1581614195 January, 2014)

Total soluble salts (TSS) concentration in ppm (mg/L) is calculated by multiplying EC_e (dS/m) by 640 for EC readings <5.0 dS/m or by 800 for EC readings >5.0 dS/m. (Rhoades, 1996)

The report shall include at a minimum;

1. All sample identifications documented during sampling that at a minimum include, sample date, received date, test/sample number, location of composite sample (GPS, station, lift, , elevation, offset)
2. USCS visual description
3. Make/Model and Serial # of conductivity meter
4. Notes should include any deviations from this test method.
5. The Soil Electronic Conductivity (EC) shall be reported in decisiemens per metre (dS/m).
6. Total Soluble Salt shall be reported as Total Salinity in parts per million (ppm).

9. Field and Laboratory Determination of Non-Soil Volume for Levee Fill:

- A. The field excavation testing shall be performed by excavating a 10' wide x 10' long and to a depth of the lift thickness for each lift that is in question. The volume of the excavation shall be verified using the end area method through measuring the dimensions of the excavation with the use of survey equipment at each corner of the hole. A difference of +/- 10% of the theoretical excavation is allowed. The Contractor shall bring all material excavated to the lab in sealed airtight containers. All excavations shall be completely backfilled by the Contractor within 72 hours of inspection unless directed otherwise by the COR. All backfill shall be in accordance with the existing contract documents, especially EMBANKMENT.
- B. The unit weight of the soil shall be determined by ASTM D 6938 Field Density – Nuclear Method, ASTM D 1556 Field Density – Sand Cone Method, or ASTM D 698 Compaction Characteristics of Soil. All material testing shall be performed by a Corps validated lab.
- C. Once all the excavated material is delivered to a Corps validated lab, any clay pieces adhering to the non-soil pieces that can be removed by hand without damaging the non-soil piece shall be removed.
- D. All non-soil pieces shall be weighed in their existing conditions immediately prior to testing (wet weight as excavated). If all non-soil pieces do not fit in the Measure Box, then the non-soil pieces may be split into smaller sampling sizes for testing purposes and the cumulative volume reported.

E. Sturdy Measure Box containers shall be used for the non-soil volume determination processes. The minimum volume of the Measure Box is 0.8 cubic feet. This volume dimension is a minimum and may be enlarged if desired. The weight of the empty containers shall be determined using a calibrated scale and with the weight recorded to the nearest 0.1 lb. The container shall be filled in two layers with silica sand. The first layer of sand shall be densified by use of a Shake Table and vibrated such that the Silica sand achieves its maximum density. The second layer of silica sand shall be added and vibrated, with additional sand added as needed to “top off” the container as the sand achieves a greater density. The weight of the container filled with densified Silica sand shall be recorded to the nearest 0.1 lb using a calibrated scale. Determine the weight of the measure container plus sand three times to determine the average value. The maximum unit weight of the silica sand is the weight of the measure plus sand minus the weight of the measure divided by the known volume of the container and reported to the nearest 0.1 lb/ft³.

F. The volume of the non-soil shall be determined by the following USACE MVN developed procedure, Non-Soil Volume Determination.

- 1) **Volume and Weight Determination of Measures (annual):** The volume of the Measure Box shall be determined and verified on an annual basis by the water filled method as specified in ASTM C29/C29M paragraph 8 and recorded to the nearest 0.1 ft³.
- 2) **Density Sand:** Obtain silica sand also known as US Silica Sand. Verify that the quality of the silica or “Silica” sand meets the requirements specified in ASTM D1556 paragraph 6.2. The sand can be re-used, but it should be cleaned to comply with the previously referenced standard by sieving and/or rinsing, and oven drying prior to reuse.
- 3) **Determining Densified Sand within a Measure Box:** Before any tests determining non-soil volume content, a calibration test shall be run each day that testing is to be performed, to determine the standard weight of the sand in the Measure Box as discussed in section E. The three repeated determinations of densified sand weight per unit volume shall be within 2.0 pcf of each other.

A Measure Box shall be used to determine the densified sand and will be based upon use of a Shake Table and placement within layers. Clean and dry silica sand is placed loosely within each layer using a large scoop or the edge of a bucket by flowing and distributing the sand evenly across the surface area. The Shake Table is then to be used. The number and duration of vibrations will be determined as noted in the following trial. These times are approximate and should be modified by each laboratory to fit the Shake Table being used to achieve a consistent sand weight per unit volume.

MEASURE BOX – (1) Position measure over a large catch pan for collecting excess sand. Place loose Silica sand in one layer (half height of measure); (2) Using the Shake Table, vibrate the sand for 4-8 seconds; (3) Place loose silica sand in a second layer (full height of measure); (4) Vibrate the sand for 4-8 seconds. The sand should consolidate below the top rim of the measure; (5) Place additional (excess) sand above the top of the measure. It should appear to overflow. Vibrate for the sand for an additional 3-4 seconds. It is desired to have excess sand above the top of the rim after vibration of about 1/8 inch; (6) Using a straight metal bar, strike off the excess sand, leaving the sand flush with the top rim of the measure; (7) Weigh the measure and densified sand recorded to the nearest 0.1 lb; (8) Determine the weight per unit volume of the measure by subtracting the weight of the measure plus sand minus the weight of the measure then dividing by the known volume of the container and report to the nearest 0.1 lb/ft³; (9) Repeat steps 1 thru 8 for a total of three determinations of densified sand weight per unit volume, and calculate the average weight per unit volume to the nearest 0.1 lb/ft³.

- 4) **Standard Wood or Metal for Verification (annual):** Eight pieces of wood or metal, labeled A thru G, measuring 5 inches by 1 inch by 2 inches are to be used to verify the volume determination by the densified sand method as detailed in 5) below. Determine the weight and linearly measured volume of the eight standard pieces of wood or metal to verify the calculated non-soil content from the use of densified silica sand within Measure Boxes of known volume.
- 5) **Non-soil Verification (annual):** Wood or metal pieces measured in Step 4) above will be used in each measure by densifying sand and four wood or metal pieces in each layer, for a total of eight wood or metal pieces within each measure. The same procedures outlined in Step 4) above are used to place and densify the sand and wood or metal within the measures. The wood or metal is placed within each layer with at least ½ inch of loose sand beneath and around the wood or metal pieces. The weight of the densified sand, measure, and wood or metal is used to determine the density and subsequent volume of the wood or metal. The calculated volumes shall be compared to the known volumes of the wood or metal pieces to see if any change in shaking time or sand type is needed. If the calculated and known volumes are within +/- 2% of each other, the test verification is successful. See below for the step by step procedures for this:

MEASURE BOX - (1) Determine the volume and weight of the measure as noted in Step 1) above; (2) Determine the average densified sand weight per unit volume as noted in Step 3) above; (3) Determine volume and weight of pre-cut pieces of wood or metal as noted in Step 4) above; (4) Densify wood or metal in layers following the similar method noted in Step 3) above; (5) Determine the densified sand and wood or metal weight in the unit measure; (6) Calculate the volume of wood or metal as shown below:

- (a) Volume of Measure Box (ft³)
- (b) Weight of Measure Box (lb)

- (c) Average weight per unit volume of densified sand (lb/ft³)
- (d) Wood or metal Pieces total weight (lb)
- (e) Wood or metal Pieces total volume (ft³)
- (f) Average determined densified sand, wood or metal, & measure weight (lb)
- (g) Densified sand only weight (no wood or metal) = (c) x (a)
- (h) Densified sand only weight (with wood or metal) = (f) – (b) – (d)
- (i) Volume of wood or metal (from densified sand test) = [(g) – (h)] / (c)
- (j) % actual volume wood or metal = 100 x (e) / (a)
- (k) % tested volume wood or metal = 100 x (i) / (a)

- 6) **Non-soil Volume Determination:** Determination of non-soil volume for a test sample is as follows. Determine the wet weight of the sample prior to placement into the loose sand layers. Cleaned non-soil pieces from a sample are placed in one of the tested measures above by following procedures as outlined in Step 3). The non-soil pieces are placed within each layer of loose sand with at least ½ inch of loose sand beneath and around the various non-soil pieces. The non-soil piece may be cut to fit into the measure but care should be used to ensure that all pieces of the sample are measured. The weight of the combined densified sand, measure, and non-soil shall be recorded to the nearest 0.1 lb. To determine the density and subsequent volume of the non-soil pieces, see calculations below.

MEASURE BOX - (1) Determine the volume and weight of the measure as noted in Step 1) above; (2) Determine the average densified sand weight per unit volume as noted in Step 3) above; (3) Determine weight of sample pieces of non-soil; (4) Densify non-soil pieces in layers following the similar method noted in Step 3) above; Determine the densified sand and non-soil pieces weight in the unit measure; (5) Calculate the volume of non-soil pieces as shown below:

- (a) Volume of Measure Box (ft³)
- (b) Weight of Measure Box (lb)
- (c) Average weight per unit volume of densified sand (lb/ft³)
- (d) Weight of Sample Non-soil Pieces (lb)
- (e) Determined densified sand, non-soil pieces, & measure weight (lb)
- (f) Densified sand only weight (no non-soil pieces) = (c) x (a)
- (g) Densified sand only weight (with non-soil pieces) = (e) – (b) – (d)
- (h) Volume of non-soil pieces (from densified sand test) = [(f) – (g)] / (c)
- (i) Volume of excavation (ft³)
- (j) % tested volume non-soil pieces = 100 x (h) / (i)

- 7) **Documentation:** As a minimum, calibrations of Measure Boxes should be documented annually on the Unit Weight Measure Volume Determination Record. The Densified Sand unit weight shall be documented on the Densified Sand Calibration Record. Test records for samples shall be documented on the Non-soil pieces Volume Determination Record. Contact MVN-CD-Q for latest test forms.

- G. The percent volume determined in Step 6) (j) above shall be compared versus the acceptable value listed in the specifications. If the test shows the percent volume is greater than the acceptable value, the Contractor shall follow the corrective actions as noted in the contract specifications.

10. Additional Testing:

In addition to the above frequency of tests, additional tests may be required as follows:

- a. Where the Administrative Contracting Officer (ACO) or Contracting Officer's Representative (COR) has reason to doubt the adequacy of the compaction, moisture content, or organic content control.
- b. Where the Contractor is concentrating fill operations over a relatively small area.
- c. When embankment materials change substantially, the Administrative Contracting Officer or Contracting Officer's Representative (COR) may direct additional testing.
- d. Where special compaction procedures are being used.
- e. When the contract specifications require additional testing.
- f. When areas are found not meeting the specified in-place density, Atterberg limits, moisture content, and/or in-place organic content requirements; the Contractor shall retest, at no additional costs to the Government, after corrective measures have been applied.

Chapter 3 Concrete

1. Scope:

This chapter specifies methods and procedures for the Contractor Quality Control (QC) and Government Quality Assurance (QA) methods and procedures for the testing of fresh concrete and concrete aggregate. The Government will also perform checks, and assurance testing of control testing required by the Contractor.

2. Samples:

Fresh concrete samples shall be secured in accordance with ASTM C 172. Concrete aggregates shall be sampled in accordance with ASTM D 75. Sampling locations shall be randomly selected.

3. Testing Personnel:

The individuals who inspect, monitor, sample and test Concrete construction as required in this specification shall meet the following minimum criteria of certification and/or documented experience. Work experience shall be related to the field for which the inspector is being qualified and may be obtained by working either for an inspection/testing agency or engineering firm as a technician, inspector or engineer.

- Current ICC Reinforced Concrete Certificate with 1 year related experience, or
- ACI Concrete Construction Special Inspector Certificate, or
- Engineer Intern with one year related experience, or
- Registered Professional Engineer.

The individuals who perform testing of concrete or the constituents of concrete as required in this specification shall have an applicable and current ACI certification for testing being performed; ACI Concrete Strength Testing, ACI Concrete Laboratory Testing – Level 1, ACI Aggregate Testing Technician – Level 1, ACI Concrete Field Grade I.

The Contractors' QC laboratory shall submit certification and/or documentation to provide evidence of qualification. The appointed Registered Professional Civil Engineer, identified in Chapter 1, Section 3.b to certify inspections and test results, remains responsible for compliance of all inspection and testing activities.

All Laboratory facilities, personnel and equipment used to test soils as required in this specification shall be part of a Laboratory that has been validated by the USACE Materials Testing Center, Vicksburg, MS.

4. Typical Test Requirements:

Test requirements specified in the contracts documents may be more stringent than those listed below in Tables 3-2, 3-3 and 3-4. All test results will be entered into the MVN material testing database as described in Appendices by the laboratory performing the testing. Acceptable test values are contained in the contract documents.

The laboratory performing the tests shall be validated by the Materials Testing Center, Vicksburg, MS. and conform to ASTM C 1077.

**Table 3-1
ASTM References**

Concrete Lab Testing	
ASTM C 33	Specification for Concrete Aggregates
ASTM C 39	Compressive Strength of Cylindrical Concrete Specimens
ASTM C 117	Materials Finer than No. 200 Sieve in Mineral Aggregates by Washing
ASTM C 136	Sieve Analysis of Fine and Course Aggregates
ASTM C 511	Mixing Rooms, Moist Cabinets, Moist Rooms, and Water Storage Tanks Used in the Testing of Hydraulic Cements and Concretes
ASTM C 566	Total Evaporable Moisture Content of Aggregate by Drying
ASTM C 617	Capping Cylindrical Concrete Specimens
ASTM C 702	Reducing Samples of Aggregate to Testing Size
ASTM C 1231	Practice for Use of Unbonded Caps in Determination of Compressive Strength of Hardened Concrete Cylinders
CRD-C 104	Calculation of Fineness Modulus of Aggregate
Concrete Field Testing	
ASTM C 31	Making and Curing Concrete Test Specimens in the Field
ASTM C 138	Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete
ASTM C 143	Slump of Hydraulic-Cement Concrete
ASTM C 172	Sampling Freshly Mixed Concrete
ASTM C 173	Air Content of Freshly Mixed Concrete by the Volumetric Method
ASTM C 231	Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C 1064	Temperature of Freshly Mixed Hydraulic-Cement Concrete
ASTM D 75	Sampling Aggregates

5. Compilation of Test Data for Submittal:

The results of the test and inspections shall be recorded in the MVN database as directed in Appendix. Samples of the reporting forms and instruction for each form are provided on the MVN SharePoint site; **Test Form Examples** and are described as follows. The latest forms should be referenced on the MVN SharePoint site and described in Appendices. All data is to be submitted electronically **within 24 hours of completion of the tests by the laboratory performing the testing.**

- a. **MVNQC01** (Concrete Compression Test Data – ASTM C 39). This form is to be used in reporting the results of laboratory concrete compression testing. Contract specifications shall govern the required concrete compressive strength.
- b. **MVNQC02** (Concrete Field Data). This form is to be used in reporting the data collected by the laboratory while monitoring and testing concrete during placement. Contract specifications shall govern the required concrete properties during placement.
- c. **LMN FORM 853-R** (Concrete Compression Test Specimen Data). This form is to be filled out and provided to the QA laboratory for each set of cylinders delivered. This form should be filled out with information documented during concrete placement. The information on this form should match the information provided on the associated MVNQC01 and MVNQC02 test forms. The Order number on this form shall match the Batch Ticket number on the associated concrete supplier batch ticket, the MVNQC01 and the MVNQC02 test form for sample tracking purposes. The form also serves as a bill of lading for the delivered concrete samples.

Table 3-2
Test Requirements
AGGREGATE, FINE

Property	Method	Frequency	Remarks
Deleterious Substances	ASTM C 33	1 per week	
Fineness Modulus	CRD-C 104	1 per shift per batch plant when concrete plant is operating	Calculation based on gradation test results
Gradation	ASTM C 117 ASTM C 136	1 per shift per batch plant when concrete plant is operating.	Tests selected randomly.
Moisture Content	ASTM C 566	If moisture meter is working properly, 2 per week to verify If moisture meter is not working, 4 every 8 hours of mixing plant operation Additional tests if slump is out of control or variability is excessive	Tests selected randomly for each aggregate size.
Sampling Method	ASTM D 75	As specified for the individual material property.	

Table 3-3
Test Requirements
AGGREGATE, COURSE

Property	Method	Frequency	Remarks
Deleterious Substances	ASTM C 33	1 per week per batch plant, or as directed by COR	
Gradation	ASTM C 117 ASTM C 136	1 per shift per batch plant when concrete plant is operating	Tests selected randomly.
Moisture Content	ASTM C 566	If moisture meter is working properly, 2 per week to verify If moisture meter is not working, 4 every 8 hours of mixing plant operation Additional tests if slump is out of control or variability is excessive	Tests selected randomly for each aggregate size.
Sampling Method	ASTM D 75	As specified for the individual material property.	

Table 3-4
Test Requirements
FRESH CONCRETE

Property	Form	Method	Frequency	Remarks
Compression Cylinders (Quality Assurance)	MVNQC01 MVN 835	ASTM C 31 ASTM C 39	To be molded by the Contractor Quality Control Laboratory and tested by the Quality Assurance Laboratory. Mold one set of cylinders per 8 hour shift or for every 150 cubic yards placed.	Quality Assurance Cylinders shall be molded from the same sample of concrete that the Quality Control cylinders are molded.
Compression Cylinders (Quality Control)	MVNQC01 MVNQC02	ASTM C 31 ASTM C 39	To be molded and tested by the Contractor Quality Control Laboratory. Mold one set of cylinders per 8 hour shift or for every 150 cubic yards placed.	On randomly selected batches for each separate concrete mix produced. Cylinders used shall conform to paragraph 6.1 of ASTM C 31. Initial Cure in accordance with paragraph 10.1.2 of ASTM C 31.
			As a minimum; A set of test specimens for concrete with a 28-day specified strength shall consist of two cylinders to be tested at 7 days and two 6-inch by 12-inch cylinders or three 4-inch by 8-inch cylinders at 28 days. A set of test specimens for concrete with a 56-day or 90-day specified strength shall consist of two cylinders to be tested at 7 days, two 6-inch by 12-inch cylinders or three 4-inch by 8-inch cylinders at 28 days and two 6-inch by 12-inch cylinders or three 4-inch by 8-inch cylinders at 90 days.	
			Additional sets when mix proportions change or low strengths are detected.	
Compression Cylinders (QC- for putting concrete into service or other purposes indicated in paragraph 4.3 of ASTM C 31)	MVNQC01 MVNQC02	ASTM C 31 ASTM C 39	1 set of multiple pairs of QC cylinders per item to be evaluated.	Cylinders used shall conform to paragraph 6.1 of ASTM C 31. Initial Cure in accordance with paragraph 10.1.2 of ASTM C 31. Cylinders to be field cured shall conform to 10.2 of ASTM C 31.
Air Content Slump Temperature	MVNQC01 MVNQC02	ASTM C 231 ASTM C 143 ASTM C 1064	1 every time concrete cylinders are molded	On randomly selected batches for each separate concrete mix produced
			Plus 2 additional during each 8 hours of concrete production	
			Additional tests if workability variation is excessive.	

Chapter 4 Welding Inspection

1. Scope:

This chapter specifies methods and procedures for the Contractor Quality Control (QC) weld inspection for Group 1 and Group 2 carbon steels as defined by AWS D1.1, Table 3.1 and their ASTM A709 counterparts. Welding of sheet metal, reinforcement bars, castings, stainless steel, aluminum and other non ferrous metals are not included in this document and should reference the appropriate AWS or ASME Code. An approved schedule of welding procedures (WPS) is required before fabrication commences (Section 05 50 03.00 12). The Government will also perform checks, and assurance testing of control testing required by the Contractor.

2. Definitions:

- a. **Fracture Critical Welds.** Fracture critical members or member component welds as defined by ER 1110-2-8157 are tension members or tension components of bending members (including those subject to reversal of stress), the failure of which would be expected to result in collapse of the hydraulic steel structure. The designation “FCM” shall mean fracture critical member or member component. Members and components that are not subject to tensile stress under any condition of live load shall not be defined as fracture critical. FCMs, in general, are dewatering components (needle girders, bulkheads, needles), lifting eyes, or other tension members. This includes any members welded to these members as cracks could propagate to these members and cause failures also. These welds should either be shown on the drawings or called out in the specifications. Tubular welds are not applicable to AWS D1.5. AWS D1.5, Section 12 is the applicable code for these welds.
- b. **Other Welds.** These welds are the remaining welds that are not considered Fracture Critical Welds. AWS D1.1 is the applicable code for these welds.

3. Testing Personnel:

- a. **Visual Inspection.** Visual inspection shall be performed by Certified Welding Inspectors (CWI) that are qualified and certified in accordance with the provisions of AWS QC1. Verification of documentation may be obtained from the AWS web site. Note: Certification number is required for this verification.
- b. **Nondestructive Testing Technicians.** All ASNT Level III personnel shall be qualified in accordance with ASNT CP-189. Only individuals qualified for NDT Level II or individuals qualified for Level I and working under the direct supervision of a Level II shall perform nondestructive testing. Level I and Level II personnel shall be qualified in accordance with either ASNT CP-189 or ASNT SNT-TC-1A. Level III NDT Inspectors shall possess a currently valid ASNT Level III certificate in each of the processes they are qualifying inspectors to. Copies of the certifications, including the Level III NDT Technician that certified the Level I and Level II Technicians shall be included in the submittals. Verification of Level III documentation may be obtained from the ASNT web site. Note: Either Certification number or name is required for this verification.

4. Visual Inspection Requirements:

Visual inspection of welds shall conform to the requirements of AWS D1.1, Section 6, or AWS D1.5, Section 12, as applicable.

5. Nondestructive Testing Requirements:

- a. **Ultrasonic Testing.** Ultrasonic testing of welds shall conform to the requirements of AWS D1.1, Section 6, Part F or AWS D1.5, Subsection 12.16, as applicable.
- b. **Radiographic Testing.** Radiographic testing of welds shall conform to the requirements of AWS D1.1, Section 6, Part E or AWS D1.5, Subsection 12.16, as applicable. Only film types designated as “fine grain” or “extra fine” shall be employed.
- c. **Magnetic Particle, Liquid Penetrant Testing.** Magnetic particle and liquid penetrant testing of welds shall conform to the applicable provisions of ASTM E 709 or AWS D1.5 Subsection 12.16, as applicable and in addition all magnetic particle testing of welds shall be made using the Wet Contrasting Black on White Method.

6. Acceptance Criteria:

- a. **Visual, Magnetic Particle and Liquid Penetrant Testing.** Welds shall be unacceptable if shown to have defects prohibited by AWS D 1.1/D 1.1M, Section 6, Part C. Visual, magnetic particle and liquid penetrant testing acceptance criteria shall be for the applicable criteria for either “Cyclically Loaded Nontubular Connections” or “Tubular Connections” per AWS D 1.1/D 1.1M, Table 6.1. Fracture critical welds shall be unacceptable if shown to have defects prohibited by AWS D 1.5/D 1.5M, Section 12. All welds shall be assumed in tension for the acceptance criteria for visual and the appropriate nondestructive testing method.
- b. **Ultrasonic Testing.** Ultrasonic acceptance criteria shall be the applicable criteria for either “Cyclically Loaded Nontubular Connections” or “Tubular Connections, Class R”. Fracture critical welds shall be unacceptable if shown to have defects prohibited by AWS D 1.5/D 1.5M, Section 12. All welds shall be assumed in tension for the acceptance criteria for visual and the appropriate nondestructive testing method.
- c. **Radiographic Testing.** Radiographic acceptance criteria shall be the applicable criteria for either “Cyclically Loaded Nontubular Connections (Tensile Stress)” or “Tubular Connections”. Fracture critical welds shall be unacceptable if shown to have defects prohibited by AWS D 1.5/D 1.5M, Section 12. All welds shall be assumed in tension for the acceptance criteria for visual and the appropriate nondestructive testing method.

7. Frequency of Testing:

The frequency specified is the minimum required. The design engineer shall determine the required frequency and include this information in the specifications and/or drawings. The design engineer shall also specify the locations of radiographic testing.

- a. **Visual Inspection.** All welds shall be visually inspected by a CWI to insure compliance with the requirements of the applicable AWS Welding Code. Prior to any welding, a CWI shall visually inspect the preparation of material for welding to assure compliance with the applicable AWS Code (D1.1 or D1.5) and approved WPS. The CWI shall also perform VT inspection throughout the welding process to assure compliance with the applicable AWS Code (D1.1 or D1.5) and approved WPS. All completed welds shall be cleaned free of oxide, flux, scale, or other foreign matter before inspection.
- b. **Full Penetration Welds.** Full penetration welds shall be examined by the Contractor using ultrasonic testing (UT) procedures described above. In addition to the full penetration welds specified for testing, a randomly chosen twenty-five percent (25%) of the remaining full penetration welds shall be ultrasonically tested to ensure the quality of the procedure and process. The random testing shall include a representative sample of welds from all welders and each of the processes each welder used. The random testing shall be spread throughout the project.
- c. **Full Penetration Butt Splice Welds.** All full penetration butt splices shall be examined using ultrasonic testing (UT) and radiographic testing (RT) procedures described above. These welds shall be defined in the specification or noted on the drawings.
- d. **Fillet Welds and Partial Penetration Groove Welds.** Fillet welds and partial penetration groove welds shall be examined by the Contractor using magnetic particle testing (MT) procedures described above. In addition to the fillet and partial penetration welds specified for testing, a randomly chosen twenty-five percent (25%) of the remaining fillet and partial penetration welds shall be magnetic particle tested to ensure the quality of the procedure and process. The random testing shall include a representative sample of welds from all welders and each of the processes each welder used. The random testing shall be spread throughout the project.

8. **Compilation of Test Data for Submittal:**

The results of the test and inspections shall be recorded in the MVN database as directed in Appendix. Samples of the reporting forms and instruction for each form are provided on the MVN SharePoint site; **Test Form Examples** and are described as follows. The latest forms should be referenced on the MVN SharePoint site and described in Appendices. All data is to be submitted electronically **within 24 hours of completion of the tests by the laboratory performing the testing.**

- a. **MVNQW06** (Combined Weld Examinations). This form is to be used in reporting the inspection and testing of welded steel connections. Contract specifications shall govern the required compaction effort. The results shall be submitted electronically within 24 hours of the test.

Appendix A – Test Form Management

1. Report Numbering:

Each soil sample (location) is identified with a unique Test ID created by concatenating the Report No and Test No.

All soil sample locations will be reported on test forms with the same Report No and Test No throughout entire range of tests performed on that sample location. This is particularly important when reporting tests that contain 1 test per test form such as MVNQS03 (Sand Cone tests) and MVNQS02 (Compaction-Moisture Density Relationship).

It is also necessary to give the same Report No and Test No to each sample location for test form MVNQS06 (Unified Soil Classification System), MVNQS07 (Organic Content), and MVNQS10 (Field Density-Nuclear) which allow for entry of up to 5 soil samples. The soil tests included in a suite of tests allows for entry of 5 samples.

Examples of all forms are available on the SharePoint site for review.

2. Naming the Test Form Files:

Each file shall be named using the following convention:

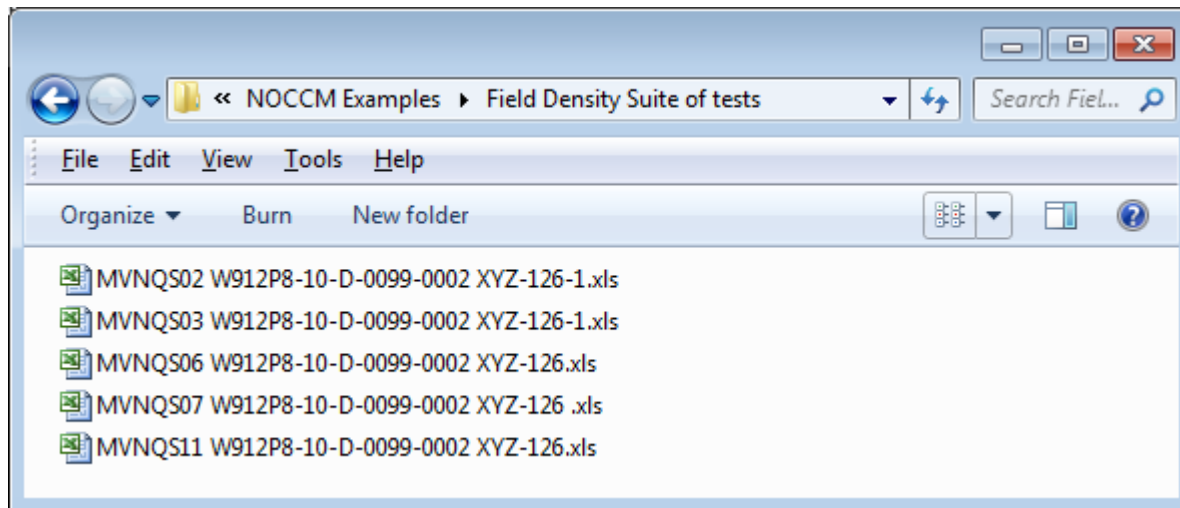
[Test Form Name][USACE Contract No][Report No][Test No (if necessary)]

Each part of the filename should be separated by a single space only, not a dash or other delimiter. Details of each portion of the filename convention are given below.

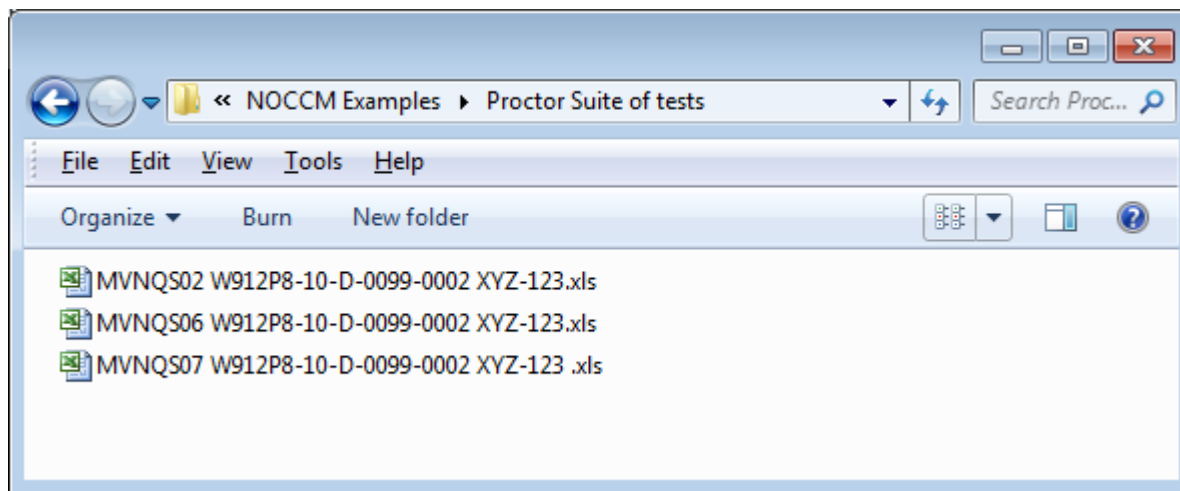
- **[Test Form Name]** is the name of the template MVNQ(C, S or W)##, for example MVNQS02. The variable letter are related to the type of test; C is for concrete, S is for soil and W is for Welds.
- **[USACE Contract No]** is the construction contract number. This must be the complete contract number including the task order if applicable. The contract numbers that contain a C or Z do not have task order numbers, whereas all contracts that contain a D have a task order number.
- **[Report number]** will be dependant on the labs report number system.
- **[Test No]** is only included in filename when necessary. This is applicable for tests reported 1 per form, as in the case of the exception listed below.

The exception to naming convention is dealing with forms that contain 1 test per test form such as MVNQS03 (Sand Cone tests), MVNQS02 (Compaction-Moisture Density Relationship) and MVNQS01 (Sieve Analysis). For these 3 Test Form types the file name will end with the Test No. The Test No is determined by the lab but should be unique to each soil sample and field location per Report No. See the Examples below for illustration of this.

Below is an example of file names for a suite of soil tests including 5 samples locations reported on a MVNQS11 (Field Density Nuclear) form, a MVNQS07 (Organic Content) form, a MVNQS06 (Unified Soil Classification System) form and a MVNQS03 (Field Density Sand Cone) form. These files are also the files included as example test forms on the SharePoint site. In this example The Report No is XYZ-126.



Below is another example for proctor data containing a MVNQS02 (Compaction-Moisture Density Relationship) test form, a MVNQS06 (Unified Soil Classification System) form and a MVNQS07 (Organic Content) form.



3. Submitting Test Forms:

All forms are to be submitted electronically **within 24 hours of completion of the tests by the laboratory performing the testing**. This is necessary since contract specifications require laboratory results to confirm compliance or failure before Contractor construction work can continue. Delays in submitting test results may result in construction delays that are to be avoided. Supporting documentation for tests should be submitted in PDF format with the same file naming convention. This completes the documentation record of data transfer to all parties.

4. How to Access USACE QACC SharePoint Site

In order to access the MVN Quality Assurance Control Center (QACC) SharePoint site each user must have an account. If a user does not have an account contact the MVN-CD Branch to receive a **USACE External Network Access Request** form. Once your account has been established you will have access to the QACC SharePoint site that contains MVNQ Test Form Templates, a QA Wiki, a QA Discussion area, a Shared Documents library, a Discrepancy Report library, and a library where completed test forms are to be uploaded; **Test Form**. The following is contact information for MVN-CD.

MVN-CD Phone;	(504) 862-2235
MVN-CD Email;	CEMVN-CD@Usace.Army.Mil
MVN-CD Public Webpage;	http://www.mvn.usace.army.mil/About/Offices/Construction.aspx
MVN QACC SharePoint;	https://partners.usace.army.mil/sites/MVN/QACC/default.aspx

5. Uploading test forms.

The exact procedure for uploading test forms to the MVN Quality Assurance Control Center (QACC) SharePoint is dependent on the computer system the user is using. Once an account has been established MVN-CD can assist each user individually by introducing the QACC system and going through the processes needed to upload test files and supporting documents. If at any point a user has questions please contact the MVN-CD-Q Branch for assistance.

Important notes before beginning the upload process; The QACC SharePoint site does have restrictions on characters (delimiters) that can be used for a file name. The following characters are not accepted by the QACC SharePoint site; \ / : * ? " < > | # { } % ~ &. If these characters are used, the QACC SharePoint site may lock up or give an error that indicates 'a nonexistent file'. If this occurs remove the delimiters used in the file name, upload the files again and verify that all files upload because this will stop the upload process for all the files. If it is determined that certain files did not upload repeat the upload process.

The second note is that the QACC SharePoint site will time out and requires logging back in after an extended time of inactivity. The QACC SharePoint site will not indicate it timed out until attempting to perform a function on the site. The site will return to the Log in screen. If this happens, the function previously being performed may not have been performed completely.

6. Revisions and Special Naming Considerations:

When resubmitting files with revisions, the same filename is to be used if possible. If a file is to be submitted to the QACC SharePoint site it is not necessary to include a revision indication in the file name, such as R1 or R2 in the filename. The test form revision should be judicated in the appropriate revision field on the form. The QACC SharePoint site does allow files to be uploaded to the **Test Form** library when the same file name exists. In general, the idea is to keep the filename as simple as possible and the same throughout the submittal and revision process. Do not add unnecessary details to the filename.

If more information is needed, see the Wiki or Discussion board on the MVN Quality Assurance Control Center (QACC) SharePoint. The QA managers are also available if there are other questions.

Appendix B - Material Testing Laboratory Requirements

1. Purpose:

All construction material testing laboratories used in support of the Contractor's Quality Control (QC) testing and the Government's Quality Assurance (QA) testing must receive validation by the Material Testing Center (MTC), Engineering and Research Development Center (ERDC), in Vicksburg Mississippi. This includes all Contractor and government on-site laboratories or commercial laboratories used either for QC or QA testing.

2. Applicability:

This procedure applies to all projects being managed by the New Orleans District for which testing of construction materials is conducted

3. References:

[ASTM E 329-06a, Agencies Engaged in the Construction Inspection and/or Testing](#)

[ER 1110-1-261 \(28 April 99\), Quality Assurance of Laboratory Testing Procedures](#)

[ER 1110-1-8100 \(31 Dec 97\), Laboratory Investigations and Testing](#)

Corps of Engineers Validated Laboratories;

Engineering Research and Development Center - Material Testing Center

<http://www.erd.c.usace.army.mil/Media/FactSheets/FactSheetArticleView/tabid/9254/Article/476661/materials-testing-center.aspx>

New Orleans Construction Division Operating Manual (CDOM), 1 March 2002

4. Responsibilities:

The Administrative Contracting Officer (ACO) / Contracting Officer's Representative (COR) is responsible for ensuring that all testing laboratories used for QC or QA testing are on the electronic validated list for the tests to be performed and for requesting that New Orleans District coordinate as necessary to pursue validation of a desired laboratory.

5. Procedures:

After award, the Contractor submits a QC Plan which delineates the scope of the testing program and identifies the testing laboratory (s) proposed specific tests. Contract specific Quality Assurance Plans will include requirements for QA verification testing by a Corps validated laboratory.

The Administrative Contracting Officer (ACO) / Contracting Officer's Representative (COR) will ensure that the QC laboratory is independent of the QA laboratory and will work with the Contractor if necessary to select another laboratory for QC or QA testing. The QC plan will

reflect the selected laboratories. If the laboratory proposed by the Contractor is not a currently validated lab, then the Administrative Contracting Officer (ACO) / Contracting Officer's Representative (COR) will notify the Contractor and request an inspection of the selected laboratory coordinated by MVN-CD-Q in accordance with the procedures described in Construction Division's Operating Manual (CDOM). For planning purposes, the validation process may require a period of six months to complete.

Briefly, the MTC validation process is described as follows:

Validation of a laboratory may consist of either (1) an inspection of the laboratory and their processes or (2) an audit of inspection reports and other documentation furnished by other validating agencies or organizations.

MTC will perform inspections in accordance with ASTM E 329 and applicable tests in ER 1110-2-1906 or tests required by project specifications.

The MTC may validate a laboratory if it has been accredited by the Concrete and Cement Reference Laboratory (CCRL) or AASHTO Materials Reference Laboratory (AMRL) within the past two years using ASTM E 329. Inspection by the MTC may be required after auditing if one or more of the critical testing procedures required in the project specifications were not included in the CCRL or AMRL inspection report or if there is any question that the laboratory may not be able to provide the required services for the specified tests.

More information about the validation process is available at the following:

Phone; (601) 634-3123

Email; MTC-info@usace.army.mil

Public Website;

<http://www.erdc.usace.army.mil/Media/FactSheets/FactSheetArticleView/tabid/9254/Article/476661/materials-testing-center.aspx>

6. Records:

Records demonstrating laboratory validation will be maintained by MTC web site for the most current laboratory listing.

Appendix C - Filling Out Test Form Templates

1. Test Form Templates:

The latest Construction Material Testing report forms are located at the following locations:

Navigate to the **MVNQ Test Form Template** library to download the most up to date forms.

<https://partners.usace.army.mil/sites/MVN/QACC/TFT/Forms/AllItems.aspx>

For access to the Extranet SharePoint site, follow procedures in Appendix A or contact the MVN Construction Division Quality Branch. Once access is granted, reference the Extranet SharePoint site to download the latest test form templates in the MVNQ Test Form Template library, as they are periodically updated and/or revised. Failure to submit the latest version of the test form template will prevent data from being loaded into the QACC database. The rejected form will be required to be resubmitted on the proper test form template.

The Test Form Examples library on the SharePoint site provides guidelines for completing several of the test form templates. Further information, definition and updates can be found in the MVNQTERMS documents and the QACC Wiki located on the Extranet SharePoint site.

On all forms, the Sample Date is defined as the date the test was performed in the field and not the date the sample was tested in the lab. Please use the Remarks section on each form for any comments that pertain to the tests performed. Comments may include items such as: meet specs, meet specs of xx% (for different types of material, say embankment is 90% compaction and trench is only 85%), in-situ material, failing tests reported to John Smith, etc. There is no such thing as too much detail or information.

This reporting and submittal system is to be used for all Corps of Engineers work in the MVN division.

2. MVNQ Terms Document:

The MVNQ Terms document located on the Extranet SharePoint site provides a list of terms that are referenced directly from the MVNQ Test Forms. This document will define the terms used on the MVNQ Test Forms, and in some cases, provide examples of the information needed in associated cells. If there is a term that is not provided, an error or a term that is not defined clearly please contact a QA Manager.

C-1 List of Forms

Form Name	Procedure(s)	Form ID
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Soil Testing Forms

#200 Wash and Sieve Analysis	ASTM C 117- C136	MVNQS01
Lab Compaction of Soil Standard Effort	ASTM D 698	MVNQS02
Density by Sand Cone	ASTM D 1556	MVNQS03
Unconfined Compression Strength	ASTM D 2166	MVNQS05
Classification of Soils – USCS	ASTM D 2487	MVNQS06
Moisture, Ash and Organic Matter of Soils	ASTM D 2974	MVNQS07
Moisture Content Determination	ASTM D 2216-4643	MVNQS09
In-place Density and Moisture of Soils	ASTM D 6938	MVNQS11
Field Density (Relative Density) - Nuclear Method	ASTM D 6938	MVNQS12

Concrete Forms

Concrete Compression Test	CCT	MVNQC01
Concrete Field Test	CFD	MVNQC02

Welding Forms

Welds – LIQUID	MVNQW06
Welds - MAGNETIC	MVNQW06
Welds - RADIO	MVNQW06
Welds - UT	MVNQW06
Welds - VISUAL	MVNQW06

Contractor Quality Control

attachments

....follow this page....

CONTRACTOR QUALITY CONTROL PLAN

Contract No. W912P8-__ - __ - __

Project Name: _____

Contractor: _____

TABLE OF CONTENTS

- 1.0 Company Policy
- 2.0 Plan Purpose
- 3.0 Quality Control Organization
 - 3.1 CQC System Manager
 - 3.2 CQC System Manager Alternate
 - 3.3 Manager of Construction
- 4.0 Resume' of personnel
 - 4.1 CQC System Manager
 - 4.2 CQC System Manager Alternate
 - 4.3 Manager of Construction
- 5.0 Designation of CQC System Manager
- 6.0 Designation of CQC Systems Manager Alternate
- 7.0 Procedures
 - 7.1 Scheduling and Managing Submittals
 - 7.2 Control Testing
 - 7.2.1 Test List
 - 7.2.2 Testing Facilities
 - 7.2.3 Test Records
 - 7.3 Inspection
 - 7.3.1 Materials
 - 7.3.2 Off-Site Inspection
 - 7.3.3 On-Site Inspection
 - 7.3.4 Completion Inspection
 - 7.4 Control Procedures
 - 7.4.1 Preparatory Phase
 - 7.4.2 Initial Phase
 - 7.4.3 Follow-Up Phase
 - 7.5 Reporting and Documentation
 - 7.5.1 Daily Record
 - 7.5.2 Control Phase Checklists
 - 7.5.3 Tracking Construction Deficiencies
 - 7.5.4 Contract Document Control
 - 7.5.5 Changes to the CQC Plan

8.0 Quality Control Program

9.0 Forms

1.0 COMPANY POLICY

_____ Construction, Corp. considers quality control to be an inherent safeguard to ensure quality work and to guarantee that all work is done according to the contract documents in a professional manner. Noncompliance with plans and specifications must be detected promptly, and proper action taken to assure that this policy is a viable tool in monitoring the work.

2.0 PLAN PURPOSE

It is the intent of this Quality Control Plan (QCP) to establish and explain how this construction corporation plans to organize, control, and review all activities according to the plans and specifications provided by the U. S. Army Corps of Engineers with regard to quality for the above reference project. The plans primary purposes are to provide for the level of construction quality required by strict accordance with the plans and specifications.

3.0 QUALITY CONTROL ORGANIZATION

3.1 CQC System Manager

The CQC System Manager (CQCM) has front line responsibility for quality control. He will become thoroughly familiar with all aspects of the project and ultimately inspect all work to ensure quality is being maintained by all craftsmen, vendors and subcontractors. The CQCM is ultimately responsible for inspecting, documenting, and reporting to the contracting officer all aspects of the work described and detailed in the plans and specifications. He is responsible for implementing and enforcing the Quality Control Plan. His duties include, but are not limited to:

- a. Implementation of the 3-phase control system for all definable features of work.
- b. Day-to-day inspection of the work.
- c. Daily on site documentation
- d. Ensure that all in-place work meets or exceeds all minimum standards set forth in the plans and specifications.
- e. Detect discrepancies or problems on site and immediately bring the same to

the attention of the Contracting Officer's Representative, as should be necessary.

f. Preparation and review of submittals and certification of submittals prior to submission.

g. Maintain document control.

h. Maintain As-built conditions.

i. Interface with the owner and outside agencies as required.

The CQCM proposed for this project is _____. See section 4.0 for a copy of his resume'.

3.2 CQC System Manager Alternate

The CQC system manager alternate will assume responsibilities for all aspects of quality control as required by our Quality Control Plan and the Contract Documents should the CQCM not be able to perform his duties. The CQC system manager alternate for this project is _____.

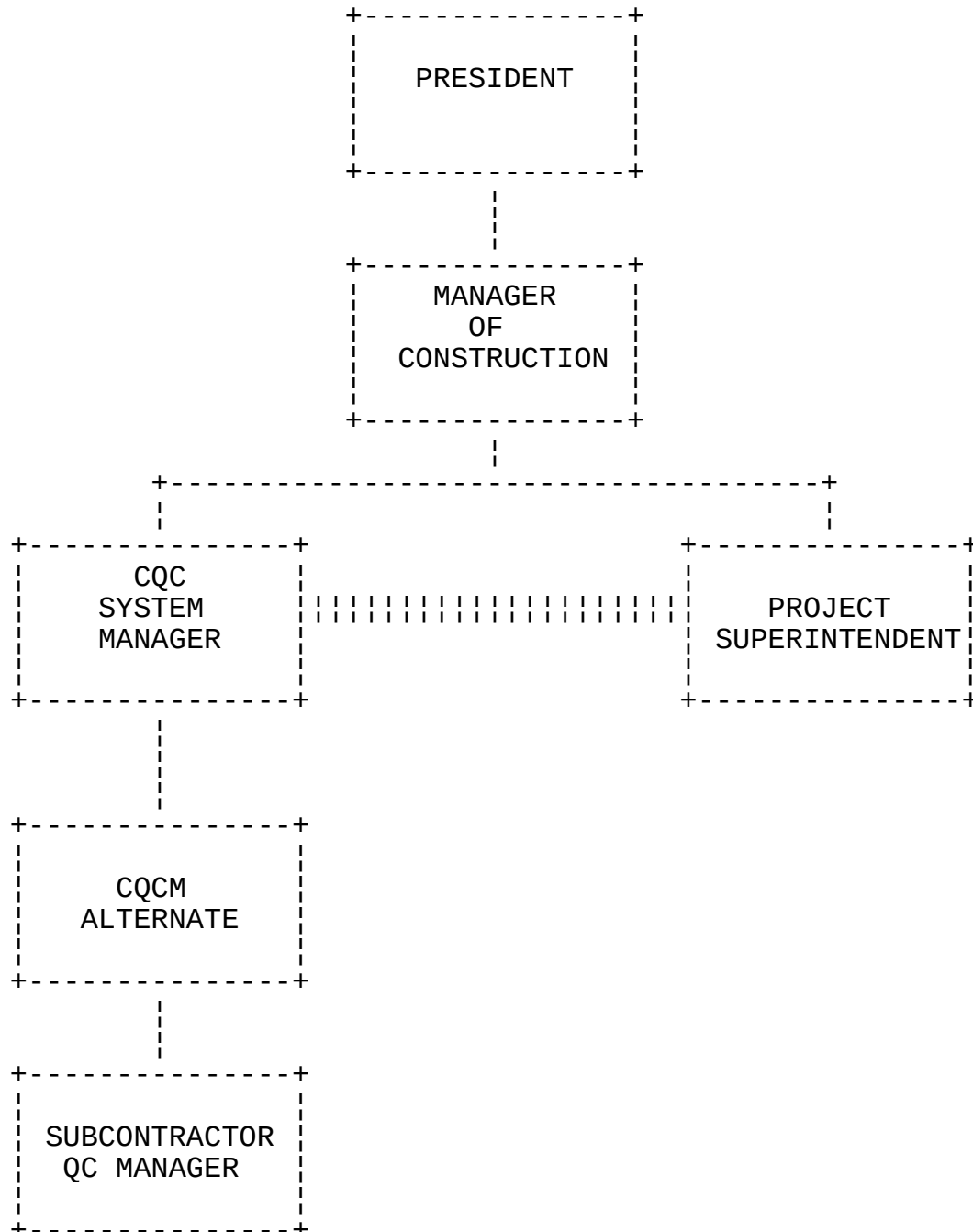
3.3 Manager of Construction

The Manager of Construction for this corporation is based in the home office in _____ and has a major responsibility for quality control through a supervisory role of the CQCM. The Manager of Construction will at all times keep the field forces focused on the company's commitment to quality in all phases of the work. The Manager of Construction will make routine visits to the site of work. The Manager of Construction for this company is _____.

4.0 RESUME OF PERSONNEL

Attached are resumes of all personnel in the above described organization. The Contracting Officer's approval will be requested before any staff changes occur, if they should become necessary.

**QUALITY CONTROL
ORGANIZATIONAL CHART
CONTRACT NO. W912P8-__-__-__**



4.1 Resume' of _____, CQC System Manager

Personal Data and Education

Date of Birth:

Residence:

Graduate of:

Completed courses in:

Professional Experience

4.2 Resume' of _____, CQCM Alternate

Personal Data and Education

Date of Birth:

Residence:

Graduate of:

Completed courses in:

Professional Experience

4.3 Resume' of _____, Manager of Construction

Personal Data and Education

Date of Birth:

Residence:

Graduate of:

Completed courses in:

Professional Experience

5.0 DESIGNATION OF CQC SYSTEMS MANAGER

(Contractor)

Date

Mr. _____

(Mailing Address)

SUBJECT: Contract No. W912P8-__-__-____
(Project Name)

Mr. _____:

This letter is to designate you as the Contract Quality Control Systems Manager for the subject project. In this capacity, you will be responsible for all aspects of quality control as required by our Quality Control Plan and the Contract Documents. You have complete authority to implement these programs including authorization to stop work which fails to comply with the requirements of the Contract Documents.

Sincerely,

_____, President

6.0 DESIGNATION OF CQC SYSTEM MANAGER ATERNATE

(Contractor)

Date

Mr. _____

(Mailing Address)

SUBJECT: Contract No. W912P8-__-__-____
(Project Name)

Mr. _____

This letter is to designate you as the Quality Control System Manager Alternate for the subject project. Should for any reason Mr. _____ not be able to perform his duties as CQCM, you will assume responsibility for all aspects of quality control as required by our Quality Control Plan and Contract Documents. To enable you to fulfill this responsibility, you have complete authority to implement these programs including authorization to stop work which fails to comply with the requirements of the Contract Documents.

Sincerely,

_____, President

7.0 PROCEDURES

7.1 Scheduling and Managing Submittals.

The CQCM will be the submittal manager. The CQCM has full authority to act for the firm in all submittal matters. His responsibilities include scheduling, review, updating and any submittals required from subcontractors.

Within 7 days of the Notice to Proceed, the CQCM will complete the submittal register contained in Section 01300 and submit to the Contracting Officer 4 copies for approval. Contractor schedule dates will be coordinated with the progress schedule and shall reflect 30-day minimum period for review and approval.

The CQCM will review the submittal register a minimum of every 10 days. The submittal register will be utilized to plan and monitor submittal progress so as to ensure timely approval of methods/materials prior to their scheduled need times. The submittal register will be available for inspection by the Contracting Officer at all times. An updated submittal register will be forwarded to the CO at 60-day intervals or as requested.

The CQCM will review the submittal register during preparatory phase of quality control to ensure that all submittals for the ensuing feature of work are approved and will take action to correct any deficiencies in submittal requirements.

All submittals required by the specifications or as needed for approval of deviation will be submitted by the CQCM in original and 4 copies utilizing ENG form 4025 in accordance with submittal register schedule dates or sooner. Prior to submittal, all shop drawings, data, samples, certifications, and test reports will be reviewed by the CQCM to ensure compliance with the contract requirements. Corrections and revisions will be requested where necessary.

7.2 Control Testing

7.2.1 Test List - A listing of all tests indicated in the contract specifications and additional tests as needed to establish quality control will be incorporated in the Contractor Quality Control Program found in section 8.0 of this plan. This listing will include the name of the test, specification para. number, feature of work tested, responsible person, and frequency.

7.2.2 Testing Facilities - The proposed testing lab for use on this project is:

_____ Testing Laboratories
PO Box _____
_____, LA _____

If required, a resume' of _____ facilities and personnel qualifications will be furnished to the Contracting Officer.

7.2.3 Test Records - All testing activities will be recorded on the CQC report, indicating the name of the test performed, specification paragraph reference, and

location performed. Results of the tests will be recorded on the daily CQC report or attachments. Actual test reports will be furnished promptly to the Contracting Officer as directed by the specifications.

7.3 Inspection

7.3.1 Materials - The CQCM will inspect all material/equipment deliveries for: (1) compliance with approved submittals, (2) damage, (3) correct dimensions and quantities, and (4) required labeling and documentation. The Contracting Officer will be notified of any materials/equipment failing to meet requirements. A record of inspection will be noted in the CQC report and any necessary corrective action will be initiated. Proper storage will be checked.

7.3.2 Off-Site Inspection - The CQCM will inspect manufacturing facilities and material sources as specifically directed by the specifications. Additional inspections will be conducted as necessary to ensure compliance with the specifications. The CQCM will record off-site surveillance activities in the CQC report. Where instances of noncompliance are observed, corrective action will be initiated.

7.3.3 On-Site Inspection - Each craftsman will be charged with the responsibility of performing his or her work in a workman like manner and continually striving for the highest degree of quality. Only craftsman who exhibit an ability to perform and desire to achieve quality will be employed.

The CQCM will routinely and continually inspect the work for compliance with contract documents. His duties, as outlined in 3.1 above, are for the purpose of maintaining and documenting the work as required to achieve a high degree of quality.

The Contract Quality Control Program outlined in paragraph 8.0 of this plan will provide an outline for the CQCM with regard to all definable features of the work. The CQCM's inspection of these work features will be accomplished through implementation of the 3-phase control procedure outline in para 7.4.

7.3.4 Completion Inspection - After completion of all work, the CQCM will conduct a completion inspection of all work features. A punchlist will be developed to identify all items which are not in compliance with the specifications and drawings. The CQCM will establish a date by which each deficiency will be corrected and note such date on the punchlist. A follow-up inspection will be conducted to verify completion of all punchlist items. The completion inspection and any resulting corrective action will be accomplished within the contract performance period. The Contracting Officer will be notified upon completion of the punchlist and corrective work. The punchlist will be made part of the Quality Control documentation by attachment to the CQC report.

7.4 Control Procedures

A 3-phase control system shall be implemented by the Quality Control staff to ensure that construction, including subcontractors and suppliers, complies with the requirements of the contract documents. This system of management will address each definable feature of work beginning with early planning stage requirements and ending with the finished work. Each phase will allow the opportunity to prevent problems and deficiencies and ensure that the accident prevention program is implemented. The 3 control phases are outlined in para 7.4.1 thru 7.4.3.

7.4.1 Preparatory Phase - This phase will be performed prior to beginning work on each definable feature of work. This phase will be conducted at a meeting involving the CQCM/Project Superintendent, QA personnel, and the foreman involved in the particular work feature. The Contracting Officer will be notified 48 hours in advance of the preparatory phase. This phase will include:

- a. A review of the applicable section of the specifications and contract drawings. (review specs)
- b. A review of the submittal register to ensure that all required submittals are submitted and approved. Take corrective action when necessary. Submittal data will be discussed to acquaint all team members with technical aspects and points particular to the work feature. (review submittals)
- c. A check to ensure that materials and equipment are in compliance with approved submittals and specifications. Verify that required materials/equipment are on hand and properly stored. (check material)
- d. Verify that preliminary work is completed.
- e. Review control testing requirements and verify that testing facilities are approved. Verify that necessary provisions are made for testing. (review testing)
- f. A consensus will be reached on planned construction procedures and the required level of quality expected from the CQCM in order to meet contract specifications. (set standards)
- g. Review appropriate Activity Hazard Analysis to assure safety requirements are met. The CQCM will inspect all equipment to ensure that minimum requirements for safety provisions in accordance with EM 385-1-1 and applicable regulations are met. (safety check)
- h. The above described activities will be documented on the COE form

"Preparatory Phase Checklist". This form will be attached to the CQC report and furnished to the Contracting Officer. Problems and deficiencies apparent during the preparatory phase and corrective action initiated will be noted in this report.

7.4.2 Initial Phase - This phase is performed once a representative portion of work has taken place for each definable feature of work and will be conducted at a meeting involving the CQCM/Project Superintendent and foreman involved in the particular work feature. The Contracting Officer will be notified 48 hours in advance of this phase. Initial phase will include:

- a. A check to ensure that preliminary work is completed.
- b. Verify that materials/equipment and construction procedures are in compliance with the contract documents.
- c. Review control testing requirements.
- d. Set standards of quality required to meet contract specifications.
- e. Review the Activity Hazard Analysis to ensure safety requirements are met. Check equipment for safety provisions.
- f. The above described activities will be documented on the COE form "Initial Phase Checklist". This form will be attached to the CQC report and furnished to the Contracting Officer. Problems and deficiencies apparent during the initial phase and corrective actions initiated will be noted in this report. The initial phase will be repeated any time the CQCM feels that quality standards and safety requirements must be reinforced.

7.4.3 Follow-Up Phase - This phase is accomplished through the daily inspections by the CQCM, also through performance of the required control testing. Follow-up phase efforts will ensure a continuation of quality and safety standards established during preparatory and initial phases until completion of the work feature. The CQCM's follow-up phase activities, including deficiencies noted, corrective action taken, and control testing results will be recorded in the daily CQC report.

7.5 Reporting and Documentation

The CQCM will maintain records of all quality control activities including documentation of control testing and inspection, and maintain integrity of the contract documents through use of the following described forms and procedures. Additional reports will be formulated or added as needed.

7.5.1 Daily Record - The CQCM will utilize the COE furnished forms titled "*Contractor Quality Control CQC Form*" to record daily control activities and resources used, work performed, and other data indicated on this form. The original and two copies will be furnished to the Contracting Officer within 12 hours of the reporting date. The CQCM will maintain copies for his files. Test reports will be included in the CQC report.

7.5.2 Control Phase Checklists - The CQCM will utilize the COE furnished forms entitled "*Preparatory Phase Checklist*" and "*Initial Phase Checklist*" to document these control phase activities. Original and two copies will be attached to the CQC report for the date on which the control phase is completed. A log will be posted at the jobsite office in chart form to record the dates on which preparatory and initial phases were completed for each definable feature of work so as to allow easy verification of control activities.

7.5.3 Tracking Construction Deficiencies - The form for tracking construction deficiencies is the Deficiency Report (DR). A DR can be issued by the CQCM/Project Superintendent or Manager of Construction. All DR's shall be kept and updated by the CQCM. The DR log will be available for inspection by the Contracting Officer. See attached forms for the example of a DR.

The DR tracking log will be in chart form and bound in a log book maintained on site. See attached example of the deficiency report tracking log. The DR log book is available for inspection by the Contracting Officer at all times.

A construction deficiency for the purposes of this plan is defined as:

1. An occurrence in which defective work or work lacking some essential part has been covered or is otherwise left as complete.
2. Products are furnished to the site or incorporated into the work which do not meet the conditions of the contract documents.
3. Inspection points or contract requirements affecting quality of the work that have not been met. Minor defects in work on which construction is underway is not to be considered a Construction Deficiency.

7.5.4 Contract Document Control - The CQCM will maintain a record in log form of the most up-to-date documents issued for construction and adjustments. No contract documents will be replaced or revised without receipt of a modification or direction from the Contracting Officer. The CQCM will maintain As-Built contract drawings.

7.6 Changes to the CQC Plan - Periodically, and at least once weekly, the CQCM

will review the CQC plan with the possible need for changes in mind. During the course of work on this contract, it is reasonable to expect the need for some changes to arise. When they do, the QC Manager will incorporate these changes in the form of written amendments and copies will be furnished to the Contracting Officer.

8.0 QUALITY CONTROL PROGRAM

(Sample only, this is done for each definable feature of work)

**Q. C. ACTIVITIES & TESTING REQUIREMENTS
FOR DEFINABLE FEATURES OF WORK**

CONTRACT W912P8-__-__-__

**Definable Feature: Cast In-Place Structural Concrete
Section 03301**

Definable Feature		Submittal	Quality Control Activities		
Description	Spec. Para.	Req'd	Description of Observation Procedure or Test Required	Freq.	Remarks
Concrete - Grout, Water	03301-15.1.4.1 03301-15.5.1.6	Submittal Register	Grout certificate, equipment & method used, & source of mixing & curing water	Once prior to placement	
Concrete - Finishing Formed Surfaces	03301-12.2	QC Report	Visually inspect all finishing is started within 24 hours of form removal, tie rod holes & defective concrete voids and honeycombs are filled properly, smooth surface	After each placement	
Concrete - Fine Aggregate	03301-15.2.1.1	QC Report	Sieve analysis and fineness modulus determination	At least once each delivery	Testing by _____ Laboratory
Concrete - Coarse Aggregate	03301-15.2.2	QC Report	Sieve Analysis	At least once each delivery	Testing by _____ Laboratory
Concrete - Moisture Test	03301-15.2.2.2	QC Report	Test for moisture content for each size coarse aggregate	At least once each delivery	Testing by _____ Laboratory
Concrete - Mixer Uniformity	03301-15.2.12	QC Report	Uniformity of concrete determined in accordance with ASTM C 94.	Prior to concrete placement & 1/ 6 mo	

**Q. C. ACTIVITIES & TESTING REQUIREMENTS
FOR DEFINABLE FEATURES OF WORK**

CONTRACT W912P8-__ -__ -__

Definable Feature: Cast In-Place Structural Concrete
Section 03301

Definable Feature		Submittal	Quality Control Activities		
Description	Spec. Para.	Req'd	Description of Observation Procedure or Test Required	Freq.	Remarks
Concrete	03301-5.3	Submittal Register	Submit batch plant details, mixer details, conveying methods and equipment, placing, joint clean-up, curing, and weather requirements	once 14 days prior to placement	
Concrete	03301-3.1.1	Submittal Register	Submit 500 lb sample of aggregate to Waterway Experiment Station for testing if an approved supplier is not used	Once prior to concrete placement	
Concrete	03301-5.1.1	Submittal Register	Submit concrete mixture proportion	Once	
Concrete - Materials	03301-5.1.2	Submittal Register	Submit cement cert. of compliance	Once	Testing by supplier or _____ Lab
	03301-5.1.5		Submit sieve analysis for aggregates		
	03301-5.2.3		Submit air-entraining agent cert. of compliance		
	03301-5.2.5		Submit curing compound cert. of compliance		

**Q. C. ACTIVITIES & TESTING REQUIREMENTS
FOR DEFINABLE FEATURES OF WORK
CONTRACT W912P8-__-__-__**

Definable Feature: Cast In-Place Structural Concrete
Section 03301

Definable Feature		Submittal	Quality Control Activities		
Description	Spec. Para.	Req'd	Description of Observation Procedure or Test Required	Freq.	Remarks
Concrete - Placement Preparations	03301-15.2.7	QC Report & LMV Form 1246	Visually & Measure as needed, prior to placement, foundations, const. joints, forms, embedded items, etc., to verify that concrete placement OK	Prior to placement	
Concrete - Air Content	03301- 15.2.6.1	QC Report	Test Concrete for air content	Twice per day	Use _____ Laboratory
Concrete - Slump	03301- 15.2.5.2	QC Report	Test Concrete slump	Twice per day	Use _____ Laboratory
Concrete - Placement	03301-15.2.8	QC Report	Visually inspect placement operations to verify proper equipment, methods, time interval, temp., yardage placed, & placement method	Each Placement	
Concrete - Curing	03301-13.2,4	QC Report	Inspect all surfaces subject to moist curing & impervious sheet curing	At least once/day	Including weekend/hol.
Concrete - Curing	03301-13.3	QC Report	Assure that curing compound is mixed properly, & meets minimum pressure and coverage requirements	After removal of forms	Measure & visual
Concrete - Vibration	03301-15.2.9	QC Report	Test frequency and amplitude of vibrator	Prior to 1st use & 1/month	

9.0 FORMS

CONTRACTOR QUALITY CONTROL (CQC) FORM

Contractor's Name

Daily Report No: _____

Date: _____

Contract No: W912P8-__-__-__

Project Title and Location: _____

Weather: _____ Rain: ____ in. Temp: ____ Min. ____ Max.

1. Contractor/Subcontractors and Area of Responsibility:

NUMBER	TRADE	HOURS	EMPLOYER	LOCATION/DESCRIPTION

2. Operating Plant of Equipment. (Not hand tools)

PLANT/ EQUIPMENT	DATE OF ARRIVAL/ DEPARTURE	LEASED/ OWNED L OR O	DATE OF SAFETY CHECK	HOURS USED	HOURS IDLE	HOURS REPAIR

CQC Report Form (Cont'd)

3. Work performed today: (Indicate location and description of work performed by prime and/or subcontractor by letter in table above.)

4. Results of control activities: (Indicate whether P - preparatory, I - Initial, or F - Follow-up Phase. When a P or I meeting is conducted, complete appropriate forms, attached.)

5. Test performed as required by plans and/or specifications:

6. Materials received:

CQC REPORT FORM (CONT'D)

7. Submittals Reviewed:

(a) Submittal No.	(b) Spec/Plan Reference	(c) By Whom	(d) Action

8. Off-site surveillance activities, including action taken:

9. Job Safety: (Report violations; Corrective instructions given, taken.)

10. Environmental Protection: (Report violations; Corrective instructions given, taken.)

11. Remarks: (Instructions received or given. Conflicts in Plans and/or Specifications.)

Contractor's Verification: On behalf of the contractor, I certify this report is complete and correct, and all materials and equipment used and work performed during this reporting period are in compliance with the plans and specifications, to the best of my knowledge, except as noted above.

Authorized CQC System Manager

Date

PREPARATORY PHASE CHECKLIST FORM

Contract No.: W912P8-__ - __

Date: _____

Definable Feature:

Government Representative Notified 48 Hours in Advance

Yes____ No ____

I. Personnel Present:

Name	Position	Company/Government

(List Additional Personnel on reverse side)

II. Submittals

1. Review Submittals and/or submittal log 4288. Have all submittals been approved? Yes____ No____

If No, what items have not been submitted?

a.

b.

c.

2. Are all materials on hand? Yes____ No____

If No, what items are missing?

a.

b.

c.

PREPARATORY PHASE CHECKLIST FORM (CONT'D)

3. Check approved submittals against delivered material. (This should be done as material arrives.) Comments:

III. Material storage

Are materials stored properly? Yes___ No___
If No, what action will be taken?

IV. Specifications:

1. Review each paragraph of specifications.

2. Discuss procedure for accomplishing the work. (Include labor and equipment to be used)

3. Clarify any differences from specifications.

V. Preliminary Work - Ensure preliminary work is correct.

If not, what action will be taken?

PREPARATORY PHASE CHECKLIST FORM (CONT'D)

VI. Testing

1. Identify test to be performed, frequency and by whom.

2. When required?

3. Where required?

4. Review Testing Plan.

VII. Safety

1. Review applicable portion of COE EM 385-1-1.

2. Activity Hazard Analysis Approved? Yes___ No___

3. All equipment checked and checklists recorded? Yes___ No___
If not, what action will be taken?

VIII. Corps of Engineers comments during meeting.

CQC Representative

INITIAL PHASE CHECKLIST FORM

Contract No.: W912P8-__-__-__

Date: _____

Definable Feature:

Government Representative Notified 48 Hours in advance Yes___ No___

I. Personnel Present:

Name	Position	Company/Government

(List Additional Personnel on Reverse Side)

II. Is work in full compliance with plans, specifications and submittals. Are procedures and quality control measures being used acceptable.

INITIAL PHASE CHECKLIST FORM (CONT'D)

III. Preliminary work. Ensure preliminary work is complete and correct. If not, what action will be taken?

IV. Establish Level of Workmanship.

1. Where is work located?

2. Quantity of work performed?

3. Is a sample panel required? Yes___ No___

4. Will the initial work be considered as a sample? Yes___ No___

V. Are standards of acceptance mutually agreed upon? Resolve any differences.

VI. Check Safety.

Review job condition using COE EM 385-1-1 and job hazard analysis. Comments:

CQC Representative

DEFICIENCY REPORT

Contract No.: W912P8-__-__-__

DCR NO.: _____

Project

Name: _____

Contractor:

Description of Deficiency:

Sketch Attached: Yes__ No__

Issued By: _____ Date: _____

Approved and Logged By: _____ Date: _____

CQCM

~~~~~

Planned Corrective Action:

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

CO or Representative: \_\_\_\_\_ Date: \_\_\_\_\_

CQCM: \_\_\_\_\_ Date: \_\_\_\_\_

Corrective Action Implemented: \_\_\_\_\_ Date: \_\_\_\_\_

Project Super.

Corrective Action Inspected: \_\_\_\_\_ Date: \_\_\_\_\_

CQCM

# DEFICIENCY REPORT TRACKING LOG

Contract No. W912P8-\_\_-\_\_-\_\_\_\_

| DR<br>NUMBER | DATE<br>ISSUED | ISSUED<br>BY<br>(Initial) | WORK FEATURE<br>(See DR Report<br>for details) | DATE<br>CORRECTED | DATE<br>INSPECTED | INSPECTOR<br>(INITIAL) |
|--------------|----------------|---------------------------|------------------------------------------------|-------------------|-------------------|------------------------|
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |
|              |                |                           |                                                |                   |                   |                        |

## SECTION TABLE OF CONTENTS

## DIVISION 01 - GENERAL REQUIREMENTS

## SECTION 01 57 20.01 12

## ENVIRONMENTAL PROTECTION

03/14

## PART 1 GENERAL

- 1.1 SCOPE
- 1.2 APPLICABLE REGULATIONS
- 1.3 SUBMITTALS
- 1.4 MEASUREMENT AND PAYMENT
  - 1.4.1 Environmental Protection
  - 1.4.2 Bird Nesting Prevention and Avoidance Measures
  - 1.4.3 Non-Regulated Waste
  - 1.4.4 Hazardous/Regulated Waste
- 1.5 QUALITY CONTROL
  - 1.5.1 General
  - 1.5.2 Reporting
- 1.6 NOTIFICATION
- 1.7 SUBCONTRACTORS
- 1.8 IMPLEMENTATION
- 1.9 REFERENCES
- 1.10 ENVIROMENTAL ASSESSMENT OF CONTRACT DEVIATIONS

## PART 2 PRODUCTS

## PART 3 EXECUTION

- 3.1 PROTECTION OF LAND RESOURCES
  - 3.1.1 General
  - 3.1.2 Prevention Of Landscape Defacement
  - 3.1.3 Temporary Excavation And Embankments
  - 3.1.4 Post-Construction Cleanup Or Obliteration
  - 3.1.5 Discovery of Previously Unknown Archaeological Remains and Artifacts
- 3.2 PROTECTION OF WATER RESOURCES
  - 3.2.1 Contamination Of Water
  - 3.2.2 Disposal Of Materials
  - 3.2.3 Erosion Control
  - 3.2.4 Fuel and Lubricants
- 3.3 PROTECTION OF FISH AND WILDLIFE
  - 3.3.1 Nesting Birds
    - 3.3.1.1 Nesting Birds
    - 3.3.1.2 Reporting
    - 3.3.1.3 No Work Distances
    - 3.3.1.4 Bird Nesting Prevention and Avoidance Measures
    - 3.3.1.5 Discovery of Bird Nest at the Work Site
  - 3.3.2 Manatees
    - 3.3.2.1 Special Operating Conditions If Manatees Are Present
    - 3.3.2.2 Manatee Sighting Reports



- 3.4 JANITOR SERVICES
- 3.5 DISPOSAL OF REGULATED SOLID WASTES
- 3.6 DISPOSAL OF NON-REGULATED DEBRIS AND OTHER WASTE
- 3.7 MAINTENANCE OF POLLUTION CONTROL FACILITIES
- 3.8 REPORTING OF POLLUTION SPILLS
- 3.9 TRAINING OF CONTRACTOR PERSONNEL

-- End of Section Table of Contents --

## SECTION 01 57 20.01 12

## ENVIRONMENTAL PROTECTION

03/14

## PART 1 GENERAL

## 1.1 SCOPE

The work covered by this section consists of furnishing all labor, materials and equipment, and performing all work required for the prevention of environmental pollution during and as the result of construction operations under this contract except for those measures set forth in other provisions of these specifications. For the purpose of this specification, environmental pollution is defined as the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to man; or degrade the utility of the environment for esthetic and recreational purposes. The control of environmental pollution requires consideration of air, water, and land, and involves noise, solid waste-management and management of radiant energy and radioactive materials, as well as other pollutants.

## 1.2 APPLICABLE REGULATIONS

In order to prevent, and to provide for abatement and control of any environmental pollution arising from construction activities in the performance of this contract, the Contractor and his/her subcontractors shall comply with all applicable Federal, State, and Local laws, and regulations concerning environmental pollution control and abatement.

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-01 Preconstruction Submittals

## Environment Pollution Control Plan; G, CD

The Contractor shall prepare and submit to the Contracting Officer's Representative, for approval, an Environment Pollution Control Plan, detailing the efforts that will be undertaken to prevent on-site activities that pose a risk to the environment. The plan must include an Oil Spill Reporting and Response Plan. Contractor shall submit two (2) printed copies of the plan for review prior to approval of the document by the Contracting Officer. (See paragraph QUALITY CONTROL, paragraph "General".

## Bird Nesting Prevention Plan; G, ODT

The Contractor shall prepare and submit to the Contracting Officer's Representative, for approval, a plan, detailing the efforts that will be undertaken to prevent birds from nesting

within the minimum distances. Contractor shall submit two printed copies of the plan for review prior to approval of the document by the Contracting Officer. (See subparagraph "Bird Nesting and Avoidance Measures" of paragraph PROTECTION OF FISH AND WILDLIFE).

#### 1.4 MEASUREMENT AND PAYMENT

##### 1.4.1 Environmental Protection

No separate measurement or payment will be made for environmental protection, except for "Bird Nesting Prevention and Avoidance Measures. Payment for the work covered under this section shall be distributed throughout the existing bid items.

##### 1.4.2 Bird Nesting Prevention and Avoidance Measures

Measurement for providing bird nesting prevention and avoidance measures will be per day satisfactorily performed. Payment will be made at the contract unit price per day for "Bird Nesting Prevention and Avoidance Measures-OW" (if Option Work is exercised). Price and payment shall constitute full compensation for furnishing all equipment, labor, materials, supplies, supervision, and all operations incidental thereto.

##### 1.4.3 Non-Regulated Waste

No separate measurement or payment will be made for the work associated with the disposal of non-regulated debris not specifically covered elsewhere. Payment for the work associated with the disposal of non-regulated debris not specifically covered elsewhere shall be distributed throughout the existing bid items.

##### 1.4.4 Hazardous/Regulated Waste

If the Contractor uncovers an existing hazardous/regulated waste not Contractor generated, not shown on the drawings, and not specified herein, the Contractor shall notify the Contracting Officer's Representative immediately. Payment for handling, removal, transportation and disposal of hazardous and/or regulated solid wastes not Contractor generated, not shown on the drawings, and not specified herein will be made as an equitable adjustment in contract price under the Contract Clause in Section 00700, entitled CHANGES (FAR 52.243-4).

#### 1.5 QUALITY CONTROL

##### 1.5.1 General

The Contractor shall establish and maintain quality control for environmental protection to assure compliance with contract specifications and maintain records of his quality control for all construction operations including but not limited to the following:

- (1) Submit plan of Environment Pollution Control Plan.
- (2) Procure applicable Federal, State, and Local regulations on pollution control.
- (3) Air Pollution - Checks made on dust, smoke, noise.
- (4) Water Pollution - Checks made on disposal of water, oil, etc.

(5) Land Pollution - Checks made on disposal of materials, restoration of temporary construction sites, etc.

(6) Training course for employees.

#### 1.5.2 Reporting

The original and two copies of these records, as well as the records of corrective action taken, shall be furnished the Government daily. Format of report shall be as prescribed in Section 01 45 04.00 10 CONTRACTOR QUALITY CONTROL.

#### 1.6 NOTIFICATION

The Contracting Officer will notify the Contractor in writing of any non-compliance with the foregoing provisions and the action to be taken. The Contractor shall, after receipt of such notice, immediately take corrective action. Such notice, when delivered to the Contractor or his/her authorized representative at the site of the work, shall be deemed sufficient for the purpose. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to any such stop orders shall be made the subject of a claim for extension of time or for excess cost of damages by the Contractor.

#### 1.7 SUBCONTRACTORS

Compliance with the provisions of this section by subcontractors will be the responsibility of the Contractor.

#### 1.8 IMPLEMENTATION

Within 10 days after receipt of Notice of Award, the Contractor shall:

(1) Submit, in writing, proposals for environmental pollution control, disposal of debris, non-hazardous wastes and hazardous wastes generated at the project site as well as storage and management of regulated materials, and an Environmental Protection Plan. Provide 1 copy each to the Contracting Officer and 1 copy to ODT; Attention Jeff Corbino for review prior to approval of the document by the Contracting Officer.

(2) Upon review and approval of requirements above, the Contractor shall meet with representatives of the Contracting Officer to develop mutual understandings relative to compliance with these provisions and administration of the environmental pollution control program and Environmental Protection Plan.

#### 1.9 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(2024) Safety and Health Requirements

## U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA ESA (1973) Endangered Species Act

EPA MMPA (1972) Marine Mammal Protection Act

## U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

33 CFR 153.203 Procedure for the Notice of Discharge

## 1.10 ENVIRONMENTAL ASSESSMENT OF CONTRACT DEVIATIONS

The Contractor is advised that deviations from the drawings or specifications (e.g., proposed alternate borrow areas, disposal areas, staging areas, alternate access routes, etc.) could result in the requirement for the Government to reanalyze the project from an environmental standpoint. Deviations from the construction methods and procedures indicated by the plans and specifications, which may have an environmental impact will require an extended review, processing, and approval time by the Government. The Contracting Officer reserves the right to disapprove alternate methods, even if they are more cost effective, if the Contracting Officer determines that the proposed alternate method will have an adverse environmental impact.

## PART 2 PRODUCTS

Not used.

## PART 3 EXECUTION

## 3.1 PROTECTION OF LAND RESOURCES

## 3.1.1 General

The land resources within the project boundaries and outside the limits of permanent work performed under this contract shall be preserved in their present condition or be restored to a condition after completion of construction that will appear to be natural and not detract from the appearance of the project. The Contractor shall confine his/her construction activities to areas defined by the plans and/or specifications. The following additional requirements are intended to supplement and clarify the requirements of the Contract Clauses in Section 00700, entitled PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS (FAR 52.236-9), OPERATIONS AND STORAGE AREAS (FAR 52.236-10), and CLEANING UP (FAR 52.236-12).

## 3.1.2 Prevention Of Landscape Defacement

The Contractor shall not deface, injure, or destroy trees or shrubs, nor remove or cut them without the approval of the Contracting Officer. Felling of trees shall be performed in such a manner as to avoid damage to trees to be left standing. Where trees may possibly be defaced, bruised, injured, or otherwise damaged by the Contractor's operations or equipment; he/she shall protect adequately such trees. All monuments and markers shall be protected before beginning operations near them. Any trees or other landscape feature scarred or damaged by the Contractor's equipment or operations shall be restored as nearly as possible to its original condition at the Contractor's expense. Trees that are scarred shall be immediately painted with an acceptable tree wound paint. Any trees which

are damaged beyond restoration shall be removed and disposed of as directed by the Contracting Officer.

### 3.1.3 Temporary Excavation And Embankments

If the Contractor proposes to construct temporary roads or embankments and excavation for plant and/or work areas, he shall obtain approval of the Contracting Officer prior to start of such temporary work.

### 3.1.4 Post-Construction Cleanup Or Obliteration

The Contractor shall obliterate all signs of temporary construction facilities such as work areas, structures, foundations of temporary structures, and stockpiles of excess or waste materials upon completion of construction. The Contractor will be required to restore the construction area to near natural conditions which will permit the growth of vegetation.

### 3.1.5 Discovery of Previously Unknown Archaeological Remains and Artifacts

If during the course of work, archaeological artifacts (prehistoric or historic), unmarked graves, burials, human remains, or items of cultural patrimony are discovered, the Contractor shall stop work in the vicinity of the discovery and take all reasonable measures to avoid or minimize harm to the finds. The Contractor shall immediately report the find to the Contracting Officer at CEMVN, who will in turn contact CEMVN Historic Preservation (HP) staff. The Contractor shall not proceed with work in the area until CEMVN HP complete consultation with the State Historic Preservation Office, and others as appropriate.

## 3.2 PROTECTION OF WATER RESOURCES

### 3.2.1 Contamination Of Water

The Contractor shall not pollute lakes, ditches, rivers, bayous, canals, groundwater, waterways, or reservoirs with fuels, oils, bitumen, calcium chloride, insecticides, herbicides, or other similar materials harmful to fish, shellfish, or wildlife, or materials which may be a detriment to outdoor recreation.

### 3.2.2 Disposal Of Materials

The methods and locations of disposal of materials, wastes, effluent, trash, garbage, oil, grease, chemicals, etc., within the right-of-way limits shall be such that harmful debris will not enter lakes, ditches, rivers, bayous, canals, groundwater, waterways, or reservoirs by erosion, and thus prevent the use of the area for recreation or present a hazard to wildlife.

### 3.2.3 Erosion Control

Surface drainage from cuts and fills within the construction limits, whether or not completed, and from waste disposal areas, shall, if turbidity producing materials are present, be held in suitable sedimentation ponds or shall be graded to control erosion within acceptable limits. Temporary erosion and sediment control measures such as berms, dikes, drains, or sedimentation basins, if required to meet the above standards, shall be provided and maintained until permanent drainage and erosion control facilities are completed and operative. Waste areas shall be constructed by selective placement to eliminate silts or clays on the

surface that will erode and contaminate adjacent streams.

#### 3.2.4 Fuel and Lubricants

Storage, fueling, and lubrication of equipment and motor vehicles must be conducted in a manner that affords the maximum protection against spill and evaporation. Manage and store fuel, lubricants, and oil in accordance with all Federal, State, and local laws and regulations. Used lubricants and used oil to be discarded must be stored in marked corrosion-resistant containers and recycled or disposed in accordance with State and local laws and regulations.

### 3.3 PROTECTION OF FISH AND WILDLIFE

The Contractor shall at all times perform all work and take such steps required to prevent any interference or disturbance to fish and wildlife. The Contractor is not permitted to alter water flows or otherwise disturb native habitat adjacent to the project area which are critical to fish or wildlife.}

#### 3.3.1 Nesting Birds

##### 3.3.1.1 Nesting Birds

Colonial nesting wading birds (including but not limited to, herons, egrets, and Ibis) and seabirds/waterbirds (including, but not limited to terns, gulls, black skimmers, and brown pelicans) may nest in the project area. The nesting birds and their nests must not be disturbed or destroyed. The nesting activity period extends from February 15 through September 15. Dredging activity during this period may be subject to additional requirements as stated below.

##### 3.3.1.2 Reporting

The presence of nesting wading birds and/or seabirds/waterbirds within the minimum distances from the work area as specified in paragraph "No Work Distances" must be immediately reported to Mr. Jeff Corbino, of the U.S. Army Corps of Engineers at (504) 862-1958 or email: Jeffrey.m.Corbino@usace.army.mil.

##### 3.3.1.3 No Work Distances

No-work distance restrictions are as follows:

- " Terns, gulls, and Black Skimmers - 650 feet;
- " Colonial nesting wading birds - 1000 feet; and,
- " Brown Pelicans - 2000 feet.

Coordination by the New Orleans District personnel with the U.S. Fish and Wildlife Service may result in a reduction of these no-work distance restrictions depending on the species of birds found nesting at the work site.

##### 3.3.1.4 Bird Nesting Prevention and Avoidance Measures

The Contractor shall prepare and submit to the Contracting Officer's Representative, for approval, a plan, detailing the efforts that will be undertaken to prevent birds from nesting within the minimum distances, as specified in paragraph "No Work Distances", from any work activity. The

plan shall be submitted in accordance with paragraph titled IMPLEMENTATION. Additionally, one copy of this plan shall be submitted to:

U.S. Army Corps of Engineers, New Orleans District  
Attn: CEMVN-ODT, Mr. Jeff Corbino  
7400 Leake Avenue  
New Orleans, LA 70118

Nest prevention measures shall be intended to deter birds from nesting on the disposal area(s) and access corridor(s) without physically harming birds during the nesting activity period, as specified in paragraph "Nesting Birds". Nest prevention measures may be used in combination and or adjusted to be most effective. The use of any harassment measures shall be in accordance with EM 385-1-1. At minimum, nest prevention measures shall include the following:

Flagging/Streamers - Flagging and or streamers at least 2 feet in length and which consist of reflective plastic/mylar type material shall be attached to the top of stakes at least 3 feet in height. The stakes shall be driven into the ground at approximately 20-foot intervals. Flagging and or streamers shall be placed such that the flags/streamers move in a light wind.

Vehicular/Pedestrian Traffic - At a minimum, one terrain vehicle (ATV) and/or one person shall travel throughout the entire disposal area at least once per hour from dawn to dusk.

Specific nest prevention measures used during the work shall be monitored for effectiveness and may require adjustment and/or modification. All equipment/supplies used for nest prevention shall be removed from the work site upon the completion of the work and as directed by the Contracting Officer.

#### 3.3.1.5 Discovery of Bird Nest at the Work Site

If bird nests are discovered at the work site, immediate notification shall be made, as specified in paragraph "Reporting". The Contractor shall immediately mark the bird nests with flagging on stakes 3-feet above the ground surface and no closer than 3 feet from the nest. The contractor shall immediately implement safe work distances from the nest(s), as specified in paragraph "No Work Distances", and place flagging to create exclusion zone(s) around the nest(s), and advise all equipment operators of the bird nest(s) and exclusion zone(s).

#### 3.3.2 Manatees

The West Indian manatee may be present in the project vicinity. The Contractor shall instruct all personnel associated with the project of the potential presence of manatees in the area, and the need to avoid collisions with these animals. All construction personnel shall be advised that there are civil and criminal penalties for harming, harassing, or killing manatees, which are protected under the EPA MMPA and the EPA ESA. The Contractor will be responsible for any manatee harmed, harassed, or killed as a result of construction activities not conducted in accordance with these specifications.

##### 3.3.2.1 Special Operating Conditions If Manatees Are Present

- (1) If a manatee(s) is sighted within 100 yards of the project area,



all appropriate precautions shall be implemented by the Contractor to ensure protection of the manatee. These precautions shall include the operation of all moving equipment no closer than 50 feet of a manatee. If a manatee is closer than 50 feet to moving equipment or the project area, the equipment shall be shut down and all construction activities shall cease to ensure protection of the manatee. Construction activities shall not resume until the manatee has departed and the 50-foot buffer has been re-established.

(2) If a manatee(s) is sighted in the project area, all vessels associated with the project shall operate at "no wake/idle" speeds at all times, and vessels shall follow routes of deep water whenever possible, until the manatee has departed the project area. Boats used to transport personnel shall be shallow-draft vessels, preferably of the light-displacement category, where navigational safety permits.

(3) If siltation barriers are used, they shall be made of material in which manatees cannot become entangled, are properly secured, and are regularly monitored to avoid manatee entrapment.

(4) Manatee Signs. Prior to commencement of construction, each vessel involved in construction activities shall display at the vessel control station or in a prominent location, visible to all employees operating the vessel, a temporary sign at least 8-1/2-inch x 11-inch reading, "CAUTION: MANATEE HABITAT/IDLE SPEED IS REQUIRED IN CONSTRUCTION AREA." In the absence of a vessel, a temporary 3-foot x 4-foot sign reading "CAUTION: MANATEE AREA" shall be posted adjacent to the issued construction permit. A second temporary sign measuring 8-1/2-inch x 11-inch reading "CAUTION: MANATEE HABITAT. EQUIPMENT MUST BE SHUTDOWN IMMEDIATELY IF A MANATEE COMES WITHIN 50 FEET OF OPERATION" shall be posted at the dredge operator control station and at a location prominently adjacent to the issued construction permit. The Contractor shall remove the signs upon completion of construction.

### 3.3.2.2 Manatee Sighting Reports

Any sightings of manatees, or collisions with a manatee, shall be reported immediately to the Corps of Engineers. The point of contact within the Corps of Engineers will be the USACE Environmental Manager, Mr. Jeff Corbino, phone (504)862-1958 or email to Jeffery.m.Corbino@usace.army.mil. In addition, collisions with, injury to, or sightings of manatees should be immediately reported to the U.S. Fish and Wildlife Service's Louisiana Ecological Services Office (337)291-3100 and the Louisiana Department of Wildlife and Fisheries, Natural Heritage Program (225)765-2821. Please provide the nature of the call (i.e., report of an incident, manatee sighting, etc.); time of incident/sighting; and the approximate location, including the latitude and longitude coordinates, if possible.

### 3.4 JANITOR SERVICES

The Contractor shall furnish daily janitorial services for all the offices, shops, or other facilities being used by the Contractor or Government employees, whether existing or Contractor furnished, and perform any required maintenance of the facilities and grounds during the life of the contract. Toilet facilities shall be kept clean and sanitary at all times. Services shall be performed at such a time and in such a manner to least interfere with the operations but will be accomplished only when the buildings are in daily use. Services shall be accomplished to the satisfaction of the Contracting Officer. The Contractor shall also provide

daily trash collection and cleanup of the buildings and adjacent outside areas, and shall dispose of all discarded debris in a manner approved by the Contracting Officer.

### 3.5 DISPOSAL OF REGULATED SOLID WASTES

If any hazardous or regulated solid wastes will be generated as a result of the Contractor's operations, the Contractor shall submit a plan that details the proper handling, removal, transportation and disposal of such wastes. The plan shall identify what types of hazardous and/or regulated solid wastes will be generated and shall list the hazards involved with each waste. All waste generated on-site by the Contractor shall be properly identified within 30 days of generation. No regulated wastes shall be allowed to accumulate on-site for more than 90 days. If the Contractor discovers or comes into contact with any hazardous chemicals or other materials other than those addressed in this section, the Contractor shall immediately notify the Contracting Officer (CO) who will make a determination as to the course of action.

### 3.6 DISPOSAL OF NON-REGULATED DEBRIS AND OTHER WASTE

All debris and other wastes resulting from construction operations on this contract shall be disposed of by removal from the site.

### 3.7 MAINTENANCE OF POLLUTION CONTROL FACILITIES

During the life of this contract the Contractor shall maintain all facilities constructed for pollution control under this contract as long as the operations creating the particular pollutant are being carried out or until the material concerned has become stabilized to the extent that pollution is no longer being created. Early in the construction period the Contractor shall conduct a training course that will emphasize all phases of environmental protection.

### 3.8 REPORTING OF POLLUTION SPILLS

In the event that an oil spill or chemical release occurs during the performance of this contract, the Contractor is required to contact the National Response Center, telephone number 1-800-424-8802 as soon as possible, or if telephone communication is not possible, the nearest U.S. Coast Guard office may be contacted by radio to report the spill, (33 CFR 153.203). The Contractor shall comply with any instructions from the responding agency concerning containment and/or cleanup of the spill.

### 3.9 TRAINING OF CONTRACTOR PERSONNEL

Train the personnel in environmental protection and pollution control. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda:

- a. methods of detecting and avoiding pollution
- b. familiarization with statutory and contractual pollution standards
- c. installation and care of devices, vegetation covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control

d. anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants

e. recognition and protection of archaeological sites, artifacts, and wetlands

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 03.00 10

CLOSEOUT SUBMITTALS - DREDGING

**06/2022**

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

1.2 SUBMITTALS

1.3 PROJECT RECORD DOCUMENTS

1.3.1 "As-Built" Drawings

1.3.1.1 Government Furnished Materials

1.3.1.2 Working "As-Built" and Final "As-Built" Drawings

1.3.1.3 Drawing Preparation

1.3.1.4 Qualifications of CADD personnel

1.3.1.5 Computer Aided Design and Drafting (CADD) Drawings

1.3.1.6 Final "As-Built" Drawings

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

## SECTION 01 78 03.00 10

CLOSEOUT SUBMITTALS - DREDGING  
06/2022

## PART 1 GENERAL

## 1.1 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for providing Closeout Submittals, including "As-Built" drawings required under this section. All costs associated therewith shall be included in the applicable contract unit or job prices contained in the Bidding Schedule.

## 1.2 SUBMITTALS

Government approval is required for the submittal specified herein. The designation following the "G" identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

"As-Built" Drawings; G, EDC

The Contractor shall submit full-scale drawings showing final "As-Built" conditions of the project. The final (red-lined) "As-Built" drawings shall consist of 3 sets (an original and two color copies) and 3 electronic (.PDF) copies on CD-R of the approved working as-builts.

## 1.3 PROJECT RECORD DOCUMENTS

## 1.3.1 "As-Built" Drawings

This paragraph covers "As-Built" drawings complete, as a requirement of the contract. The terms "drawings," "contract drawings," "drawing files," "working "As-Built" drawings" and "final as-built drawings" refer to contract drawings which are revised to be used for final "As-Built" drawings.

## 1.3.1.1 Government Furnished Materials

The Contractor will be furnished "as-designed" drawings in Bentley OpenRoads Designer format compatible with a Windows XP operating system. The electronic files will be supplied on compact disc, read-only memory (CD-ROM). The plan drawings provided shall be the basis for resulting "As-Built" sheet scale, coverage and layout. If the Contractor submits a CADD project master overview file, it does not eliminate the need for the specific individual plates matching the scale and coverage as those provided in the Government furnished CD-ROM.

## 1.3.1.2 Working "As-Built" and Final "As-Built" Drawings

The Contractor shall revise 4 full size 22"x34" sets of paper drawings by

red-line process to show the "As-Built" conditions during the prosecution of the project. These working "As-Built" marked drawings shall be kept current on a weekly basis and at least one set shall be available on the jobsite at all times. Changes from the contract plans which are made in the work or additional information which might be uncovered in the course of construction shall be accurately and neatly recorded as they occur by means of details and notes. The working "As-Built" marked prints and final "As-Built" drawings will be jointly reviewed for accuracy and completeness by the Contracting Officer and the Contractor prior to submission of each monthly pay estimate. If the Contractor fails to maintain the working and final "As-Built" drawings as specified herein, the Contracting Officer will deduct from the monthly progress payment an amount representing the estimated cost of maintaining the "As-Built" drawings. This monthly deduction will continue until an agreement can be reached between the Contracting Officer and the Contractor regarding the accuracy and completeness of updated drawings. The working and final "As-Built" drawings shall show, but shall not be limited to, the following information:

- a. All information shown on the contract drawings and a record of all completed work, deviations, modifications, or changes from those drawings, however minor, which may have been incorporated into the work.
- b. After dredging channel sections and acceptance profiles.
- c. Gross and net yards dredged distinguished by dates of dredging and reach of channel dredged.
- d. Disposal locations used, including gross yardage placed by site.
- e. Period of disposal placement for each disposal site utilized.
- f. Vertical control utilized including any applicable conversions.
- g. Utility locations as verified by owners, including station, C/L XY-coordinate, and minimum elevation.
- h. Channel markers, buoys, daymarkers, etc. surveyed during required preliminary survey efforts.
- i. Changes or modifications which result from the final inspection.
- j. Modifications shall be shown in accordance with the following procedures.
  - (1) Directions in the modification for posting descriptive changes shall be followed.
  - (2) A Modification Triangle shall be placed at the location of each deletion.
  - (3) For new details or sections which are added to a drawing, a Modification Triangle shall be placed by the detail or section title.
  - (4) For minor changes, a Modification Triangle shall be placed by the area changed on the drawing (each location).
  - (5) For major changes to a drawing, a Modification Triangle shall be placed by the title of the affected plan, section, or detail at

each location.

(6) The Modification Triangle size shall be 1/2 inch on a side unless the area where the circle is to be placed is crowded. Smaller size circle shall be used for crowded areas.

#### 1.3.1.3 Drawing Preparation

a. Each various feature of work performed shall be distinguishable by color coding and/or symbology. The As-Built drawings shall be modified as may be necessary to correctly show the features of the project as it has been constructed by bringing the contract set into agreement with approved working as-built prints, and adding such additional drawings as may be necessary. These working as-built marked prints shall be neat, legible and accurate. These drawings are part of the permanent records of this project and shall be returned to the Contracting Officer after approval by the Government. Any drawings damaged or lost by the Contractor shall be satisfactorily replaced by the Contractor at no expense to the Government.

b. Additions and corrections to the contract drawings shall be equal in quality and detail to that of the originals. Line colors, line weights, lettering, layering conventions, and symbols shall be the same as the original line colors, line weights, lettering, layering conventions, and symbols. If additional drawings are required, they shall be prepared using the specified electronic file format applying the same graphic standards specified for original drawings. The title block and drawing border to be used for any new final as-built drawings shall be identical to that used on the contract drawings. The Contractor is allowed to use a red line markup system to produce hard copy A-Built. This will require scanning of the documents to create .pdf drawings for submittal. If additions and corrections to the contract drawings are accomplished using CADD files, the Contractor shall be responsible for providing all program files and hardware necessary to prepare final As-Built drawings. The Contracting Officer will review final As-Built drawings for accuracy and the Contractor shall make required corrections, changes, additions, and deletions.

#### 1.3.1.4 Qualifications of CADD personnel

Only personnel proficient in the preparation of CADD drawings shall be employed to modify the contract drawings if required, or prepare additional new drawings.

#### 1.3.1.5 Computer Aided Design and Drafting (CADD) Drawings

a. CADD colors shall be the "base" colors of red, green, and blue. Color code for changes shall be as follows.

(1) Deletions (red) - Deleted graphic items (lines) shall be colored red with red lettering in notes and leaders

(2) Additions (Green) - Added items shall be drawn in green with green lettering in notes and leaders

(3) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes shall be in blue.

b. The Contract Drawing files shall be in Microstation (DGN) format, properly geo-referenced, and renamed in a manner related to the contract

number (i.e., 98-C-10.DGN) as instructed in the Pre-Construction conference. Marked-up changes shall be made only to those renamed files. All changes shall be made on the layer/level as the original item. There shall be no deletions of existing lines; existing lines shall be over struck in red. Additions shall be in green with line weights the same as the drawing. Special notes shall be in blue on layer "?-ANNO-NOTE" where "?" represents the discipline designator (for example; "C" for Civil, "E" for electrical, "S" for Structural, etc.).

c. Within twenty (20) days after Government approval of all of the working as-built drawings, the Contractor shall prepare the final CADD as-built drawings and submit two (2) sets of hard-copy prints of these drawings for Government review and approval. The Government will promptly return one (1) set of prints annotated with any necessary corrections. Within 10 days the Contractor shall revise the CADD files accordingly at no additional cost and submit the final as-built drawing package for the entire project. The submittal shall consist of one (1) set of electronic files on 4 3/4" compact disc, read-only memory (CD-ROM), three (3) sets of hard-copy prints and one (1) set of the approved working as-built drawings. They shall be complete in all details and identical in form and function to the contract drawing files supplied by the Government. Any transactions or an adjustment necessary to accomplish this is the responsibility of the Contractor. The Government reserves the right to reject any drawing files it deems incompatible with the customer's CADD system.

#### 1.3.1.6 Final "As-Built" Drawings

a. When final revisions have been completed, the cover sheet drawing shall show the wording "RECORD DRAWING AS-BUILT" followed by the name of the Contractor in letters at least 3/16 inch high. All other contract drawings shall be marked either "AS-Built" drawing denoting no revisions on the sheet or "Revised As-Built" denoting one or more revisions. Original contract drawings shall be dated in the revision block.

b. In lieu of revising contract cross section drawings to depict before and after conditions, the Contractor is allowed to submit electronically plotted cross sections depicting before and after dredging cross sections overlayed with the required dredge template. The submission shall be in pdf format, such that each individual cross section is paged and scaled to 8.5" x 11" paper.

c. Paper prints, drawing files and storage media submitted will become the property of the Government upon final approval. Failure to submit final "As-Built" drawing files and marked prints as specified shall be cause for withholding any payment due the Contractor under this contract. Approval and acceptance of final "As-Built" drawings shall be accomplished before final payment is made to the Contractor.

d. The Contractor shall submit a Utility Summary Sheet listing every utility encountered for both the dredging channel and disposal areas. For every utility, the following verified information shall be provided: C/L station, C/L XY-coordinate, right descending bank toe XY-coordinate, left descending bank toe XY-coordinate, minimum elevation, all horizontal information required to determine actual alignments of all utilities, utility size, utility type, and point of contact information. The Contractor shall also submit a copy of all utility as-builts, surveys, etc. received from the owner. All utility as-builts shall be labeled such that each asbuilt has the current owner's name and pipeline location (C/L Station and Mile #) on it."



PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

Dredging Region: \_\_\_\_\_

Date submitted: \_\_\_\_\_

Note: The Contractor shall fill out one sheet for each dredging region.

| Utility Summary Sheet |                        |                            |                            |                      |                    |                    |                                                                    |            |                                                                                |                                                                                                                    |
|-----------------------|------------------------|----------------------------|----------------------------|----------------------|--------------------|--------------------|--------------------------------------------------------------------|------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| C/L Station/Mile #    | C/L<br>X, Y Coordinate | RDB Toe<br>X, Y Coordinate | LDB Toe<br>X, Y Coordinate | Minimum<br>Elevation | Size of<br>Utility | Type of<br>Utility | Is utility line properly<br>marked (yes/no)? If no,<br>reason why. | Owner Name | Owner Point of Contacts<br>(email address, phone number, and physical address) | Notes:<br>(e.g. - was line dredged over, disposed on, skipped (discuss reason why), any issues with<br>owner, etc) |
|                       |                        |                            |                            |                      |                    |                    |                                                                    |            |                                                                                |                                                                                                                    |
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|                       |                        |                            |                            |                      |                    |                    |                                                                    |            |                                                                                |                                                                                                                    |

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 21 10.00 12

SURVEYS

03/14

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 MEASUREMENT AND PAYMENT
- 1.3 SUBMITTALS
- 1.4 QUALITY CONTROL
- 1.5 PROFESSIONAL CERTIFICATION
- 1.6 SURVEY EQUIPMENT
  - 1.6.1 Fathometer
    - 1.6.1.1 General
    - 1.6.1.2 Calibration Checks
    - 1.6.1.3 Low Frequency Signal Parameters
  - 1.6.2 Lead Line/Level Rod
  - 1.6.3 Positioning Equipment
    - 1.6.3.1 Electronic Surveys
    - 1.6.3.2 Data
  - 1.6.4 Survey Boat
  - 1.6.5 Automatic Recording Tide Gage
- 1.7 REFERENCES

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 COMMENCEMENT, PROSECUTION, AND COMPLETION
  - 3.1.1 General
  - 3.1.2 Positioning Surveys
  - 3.1.3 Preliminary Field Surveys
  - 3.1.4 "Before and After" Dredging Cross Sections
  - 3.1.5 "Before and After" Disposal Area Surveys
  - 3.1.6 Acceptance Profiles
  - 3.1.7 Survey Data

ATTACHMENTS:

daily to the Contracting Officer's Representative (See paragraph  
COMMENCEMENT, PROSECUTION, AND COMPLETION, subparagraph "General").

-- End of Section Table of Contents --

## SECTION 02 21 10.00 12

## SURVEYS

03/14

## PART 1 GENERAL

## 1.1 SCOPE

The work provided for herein consists of furnishing all plant, labor, equipment, and materials, and performing all operations necessary for surveying as specified herein and as indicated on the contract drawings.

## 1.2 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for surveys required by this section. The cost of survey work will be included in the contract unit price per cubic yard for "Dredging".

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

## SD-02 Shop Drawings

Preliminary Field Surveys; G, CD

Preliminary Field Surveys; G, EDC

Prior to taking any original cross-sections, the Contractor shall run a preliminary electronically controlled field survey of the base dredging reach. (See paragraph "Preliminary Field Surveys")

"Before and After" Dredging Cross Sections; G, CD

"Before and After" Dredging Cross Sections; G, EDC

"Before and after" dredging cross sections of the Tiger Pass Bar channel shall be taken and submitted (See paragraph "Before and After" Dredging Cross Sections").

## Disposal Area Surveys

"Before and after" surveys shall be taken at each wetland creation area used and/or anticipated to be used by the Contractor to adequately capture impacts of material disposal (See paragraph "Before and After" Disposal Area Survey").

After dredging profiles of the Tiger Pass Bar Channel shall be taken and submitted (See paragraph "Acceptance Profiles").

## SD-05 Design Data

## Compiled Survey Data CD Disk

All survey data shall be collected, recorded and processed. The Contractor shall furnish the required data on compact disk containing ASCII character set (See paragraph "Data").

## Acceptance Profiles; G

## SD-06 Test Reports

## Calibration Checks; G

## SD-09 Manufacturer's Field Reports

## Survey Equipment; G, CD

## Verification of Survey Equipment

Calibration of the Fathometer by a method approved by the Contracting Officer

Calibration checks shall be run at the beginning and completion of each day's survey or portion thereof. Results shall be furnished daily to the Contracting Officer's Representative (See paragraph COMMENCEMENT, PROSECUTION, AND COMPLETION, subparagraph "General").

## 1.4 QUALITY CONTROL

The Contractor shall establish and maintain quality control for the work specified in this section to assure compliance with the contract requirements and maintain records of his quality control for all surveying including but not limited to the following:

(1) Layouts and Surveys . Daily log of layout and surveys consisting of type and location using Contractor furnished control system as stated in the Contract Clause in Section 00700, entitled LAYOUT OF WORK (FAR 52.236-17).

(2) Reporting. The original and two copies of these reports and tests as well as the records of corrective action taken shall be furnished the Government daily. The report shall include a record of times and dates surveys were run, the horizontal control stations used and their coordinates, the vertical control points used and their elevations, and weather conditions. Format of report shall be as prescribed in Section 01 45 04.00 10 CONTRACTOR QUALITY CONTROL.

## 1.5 PROFESSIONAL CERTIFICATION

All surveys shall be performed in the presence of the Contracting Officer's Representative. All surveys shall be performed under the direction of and certified by a Professional Land Surveyor currently licensed by the state in which the work is being performed. In addition, the surveyor shall document at least three years of experience in hydrographic surveying of navigable channels and possess a current hydrographic certification from The Hydrographic Society of America (THSOA).

## 1.6 SURVEY EQUIPMENT

### 1.6.1 Fathometer

#### 1.6.1.1 General

All survey soundings shall be taken with a precision, high-resolution (0.1-ft) fathometer. The acoustic ping rate shall be, at minimum, 10-soundings per second. To limit signal attenuation in areas of unconsolidated bottom sediments, the acoustic frequency utilized shall be within the range of 24-28 kHz and low frequency low pulse width shall be in accordance with the manufacturers recommended settings for unconsolidated bottom sediments in shallow water depths (i.e., 0 to 150-ft). If a dual frequency fathometer is utilized, the low frequency shall be within the 24-28 kHz range. The equipment shall be capable of recording depths of water ranging from 3-feet under transducers to a maximum of 100-feet. Water depths shall be indicated on a vertical scale of not less than 0.1-in = 1-ft. The accuracy shall be one half of one percent from 3 to 100-foot. A manual control shall be provided to permit offsetting the electrical zero over a range not less than minus 4 to plus 22-feet to allow for draft or water level compensation. The acoustic frequency shall not be higher than 28 kHz nor lower than 8 kHz to permit penetration to the true bottom in highly sedimentated waters. The number of soundings shall be not less than 534 per minute. The Contractor shall provide a printout or electronic file of the data produced by the electronic fathometer. Chart paper speeds of 1, 2, 3, and 4-inches per minute shall be provided. Indication of stylus motor speed shall be provided by means of a panel mounted frequency meter, calibration line, or other accurate indicator of stylus speed covering the range of 57 to 63 Hz. a means of smoothly varying stylus speed to compensate for water temperature and salinity shall be provided in order that accurate calibration may be effected. There shall be supplied a fix marker switch electrically interlocked into the electronic horizontal control system as specified in paragraph "Positioning Equipment", capable of being operated to accurately mark the positions of the soundings.

#### 1.6.1.2 Calibration Checks

Calibration of the Fathometer by a method approved by the Contracting Officer shall be made daily at depths of 5-feet, 10-feet, and 15-feet, upon commencement and completion of surveys, or more often if required by the Contracting Officer's Representative. All calibration and soundings shall be made in the presence of the Contracting Officer's Representative at the job site.

#### 1.6.1.3 Low Frequency Signal Parameters

The low frequency pulse width range of cycles (i.e. 1 to 10) used shall not produce signal lengths less than 35 microseconds or greater than 125 microseconds. For example, using an Odom MKIII 24 kHz transducer with a LF Pulse Width Setting of 2, will result in a 83.3 microsecond long signal. 2 cycles (i.e. LF Pulse Width Setting =  $2/24,000 \text{ Hz} = 83.3 \text{ microseconds}$ ).

### 1.6.2 Lead Line/Level Rod

Disposal area surveys may require the use of convention survey means to obtain required data. Lead lines used for soundings shall consist of a twelve (12)-pound lead having a six (6)-inch mushroom shaped bottom attached to a low stretch line graduated in 0.1-foot increments. Level rods, if used for soundings, shall have a six (6)-inch diameter plate

attached to the bottom. Methodology to obtain additional conventional survey requirements shall be approved by the Contracting Officer's Representative. If a total station instrument is used to perform the survey, the Contractor shall submit a copy of the survey data on magnetic media, a printout of the survey data, and computer-plotted profiles and cross sections from the original data. Plotted cross sections and profiles plus duplicate notes shall be kept at the jobsite at all times and made available to the Government.

#### 1.6.3 Positioning Equipment

##### 1.6.3.1 Electronic Surveys

Positioning equipment for electronic surveys shall be capable of achieving the required accuracy stated in paragraph "Positioning Surveys". Initial calibration and subsequent checks shall be in accordance with the manufacturer's instructions as required in paragraph "General".

##### 1.6.3.2 Data

The data obtained from the electronic control system (paragraph "Positioning Surveys") and the echo depth sounding instrument shall be collected, recorded and processed. Soundings shall be taken to the nearest 0.1-foot and X & Y coordinates of the location should be to the nearest foot. The applicable gage shall be read to the nearest 0.1 foot prior to and after each day's survey. More frequent readings will be required when directed by the Government Inspector. From the processed data, the Contractor shall provide certain survey information taken for the purpose of computing the total amount of work to be paid for. The Contractor shall furnish the required data on 3-1/2 inch IBM compatible diskettes containing ASCII character set. The information received should be free of errors and in the following format: (a) The X-coordinate in feet of a recognized Louisiana Lambert grid system, (b) the Y-coordinate in feet of same recognized Louisiana grid system, (c) the sounding, and (d) remarks such as the station number (i.e. 1+00), CL (for centerline), ES (for end of station), EJ (for end of file), and direction of survey (either "L" for + or "R" for -, followed by Station No., i.e. L 30+00). The time and date the cross section was taken and the gage reading applicable to the section shall also be included. The electronically recorded data files, along with a hard copy of the track plotter surveys, plotted on the chart to show compliance with alignment, shall be presented to the Government Representative on site no later than 2 days after the survey is taken. Under no circumstances shall the information be edited for the purpose of eliminating incorrect soundings. The Contractor shall provide a separate file listing all incorrect soundings to be eliminated. Additional format requirements for these files will be discussed, and sample formatted files will be made available, at the pre-work coordination meeting. The beginning station for each file shall be the repeated ending station from the previous file (i.e., Sta. 820+00 to 840+00, Sta. 840+00 to 860+00, etc.). The format for survey file(s) to be submitted to the Government will be discussed and sample sheets will be given at the Pework Coordination Meeting.

##### 1.6.4 Survey Boat

The Contractor shall furnish diesel or gasoline-powered boat(s), twin screw, not less than 30 feet in length, enclosed cabin space for six (6) passengers, air conditioned and heated, fully operated, Coast Guard certified, radar equipped, and an operable radio (same frequency as

dredge). The survey boat(s) shall be equipped with electronic positioning equipment for conducting channel surveys and a surveying lead line as described in EM 1110-2-1003, Chapter 8-2, and shall be capable of maneuvering to permit surveying within the tolerances specified in paragraph COMMENCEMENT, PROSECUTION, AND COMPLETION. The survey boat(s) will be used solely for surveying and not providing the inspector's (Government agent's) transportation as stated in Section 35 20 23.00 12 DREDGING, paragraph entitled "Inspectors (Government Agents) Transportation" unless in the case of mechanical malfunction of Government transportation. No smoking shall be allowed inside the cabin and operator's space while occupied by Government personnel. No smoking signs shall be posted.

#### 1.6.5 Automatic Recording Tide Gage

The Contractor shall furnish a Hazen or compatible remote, automatic recording tide gage that may be monitored aboard the dredge. The Auto Gage shall be initially calibrated to the staff gage and once a week the staff gage and auto gage shall be calibrated through the completion of the job. The calibration back up data shall be submitted to the Government Inspector for daily reports.

#### 1.7 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

United States Army Corps of Engineer

EM 1110-2-1003

2013 Hydrographic Surveying

#### PART 2 PRODUCTS

Not used.

#### PART 3 EXECUTION

##### 3.1 COMMENCEMENT, PROSECUTION, AND COMPLETION

###### 3.1.1 General

Surveys for all fixed stations shall be performed in accordance with Class 1 Third Order accuracy as defined by the National Oceanic and Atmospheric administration in "Classification, Standards of Accuracy, and General Specifications of Geodetic control Surveys". Prior to initiating positioning surveys using electronic survey equipment, a calibration test of the electronic measuring device shall be performed according to the manufacturer's instructions in the presence of the Contracting Officer's Representative. Calibration checks shall be run at the beginning and completion of each day's survey or portion thereof. Results shall be furnished daily to the Contracting Officer's Representative. All electronic surveys shall be tied to a fixed station. The station may be a Government furnished point or a temporary point established by the Contractor and approved by the Contracting Officer's Representative. Should the electronic measuring device fail to indicate the known distance within the factory defined error range for the device, the device shall not be used for determining survey positions.



### 3.1.2 Positioning Surveys

Positioning surveys shall be controlled using electronic surveying equipment. Information and data to reestablish the Government baseline, or to establish the relationship between the electronic control system and the Government baseline and the channel centerline will be furnished by the Government upon request. The electronic positioning fixed stations (antenna locations) shall be located at points that will give the greatest amount of accuracy to the dredge and survey boats using the equipment. The coordinates of the fixed stations shall be tied to the Government baseline and shall be made available to the Government upon request. A positional tolerance of +/-5-feet will be permitted for surveys performed using electronic positioning equipment installed on the dredge and/or survey boats.

### 3.1.3 Preliminary Field Surveys

Prior to taking any original cross-sections, the Contractor shall run an electronically controlled preliminary field survey(s) locating the position of all day markers, buoys, and pipeline signs/markers, within the limits of the work. In addition, controlled cross sections of the channel (Stations 330+00 to 603+00: 300-feet wide or TBK to TBK, whichever comes first; and Stations 603+00 to 740+00: 600-feet wide, or WEDS to WES in the jetty reach, whichever comes first) shall be taken approximately every 2,000-feet along the channel centerline. These surveys shall be performed in accordance with the requirements stated in the following paragraph.

The data from this survey shall be plotted on a layout sheet (Track Plotter Chart) showing the ranges plotted in plan view, along with all day marker, buoy, and pipeline locations. All data obtained in this survey shall also be furnished in ASCII format. The cross sections shall also be plotted with the theoretical dredging template superimposed upon the cross sections. The survey shall be submitted to the Contracting Officer for determination of the final channel alignment. The Contractor shall allow five (5) calendar days for determination/verification of the final channel alignment by the Contracting Officer. The Contractor shall not proceed with any dredging work until the Contracting Officer determines the final channel alignment and furnishes this information to the Contractor. Towers, buoys, channel marker locations, pipeline sign.markers, pipeline alignments across the channel, and pipeline ownership/contact information shall be noted on the required "As-Built" Drawings.

### 3.1.4 "Before and After" Dredging Cross Sections

"Before and After" dredging cross sections of the channel for measurement of work shall be taken at approximately 200-foot intervals along the channel and at all P.I.'s, P.C.'s and P.T.'s. The soundings shall be made with a depth-sounding instrument described in paragraph "Fathometer". Distances between soundings on each cross section shall not exceed 5 feet on an azimuth normal to the centerline of the cut. A tolerance of +/- 20 feet normal to and between the designated station and the actual survey line will be permitted. Data may be transmitted from the depth sounding equipment and recorded directly onto magnetic media, or picked from the depth sounding charts. If soundings are obtained from the depth sounder rolls, they shall be picked from the top of the true bottom line. Reflected lines (fluff lines) from lighter sedimentary materials shall be disregarded. Soundings shall be chosen to the nearest 0.1-feet. In areas of consolidated bottom material, the digitized and recorded depth soundings shall indicate the true channel bottom. However, in survey areas containing

unconsolidated bottom materials, acoustic measurements must be augmented with lead line measurements. These lead line measurements must be notated on the fathometer paper chart. Regardless of which data collection method is used, the depth sounder rolls shall be furnished to the Government. Data shall also be furnished to the Government in a structured IBM compatible ASCII format on magnetic media as specified in paragraph "Data". A layout of the before dredging cross sections shall be prepared and plotted on a layout sheet (Track Plotter Chart) showing the ranges plotted in plan view. In addition, the cross section information shall also be plotted on a printout sheet, 1 inch = 100 feet horizontal and 1 inch = 5 feet vertical scales, along with a section showing the required theoretical section. The time and date taken shall be shown and the gage reading applicable to the survey shall be shown. The applicable gage shall be read to the nearest 0.1-foot prior to and after each day's survey. More frequent readings may be required when directed by the Contracting Officer's Representative.

#### 3.1.5 "Before and After" Disposal Area Surveys

"Before and after" disposal area cross-sections shall be taken and plotted for each disposal areas selected for use by the Contractor. Cross-sections shall extend over the anticipated limits of disposal at 500-foot intervals centered on the discharge location. As a minimum, three (3) cross sections shall be taken in and east-west direction, and three (3) sections in the north-south direction. All cross-sections shall extend at least 1000 feet seaward of the discharge location. Soundings and/or rod readings shall be taken every 20 feet along the cross-section. Readings shall be adjusted to the nearest 0.1-foot. Data shall be furnished to the Government in a structured XYZ ASCII format on magnetic media, regardless of which data collection or survey method is used.

#### 3.1.6 Acceptance Profiles

Upon completion of dredging in an acceptance reach, as specified in Section 35 20 23.00 12 DREDGING, paragraph "Acceptance Profiles" profiles shall be taken over the same acceptance reach. The depth sounding instrument used to make the before dredging cross sections shall be used to make the after dredging profiles. No substitutions will be permitted unless approved by the Contracting Officer. Distances between soundings on each profile shall not exceed 20-feet. Controlled profiles of the channel shall be made along the channel centerline and one at each outside bottom edge of the cut. For channels greater than 300-feet wide, two additional profiles shall be made and they shall be located midway between the centerline of the channel and the bottom edge of the cut. The Government reserves the right to direct additional soundings to be taken along ranges normal to the above defined acceptance profiles.

#### 3.1.7 Survey Data

Each field book shall be given an identification number, which shall be noted on each page of the book. The information shown in the field notes shall meet the requirements of the Contracting Officer's Representative. The Contractor shall promptly plot the before dredging cross sections on a scale of 1-inch = 30-feet horizontally and 1-inch = 5-feet vertically. Plotted cross sections, profiles, and duplicate notes shall be kept at the jobsite at all times and made available to the Contracting Officer's Representative as required. The Contractor shall furnish to the Contracting Officer's Representative the plotted original cross sections for a minimum distance of 2000-feet ahead of dredging.

-- End of Section --

## SECTION TABLE OF CONTENTS

## DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

## SECTION 35 20 23.00 12

## DREDGING

**10/07**

## PART 1 GENERAL

- 1.1 SCOPE
- 1.2 MEASUREMENT
  - 1.2.1 Quantity Surveys
  - 1.2.2 Acceptance Profiles
  - 1.2.3 Quantity Computations
  - 1.2.4 Acceptance Reach
  - 1.2.5 Timely Performance Of Quantity Surveys
  - 1.2.6 Progress Payments
  - 1.2.7 Radio And Telephone Communication Equipment
- 1.3 PAYMENT
  - 1.3.1 Mobilization And Demobilization
  - 1.3.2 Dredging
  - 1.3.3 Radio and Telephone Communication Equipment
- 1.4 SUBMITTALS
- 1.5 QUALITY CONTROL
  - 1.5.1 Quality Control Reports
  - 1.5.2 Report Of Operations
  - 1.5.3 Access Plan
  - 1.5.4 Construction Schedule
- 1.6 RADIO AND TELEPHONE COMMUNICATIONS
  - 1.6.1 Maritime Radio Transceiver
  - 1.6.2 Radio Equipment For Additional Dredges
  - 1.6.3 Facsimile and Cellular Telephone
  - 1.6.4 High Speed Internet Service
  - 1.6.5 Automatic Identification System (AIS)
- 1.7 PLANT
  - 1.7.1 General Requirements
  - 1.7.2 Plant Data Sheet Submittal
  - 1.7.3 Capacity
  - 1.7.4 Inspectors (Government Agents) Transportation
- 1.8 REFERENCES

## PART 2 PRODUCTS

## PART 3 EXECUTION

- 3.1 DREDGING
  - 3.1.1 General
  - 3.1.2 Hydraulic Dredging
  - 3.1.3 Order Of Work
  - 3.1.4 Channel Dimensions
  - 3.1.5 Excessive Dredging
  - 3.1.6 Slides

- 3.1.7 Character Of Materials
- 3.2 WORKING IN THE VICINITY OF STRUCTURES AND UTILITY CROSSINGS
  - 3.2.1 General
  - 3.2.2 Required Dimensions
  - 3.2.3 Existing Pipelines, Structures Or Utilities
  - 3.2.4 Unidentified Pipelines, Structures, Or Utilities
  - 3.2.5 Utility Pipeline Marking Required
  - 3.2.6 Disposal
- 3.3 DISPOSAL OF DREDGED MATERIAL
  - 3.3.1 Disposal Areas
    - 3.3.1.1 Government Furnished Disposal Areas
    - 3.3.1.2 Access Corridors
    - 3.3.1.3 Wetland Development Areas
  - 3.3.2 Prevention Of Damage
  - 3.3.3 Excessive Discharge Pipe Leakage
  - 3.3.4 Effluent Sedimentation
  - 3.3.5 Deposition In Non-Approved Areas
  - 3.3.6 Protection Of Government Monuments, Markers, Or Towers
  - 3.3.7 Submerged Discharge Lines
  - 3.3.8 Work In The Vicinity Of Other Government Contractors
  - 3.3.9 Protection Of Bird Nesting Colonies

-- End of Section Table of Contents --

## SECTION 35 20 23.00 12

DREDGING  
10/07

## PART 1 GENERAL

## 1.1 SCOPE

The work provided for herein consists of furnishing all plant, labor, materials, and equipment and performing all operations necessary for the removal of all materials to the required dimensions; satisfactory disposal thereof, and all operations incidental thereto; as specified herein and as shown on the drawings.

## 1.2 MEASUREMENT

## 1.2.1 Quantity Surveys

A survey of the sites for dredging shall be made in accordance with the provisions of Section 02 21 10.00 12 SURVEYS and all measurements to determine dredging quantities will be based on the "before dredging" cross sections described therein. The soundings and contours shown on the contract drawings are representative of conditions that existed November 2023 and April 2024 and do not necessarily represent existing conditions. The elevations shown thereon shall be verified or corrected by the "before dredging" surveys. Determinations of quantities to be paid for in the area specified, after having once been made, will not be reopened.

## 1.2.2 Acceptance Profiles

Should the acceptance profiles and/or soundings described in Section 02 21 10.00 12 SURVEYS, paragraph "Acceptance Profiles", disclose any lack of the required dimensions, the Contractor shall redredge the area at no additional expense to the Government. After redredging, the required dimensions shall be verified by rerunning the profiles. Redredging shall be performed until the acceptance profiles and/or soundings show that the required dimensions have been obtained.

## 1.2.3 Quantity Computations

The quantity of dredged material removed and to be paid for will be measured by the cubic yard by computing the volume along the centerline between the "before-dredging" cross sections and the required dimensions as shown on the contract drawings using the average end area method. No allowance will be made for excessive dredging except as provided in paragraph "Excessive Dredging". Dredging quantities will be determined by the Government. In order for the Government to make dredging volume computations, the Contractor shall furnish the track plotter charts described in Section 02 21 10.00 12 SURVEYS, paragraph "Preliminary Field Surveys", and the "before dredging" data described in Section 02 21 10.00 12 SURVEYS, paragraph "Before and After" Dredging Cross Sections".

## 1.2.4 Acceptance Reach

For the purpose of acceptance, the completed work will be accepted in reaches of 2,000 linear feet along the channel centerline over the full

bottom width of the required dimensions specified on the drawings. The Contractor may be allowed to take "after dredging" cross sections as soon as a station is completed behind the dredge until the entire 2000-foot acceptance reach has been completed, subject to approval by the Contracting Officer's Representative. All after dredging cross-section data for that 2,000-foot acceptance reach shall be submitted as one file.

#### 1.2.5 Timely Performance Of Quantity Surveys

"Before-dredging" quantity surveys shall be made at least 2,000-feet in advance of dredging operations but not more than two weeks prior to commencement of work in the acceptance reach or channel reach to be covered by the survey unless an exception is granted by the Contracting Officer. Acceptance profiles shall be made within 2,000-feet behind the dredge but not more than two weeks after completion of work. Deviations from this limiting time element may be necessary because of unusual job conditions or adverse weather. Any deviation is subject to approval by the Contracting Officer. Before and after dredging survey data shall be plotted and submitted to the Government in 2,000 foot reaches within three (3) calendar days after the surveys are taken.

#### 1.2.6 Progress Payments

Monthly progress payments will be made based on quantities determined using "before dredging" surveys and the required dimensions, provided the acceptance profiles and any additional soundings that may be directed indicate that the Contractor has met the required dimensions and the acceptance reach length.

#### 1.2.7 Radio And Telephone Communication Equipment

No separate measurement will be made for furnishing and maintaining radio and telephone communication equipment.

### 1.3 PAYMENT

#### 1.3.1 Mobilization And Demobilization

An item to cover the cost of initial mobilization and final demobilization of the Contractor's plant and equipment under this contract is included and will be paid for as stipulated in the Contract Clause in Section 00700, entitled PAYMENT FOR MOBILIZATION AND DEMOBILIZATION (DFARS 252.236-7004).

#### 1.3.2 Dredging

Payment for dredging will be made at the contract unit price per cubic yard for "Dredging". Price and payment shall constitute full compensation for furnishing all plant, labor, materials and equipment required for dredging; including performing all surveying, any additional soundings that may be directed, excavation, disposal of materials, furnishing diesel powered crew boats fully operated, aluminum skiffs fully operated, pollution control, furnishing and maintaining radio and telephone communications equipment, and all operations incidental thereto; as specified herein and as shown on the drawings.

#### 1.3.3 Radio and Telephone Communication Equipment

No separate payment will be made for furnishing and maintaining radio and telephone communication equipment. All costs shall be distributed amongst

the existing Bidding Schedule items.

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

###### Dredge Plant Data

Plant Data Sheets. The Contractor shall complete the plant data sheets attached at the end of this section for the dredge and attendant plant the Contractor intends to use to perform the work under this contract. Dredge physical data is required for submittal in the data sheets. If any of the data submitted in the plant data sheets changes during the execution of the contract the Contractor shall submit new data to the Government within 48 hours thereafter, showing the changes made to the equipment, along with the date(s) the changes were made. The completed data sheets shall be submitted within 5 calendar days after Contractor receipt of Government Notice to Proceed.

###### Plan of Operation

Submit a Plan of Operation at each pipeline and utility crossing.

###### Access Plan; G, EDC

See paragraph QUALITY CONTROL, subparagraph "Access Plan" and the Section 01100 GENERAL PROVISIONS entitled ACCESS PLAN.

###### Construction Schedule; G, EDC

See paragraph QUALITY CONTROL, subparagraph "Construction Schedule" and Section 00700 paragraph SCHEDULES FOR CONSTRUCTION CONTRACTS (Far 52.236-15 - APR 1984).

###### Construction Schedule; G, CD

See paragraph QUALITY CONTROL, subparagraph "Construction Schedule" and Section 00700 paragraph SCHEDULES FOR CONSTRUCTION CONTRACTS (Far 52.236-15 - APR 1984).

###### Disposal of Dredged Material; G, CD

Submit a Disposal Of Dredging Material Plan in Government furnished disposal areas no later than 7 days prior to commencement of the work.

##### SD-07 Certificates

###### Certified Letters; G, EDC

Copy of Certified Letters to Pipelines and Utilities Owners



## SD-11 Closeout Submittals

Report of Operations; G, EDC  
See paragraph "Report of Operations".

## 1.5 QUALITY CONTROL

The Contractor shall establish and maintain quality control for dredging operations to assure compliance with contract requirements and maintain records of his/her quality control for all dredging operations including but not limited to the following:

- (1) Dredging. Visual classification of material; limits of dredging as to bottom grades and widths; side slopes; alignment of channel as to distance from baseline.
- (2) Access Corridors and Discharge Pipeline. Location, alignment, and dimensions of access corridors used and discharge pipeline locations and lengths.
- (3) Dredged Material Disposal. Limits of dredged material as deposited in a disposal area, and location. Surveillance and location of the dredge discharge outlet and distance from the channel centerline.
- (4) Discharge Effluent. Any shoaling or leak in pipeline.

## 1.5.1 Quality Control Reports

The original and two copies of these reports and tests, as well as the records of corrective action taken, shall be furnished the Government daily. Format of this report shall be as prescribed in Section 01 45 04.00 10 CONTRACTOR QUALITY CONTROL.

## 1.5.2 Report Of Operations

- (1) The Contractor shall prepare and submit a Report of Operations (MVNForm 4267) for each dredge working. This report shall be submitted on a daily basis and not in groups, (e.g., several daily reports packaged together at one time). A sample of ENG Form 4267 is included at the end of this section.
- (2) The Contractor shall also prepare a report of operations for each month or partial month's work on MVN Form 322 (Work Sheet for Preparing Consolidated Form 4267). The monthly report shall be submitted on or before the 7th of each month, consolidating the previous month's work. MVN Form 322 can be computer generated and shall be approved by the Contracting Officer's Representative on site. A sample of MVN Form 322 is attached at the end of this section.
- (3) The Contractor shall distribute one copy of each report to each of the following:
  - (a) U.S. Army Engineer District, New Orleans  
ATTN: Mr. Jeffrey Falati, CEMVN-CD-NO  
P.O. Box 60267  
New Orleans, LA 70160-0267
  - (b) Government Inspector

(4) One copy of each Report of Operations shall be maintained by the Contractor on the dredge(s).

(5) Further instructions on the preparation of the reports will be furnished at the Preconstruction Conference.

(6) Upon completion of work a comprehensive compilation of all prepared reports shall be supplied on CD ROM, grouped by report type, and organized chronologically to CEMVN-EDC, ATTN: Matthew Barbier. Submittals shall include one PDF file of all Form 4267 in chronological order with the Form 322(s) combined into one file.

#### 1.5.3 Access Plan

The Contractor shall submit an Access Plan to be reviewed and approved by the Contracting Officer. The requirement is specified and further detailed in Section 01100 GENERAL PROVISIONS, paragraph ACCESS PLAN.

#### 1.5.4 Construction Schedule

The Contractor shall submit a practicable schedule showing the order in which the Contractor proposes to perform the work, and the dates on which the Contractor contemplates starting and completing the several salient features of the work. The requirement is specified and further detailed in the Section 00700 Contract Clause entitled SCHEDULES FOR CONSTRUCTION CONTRACTS (FAR 52.236-15).

### 1.6 RADIO AND TELEPHONE COMMUNICATIONS

#### 1.6.1 Maritime Radio Transceiver

The Contractor shall furnish and maintain throughout the contract, one FM ship's radio transceiver with power not in excess of 25 watts, and at least 15 watts output on the maritime frequencies of 156.800 (Channel 16) and 156.375 (Channel 67) MHz 16F3 emission, with a tolerance of plus or minus 5 kHz deviation at 100 percent modulation for communication concerning navigation in the vicinity of the dredge. The radio shall be operated in accordance with FCC rules and regulations.

#### 1.6.2 Radio Equipment For Additional Dredges

In the event that the Contractor has more than one dredge operating simultaneously under this contract, the above-specified radio equipment shall be furnished and maintained on each dredge. The radio transceivers provided for in paragraph Maritime Radio Transceiver shall be continuously monitored by qualified Contractor personnel aboard each dredge.

#### 1.6.3 Facsimile and Cellular Telephone

The Contractor shall provide at least one cellular telephone and facsimile aboard the dredge. Final approval of the plant will not be made until this equipment is installed and in good working order. Facsimile and cellular phone service shall be available to Government personnel for conducting official Government business 24 hours per day, 7 days per week.

#### 1.6.4 High Speed Internet Service

The Contractor shall provide the Government inspector with high speed

internet service and associated equipment for the duration of the contract. The service and equipment must be capable of providing adequate connection to allow the inspector to import/export files through RMS and shall have a minimum download speed of 10 Mbps and a minimum upload speed of 1.5 Mbps. The Contractor must field verify that the service provider chosen has adequate continuous coverage at the construction site. The Contractor shall be responsible for the installation, maintenance, and monthly service fees necessary to provide continuous cellular telephone and high-speed internet service for the duration of the contract. Final approval of the plant will not be made until the equipment is installed and in good working order.

#### 1.6.5 Automatic Identification System (AIS)

The dredge shall be equipped with an Automatic Identification System (AIS) in accordance with the Code of Federal Regulations, 33 CFR 164.46. The system shall be operable at all times throughout the contract period. No separate payment will be made for AIS equipment. The Contractor shall include any and all costs for AIS equipment in the contract prices for items of work to which the equipment is incidental. For further information, go to:

<https://www.ecfr.gov/cgi-bin/text-idx?SID=ca7d648a1ef7e30fba8dda8d8c76425c&mc=true&node=>

#### 1.7 PLANT

##### 1.7.1 General Requirements

The Contractor shall keep on the job the necessary dredge equipment and attendant plant to meet the requirements of the work. The dredge equipment and attendant plant shall be in satisfactory operating condition and capable of safely and efficiently performing the work as set forth in specifications and shall be subject to inspection by the Contracting Officer's representative at all times.

##### 1.7.2 Plant Data Sheet Submittal

The Contractor shall complete the Dredge Plant Data sheets attached at the end of this section for the dredge and attendant plant the contractor intends to use to perform the work under this contract. Dredge physical data is required for submittal in the data sheets. If any of the data submitted in the plant data sheets changes during the execution of the contract the Contractor shall submit new data to the Government within 48 hours thereafter, showing the changes made to the equipment, along with the date(s) the changes were made. The completed data sheets shall be submitted within five (5) calendar days after Contractor receipt of the Government's Notice to Proceed. The dredge plant data shall be submitted to the ACO at the address listed in paragraph "Report of Operations".

##### 1.7.3 Capacity

No reduction in the capacity of the dredge equipment and attendant plant employed to execute the work shall be made except by written permission of the Contracting Officer. The measure of the "Capacity of the Dredge and Attendant Plant" shall be its actual performance on the work to which these specifications apply.

##### 1.7.4 Inspectors (Government Agents) Transportation

Crewboats. The Contractor shall furnish, throughout the contract period,

for the exclusive use of the Government, one diesel or gasoline powered crew boat, with twin propellers, capable of maintaining and traveling no less than the minimum speed of 25 miles per hour / 22 knots, not less than 30 feet in overall length, with enclosed space for at least six passengers after installation of all required equipment. The crew boat shall meet or exceed all applicable US Coast Guard regulations for vessels 65 feet or less in length, and shall be certified by the US Coast Guard. The crew boat shall be equipped with the following:

- (1) air conditioning and heating;
- (2) VHF radio for ship-to-ship communications, as described in paragraph RADIO AND TELEPHONE COMMUNICATIONS.
- (3) Navigation radar for travel at night and in inclement weather; and
- (4) Safety equipment meeting the inspection requirements of the US Coast Guard Auxiliary and EM 385-1-1.
- (5) Global Positioning System (GPS) Navigation system which constantly provides latitude and longitude position, compass heading, and speed.

The crew boat will be used solely for providing the Government Inspectors and Government Agents transportation, and not for conducting surveys, providing transportation for the Contractor's personnel, or transporting the Contractor's supplies. The Contractor shall make provisions to obtain a separate crew boat or vessel for conducting Contractor-related operations. The minimum full time crew for the crew boat shall be one operator. The crew boat shall be operated and available 24 hours a day. Refer to EM 385-1-1, Chapter 19-3 for additional crew and equipment requirements. Smoking shall not be allowed inside the cabin and operator's space while occupied by Government personnel. NO SMOKING signs shall be posted.

Skiff. Aluminum 18-foot aluminum skiff, fully operated, with a minimum 25 horsepower outboard motor.

Airboat. In addition to the above services, the Contractor shall provide the services of a fully operated airboat, capable of accommodating a minimum of four (4) passengers for the use of the Government to access the disposal sites.

## 1.8 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(2024) Safety and Health Requirements

## PART 2 PRODUCTS

Not Used.

## PART 3 EXECUTION

## 3.1 DREDGING

## 3.1.1 General

Dredging shall consist of the removal and satisfactory disposal of all material encountered to achieve the design section shown on the contract drawings and as specified herein.

## 3.1.2 Hydraulic Dredging

Hydraulic dredging operations shall be performed between Station 330+00 and Station 690+00. All work shall be performed as specified herein and as shown on the contract drawings.

The dredging limits were developed from channel hydrographic surveys available at the time of solicitation preparation, and may be adjusted based on Contractor's before dredging surveys. All work shall be performed as specified herein and as shown on the contract drawings.

## 3.1.3 Order Of Work

Due to excessive shoaling within reaches of Tiger Pass, dredging operations may commence from upstream to downstream. This will allow for Contractor and plant to take advantage of the deeper waters north of the upper limit and coming from the Mississippi River. The dredge plant must have the capability to enter Tiger Pass from both the Mississippi River and the Gulf Of Mexico. Dredging in protected areas shall be performed when in the Contractor's opinion weather conditions make offshore operations hazardous.

## 3.1.4 Channel Dimensions

The channel shall be dredged to the required dimensions shown on the contract drawings. A box cut will be permitted. The extent of required dredging and control of the box cut is illustrated in the Box Cut detail attached at the end of this section. The required dimension are as follows:

(a) (-)15' MLLW x 150 foot bottom width with 4H:1V side slopes from C/L Station 330+00 to C/L Station 603+00.

(b) (-)15' MLLW x 250 foot bottom width with 4H:1V side slopes from C/L Station 613+00 to C/L Station 690+00.

(c) A uniform dredge template transition, horizontally, is required between C/L Stations 603+00 and 613+00.

## 3.1.5 Excessive Dredging

Excessive dredging is defined to be any dredging which is in excess of the required dimensions, as shown on the drawings. Any material removed in excess of the required dimensions will not be measured for payment. The Contractor shall be responsible for damages caused by excessive dredging. Nothing herein shall be construed to prevent payment for removal of shoals performed in accordance with the applicable provisions of Section 01100 GENERAL PROVISION entitled SHOALING.

### 3.1.6 Slides

In the event sliding occurs in any part of any excavation after its completion but prior to its acceptance, the Contractor shall remove such portions of the slide as the Contracting Officer may direct. In the event, the slide is caused through the fault of the Contractor as determined by the Contracting Officer, the slide shall be removed by the Contractor at no additional expense to the Government. In the event the slide is not due to the fault of the Contractor, payment for removal will be made at the contract unit price for "Dredging". A box cut will be permitted. The extent of required dredging and control of the box cut is illustrated in the Box Cut detail attached at the end of this Section.

### 3.1.7 Character Of Materials

The material to be removed within the required dimensions specified consists of shoaling that has occurred since the channel was last dredged and insitu material. Sand, silt, clay, shell, logs, stumps, snags, debris, and other obstructions may be encountered. Bidders are expected to examine the site of the work and, after investigation, decide for themselves the character of the materials. The Contractor shall note the following dredging jobs previously performed.

| Limits             | Date<br>Dredged | Design<br>Elevation<br>Depth (ft-MLG) | Width<br>Dimensions |
|--------------------|-----------------|---------------------------------------|---------------------|
| C/L Sta. 348+30+00 | October 2021    | -12.0                                 | 150'                |
| to                 |                 |                                       |                     |
| C/L Sta. 517+09    | October 2021    | -12.0                                 | 150'                |

NOTE: The table above indicates the last maintenance dredging event(s) performed. Channel transitions horizontally between Stations 603+00 and 613+00. Contractor shall note that due to lengthy time since last dredging, the Contractor shall expect hard packed materials and debris. Should the Contractor encounter hard packed materials or debris, no claim shall be made against the Government.

## 3.2 WORKING IN THE VICINITY OF STRUCTURES AND UTILITY CROSSINGS

### 3.2.1 General

The Contractor shall exercise caution when working in the vicinity of structures and utilities adjacent to the channel or disposal areas and/or pipeline crossings in the channel or disposal areas. Repair of any damage resulting from excessive or improper excavation in the bottom or on the side slopes of the channel shall be the responsibility of the Contractor. Where dredging to obtain the required dimensions might endanger any structure, the Contracting Officer, upon request, may reduce the required excavation in the vicinity of such structure.

### 3.2.2 Required Dimensions

The Contractor shall provide the following dimensions over all utility crossings by whatever approved method the Contractor elects to use. The Contractor shall submit for approval by the Contracting Officer a detailed Plan of Operation at each pipeline or utility crossing where surveys

indicate the above dimensions do not exist. The plan shall contain emergency measures to be taken in the event of an accident. The contractor shall notify the owners of pipelines or utilities by certified mail at least 7 days prior to operating within 500-feet of a pipeline or utility referencing the following information:

- \* The anticipated date the dredge will work over pipeline/utility;
- \* The depth and width of the dredge cut over the pipeline/utility;
- \* Points of contact and telephone numbers;
- \* Any other information from the Contractor that the pipeline/utility owner may consider significant for safe dredging operations over the crossing.

A copy of the certified letters shall be submitted for information to the Contracting Officer concurrent with owner notification. The Contractor shall also provide transportation, meals and/or lodging for any pipeline/utility representatives who are deemed necessary to remain aboard the dredge while working in the vicinity of these crossings.

The Government will not be responsible for any damages to structures or pipelines/utilities due to the Contractor's deviation from the approved plan.

### 3.2.3 Existing Pipelines, Structures Or Utilities

The vertical clearances listed below were furnished by the utility owners at the time permits were issued and/or from "AS-BUILTS" and are not the result of a survey made by the Government. In addition, various pipeline databases available here at the New Orleans District, as well as nautical charts published by NOAA, have been used in verifying the existence and locations of utilities within the vicinity of the project. The Contractor contact "LA One Call" at either 811 or 800-272-3020 and GULFSafe at 1-888-910-4853 prior to dredging operations. Finally, prior to initiating dredging operations, the Contractor shall investigate the NOAA Charts OnLine Viewer to verify any chart and utility updates in the project vicinity. The specific chart relating to this project can be found at:

<http://www.charts.noaa.gov/OnLineViewer/11361.shtml>

The following pipelines, structures or utilities are located within the project area. Also, The Contractor should be aware that there are other pipelines located within the general area:

| Utility/Structure  | Approximate Station | Elevation MLLW | Name and Address of Owner                                                                                                           |
|--------------------|---------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 5 - 4.5" Flowlines | 318+33              | ---            | Hilcorp Energy-Harvest<br>(see below)                                                                                               |
| 24" Discharge Line | 321+05              | ---            | Hilcorp Energy-Harvest<br>(see below)                                                                                               |
| 10" P/L            | 423+01              | -25.6'         | Harvest (Old Chevron Line)<br>Mr. Martin Mistich<br>1804 Tidewater Rd<br>Venice, LA 70091<br>(281) 796-9327<br>mmistich@hilcorp.com |

|                                 |        |        |                                                                                                                                                                                      |
|---------------------------------|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |        |        | Alternate Contact<br>Shelby<br>(225) 250-6632                                                                                                                                        |
| 6" P/L                          | 446+94 | -28.0' | American Midstream<br>Robin Asevedo<br>2402 Bayou Road<br>St. Bernard, LA 70085<br>Office- (504) 681-6925<br>ext. 2028<br>Cell- (504) 812-7151<br>RAsevedo@american<br>midstream.com |
|                                 |        |        | Alternate Contact<br>Mike Barbier<br>(601) 569-9123                                                                                                                                  |
| 2.5" P/L                        | 450+36 | -23.6' | Hilcorp Energy-Harvest<br>Allan Pringle<br>1111 Travis Street<br>Houston, TX 77002<br>346-237-2232<br>apringle@hilcorp.com                                                           |
|                                 |        |        | William Riley<br>1111 Travis Street<br>Houston, TX 77002<br>713-209-2406                                                                                                             |
|                                 |        |        | Ronald DarDar<br>15211 Old Spanish Trl.<br>Paradis, LA 70080<br>Phone: 985-785-3050<br>Cell: 985-258-1421                                                                            |
|                                 |        |        | Martin Hargrave<br>141 Phillips Rd.<br>Port Sulphur, LA 70083<br>Phone: 504-354-5101<br>Cell: 985-759-2224                                                                           |
|                                 |        |        | Brent Tompkins<br>141 Phillips Rd<br>Port Sulfur, LA 70083<br>Phone: 504-354-5101<br>Cell: 504-289-4416                                                                              |
|                                 |        |        | James McMicheal<br>1804 Tide Water Rd.<br>Venice, LA 70091<br>Phone: 504-592-3991<br>Cell: 504-416-2615                                                                              |
| 10" P/L<br>(Abandoned in Place) | 483+94 | -26.5' | Gulf South Pipeline Co.<br>Harold Sylve<br>520 Alliance St.<br>Kenner, LA 70062                                                                                                      |



|                                                       |        |         |                                                                                                                                        |
|-------------------------------------------------------|--------|---------|----------------------------------------------------------------------------------------------------------------------------------------|
|                                                       |        |         | (504) 415-1864<br>(504) 469-5903, x-236<br>harold.sylve@bwpmlp.com                                                                     |
| Submerged Cable<br>(Abandoned in Place)               | 486+50 | -26.0'  | Entergy Louisiana<br>Mr. Jeremy Rich<br>3714 Tulane Ave.<br>L-TUL-113<br>New Orleans, LA 70119<br>(504) 595-3816<br>jrich1@entergy.com |
| 1-20" P/L                                             | 498+60 | -27.0'  | American Midstream<br>(see above)                                                                                                      |
| 26" P/L                                               | 499+35 | -25.0'  | American Midstream<br>(see above)                                                                                                      |
| 2" P/L                                                | 515+60 | Unknown | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| 3" P/L                                                | 517+25 | Unknown | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| 6" P/L & (15)-2.5"<br>Flowlines<br>(50' wide trench)  | 519+10 | -24.2'  | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| Possible P/L                                          | 520+00 | Unknown | Unknown                                                                                                                                |
| 4" P/L                                                | 528+01 | -23.5'  | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| 3" P/L                                                | 533+00 | Unknown | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| 12-Flowlines                                          | 533+34 | -25.2'  | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| (3)-3.5" & (3)-2.5"<br>Pipelines<br>(50' wide trench) | 536+94 | -26.5'  | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| Possible P/L                                          | 515+60 | Unknown | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| 8" P/L                                                | 550+62 | -28.9'  | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| 6" Oil & 12" Gas<br>Pipelines<br>(50' wide trench)    | 551+62 | -28.8'  | Hilcorp Energy-Harvest<br>(see above)                                                                                                  |
| 12" Gas P/L                                           | 615+26 | -29.4'  | Kinetica Partners, LLC<br>224 Aviation Road<br>Houma, LA 70363<br>(985) 879-3516<br>Jarus.LebLANC@kineticallc.com                      |

Shaun Harwood  
Shawn.Harwood@kineticallc.com

|          |        |        |                                       |
|----------|--------|--------|---------------------------------------|
| 2-3" P/L | 659+19 | -29.4' | Hilcorp Energy-Harvest<br>(see above) |
| 2.5" P/L | 661+24 | -29.4' | Hilcorp Energy-Harvest<br>(see above) |

Note: Elevations shown are the shallowest top of pipe elevation from within the channel template.

### 3.2.4 Unidentified Pipelines, Structures, Or Utilities

Any unidentified pipelines, structures, or utilities that may be found within the limits of the work, during the course of dredging shall not be disturbed nor shall excavation be performed at this location unless approved by the Contracting Officer.

### 3.2.5 Utility Pipeline Marking Required

If/when the dredge(s) is within 1,000 feet of the location of any utility or pipeline, the Contractor shall mark the established location of each utility or pipeline by placing one (1) Coast Guard approved yellow flashing buoy on each side of the channel to accurately mark the location of the pipeline or utility within easy view of the dredge leverman. The Contractor shall immediately remove such buoys upon completion of dredging in the area of the utility or pipeline.

### 3.2.6 Disposal

No dredged material shall be placed within 300 feet of any structure or utility.

## 3.3 DISPOSAL OF DREDGED MATERIAL

### 3.3.1 Disposal Areas

The dredged material shall be transported and deposited in the disposal areas provided by the Government as shown on the contract drawings and as specified herein. In no case, where open water disposal is used, shall the excavated material be discharged closer to the channel than the channelward limit of the disposal area as shown on the drawings. The Contractor shall verify that all disposal areas have adequate capacity. The use of marsh buggies or other wheeled or tracked vehicles shall not be permitted over vegetated marsh.

#### 3.3.1.1 Government Furnished Disposal Areas

The Contractor shall submit to the Contracting Officer for review and approval a plan for disposal of dredged material in Government furnished disposal areas a minimum of 7 days prior to commencement of the work. This is to ensure that the retention of material is maximized. This plan shall include dredge discharge locations.

#### 3.3.1.2 Access Corridors

Access corridors are available to the Contractor to access disposal areas.

The Contractor shall restrict all movement of pipe, marsh buggies, personnel, and other equipment to the access corridors. Access corridors may be field adjusted in the immediate area of the allowable stationing to avoid or minimize adverse impact to existing wetland vegetation, upon approval of the Contracting Officer's Representative. All access corridors and disposal areas shall be inspected for nesting birds prior to marsh buggy intrusion or discharge pipe placement. The Contractor shall take precautions to avoid/minimize damages to existing wetlands during access operations. If required, the Contractor is allowed floatation excavation through access corridor as long as existing marsh is not impacted by either digging through it or placing floatation channel material on it. No floatation dredging is allowed within 50 feet of any existing pipeline crossings. Any floatation dredging in the general vicinity of a pipeline must be coordinated with the utility owner. All costs associated with access corridors shall be included in the contract unit price per cubic yard for "Dredging".

### 3.3.1.3 Wetland Development Areas

Seven (7) wetland development sites, as described below, have been designated for unconfined disposal of dredged material. Material deposited in these areas shall be placed in accordance with the initial elevations specified. Dredged material slurry will be allowed to flow onto existing wetland vegetation. However, at no time shall the elevation of dredged material flowing through or onto existing wetland vegetation be allowed to exceed +1 feet NAVD 88. The Contractor shall establish and maintain a staff gage at each discharge location, and additional gages throughout each wetlands development area as needed, in order to monitor the elevation of material for compliance with the contract documents. The staff gage shall be marked at one-half foot intervals, from elevation +3.0 feet NAVD 88 to the existing water surface. The limits of disposal shall be marked by the Contractor with conspicuous buoys or stakes.

#### (1) Wetland Development Area #1 (WDA-1)

The first wetland development area is located along the right descending bank at approximate C/L Station 420+00, and is designated on the drawings as "UNCONFINED DISPOSAL AREA WDA-1". Material dredged shall be deposited as unconfined material into this designated site at an initial elevation of approximately +2.5-feet NAVD 88. In open water portions of the disposal area adjacent to established marshlands, dredged material discharge should be directed at or perpendicular to the existing marshlands.

#### (2) Wetland Development Area #2 (WDA-2)

The second wetland development area is also located on the right descending bank at approximate C/L Station 443+00, and is designated on the drawings as "UNCONFINED DISPOSAL AREA WDA-2". Material dredged shall be deposited as unconfined material into this designated site at an initial elevation of approximately +2.5 feet NAVD 88. In open water portions of the disposal area adjacent to established marshlands, dredged material discharge should be directed at or perpendicular to the existing marshlands.

#### (3) Wetland Development Area #3 (WDA-3)

The third wetland development area is also located on the right descending bank at approximate C/L Station 475+00, and is designated on

the drawings as "UNCONFINED DISPOSAL AREA WDA-3". Material dredged shall be deposited as unconfined material into this designated site at an initial elevation of approximately +2.5 feet NAVD 88. In open water portions of the disposal area adjacent to established marshlands, dredged material discharge should be directed at or perpendicular to the existing marshlands.

(4) Wetland Development Area #4 (WDA-4)

The fourth wetland development area is also located along the right descending bank at approximate C/L Station 588+20, and is designated on the drawings as "SINGLE POINT DISCHARGE (SPD) UNCONFINED DISPOSAL AREA WDA 4". Material dredged shall be deposited unconfined at the Single Point Discharge (SPD) location shown on the drawings. The dredged material shall be discharged in a manner such that the material is directed away from the channel. Dredged material in this area shall be deposited at an initial elevation of approximately +2.5 feet NAVD 88. Retaining dikes will not be required to prevent dredge material from entering Alphonse Bayou and/or the adjacent pipeline canal, nor will "before and after" disposal surveys of these water bodies be required.

(5) Wetland Development Area #5 (WDA-5)

The fifth wetland development area is located along the left descending bank at approximate C/L Station 635+00, and is designated on the drawings as "UNCONFINED DISPOSAL AREA WDA-5". Material dredged shall be deposited as unconfined material in this area at an initial elevation of approximately +2.5 feet NAVD 88.

(6) Wetland Development Area #6 (WDA-6)

The sixth wetland development area is located along the right descending bank at approximate C/L Station 485+00, and is designated on the drawings as "UNCONFINED DISPOSAL AREA WDA-6". Material dredged shall be deposited as unconfined material in this area at an initial elevation of approximately +2.5 feet NAVD 88. Discharge locations shall be limited to locations "SPD-1" and "SPD-2" as shown on the contract drawings. To increase the amount of emergent wetlands developed, material deposited in open water shall be pumped toward the shoreline in an effort to increase the existing subaerial footprint. The dredged material shall be discharged in a manner such that the material is directed away from the channel. Contractor shall monitor during disposal and terminate pumping of dredged material in these locations if backflow of dredged material into the Tiger Pass Channel is observed.

(7) Wetland Development Area #7 (WDA-7)

The seventh wetland development area is also located along the right descending bank at approximate C/L Station 550+00, and is designated on the drawings as "SINGLE POINT DISCHARGE (SPD) UNCONFINED DISPOSAL AREA WDA 7". Material dredged shall be deposited unconfined at the Single Point Discharge (SPD) location shown on the drawings. The dredged material shall be discharged in a manner such that the material is directed away from the channel. Dredged material in this area shall be deposited at an initial elevation of approximately +2.5 feet NAVD 88. Retaining dikes will not be required to prevent dredge material from entering Alphonse Bayou and/or the adjacent pipeline canal, nor will "before and after" disposal surveys of these water bodies be required.

### 3.3.2 Prevention Of Damage

The dredged material shall be transported and deposited in such a manner as to ensure that no damage will occur to growing crops, highways, levees, drainage systems, pipelines, utility lines, structures, or other marked by the Contractor with conspicuous buoys or stakes.

### 3.3.3 Excessive Discharge Pipe Leakage

Excessive leakage of the discharge pipe shall be sufficient cause for the Contracting Officer to require the Contractor to cease pumping until corrective measures can be taken at no additional cost to the Government.

### 3.3.4 Effluent Sedimentation

In areas not yet accepted for payment or in areas not yet dredged, the Contractor shall remove all effluent sedimentation from the channel at no additional expense to the Government. In areas previously accepted for payment, and in adjacent areas of the channel where no work is required, the Contractor shall remove all effluent sedimentation from the channel within 500-feet of the point where the wastewater is returned to the channel at no additional expense to the Government.

### 3.3.5 Deposition In Non-Approved Areas

Any material that is deposited elsewhere than in disposal areas shown on the contract drawings or approved Contractor furnished disposal areas may be required to be removed and deposited in approved areas at no additional expense to the Government. Should the Contractor refuse, or delay compliance with the above requirement, such material may be removed by the Contracting Officer, and the cost of such removal may be deducted from any money due or to become due the Contractor.

### 3.3.6 Protection Of Government Monuments, Markers, Or Towers

No dredged material shall be deposited on or near any Government monuments, markers, or towers such that they may become covered or destroyed. At no time shall dredging plant be anchored, moored, or attached to any Government pilings or towers along the route of work. Benchmarks in the disposal areas shall be protected by ring levees, and if material seeps through the levee, the benchmark shall be uncovered without disturbance and exposed for use.

### 3.3.7 Submerged Discharge Lines

If a submerged discharge line is used, it shall at no time reduce the depth and width of the existing channel in which it is placed. When the submerged pipeline is placed in shallow water and where the possibility exists for motorboats to cross the pipeline, the pipeline shall be marked at 150ft intervals for the entire length of the pipeline with Coast Guard approved fluorescent orange buoys and signs stating "DANGER SUBMERGED PIPELINE".

### 3.3.8 Work In The Vicinity Of Other Government Contractors

The Contractor shall coordinate his/her operations, through the Contracting Officer's Representative, with any other Government Contractors who may be working in the vicinity (i.e. locks, control structures, revetment, jetty

repairs, and dike construction, etc.).

### 3.3.9 Protection Of Bird Nesting Colonies

The Contractor shall conduct his/her operations in such a manner as to cause minimum disturbance to bird nesting colonies.

-- End of Section --

## DREDGE AND ATTENDANT PLANT DATA SHEET

Submittal Date: \_\_\_\_\_

In compliance with the contract requirements to submit plant data and subject to all conditions thereof, the undersigned \_\_\_\_\_ a corporation/joint venture/individual (indicate appropriate status) organized and existing under the laws of the City of \_\_\_\_\_ and the State of \_\_\_\_\_, hereby correctly describes the Contractor's plant to the Government, which is performing work under the following named contract \_\_\_\_\_ and

Contract No: \_\_\_\_\_.

Signed: \_\_\_\_\_

Certifying Officer of the Contractor's Firm

Title: \_\_\_\_\_

One cutterhead dredge and attendant plant with the following characteristics (in English units of measurement):

### 1. DREDGE INFORMATION:

(a) Bid Lot Number: N/A

(b) Dredge Name: \_\_\_\_\_ Dredge USCG Official Number: \_\_\_\_\_

(c) Minimum width of channel in which dredge can successfully operate and make a 180 degree turn around: \_\_\_\_\_

(d) Maximum draft of dredge: \_\_\_\_\_

(e) Depth range to which dredge will dig:

Maximum: \_\_\_\_\_ Minimum: \_\_\_\_\_

- (f) Maximum effective dredge swing, in degrees: \_\_\_\_\_
- (g) Length of dredge spuds: \_\_\_\_\_
- (h) Length and beam of dredge hull: \_\_\_\_\_
- (i) Length of dredge ladder: \_\_\_\_\_
- (j) Length of suction and boat lines: \_\_\_\_\_
- (k) Inside diameter of pump discharge: \_\_\_\_\_
- (l) Inside diameter of pump suction inlet: \_\_\_\_\_
- (m) Suction lift (Elevation of main dredge pump relative to the water surface level): \_\_\_\_\_
- (n) Diameter of main pump impeller eye: \_\_\_\_\_
- (o) Outside diameter of main pump impeller: \_\_\_\_\_
- (p) Brake horsepower and corresponding engine RPMs (during dredging operations) applied to main pump impeller at rated drive of the prime mover, during dredging operations: \_\_\_\_\_
- (q) Inside diameter of submerged pump discharge: \_\_\_\_\_
- (r) Inside diameter of submerged pump suction inlet: \_\_\_\_\_
- (s) Suction lift (Elevation of submerged dredge pump relative to the water surface level): \_\_\_\_\_  
\_\_\_\_\_
- (t) Diameter of submerged pump impeller eye: \_\_\_\_\_
- (u) Outside diameter of submerged pump impeller: \_\_\_\_\_
- (v) Brake horsepower and corresponding engine RPMs (during dredging operations) applied to submerged pump impeller at rated drive of the prime mover, during dredging operations: \_\_\_\_\_  
\_\_\_\_\_
- (w) Cutterhead type and diameter: \_\_\_\_\_
- (x) Brake horsepower applied to cutterhead during dredging operations: \_\_\_\_\_
- (y) Pump engine(s) horsepower and corresponding RPM: \_\_\_\_\_
- (z) Completion date of each dredge pump engine re-build: \_\_\_\_\_



(aa) Type(s) of production rate monitoring equipment on-board the dredge (measuring cy/hr of material dredged): \_\_\_\_\_

(ab) Indicate type of dredge advance mechanism (e.g. walking spuds, spud carriage or wires, etc.): \_\_\_\_\_

## 2. BOOSTER PLANT INFORMATION:

(a) Length of suction and boat lines \_\_\_\_\_

(b) Inside diameter of pump discharge: \_\_\_\_\_ (This item must agree with the bid lot used.)

(c) Inside diameter of pump suction inlet: \_\_\_\_\_

(d) Suction lift (Elevation of booster pump relative to the water surface level): \_\_\_\_\_

(e) Diameter of pump impeller eye: \_\_\_\_\_

(f) Outside diameter of pump impeller: \_\_\_\_\_

(g) Brake horsepower and corresponding engine RPMs (during dredging operations) applied to pump impeller at rated drive of the prime mover, during dredging operations: \_\_\_\_\_

(h) Pump engine(s) horsepower and corresponding RPM: \_\_\_\_\_

(i) Completion date of booster pump engine re-build: \_\_\_\_\_

## 3. TENDER NO. 1:

(a) Name: \_\_\_\_\_

(b) Length and Beam: \_\_\_\_\_

(c) Owner, Name and Address: \_\_\_\_\_

(d) Total Propulsion Horsepower: \_\_\_\_\_

(e) The Tender may be inspected at the following location: \_\_\_\_\_

**4. TENDER NO. 2:**

(a) Name: \_\_\_\_\_

(b) Length and Beam: \_\_\_\_\_

(c) Owner, Name and Address: \_\_\_\_\_

\_\_\_\_\_

(d) Total Propulsion Horsepower: \_\_\_\_\_

(e) The Tender may be inspected at the following location: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**5. SURVEY BOAT:**

(a) Name: \_\_\_\_\_

(b) Length and Beam: \_\_\_\_\_

(c) Owner, Name and Address: \_\_\_\_\_

\_\_\_\_\_

(d) Total Propulsion Horsepower: \_\_\_\_\_

**6. SURVEY SKIFF:**

(a) Name: \_\_\_\_\_

(b) Length and Beam: \_\_\_\_\_

(c) Owner, Name and Address: \_\_\_\_\_

\_\_\_\_\_

(d) Total Propulsion Horsepower: \_\_\_\_\_

**7. PICKET BOAT:**

(a) Name: \_\_\_\_\_

(b) Length and Beam: \_\_\_\_\_

(c) Owner, Name and Address: \_\_\_\_\_  
\_\_\_\_\_

(d) Total Propulsion Horsepower: \_\_\_\_\_

**8. CREW BOAT:**

(a) Name: \_\_\_\_\_

(b) Length and Beam: \_\_\_\_\_

(c) Owner, Name and Address: \_\_\_\_\_  
\_\_\_\_\_

(d) Total Propulsion Horsepower: \_\_\_\_\_

**9. ELECTRONIC POSITIONING EQUIPMENT:** (Model and Type): \_\_\_\_\_  
\_\_\_\_\_

**10. DEPTH SOUNDER:** (Model and Type): \_\_\_\_\_  
\_\_\_\_\_

**11. DREDGE OWNER INFORMATION:**

Firm Name: \_\_\_\_\_

Point of Contact: \_\_\_\_\_

Title: \_\_\_\_\_

Business Address:

Street: \_\_\_\_\_

City: \_\_\_\_\_

Parish/County: \_\_\_\_\_

State: \_\_\_\_\_ Zip+4 \_\_\_\_\_

Telephone No: ( \_\_\_\_\_ ) \_\_\_\_\_

Facsimile No: ( \_\_\_\_\_ ) \_\_\_\_\_

Additional signature blocks to be used for joint venture partner(s):

Firm Name: \_\_\_\_\_

Point of Contact: \_\_\_\_\_

Title: \_\_\_\_\_

Business Address:

Street: \_\_\_\_\_

City: \_\_\_\_\_

Parish/County: \_\_\_\_\_

State: \_\_\_\_\_ Zip+4 \_\_\_\_\_

Telephone No: ( \_\_\_\_\_ ) \_\_\_\_\_

Facsimile No: ( \_\_\_\_\_ ) \_\_\_\_\_

Firm Name: \_\_\_\_\_

Point of Contact: \_\_\_\_\_

Title: \_\_\_\_\_

Business Address:

Street: \_\_\_\_\_

City: \_\_\_\_\_

Parish/County: \_\_\_\_\_

State: \_\_\_\_\_ Zip+4 \_\_\_\_\_

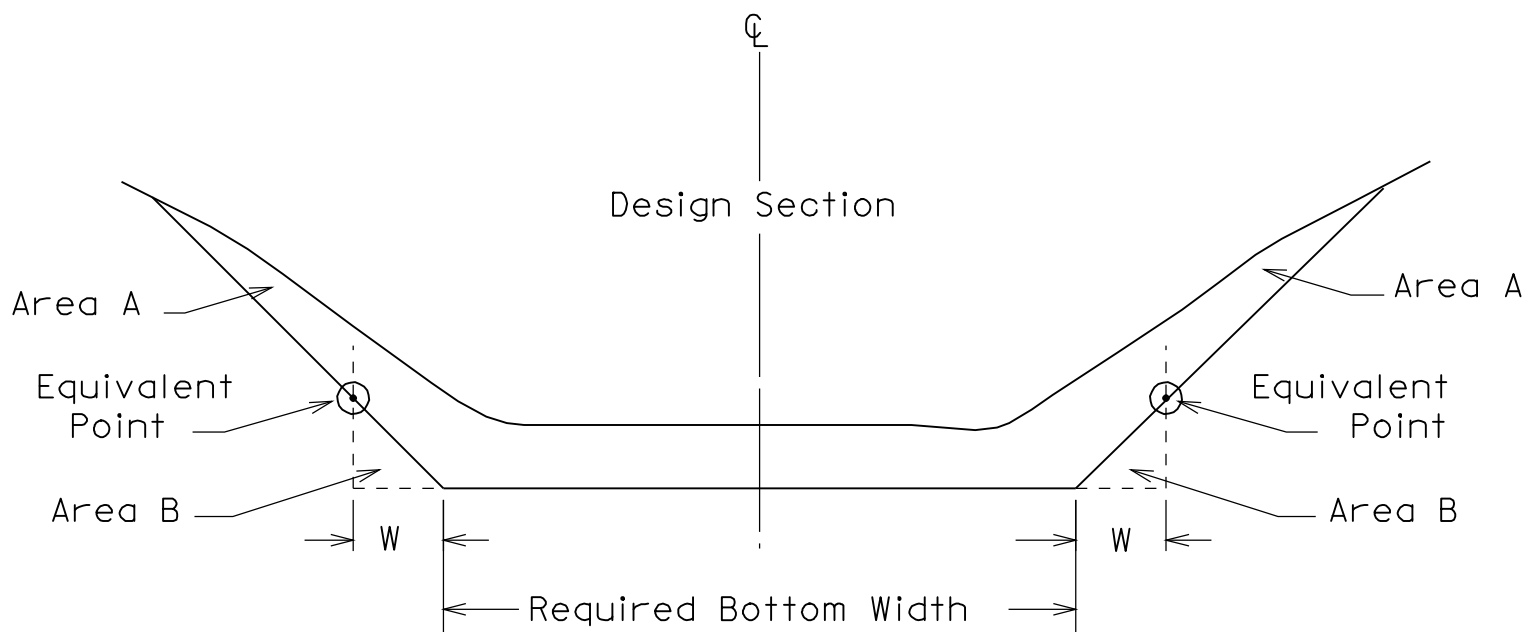
Telephone No: ( \_\_\_\_\_ ) \_\_\_\_\_

Facsimile No: ( \_\_\_\_\_ ) \_\_\_\_\_



| REPORT OF OPERATIONS -- PIPELINE, DIPPER OR BUCKET DREDGES          |                                                                     |             |                 |        |                                |                      |                                                                        |                     |                 |                                              |                | REPORTS CONTROL SYMBOL<br>ENG CW-0-13 |            |
|---------------------------------------------------------------------|---------------------------------------------------------------------|-------------|-----------------|--------|--------------------------------|----------------------|------------------------------------------------------------------------|---------------------|-----------------|----------------------------------------------|----------------|---------------------------------------|------------|
| THRU:                                                               |                                                                     | TO:         |                 |        |                                | FROM:                |                                                                        |                     |                 | DATE OR PERIOD                               |                | REPORT NO.                            |            |
| [ ]                                                                 |                                                                     | [ ]         |                 |        |                                | [ ]                  |                                                                        |                     |                 | [ ]                                          |                | [ ]                                   |            |
| CHARACTER OF REPORT                                                 | [ ]                                                                 | MAINTENANCE | [ ]             | DAILY  | [ ]                            | COMPLETION           | DREDGE PUMPS OPERATING & CORRESPONDING RPM                             | [ ]                 | MAIN 1          |                                              | RPM            | DREDGE TYPE:                          |            |
|                                                                     | [ ]                                                                 | NEW WORK    | [ ]             | ANNUAL | [ ]                            |                      |                                                                        | [ ]                 | MAIN 2          |                                              | RPM            | [ ]                                   | BUCKET     |
| DREDGE                                                              | NAME                                                                |             |                 |        | DREDGE ADVANCE MECHANISM       |                      |                                                                        | [ ]                 | SUBMERGED       |                                              | RPM            | [ ]                                   | CUTTERHEAD |
|                                                                     | DREDGE DISCHARGE SIZE (ID), IN                                      |             |                 |        | CUTTER, HP                     |                      |                                                                        | [ ]                 | BOOSTER 1       |                                              | RPM            | [ ]                                   | DIPPER     |
|                                                                     | DREDGE                                                              |             |                 |        | SHORE                          |                      |                                                                        | [ ]                 | BOOSTER 2       |                                              | RPM            | [ ]                                   | DUSTPAN    |
|                                                                     | DREDGE                                                              |             |                 |        | OTHER                          |                      | TOTAL                                                                  |                     | WORK SCHEDULE   |                                              | SHIFTS PER DAY |                                       |            |
| NUMBER OF CREWMEMBERS                                               |                                                                     |             |                 |        |                                |                      |                                                                        |                     |                 |                                              |                | DAYS PER WEEK                         |            |
|                                                                     |                                                                     |             |                 |        |                                |                      |                                                                        |                     |                 |                                              |                | cu. yds. cap.                         |            |
| PROJECT AND ASSIGNMENT                                              | ASSIGNED DIMENSIONS                                                 |             |                 |        |                                | WIDTH                |                                                                        | DEPTH               |                 | OVERDEPTH                                    |                |                                       |            |
|                                                                     | LOCATION (INCLUDE STATIONS NUMBERS)                                 |             |                 |        |                                | WIDTH OF SWING       |                                                                        | RANGES - RT, CL, LT |                 |                                              |                |                                       |            |
|                                                                     | C/L STATIONS                                                        |             |                 |        |                                | WIDTH OF SWING       |                                                                        | RANGES - RT, CL, LT |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                | WIDTH OF SWING       |                                                                        | RANGES - RT, CL, LT |                 |                                              |                |                                       |            |
| CHARACTER OF MATERIAL<br>(leave blank if grab samples aren't taken) | ABSOLUTE DENSITY                                                    |             |                 |        | IN PLACE DENSITY               |                      |                                                                        |                     | VOIDS RATIO     |                                              |                |                                       |            |
|                                                                     | GMS/liter                                                           |             |                 |        | GMS/liter                      |                      |                                                                        |                     |                 |                                              |                |                                       |            |
|                                                                     | GRAIN SIZE                                                          |             |                 |        | VISUAL MATERIAL CLASSIFICATION |                      |                                                                        |                     | TYPE            |                                              |                |                                       |            |
|                                                                     | MM                                                                  |             |                 |        | MM                             |                      |                                                                        |                     | TYPE            |                                              |                |                                       |            |
| CONTRACT OR ORDER NUMBER                                            | NUMBER                                                              |             |                 |        |                                | CONTRACTOR           |                                                                        | HIRED LABOR         |                 | TOTAL NUMBER OF DAYS ON WHICH WORK WAS DONE: |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      |                                                                        |                     |                 |                                              |                |                                       |            |
| CHANNEL CONDITION                                                   | AVERAGE DEPTH                                                       |             | BEFORE DREDGING |        | AFTER DREDGING                 |                      | MINIMUM SOUNDING                                                       |                     | BEFORE DREDGING |                                              | AFTER DREDGING |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      |                                                                        |                     |                 |                                              |                |                                       |            |
| RIVER STAGE                                                         | MINIMUM                                                             |             | TIME            |        | MAXIMUM                        |                      | TIME                                                                   |                     | GAGE LOCATION   |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      |                                                                        |                     |                 |                                              |                |                                       |            |
| WEATHER CONDITION                                                   | (clear, cloudy, rain, snow, and fog)                                |             |                 |        |                                |                      |                                                                        | VISIBILITY          |                 | WIND (maximum velocity & direction)          |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      |                                                                        | miles               |                 |                                              |                |                                       |            |
| WORK PERFORMED                                                      |                                                                     |             |                 |        |                                | DISTRIBUTION OF TIME |                                                                        |                     |                 |                                              |                |                                       |            |
| ITEM                                                                |                                                                     |             | UNIT            |        | QUANTITY                       |                      | EFFECTIVE WORKING TIME<br>(chargeable to cost of work)                 |                     |                 |                                              | AMOUNT         |                                       |            |
| AVERAGE WIDTH OF CUT (SWING)                                        |                                                                     |             | FEET            |        |                                |                      | PUMPING OR DREDGING                                                    |                     |                 |                                              |                |                                       |            |
| TOTAL ADVANCE THIS PERIOD                                           |                                                                     |             | FEET            |        |                                |                      | PCT. OF EFFECTIVE TIME                                                 |                     |                 |                                              |                |                                       |            |
| TOTAL ADVANCE PREVIOUS TO THIS PERIOD                               |                                                                     |             | FEET            |        |                                |                      | BOOSTER (in line):                                                     |                     |                 |                                              |                |                                       |            |
| TOTAL ADVANCE TO DATE                                               |                                                                     |             | FEET            |        |                                |                      | NON-EFFECTIVE WORKING TIME<br>(chargeable to cost of work)             |                     |                 |                                              | NO. OF TIMES   |                                       |            |
| TOTAL LENGTH OF DISCHARGE PIPELINE: *SEE REVERSE SIDE               |                                                                     |             |                 |        |                                |                      |                                                                        |                     |                 |                                              | AMOUNT         |                                       |            |
| AVERAGE LIFT                                                        |                                                                     |             | FEET            |        |                                |                      | HANDLING PIPELINES                                                     |                     |                 |                                              | HPL            |                                       |            |
| AVERAGE PUMP SPEED                                                  |                                                                     |             | RPM             |        |                                |                      | HANDLING ANCHOR LINES                                                  |                     |                 |                                              | HAL            |                                       |            |
| AVERAGE GROSS CY'S DREDGED PER PUMP HOUR                            |                                                                     |             | CY/HR           |        |                                |                      | CLEARING PUMP AND PIPELINE                                             |                     |                 |                                              | CPPL           |                                       |            |
| AVERAGE NET CY'S DREDGED PER PUMP HOUR                              |                                                                     |             | CY/HR           |        |                                |                      | CLEARING CUTTER OR SUCTION HEAD                                        |                     |                 |                                              | CCSH           |                                       |            |
| AVERAGE LINEAR FOOT PER HOUR                                        |                                                                     |             | LF/HR           |        |                                |                      | WAITING FOR SCOWS                                                      |                     |                 |                                              | WFS            |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | TO AND FROM WHARF OR ANCHORAGE                                         |                     |                 |                                              | TFWA           |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | CHANGING LOCATION OF PLANT ON JOB                                      |                     |                 |                                              | CLPJ           |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | LOSS DUE TO OPPOSING NATURAL ELEMENTS                                  |                     |                 |                                              | LDNE           |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | LOSS DUE TO PASSING VESSELS                                            |                     |                 |                                              | WVP            |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | SHORELINE AND SHORE WORK                                               |                     |                 |                                              | SLSW           |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | WAITING FOR BOOSTER                                                    |                     |                 |                                              | WFB            |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | MINOR OPER. REPAIRS (explain in remarks)                               |                     |                 |                                              | MOR            |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | WAITING FOR ATTENDANT PLANT                                            |                     |                 |                                              | WAP            |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | PREPARATION AND MAKING UP TOW                                          |                     |                 |                                              | PMUT           |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | TRANSFERRING PLANT BETWEEN WORKS                                       |                     |                 |                                              | TPBW           |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | LAY TIME OFF SHIFT AND SATURDAYS                                       |                     |                 |                                              | LTOSS          |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | SUNDAYS AND HOLIDAYS                                                   |                     |                 |                                              | SH             |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | FIRE DRILL                                                             |                     |                 |                                              | FD             |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | MISCELLANEOUS (explain in remarks)                                     |                     |                 |                                              | MISC           |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | TOTAL NON-EFFECTIVE WORKING TIME                                       |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | PCT. OF NON-EFFECTIVE TIME                                             |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | TOTAL EFFECTIVE AND NON-EFFECTIVE TIME<br>(chargeable to cost of work) |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | PCT. OF TOTAL TIME IN PERIOD                                           |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | LOST TIME<br>(not chargeable to cost of work)                          |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | MAJOR REPAIRS AND ALTERATIONS                                          |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | CESSATION                                                              |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | COLLISIONS                                                             |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | MISCELLANEOUS (explain in remarks)                                     |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | TOTAL LOST TIME                                                        |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | PERCENTAGE OF TOTAL TIME                                               |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      | TOTAL TIME IN PERIOD                                                   |                     |                 |                                              |                |                                       |            |
|                                                                     |                                                                     |             |                 |        |                                |                      |                                                                        |                     |                 |                                              |                |                                       |            |
| CONTRACT USE ONLY                                                   | HAS ANYTHING DEVELOPED WHICH MIGHT LEAD TO A CHANGE ORDER OR CLAIM? |             |                 |        |                                | ( )                  |                                                                        | NO                  |                 |                                              |                |                                       |            |
|                                                                     | (If "YES", explain under remarks on back)                           |             |                 |        |                                | ( )                  |                                                                        | YES                 |                 |                                              |                |                                       |            |

| SUMMARY OF DREDGE DISCHARGE PIPELINE, DISPOSAL AREAS AND TIME BREAKDOWN |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|-------------------------------------------------------------------------|------------------|---------------------------------------------|-------------------|------------------------------------------------|------------------|------------------------------------------|---------------------|-------------------|----------------|--|
| ATTENDANT PLANT                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| DREDGE DISCHARGE PIPELINE                                               |                  |                                             |                   |                                                |                  | DISPOSAL AREA EQUIPMENT                  |                     | ANCHORING SYSTEMS |                |  |
| FLOATING Pontoons                                                       | Amount           | Submerged                                   | Amount            | Shore                                          | Amount           | TOTAL COMBINED DISCHARGE, LF             | Marsh Trackhoes-30' |                   | Skidder Barges |  |
| Metal                                                                   |                  | Metal                                       |                   | Metal                                          |                  |                                          | Marsh Trackhoes-60' |                   | ANCHORS        |  |
| PVC                                                                     |                  | PVC                                         |                   | PVC                                            |                  |                                          | Marsh Buggies       |                   | Breast         |  |
| Flexifloat                                                              |                  | Other                                       |                   | Other                                          |                  |                                          | Dozers              |                   | Head           |  |
| TOTAL LF                                                                |                  |                                             |                   |                                                |                  |                                          | Front End Loaders   |                   | Line           |  |
| ELBOWS                                                                  | Stiff            |                                             |                   | LEAST INSIDE DIAMETER OF DISHARGE PIPELINE, IN |                  |                                          | Air Boats           |                   | Other          |  |
|                                                                         | Swivel           |                                             |                   |                                                |                  |                                          | ATV's               |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          | Y-Valves            |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          | Spreaders           |                   |                |  |
| DISPOSAL AREA SUMMARY                                                   |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| NAME/LOCATION COORDINATE                                                | SHORE CONNECTION | DATE START                                  | DATE FINISH       | TOTAL GROSS CYS                                | TOTAL ACREAGE    | REMARKS                                  |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| TIME BREAKDOWN                                                          |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         | 100%             | 70%                                         | 35%               | 0%                                             | 0% - MISC (Mob)  | TOTAL                                    |                     |                   |                |  |
| BASE HRS                                                                |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| OPTION HRS                                                              |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| PREVIOUS HRS                                                            |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| TOTAL HRS                                                               |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| OPERATING SUPPLIES                                                      |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  | TODAY QUANTITY                              | PREVIOUS QUANTITY |                                                | QUANTITY TO DATE |                                          |                     |                   |                |  |
| FUEL (diesel)                                                           | GAL.             |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| LUBRICANT (oil)                                                         | GAL.             |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| LUBRICANT (grease)                                                      | LBS.             |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| WATER                                                                   | GAL.             |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| REMARKS:                                                                |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| SUBMITTED BY (Name, title, and signature)                               |                  | RECOMMENDED BY (Name, title, and signature) |                   |                                                |                  | APPROVED BY (Name, title, and signature) |                     |                   |                |  |
|                                                                         |                  |                                             |                   |                                                |                  |                                          |                     |                   |                |  |
| NAME:                                                                   |                  | NAME:                                       |                   | NAME:                                          |                  |                                          |                     |                   |                |  |
| TITLE:                                                                  |                  | TITLE:                                      |                   | TITLE:                                         |                  |                                          |                     |                   |                |  |



## Box Cut Section

### Definitions:

Equivalent Point - That point on each side slope where the area above the point (Area A) equals the area below the point (Area B)

W - That additional width on each side of the required bottom width, that is determined by the location of the equivalent point, necessary to make Area A = Area B



## SECTION TABLE OF CONTENTS

## DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

## SECTION 35 20 23.33

## NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM

**August 2021**

## PART 1 GENERAL

- 1.1 DESCRIPTION
- 1.2 SUBMITTALS
- 1.3 MEASUREMENT AND PAYMENT
- 1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION
- 1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

## PART 2 PRODUCTS (Not Applicable)

## PART 3 EXECUTION

- 3.1 REQUIREMENTS FOR REPORTED DATA
  - 3.1.1 Message Bundle Data
    - 3.1.1.1 Messages
      - 3.1.1.1.1 Message Time
      - 3.1.1.1.2 Comment
    - 3.1.1.2 Dredge Events - Work Event
      - 3.1.1.2.1 Vertical Correction
      - 3.1.1.2.2 Cutter/Suction Head Location and Movement
        - 3.1.1.2.2.1 Cutter/Suction Head Horizontal Position
        - 3.1.1.2.2.2 Cutter/Suction Invert Depth
        - 3.1.1.2.2.3 Cutter/Suction Head Heading
      - 3.1.1.2.3 Dredge Activity
        - 3.1.1.2.3.1 Slurry Velocity
        - 3.1.1.2.3.2 Slurry Density
        - 3.1.1.2.3.3 Pump RPM
        - 3.1.1.2.3.4 Pump Vacuum
        - 3.1.1.2.3.5 Pump Outlet Pressure
      - 3.1.1.2.4 Outfall Information (Open Water/Spill Barge Disposal)
        - 3.1.1.2.4.1 Discharge Horizontal Position
    - 3.1.1.3 Dredge Events - State Event
      - 3.1.1.3.1 Message Time
      - 3.1.1.3.2 Contract Event
        - 3.1.1.3.2.1 Contract Number
        - 3.1.1.3.2.2 Contract Start and End
      - 3.1.1.3.3 Tide Station/River Stage Gage Event
        - 3.1.1.3.3.1 Station Name
      - 3.1.1.3.4 Length of Pipe Event
        - 3.1.1.3.4.1 Floating Pipe
        - 3.1.1.3.4.2 Submerged Pipe
        - 3.1.1.3.4.3 Shore Pipe
      - 3.1.1.3.5 Booster Pump Event
        - 3.1.1.3.5.1 Number of Booster Pumps
      - 3.1.1.3.6 Dredge Advance
      - 3.1.1.3.7 Outfall Information
        - 3.1.1.3.7.1 Discharge Location

- 3.1.1.3.7.2 Discharge Horizontal Position
- 3.1.1.3.7.3 Discharge Outfall Heading
- 3.1.1.3.7.4 Discharge Pipe Elevation
- 3.1.1.3.8 Non-effective Work Event
  - 3.1.1.3.8.1 Non-effective Work Interval
  - 3.1.1.3.8.2 Dredge Function Code
  - 3.1.1.3.8.3 Additional Comments
- 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS
  - 3.2.1 Computer Requirements
  - 3.2.2 Software
  - 3.2.3 UPS
  - 3.2.4 Internet Access
  - 3.2.5 Data Routing Requirements
- 3.3 DREDGE MONITORING DATA
  - 3.3.1 General
  - 3.3.2 Data Measurement Frequency
    - 3.3.2.1 Work Event Messages
    - 3.3.2.2 State Event Messages
  - 3.3.3 Parameter Transmission to the Web Service
  - 3.3.4 Contractor Data Backup
- 3.4 PERFORMANCE REQUIREMENTS
- 3.5 LIST OF ITEMS PROVIDED BY THE CONTRACTOR

-- End of Section Table of Contents --

## SECTION 35 20 23.33

## NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM

**August 2021**

## PART 1 GENERAL

## 1.1 DESCRIPTION

The work under this contract requires use of the US Army Corps of Engineers (USACE) National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract duration and manage data history. This section identifies the minimum required output as well as the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation. A designation following the "G" designation identifies the office responsible for review of the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURE:

SD-01, Preconstruction Submittals

Dredge Plant Instrumentation Plan Revisions; G, OM

SD-07, Certificates

Letter of National Dredging Quality Management Program  
Certification; G, EDC

## 1.3 MEASUREMENT AND PAYMENT

No separate measurement or payment will be made for the installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations. All costs in connection therewith shall be considered a subsidiary obligation of the Contractor and shall be included in the contract unit price per cubic yard for "Dredging".

## 1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

The Contractor shall have a current certification from the DQM Program for the cutter/suction head hydraulic dredge instrumentation system to be used under this contract. Standard Operating Procedures (SOP) and criteria for certification are presented on the DQM website at <https://dqm.usace.army.mil>.

## 1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor shall have a digital copy of the Dredge Plant Instrumentation Plan (DPIP) on file with the DQM Support Center. While working on site, the Contractor shall also maintain on the dredge a copy of the DPIP, which is easily accessible to Government personnel at all times.

This document shall accurately describe the sensors used, the configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how the sensors/data-reporting equipment will be calibrated and repaired if it fails. A description of the computed dredge-specific data and how the sensor data will be transmitted to the DQM database shall also be included. Prior to the start of work, the Contractor shall submit to the DQM Support Center any addendum or modifications made to the plan subsequent to its original submission. Requirements and a template for the DPIIP are available on the DQM website at <https://dqm.usace.army.mil>.

## PART 2 PRODUCTS (Not Applicable)

## PART 3 EXECUTION

### 3.1 REQUIREMENTS FOR REPORTED DATA

The Contractor shall provide, operate, and maintain all hardware and software to meet these specifications. The Contractor shall also be responsible for the replacement, repair, and calibration of the sensors and other necessary data acquisition equipment needed to supply the required data.

The procedure to complete a repair shall be documented and completed as soon as practical. If repair is not possible within two business days of any sensor failure, a plan and timeline to complete the repair shall be submitted. Upon completion of a repair, replacement, installation, modification, or calibration, the Contractor shall notify the Contracting Officer's Representative (COR). The COR may request recalibration of the sensors or other hardware components at any time during the contract as deemed necessary.

The Contractor shall keep a log of sensor repair, replacement, installation, modification, and calibration in the dredge's onboard copy of the DPIIP. The log shall contain a three-year history of sensor maintenance, including the time of the sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide the required data. It shall also contain the name of the person responsible for the sensor work.

Sensors installed shall be capable of collecting parameters within the specified accuracies and resolutions indicated in the following subparagraphs and transmit these parameters to the DQM database. All data shall be transmitted in JSON message bundles. Each bundle can contain multiple message types. Sensor data shall be transmitted as work event messages, and data which relates to the operational state of the dredge or its sensors shall be transmitted as state event messages. See paragraph entitled "Parameter Transmission to the Web Service".

#### 3.1.1 Message Bundle Data

Every message bundle shall contain descriptive data that relates the message to a given dredge plant and date/time. The start of a message bundle shall be identified by the tag "DQM\_data".

##### 3.1.1.1 Messages

Messages contain operational data that populates the DQM database for a

dredge plant. A message shall consist of an event type and its associated data (as defined in paragraph entitled "Dredge Events"), a date/time stamp indicating when the event occurred or started, and a comment providing clarification or metadata about the situation. There are multiple event types, but they all fall into one of two categories - work events and state events.

#### 3.1.1.1.1 Message Time

In a work event message, message time is the date and time that the data is collected from the sensors; in a state event message, message time is the date and time that the state event begins. The message time shall be reported to the nearest second and referenced to Coordinated Universal Time (UTC) time based on a 24-hour format (YYYY-MM-DD HH:MM:SS). In order to ensure accuracy and reliability, the time stamp shall be synchronized to UTC format from an accurate, unchangeable source (for example, a GPS National Marine Electronics Association NMEA datastring). Message time shall be identified by the tag "msg\_time".

#### 3.1.1.1.2 Comment

Comments concerning the work event or state event messages being transmitted provide descriptive information that relates to the data. An example of a comment for work event data is information about a sensor issue; an example of a comment for state event data is a description of operations. A comment shall be identified by the introductory tag "comment", and the comment shall consist of no more than 250 characters.

#### 3.1.1.2 Dredge Events - Work Event

There are two types of dredge event messages - work event messages and state event messages. Work event messages contain data that are instantaneously collected or calculated from sensors and are logged as a series of events. Work events are triggered by a time interval change (as described in paragraph entitled "Work Event Messages"). All work event messages shall be initiated by the header tag "work\_event".

Sensor values reported in a work event shall represent a weighted average with the highest and lowest values not included in the calculated average for the given interval. The averaging routine used shall be consistent across all event triggers. This information shall be documented in the DPIIP sections that say "Calculations done external to the instrumentation".

##### 3.1.1.2.1 Vertical Correction

The variation of the water level from the vertical datum for the river stage or tidal gage described in the state events shall be obtained using appropriate equipment to give the water level with an accuracy of +/- 0.1 ft. Vertical correction values above project datum described in the dredging specification shall be entered with a positive sign and those below with a negative sign. The tag for vertical correction shall be "vert\_correction".

##### 3.1.1.2.2 Cutter/Suction Head Location and Movement

The X, Y, and Z components of the cutter/suction head location shall be monitored. Additional calculations made from the observed values determine the rates of movement to track the progress of the dredge.

#### 3.1.1.2.2.1 Cutter/Suction Head Horizontal Position

The forwardmost point of the cutter/suction head shall be obtained using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). It shall be reported as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values reported as negative. Position values shall be identified by the tags "ch\_latitude" and "ch\_longitude".

#### 3.1.1.2.2.2 Cutter/Suction Invert Depth

Cutter/suction invert depth is the depth of the invert of the suction mouth relative to the surface of the water. Instrumentation shall be capable of reporting to an accuracy of +/- 0.5 foot and a resolution to the nearest 0.1 foot with no tidal adjustments. Minimum accuracies are conditional to relatively calm water. The tag "ch\_depth" shall be used to identify the cutter/suction head depth.

#### 3.1.1.2.2.3 Cutter/Suction Head Heading

The cutter/suction head heading is the angle of the centerline of the cutter/suction head and dredge ladder measured relative to true north. All headings shall be provided using industry-standard equipment. The heading shall be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention. The tag "ch\_heading" shall be used to identify the cutter/suction head heading.

#### 3.1.1.2.3 Dredge Activity

Dredge activity shall be monitored using a combination of the following parameters.

##### 3.1.1.2.3.1 Slurry Velocity

A flow-metering device, calibrated according to the manufacturer's specifications, shall be used to record the slurry velocity to the nearest 0.01 fps with an accuracy of plus 0.1 fps. If the manufacturer does not specify a frequency of recalibration, calibration shall be conducted prior to the commencement of work. The slurry velocity shall be measured for the same pipeline inside diameter as that used for the slurry density measurement. The tag "slurry\_velocity" shall be associated with this value.

##### 3.1.1.2.3.2 Slurry Density

A density-metering device, calibrated according to the manufacturer's specifications, shall be used to record the slurry density of material being moved to the nearest 0.01 g/cc. It is understood that the accuracy of this sensor can vary based on several factors, including the type of material, the magnitude of the cut, and the length of time since calibration. If the manufacturer does not specify a frequency of recalibration, calibration shall be conducted prior to the commencement of work. Continuous monitoring of this sensor ensures that drift and other factors inherent in the dredging process can be accounted for in monitoring dredge activity. The tag "slurry\_density" shall be associated with this value.

#### 3.1.1.2.3.3 Pump RPM

The pump rpm is the number of revolutions per minute measured for the slurry pump shaft. The shaft revolution rate (rev/min) shall be measured with the highest level of accuracy that is standard on the vessel's operational displays either at the bridge or in the engine room. This value shall be identified by the tag "rpm".

#### 3.1.1.2.3.4 Pump Vacuum

The vacuum pressure of the dredge pump(s) (inches of mercury) shall be measured as near to the eye as practicable in the pump's suction pipe with the highest level of accuracy that is standard on the vessel's operational displays either at the leverman's controls or in the engine room. Vacuum pressure shall be identified by the tag "vacuum".

#### 3.1.1.2.3.5 Pump Outlet Pressure

The pump outlet pressure shall be measured in the discharge line on the pump side of the flap valve in terms of pounds per square inch (psi) on a gauge. Pump outlet pressure shall be identified by the tag "outlet\_psi".

#### 3.1.1.2.4 Outfall Information (Open Water/Spill Barge Disposal)

The X and Y position of the terminal end of the outfall pipe shall be monitored continuously and the position reported as part of the work event string.

##### 3.1.1.2.4.1 Discharge Horizontal Position

The horizontal position of the outfall end of the discharge pipe shall be obtained using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). It shall be reported as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values being reported as negative. Position values shall be identified by the tags "outfall\_latitude" and "outfall\_longitude".

#### 3.1.1.3 Dredge Events - State Event

There are two types of dredge event messages - work event messages and state event messages. State event messages provide information about the current state of the dredge equipment or operations. They are created and sent only when a state changes. Since state events often cannot be collected in real time, state events are tagged with a date time stamp (referenced to Coordinated Universal Time) that indicates when the state change happened relative to the work event message tag. This data is considered to be "true" until another state event tag of the same type is received. Each type of state event message shall be indicated by a specific header tag as enumerated in the following subparagraphs. State events can be transmitted along with work event message bundles directly by the contractor using the indicated format, or they can be entered on the "State" tab in the DQM-provided software. However, they should be sent only if the state value changes.

##### 3.1.1.3.1 Message Time

The state event time is the date and time that the event starts. The leverman's time shall be entered to the nearest second as local time and automatically converted to and reported in UTC based on a 24-hour format

(YYYY-MM-DD HH:MM:SS). Message time shall be identified by the tag "msg\_time".

#### 3.1.1.3.2 Contract Event

Information concerning the contract under which dredging is being performed shall be reported at the start and completion of each contract using the header tag "contract\_event".

##### 3.1.1.3.2.1 Contract Number

The USACE-assigned contract number for the project shall be reported using the tag "contract\_number".

##### 3.1.1.3.2.2 Contract Start and End

The start and end of a contract shall be reported using the tag "event\_type" with the appropriate value of "start" or "end".

#### 3.1.1.3.3 Tide Station/River Stage Gage Event

Properties associated with the vertical correction (see paragraph entitled "Vertical Correction") for the tide station/river stage gage shall be grouped together under the header tag "station\_event". This information shall be sent at the start of the contract and each time the dredge has moved enough to change the station being used.

##### 3.1.1.3.3.1 Station Name

The station name is a concise name defining the tide station/river stage gage begin referred to. It shall be introduced by the tag "station\_name", and it shall consist of a descriptor of no more than 25 characters.

#### 3.1.1.3.4 Length of Pipe Event

The leverman's estimate of the length of pipe downflow from the dredge pump, measured to the nearest whole foot, shall be reported under the header tag "pipe\_length\_event". This information shall be sent at the start of the contract and at the completion of each 24-hour period ending at midnight local time.

##### 3.1.1.3.4.1 Floating Pipe

The total length of floating pipe shall be reported with the tag "length\_floating".

##### 3.1.1.3.4.2 Submerged Pipe

The total length of submerged pipe shall be reported with the tag "length\_submerged".

##### 3.1.1.3.4.3 Shore Pipe

The total length of shore pipe shall be reported with the tag "length\_land".

#### 3.1.1.3.5 Booster Pump Event

Information concerning the booster pumps being used shall be included under the header tag "booster\_pump\_event". A message shall be sent to indicate



any change in the status of the booster pumps being used.

#### 3.1.1.3.5.1 Number of Booster Pumps

Upon the addition or removal of a booster pump, the total number of booster pumps being used shall be reported with the tag "booster\_total".

#### 3.1.1.3.6 Dredge Advance

The dredge advance, the total forward progress of the dredge relative to the centerline of the cut, shall be measured to the nearest whole foot and cumulatively calculated over a 24-hour period from midnight to midnight local time. It shall be identified by the tag "advance\_daily". The msg\_time associated with this tag shall be reported as the first timestamp of the following 24-hour period (based on the local time) rather than as midnight of the day for which the value was calculated, and it shall be reported in Greenwich Mean Time (GMT).

#### 3.1.1.3.7 Outfall Information

The X and Y position of the terminal end of the outfall pipe shall be monitored and sent at the start of the contract and thereafter according to the following table. Discharge Heading and Pipe Elevation may be omitted if the dredge is not discharging into an upland disposal site. For beach nourishment, the horizontal X and Y position of the outfall shall be sent at the start of the contract and at the completion of each 24-hour period ending at midnight local time.

| Discharge Location | Horizontal Position   | Discharge Pipe Elevation | Discharge Outfall Heading |
|--------------------|-----------------------|--------------------------|---------------------------|
| Open Water         | Continuous Work Event | N/A                      | N/A                       |
| Scow               | Upon Change           | N/A                      | N/A                       |
| Beach              | Every 24 Hours        | N/A                      | N/A                       |
| Upland             | Upon Change           | Upon Change              | Upon Change               |

##### 3.1.1.3.7.1 Discharge Location

Information on where the slurry is being discharged shall be reported with the tag "outfall\_location". Acceptable values include "upland", "open water", "beach", and "scow".

##### 3.1.1.3.7.2 Discharge Horizontal Position

The horizontal position of the outfall end of the discharge pipe shall be obtained using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). It shall be reported as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values being reported as negative. Position values shall be identified by the tags "outfall\_latitude" and "outfall\_longitude".

##### 3.1.1.3.7.3 Discharge Outfall Heading

The discharge outfall heading is the angle relative to true north measured from the centerline of the pipe in the direction of discharge. All headings

shall be provided using industry-standard equipment. They shall be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention. The discharge heading shall be identified by the tag "outfall\_heading".

#### 3.1.1.3.7.4 Discharge Pipe Elevation

The discharge pipe elevation is the height of the outfall measured in feet and tenths of a foot relative to the project datum. The required accuracy is contingent upon contract requirements. The tag "outfall\_elevation" shall be used to identify this elevation.

#### 3.1.1.3.8 Non-effective Work Event

Delays and dredge downtime shall be reported at the conclusion of the event. The reason for the non-effective work time shall be submitted under the header tag "non\_eff\_event" within 24 hours of the event.

##### 3.1.1.3.8.1 Non-effective Work Interval

The start and end times for the non-effective work event shall be reported using the tags "msg\_start\_time" and "msg\_end\_time".

##### 3.1.1.3.8.2 Dredge Function Code

The dredge operator indication of production delays, as listed on Form 4267, shall be transmitted at the end of the non-effective interval. Dredge function event messages shall be identified by the tag "function\_code" and shall consist of one of the following standardized entries to indicate the operation:

|      |                               |
|------|-------------------------------|
| AGV  | Assisting Grounded Vessels    |
| CCH  | Change Cutterhead             |
| CCSH | Clear Cutter Suction          |
| CLPJ | Change Location Bar           |
| COLL | Collision                     |
| CPPL | Clear Pump Pipeline           |
| CPR  | Change Impeller               |
| DR   | Dike Repair                   |
| FBD  | Fire Boat Drills              |
| HPL  | Handling Pipe Line            |
| HSL  | Handling Swing Line           |
| HSP  | Handling Shore Pipe           |
| LDNE | Loss Due to Natural Elements  |
| LDPV | Loss Due to Passing Vessel    |
| LNL  | Transfer to New Location      |
| MISC | Miscellaneous                 |
| MOB  | Mobilization & Demobilization |
| MSC  | Miscellaneous/Non-pay         |
| OC   | Out of Commission             |
| OR   | Operating Repairs             |
| P    | Preparation                   |
| PREP | Preparation & Making Up Tow   |
| RPL  | Repair Pipeline               |
| SB   | Sounding & Buoying            |
| SBT  | Stand-By Time as Directed     |
| SH   | Sundays-Holidays              |
| TFS  | Taking on Fuel & Supplies     |

TOW Time on Tow  
WAP Waiting Attendant Plant

### 3.1.1.3.8.3 Additional Comments

The "comment" tag shall be used to provide additional explanation for the noted delays or downtimes. For example, when the code "LDPV" (Loss Due to Passing Vessel) is indicated, the name of the vessel and the number of tows shall be listed with the "comment" tag.

## 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system shall be capable of collecting and transmitting information to the DQM onboard computer. The applicable parameters from paragraph entitled REQUIREMENTS FOR REPORTED DATA shall be recorded as events locally and continuously transmitted to the DQM database anytime an Internet connection is available. The dredge shall be equipped with a DQM computer system consisting of a computer, monitor, keyboard, mouse, data modem, Universal Power Supply (UPS), and network hub. The computer system shall be a standalone system, exclusive to the DQM monitoring system, and shall have USACE DQM software installed on it. If a hardware problem occurs, or if a part of the system is physically damaged, then the Contractor shall be responsible for repairing it within two business days of the determination of the condition or submitting a plan and timeline to the DQM Support Center for repair if the repair will take more than two business days.

### 3.2.1 Computer Requirements

The Contractor shall provide a dedicated onboard computer for use by the Dredging Quality Management system. This computer shall run the USACE DQM software and receive data from the Contractor's data-reporting interface. This computer must meet or exceed the following performance specifications:

|                  |                                                                                             |
|------------------|---------------------------------------------------------------------------------------------|
| CPU              | Intel or AMD processor with a (non-overclocked) clock speed of at least 1.6 gigahertz (GHz) |
| Hard drive       | 250 gigabytes (GB); internal                                                                |
| RAM              | 4 gigabytes (GB)                                                                            |
| Ethernet adapter | Internal network card with a RJ-45 connector                                                |
| Ports            | 1 free serial port with standard 9-pin connectors; 1 free USB port                          |
| Other hardware   | Keyboard, mouse, monitor                                                                    |

The Contractor shall install a fully licensed copy of Windows 7 Professional Operating System or later on the computer specified above. The Contractor shall also install any necessary manufacturer-provided drivers for the installed hardware.

This computer shall be located and oriented to allow data entry and data viewing as well as to provide access to data ports for connection of external hardware.

### 3.2.2 Software

The DQM computer's primary function is to transmit data to the DQM shoreside database. No other software which conflicts with this function shall be installed on it. The DQM computer shall also have the USACE-provided Dredging Quality Management Onboard Software (DQMOBS) installed on it by DQM personnel.

### 3.2.3 UPS

The Contractor shall supply an Uninterruptible Power Supply (UPS) for the computer and networking equipment. It shall interface with the DQM computer to communicate UPS status, and it shall provide backup power at 1 kVA for a minimum of 10 minutes. The Contractor shall ensure that sufficient power outlets are available to run all specified equipment.

### 3.2.4 Internet Access

The Contractor shall maintain an Internet connection capable of transmitting real-time data to the DQM server as well as enough additional bandwidth to clear historically queued data when a connection is re-established. If connectivity is lost, unsent data shall be queued and transmitted upon restoration of connectivity. Delays in pushing real-time data to the DQM database shall not exceed twenty-four (24) hours.

The Contractor shall acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the DQM web service. The hardware and software shall be configured to allow the DQM Support Center remote access to this computer, and the telemetry system shall be capable of meeting these minimum reporting requirements in all operating conditions.

In areas with poor cellular service and at the Contracting Officers discretion , it may be required to manually download the data on a daily basis using the protocol for retrieving and submitting backup files provided by the DQM Support Center. This method of data transmission should be used only if Internet connectivity is unavailable at the dredging site, and it should be considered a temporary measure. If data must be manually downloaded, it shall be submitted to the DQM Data Acquisition and Analysis Team within 48 hours of the date it was collected.

### 3.2.5 Data Routing Requirements

Onboard sensors continually monitor dredge conditions, operations, and efficiency and route this information to the shipboard dredge-specific system (DSS) computer to assist in guiding dredge operations. Portions of this Contractor-collected information, as described in this specification, shall be routed to the DQM computer on a real-time basis. Standard sensor data shall be sent to the DQM computer via an RS-232 serial interface with a baud rate of 9600 or 19200 bps. The serial interface shall be configured as 8 bits, no parity, and no flow control.

Information regarding changes in the state of the dredge shall be digitally logged and transmitted as close to the time of the occurrence as possible. These events can either be included in a separate message bundle going to the DQM onboard computer, or they can be entered on the "State" tab in the DQM Pipeline Software.

### 3.3 DREDGE MONITORING DATA

#### 3.3.1 General

Onboard sensors continuously collect dredging data in support of the dredge Contractor's operations. Portions of this Contractor-collected information, as described in this specification, and calculations based on them shall be stored and transmitted to the DQM database on a near real-time basis. Additionally, information regarding the state of the dredge shall be digitally logged and transmitted.

#### 3.3.2 Data Measurement Frequency

The frequency of data transmission is dependent on the type of message being sent. Work Event messages contain data that are instantaneously collected or calculated from sensors and are logged as a series of events. State event messages are activated by a change in the dredge state.

##### 3.3.2.1 Work Event Messages

Data shall be logged as a series of events. Each event shall consist of a dataset containing dredge information (as defined in paragraph entitled REQUIREMENTS FOR REPORTED DATA). Each set of measurements (for example, time and position) shall be considered an event, and there shall be a 6-12 second interval between work events. This interval shall remain consistent across event types for the dredge plant.

A standard data string shall be recorded within one second of an event trigger with the time stamp and all parameters reflecting when the event happened.

##### 3.3.2.2 State Event Messages

A set of descriptive information (event name, time, description, comment) shall be considered a state event. These events shall be recorded within 24 hours of a change in state with the time stamp reflecting when the event happened.

#### 3.3.3 Parameter Transmission to the Web Service

The data shall be formatted as JSON (JavaScript Object Notation, as defined at <http://www.json.org>) strings of arbitrary length. These JSON strings represent a hierarchical data structure consisting of a message bundle which may contain 0-3 automatic data messages and any number of manual data messages.

A tag/parameter is reported only when it contains a value. No "Null" value strings shall be included in a message bundle.

\*\*\*\*\*

Message bundle

\*\*\*\*\*

```
{
  "DQM_Data": {
    "messages": {
      {
        "work_event": {
          "msg_time": <24-hour UTC time YYYY-MM-DD HH:MM:SS>,

```

```

        "vert_correction":    <floating point 100th decimal place>,
        "ch_latitude":       <decimal to 6 decimal places>,
        "ch_longitude":      <decimal to 6 decimal places>,
        "ch_depth":          <floating point 100th decimal place>,
        "ch_heading":        <integer value 000-359>,
        "slurry_velocity":   <floating point 100th decimal place>,
        "slurry_density":    <floating point 100th decimal place>,
        "pump_rpm":          <integer>,
        "vacuum":             <floating point 100th decimal place>,
        "outlet_psi":        <floating point 100th decimal place>,
        "comment":           <string>},
    },
    {
        "contract_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "contract_number": <string>,
            "event_type":    <string - "start" or "end">,
            "comment":       <string>
        }
    },
    {
        "station_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "station_name":  <string>,
            "comment":       <string>
        }
    },
    {
        "pipe_length_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "length_floating": <integer>,
            "length_submerged": <integer>,
            "length_land":    <integer>,
            "comment":       <string>
        }
    },
    {
        "booster_pump_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DDHH:MM:SS>,
            "booster_total": <integer>,
            "comment":       <string>
        }
    },
    {
        "advance_Event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "advance_daily": <integer>,
            "comment":       <string>
        }
    },
    {
        "outfall_position": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "outfall_location": <string-"upland", "beach", "scow", "open
water">
            "outfall_latitude": <decimal to 6 decimal places>,
            "outfall_longitude": <decimal to 6 decimal places>,
            "outfall_heading":  <integer value 000-359>,

```

```

        "outfall_elevation": <floating point 10th decimal place>,
        "comment":           <string>
    }
},
{
    "non_eff_event": {
        "msg_start_time":    <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
        "msg_end_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
        "function_code":     <string - 1 to 4 characters>,
        "comment":           <string>
    }
}
}
}

```

#### 3.3.4 Contractor Data Backup

The Contractor shall maintain an archive of all data sent to the DQM computer during the dredging contract. The COR may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. The data shall be provided in the same JSON format as would have been transmitted to the DQM computer. There shall be no line breaks between the parameters, and each record string shall be on separate line. The naming convention for the files shall be <dredgename>\_<StartYYYYMMddhhmmss>\_<EndYYYYMMddhhmmss>.txt.

Data submission shall be via a storage medium acceptable to the COR.

At the end of the dredging contract, the Contractor shall call the DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. The Contractor shall record the following information in a separate section at the end of the dredge's onboard copy of the DPIP:

- Person who called the DQM Support Center
- Date of the call
- DQM representative who gave permission to discard the data

#### 3.4 PERFORMANCE REQUIREMENTS

The Contractor's National Dredging Quality Management Program's data transmission shall be fully operational at the start of dredging operations. To meet contract requirements for operability, the Contractor's system shall provide an accurate data string return and be compliant with hardware requirements. Data string return is defined as the number of quality records within an event or state tag sent by the contractor's system to the DQM database. Quality data strings are considered to be those providing accurate values for all parameters reported when operating according to the specification. Repairs necessary to restore data return compliance shall be made within two business days, or a plan and timeline for repair shall be submitted to the DQM Data Acquisition and Analysis Team if the repair will take more than two business days. Failure by the Contractor to report quality data within the specified time window for dredge measurements as stated in the specifications (see Paragraphs entitled "Internet Access", "Data Measurement Frequency" and "Parameter Transmission to the Web Service") will result in withholding of up to 10% of the contract progress payment per clause 52.232-5.

3.5 LIST OF ITEMS PROVIDED BY THE CONTRACTOR

- DPIP            <https://dqm.usace.army.mil>
  - DQM System    Paragraph entitled NATIONAL DREDGING QUALITY MANAGEMENT  
                 PROGRAM SYSTEM REQUIREMENTS
  - Dredge Data   Paragraph entitled DREDGE MONITORING DATA
- End of Section --